



CH350

Bioprocess Engineering



CH350 BIOPROCESS ENGINEERING

ABET eCOURSEBOOK

AY2021-2026

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UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION I

COURSE ASSESSMENTS



TAB A AY2026 COURSE ASSESSMENT

Department of Chemical and Biological Science and Engineering
United States Military Academy
West Point, New York 10996

MADN-CBE

13 January 2026

MEMORANDUM FOR Chemical Engineering Program Director, Department of Chemical and Biological Science and Engineering, United States Military Academy, West Point, New York 10996

SUBJECT: CH350 Chemical Reaction Engineering Course Assessment (AY26-1)

1. The course assessment meeting for CH350 was conducted 17 December 2025. Attendees: Dr. Simuck F. Yuk, Dr. Enoch Nagelli, Dr. Andrew Biaglow, COL Corey James, and LTC Samuel V. Cowart. The topics covered included: 1) revision of major graded events; 2) re-emphasize general chemistry concepts related to bioprocess systems. Summary of recommended changes for AY26 are included in Section III.
2. The CH350 course assessment in accordance with the course director handbook from AY26 is attached.
3. POC is the undersigned.

CF: Dr. Biaglow, Dr. Nagelli, LTC James,
LTC Cowart

LTC Stephen Hummel
Assistant Professor
CH350 Course Director

SECTION I: COURSE DESCRIPTION

1. Current Red Book Description.

CH350 (Version: 2026 1) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	DEPARTMENT	CREDIT HOURS
CH350	BIOPROCESS ENGINEERING	2026	1	Chemical and Biological Science and Engineering	3.0 (BS=1.0, ET=2.0, MA=0.0)

SCOPE

This course is designed to develop an understanding of the field of biochemical engineering. The course covers the application of engineering principles to the study of bioprocesses and biochemical reactor design. Topics include enzymes and enzyme kinetics, cell growth and cell growth kinetics, suspension and immobilization of cultures, bioreactor design, scale-up, and control, and recovery and purification technology. Mathematical modeling is an integral part of the course.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

None

CH350 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE-REQUISITE	CH102	2019	2	1	Y
	MA205	2023	2	1	Y
	PH202	2023	1	1	Y

2. Enrollment Numbers.

- AY 2022-1: 10
- AY 2023-1: 10
- AY 2024-1: 11
- AY 2025-1: 13
- AY 2026-1: 9

3. Current Course Objectives.

This course provides a basis for understanding the application of engineering principles to problems in bioprocessing and bioprocess design. Specifically, the objectives of the course are:

(1) to introduce the biochemical and bioprocess engineering and how it relates to other disciplines of engineering and science.

(2) the develop the ability to apply mathematics, science, and engineering to solve problems of biological and physiological significance.

(3) to develop an understanding of and ability to apply engineering principles to problems involving microbial and enzymatic processes.

(4) to understand current topics within the field of bioprocess engineering.

4. Survey Questions. Below are questions to the web-based survey of questions asked to CH350 cadets at the end of the AY26-1 semester:

Academy

Q1. My fellow students contributed to my learning in this course.

Q2. My motivation to learn and to continue learning has increased because of this course.

Q3. My personal schedule allows me enough time to reflect on the material I have learned in class.

Q4. My personal schedule allows me enough time to adequately prepare for my optimum academic performance.

Academy - I

Q5. This instructor encouraged students to be responsible for their own learning.

Q6. This instructor used effective techniques for learning, both in class and for out-of-class assignments.

Q7. My instructor cared about my learning in this course.

Q8. My instructor demonstrated respect for cadets as individuals.

5. Course GPA. The GPA for the course from the last 7 years follows below.

AY22-1 (Fall 2021)	3.60
AY23-1 (Fall 2022)	3.47
AY24-1 (Fall 2023)	3.58
AY25-1 (Fall 2024)	3.79
AY26-1 (Fall 2025)	3.11

6. TEE Grade.

AY26-1 (Fall 2025) 81%

7.a. Current Textbook.

Shuler, Michael L., Kargi, Fikret, and DeLisa, Matthew P., *Bioprocess Engineering: Basic Concepts*, 3rd Edition, Prentice Hall, 2017, ISBN-13: 978-0137062706, ISBN-10: 0137062702.

K. D. Wittrup, B. Tidor, B. J. Hackel, and C. A. Sarkar., *Quantitative Fundamentals of Molecular and Cellular Bioengineering*, The MIT Press, 2019, ISBN: 978-0-262-04265-9.

7.b. Course Content.

Block I: Biomolecular Considerations
Block II: Cellular Considerations
Block III: Downstream Considerations

7.c. Lesson Sequence

LESSON	TOPICS
Block 1. Biomolecular Considerations	
1	Introduction to bioprocess engineering
2	Introduction to cells and biomolecules - Part 1
3	Introduction to cells and biomolecules - Part 2
4	Enzyme kinetics and immobilized enzyme systems - Part 1
5	Enzyme kinetics and immobilized enzyme systems - Part 2
6	Enzyme kinetics and immobilized enzyme systems - Part 3
7	Non-covalent binding interactions; Binding equilibria and kinetics
8	Central dogma of molecular biology
9	Genetic / Metabolic regulation - Part 1
10	Genetic / Metabolic regulation - Part 2
11	Recombinant DNA technology / Heterologous protein expression - Part 1
12	Recombinant DNA technology / Heterologous protein expression - Part 2
13	WPR I
Block 2. Cellular Considerations	
14	General concepts of cell growth - Part 1
15	General concepts of cell growth - Part 2
16	General concepts of cell growth - Part 3
17	Operational considerations for bioreactors: suspension and immobilized cultures

18	Bioreactor selection, scale-up and process control - Part 1
19	Bioreactor selection, scale-up and process control - Part 2
20	Block II review
21	WPR II
Block 3. Downstream Considerations	
22	Product recovery and separation - Part 1
23	Product recovery and separation - Part 2
24	Bioprocess considerations using animal cells
25	Bioprocess considerations using plant cells
26	Medical applications of bioprocess engineering
27	Traditional industrial bioprocesses
28	Guest lecture
29	Course review
30	Capstone presentation

7.d. Summary of Graded Requirements

Requirement	Number	Points	Total	%
Problem Sets	6	50	300	20%
Instructor Points	---	100	100	7%
Capstone Design Presentation	1	100	100	7%
Capstone Design Paper	1	200	200	12%
WPR	2	200	400	27%
TEE	1	400	400	27%
TOTAL			1500	100%

7.e. Areas of Special Emphasis.

Course Director's Area of Special Emphasis

My primary area of interest was Bioprocess Engineering. This course emphasized applying chemical engineering principles to biological systems. A significant portion of the material focused on designing various types of bioreactors. I aimed to help cadets establish a strong connection between chemical and bioprocess engineering by encouraging them to apply these principles when solving problems on PSs, WPRs, and Capstone design project.

The second area of focus was the design of reactors utilizing biological components. In-class problem-solving activities centered on designing reactor systems that incorporate material balances for cells, substrates, and product concentrations. The course introduced various bioreactor designs, highlighting the differences in reactor conditions for batch versus continuous cultivation.

My third area of emphasis was on effective communication skills. As part of the capstone project, cadets were required to individually submit a technical abstract and a final research paper. A portion of the grade for the written report was allocated to evaluating their communication skills.

Cadets were assessed on their technical writing and oral presentation abilities, with detailed feedback provided on different aspects of their capstone projects.

The fourth area of emphasis involved the use of modern computational techniques. Some of the kinetic problems in the course required advanced calculus and the use of an ordinary differential equation (ODE) solver, such as Mathematica, to develop design equations and produce quantitative kinetic results. For example, solving Michaelis-Menten kinetics for multiple species in a chemostat was approached using computational codes, emphasizing the role of numerical solutions in problem-solving.

8. Contribution to Student Outcomes

The following list contains the Chemical Engineering Student Outcomes. The phrases in **bold-faced red font** are the topics that this course is specifically asked to accomplish by the program. These bullets are also indicated with **red dashed boxes** in Section II.8

The Chemical Engineering Program provides the graduate with a thorough grounding and working knowledge of ...

- General, organic, and physical chemistry.
- **Material and energy balances as well as safety and environmental factors.**
- Thermodynamics of physical and chemical equilibria.
- Heat, mass, and momentum transfer.
- **Chemical reaction engineering.**
- Continuous and staged separation operations.
- Process dynamics and control.
- Modern experimental and computing techniques.
- Process design.

9. Resources and Laboratories.

9a. Laboratories. There were no wet labs conducted in this course.

9b. Computer Labs. All class lectures met in BH630. No computer labs were executed in CH350.

9c. Physical Models & Demos. There are no physical models or demos used in the course.

9d. Technician Support. IT support from the CBSE IT office for maintenance of computers in BH447 was timely and addressed any issues that arose. In future considerations for distance learning, a scanner, Epson elmo, and a second monitor are important to have at designated location of teaching.

9e. Supplies. Mathematica 14.2 and Microsoft Office (Word, PowerPoint, Excel, and Explorer).

9f. Additional Facilities. None.

9g. Unfunded Requests. None.

10. Recommendations from AY25-1 and Implementation in AY26-1:

<i>Section I.</i>	<i>Recommendation (AY25-1)</i>	<i>Implementation (AY26-1)</i>
1. Redbook	No change.	Implemented as recommended (IAR)
2. Enrollment	No change.	IAR
3. Course Objectives	No change.	IAR
3.A. Coverage	Ensure in-class activities were developed to reinforce the Cadets' understanding on kinetics and other critical bioengineering knowledges. Deriving the critical design equations on boards led to multiple discussions between Cadets and faculty.	IAR
3.B. Performance	Provide more in-class class problems to further deepen cadets' understanding of how biology / organic chemistry connects to bioengineering concepts.	There were multiple in-class problems incorporated into nearly every lecture to enable cadets to understand the linkages between bio / chem and engineering concepts.
4. Survey Questions	No change.	IAR
5. Course QPA	No change.	IAR
6. TEE Grades	No change.	IAR
7. Course Processes 7.A. Textbook	No change.	IAR
7.B. Course Content	In AY24-1, the number of PSs were increased to 6 and TEE was introduced to better access cadets' understanding of bioengineering Los introduced in the textbook.	The number of problem sets remained at 6.
7.C. Lessons and Labs	No change.	IAR
7.D. Graded Events	No change.	IAR
7.E. Areas of Special Emphasis	No change.	IAR
8. Contribution to Student Outcomes 8.A. Coverage	In AY24-1, the additional emphasis was given on some of bioprocesses in the Block 3 by introducing the PS #6.	This problem set #6 was continued.

8.B. Performance	No change.	IAR
9. Resources and Laboratories 9.A. Laboratories	No change.	IAR
9.B. Computer Labs	No change.	IAR
9.C. Physical Models & Demos	No change.	IAR
9.D. Technician Support	No change.	IAR
9.E. Supplies	No change.	IAR
9.F. Additional Facilities	No change.	IAR
9.G. Unfunded Requests	No change.	IAR

SECTION II - COURSE ASSESSMENT

1. Red Book Description. The Red Book description was followed verbatim.

2. Enrollment. The enrollment in the course is 9 for AY26-1, with the projected enrollment in AY26-1 to be 12. During Term 26-1, no cadets failed the course, and none were at serious risk of failing the course going into the Term End.

3. Course Objectives. The course objectives were covered thoroughly and achieved with proficiency. The course objectives cover the body of knowledge appropriately. The course objectives lend themselves to assessment.

3.b. Coverage

Coverage of course objectives is presented in Figure 1. As in the other courses in this program, coverage is assessed by the course director, who assigns a 1 or a 0 to a graded event or portion of a graded event. A score of 1 on a graded event or a portion of a graded event designates significant coverage, while a 0 designates an insignificant amount or no coverage. Coverage scores are then averaged across all similar graded events in the course (problem sets, exams, etc.). A score close to 1 indicates a relatively large portion of that type of graded event covered that objective. For example, a score of 0.6 on design problems under Objective 1 (mass/energy balances) indicates that 60% or 3 out of 5 of the problem sets contained coverage of that objective. This does not say that the other problem set had zero coverage of that objective, only that it was not counted by the course director for assessment. An objective with multiple bars shows that multiple types of graded events covered that objective. Raw data are included in the spreadsheet attached to the end of this document in Attachment 3. Because of this analysis, we conclude that course objectives were covered.

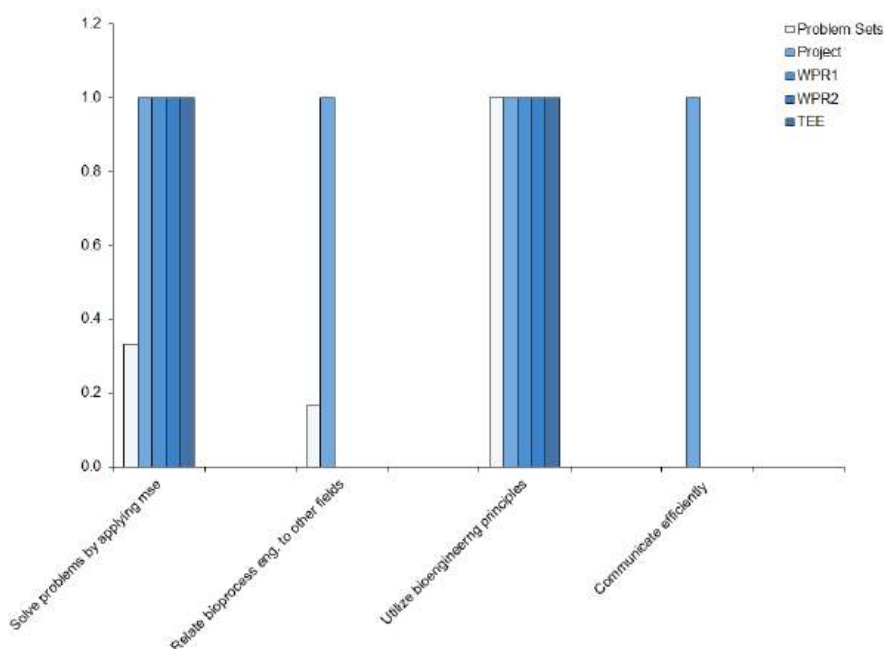


Figure 1. Course Objectives Coverage by Graded Event

3.b. Performance.

Performance on course objectives was assessed using grades taken from those graded events that included significant content in that objective area. The dotted red line shown below indicates the program standard for failure (67%). Some of the cadets still seemed to lack an understanding of how to set up and solve a series of differential or algebraic equations, which hampered their ability to get a numerical solution when analyzing the kinetic trends in biological environments, but overall, the class performed relatively well with all cadets passing the course.

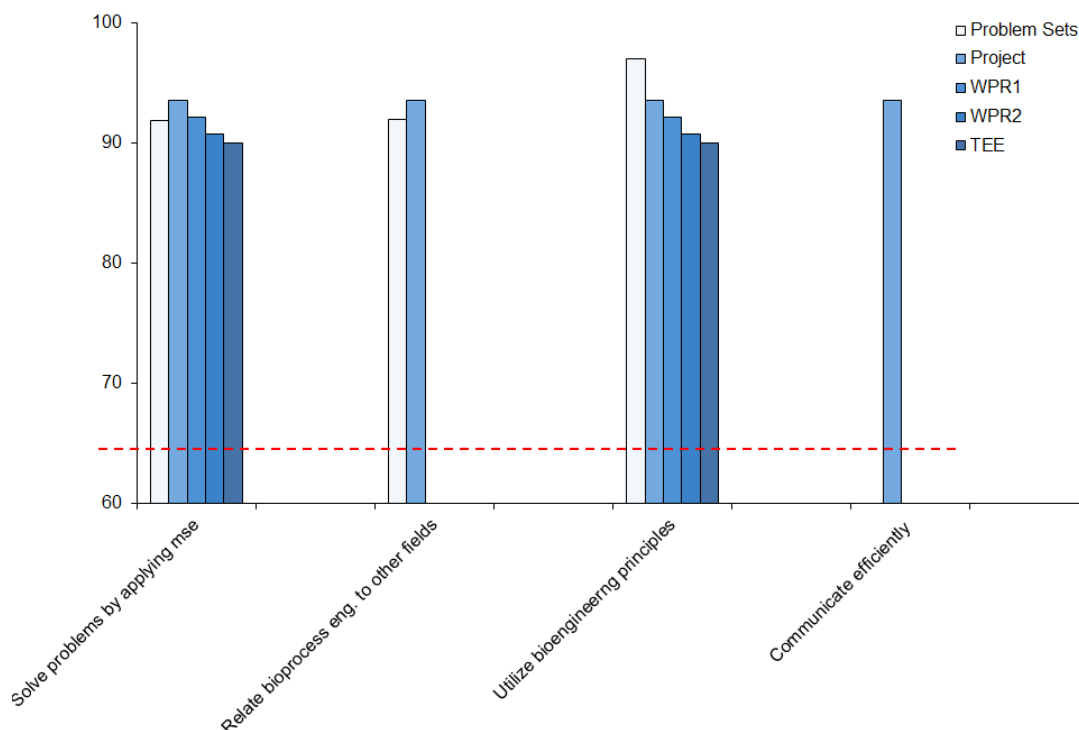


Figure 2. Course Objectives Performance by Graded Event.

4. Survey Questions

4a. Web-based:

The raw data from the web-based survey questions is captured in Section II – Attachment 1. The following graph is a summary of each of the web-based questions. For AY 26-1, the number of respondents was 4/9 cadets, or 46.15%.

4.b. Handwritten Surveys: The student outcomes survey is another method by which we assess student outcomes (SOs). No handwritten survey was conducted in AY26-1 for CH350.

5. Course GPA. The AY26-1 course GPA was 3.11 compared to AY24-1 course GPA or 3.58. AY25-1 course GPA was 3.79. One possible reason for the lower GPA is addition of more in-

depth biological and organic chemistry material in block 1 which the CEN cadets struggled with compared to the chemistry and biology major cadets.

6. TEE Grade. The class average in TEE in AY26-1 was 81%.

7. Course Processes.

7.a. Textbook. The DeLisa Bioprocess Engineering textbook is relatively suitable for this course. DeLisa's book is up to date in terms of math and engineering contents. My assessment is that most of the material is presented in a method that is not too difficult for the cadets to follow or understand. One of the noticeable disadvantages of the text is that DeLisa introduces some of core concepts without giving enough background information for cadets to follow. Also, the unique way to expressing a variety of units throughout the textbook often confuses cadets.

Also, the textbook doesn't offer any example question requiring coding. Additional recourses may be needed if the instructors want to show how the core design equations learned in CH350 can be efficiently solved using the computer programs. Mathematica and ChemCad will be good problem candidates for demonstrating coding examples in the class.

7.b. Course Content. The course content and lesson blocks were appropriate for the course. Background reading and homework problems were assigned prior to each class. After each discussion, the instructor would plan to guide the students through an in-class problem or have the students work on an in-class problem.

One major concern brought forward by the cadets is the text. The Bioprocess Engineering textbook by Shuler et al has some systemic errors. Specifically, variables defined in one chapter change in a later chapter. This creates confusion among the cadets as they read and work on problem sets. This issue discouraged cadets from completing their reading prior to class. Lecture and in-class problems attempted to alleviate confusion among the cadets and set a unifying framework for the cadets. Most of in-class problems were mainly derived from the examples from the textbook.

All administered WPRs posed both in-depth conceptual questions about biological systems as well as quantitative questions. Cadets generally worked problem sets a day before they were due or, at worst, on the day they were due (by observing the submission site on Canvas). Some of them often relied heavily on AI, their smarter colleagues, or any available external sources to solve the problem sets, which potentially degraded the learning of some cadets. This is the general trend that I observed not only from CH350, but for other STEM courses offered in the department.

There is a difference in computational comfort level between the chemical engineer major cadets and the other cadets. The ChemE cadets are very comfortable using Mathematica whereas the other cadets are hesitant to even attempt problems in Mathematica. More emphasis can be placed on the usage of computer programs in the class. Since the textbook doesn't present any actual coding problems, additional sources might be required to design the coding demo problems.

7.c. Lessons and Labs. The lessons are appropriate for this course. The sequence of lessons is arranged in a way so that cadets can see the course contents moving from molecular scale to reactor scale. No labs were conducted in AY25-1 for CH350.

7.d. Summary of Graded Requirements. The number and type of graded requirements was appropriate for the course. A large bank of problems should be developed so instructors can rotate problems from year to year.

7.e. Areas of Special Emphasis. The areas of special emphasis were appropriate for the course.

8. Contribution to Student Outcomes

8.a. Coverage. Contribution to student objectives was assessed using the same method as described in Section II.3. Figure 3 shows coverage of student curriculum outcomes. The spreadsheet containing the data for this analysis is attached to the end of this report, in the section labeled "Section II - Attachment 2 – Assessment Worksheet." The items marked with **red dashed boxes** in Figures 3 designate student directives from Section I.8.

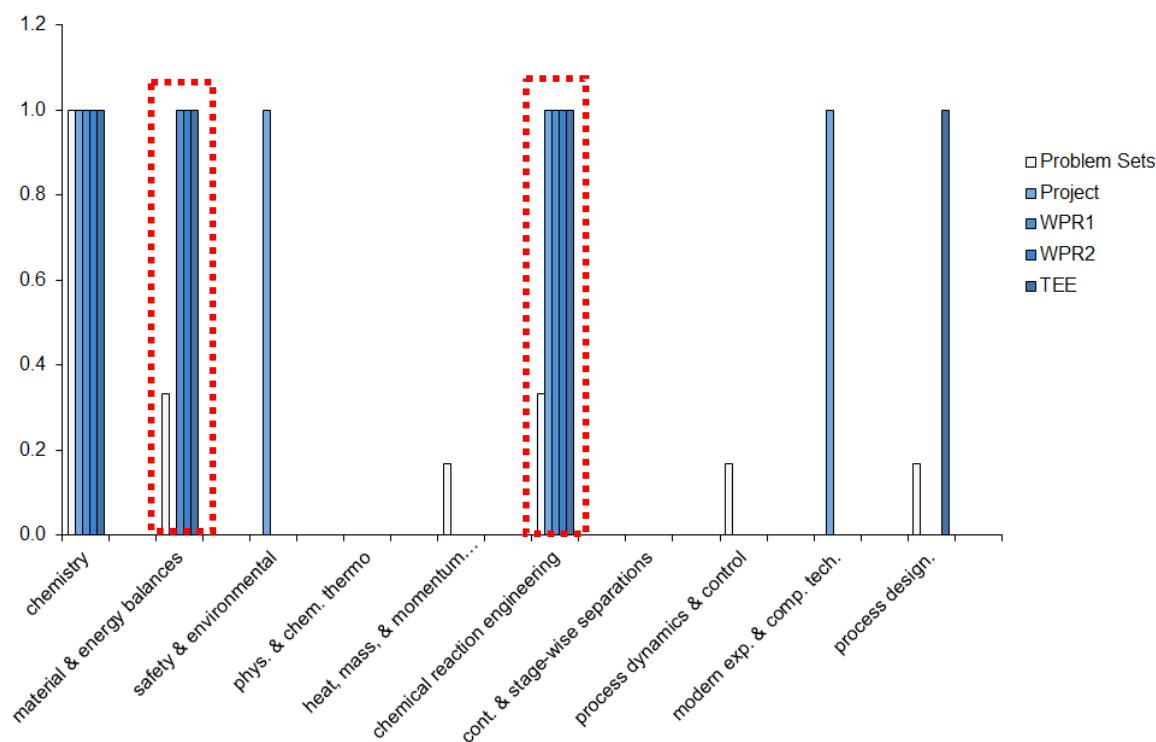


Figure 3. Student Curriculum Outcomes Coverage by Graded Event.

8.b. Performance

Performance on the student outcomes is assessed using the cadet scores from the indicators from the graded events that are outlined above. The results are shown in Figures 4 below. The average for all graded events met or exceeded the 67% standard set forth by USMA for a passing grade. Although none of cadets risked achieving less than 67% in any of area, some of them did struggle with the general chemistry-based problems (i.e., chemical equilibrium and/or kinetics).

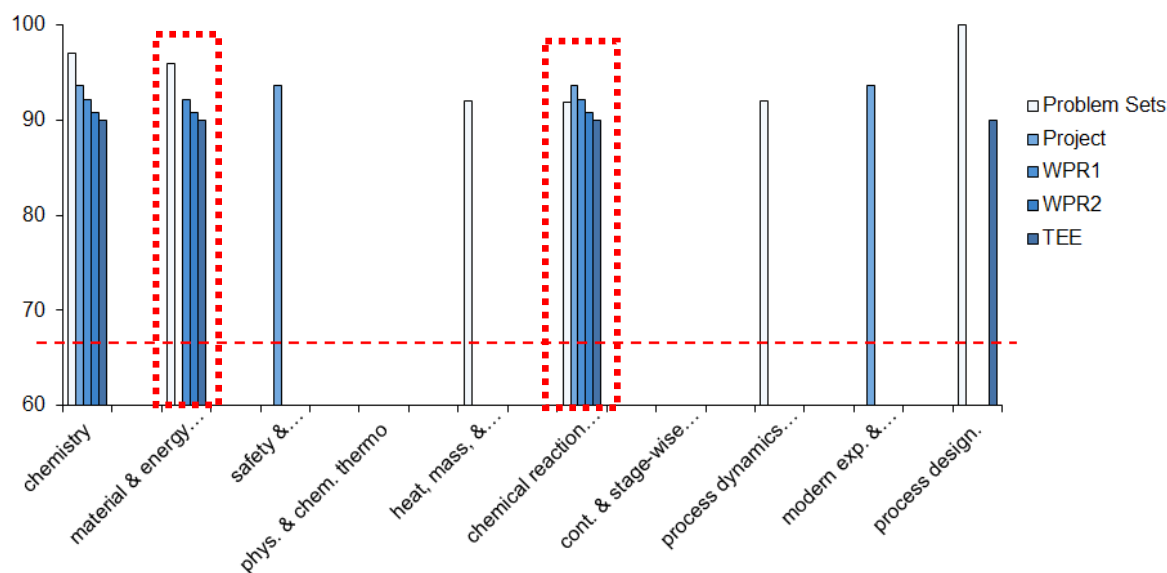


Figure 4. Student Curriculum Outcomes Performance by Graded Event.

9. Resources and Laboratories.

9a. Laboratories. There were no wet labs conducted in this course.

9b. Computer Labs. All class lectures met in BH630. No computer labs were executed in CH350.

9c. Physical Models & Demos. There are no physical models or demos used in the course.

9d. Technician Support. IT support from CLS for maintenance of computers in BH446. The IT support from CLS was adequate.

9e. Supplies. No additional supplies are required for CH350.

9f. Additional Facilities. None were used - adequate.

9g. Unfunded Requests. None - adequate.

SECTION III: RECOMMENDED CHANGES

1. Red Book Description. No change

2. Enrollment Numbers. Based on the projected enrollment of 15, the course should maintain 1 section and use both associated lab periods physically in the classroom.

3. Course Objectives. No change.

3.a. Coverage. To assess and reinforce the cadets' understanding of kinetics, additional in-class and after-class problems should be incorporated to demonstrate how ordinary differential equations (ODEs) can be coupled with reaction mechanisms. One lesson should be dedicated to revisiting the concepts of kinetics and chemical equilibria introduced in general chemistry, as many of the issues cadets struggled with during WPRs were closely related to these topics. Additional coverage should emphasize organic chemistry concepts, as Block 1 focuses on the qualitative aspects of various bioprocesses. Instructor points were placed to reflect cadets' contributions to in-class worksheets

3.b. Performance. No change.

4. Survey Questions. No change.

4.a. Survey Results. No change.

4.b. Survey Freeform Comments. Continue to include these as options.

5. Course QPA. No change.

6. TEE Grade. TEE was constructed based on the core problems from Block 1, 2, and 3, which are directly tied to the learning objectives of course. The problems from Block 1 were focused on calculating and quantitatively describing the enzyme kinetics (with and without) and metabolic pathways. Solving the material balance on different types of bioreactors was mainly associated with the contents from Block 2. The most of problems from Block 3 was designed to ask the cadets to consider various downstream conditions in bioprocess design. The average TEE score was 81%, showing that most of cadets have a good understanding of core concepts introduced in CH350.

7. Course Process

7.a. Current Textbook. Use of the 1) *Bioprocess Engineering: Basic Concepts*, by M. L. Shuler, F. Kargi, and M. DeLisa (published by Prentice-Hall in 2017) and 2) *Quantitative Fundamentals of Molecular and Cellular Bioengineering* by K. D. Wittrup, B. Tidor, B. J. Hackel, and C. A. Sarkar (published by the MIT Press in 2019) is a sustain, but needs to be supplemented to enable a better foundation of biological concepts and cellular processes that drive the cellular engineering concepts.

7.b. Course Content. To assess and reinforce the cadets' understanding on kinetics, multiple in-class and after-class problems were incorporated to show how the ordinary differential equations (ODEs) can be coupled with the reaction mechanisms. Also, in accordance with the developing the cadet's ability for technical writing, it will be beneficial to include the assignment for writing a technical abstract and developing figures/tables since these are the essential parts of constructing a scientific paper. Continued emphasis on IPRs to provide feedback for cadets on their assignments for development is always important for any type of project-based assignments. The capstone experiences still need to be fully converted into the project-based scenarios, so the cadets can spend additional time/efforts working on the engineering processes.

7.c. Lessons and Labs. Continue to have cadets work more in-class problems. Working a problem in every lesson is ideal and I recommend continuing to work through the full solution to the instructor problem to demonstrate to the cadets both setting the problem as well as the analytical solution with Mathematica. The Chemical Engineer cadets have a strong understanding of fundamental reactor mass balance concepts however the cadets from the other major need more assistance. All cadets have a good foundation in data analysis. At the same time, cadets' heavy dependency on coding without attempting to access the problems contents need to be alleviated by introducing non-computer based graded events. As mentioned in Section 3.a., revisiting some of major contents from previous course (i.e., general chemistry) may be needed to reinforce the cadets' understanding of kinetic and chemical equilibrium in biological systems.

7.d. Summary of Graded Requirements. Problem sets need to be updated to create problems that merge both biological concepts and computational reactor problems. These are discrete at this point. Some less-motivated cadets will do little more than copying the old solutions. As always, rewriting/adjusting the major portions of WPR/TEE is strongly recommended for each year.

7.e. Areas of Special Emphasis. No change.

8. Contribution to Student Outcomes

8.a. Coverage. No change.

8.b. Performance. No change.

9. Resources and Laboratories.

9a. Laboratories. No change.

9b. Computer Labs. No change.

9c. Physical Models & Demos. No change.

9d. Technician Support. No change.

9e. Supplies. No change

9f. Additional Facilities. No change.

9g. Unfunded Requests. No change.

SECTION IV: GRADED EVENTS AND SUPPLEMENTAL MATERIAL

- 1. Approved Solutions to all WPRS, Quizzes, and Labs.** Digital records are in the CH350 SharePoint graded events directory.
- 2. All supplemental material provided to cadets not included in their book issue purchase.** Digital records are in the CH350 directory.
- 3. A listing of any other resource that provided significant instructional material for cadet use. (Web sites, shared programs, etc.)**

None for CH350.

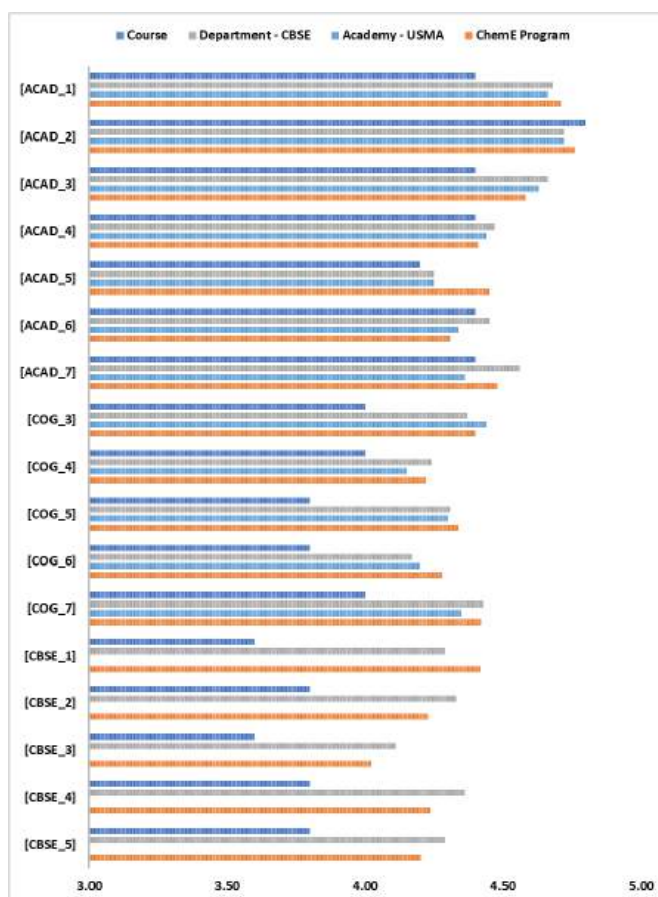
4. Other historical information

4.a. Course Director's Notes

See course director's notebook.

SECTION II: ATTACHMENT 1 – WEB-BASED SURVEY RAW DATA

Survey Questions
[ACAD_01]: My instructor cared about my learning in this course.
[ACAD_02]: My instructor demonstrated respect for cadets as individuals.
[ACAD_03]: This instructor encouraged students to be responsible for their own learning.
[ACAD_04]: The instructor provided effective feedback.
[ACAD_05]: My fellow students contributed to my learning in this course.
[ACAD_06]: The course objectives were clearly explained.
[ACAD_07]: I believe the grading was fair.
[COG_01]: This instructor used effective techniques for learning, both in class and for out of class assignments.
[COG_02]: This instructor stimulated my thinking
[COG_03]: This instructor encouraged students to think creatively and take intellectual risks.
[COG_04]: My motivation to learn and to continue learning has increased because of this course.
[COG_05]: In this course, my critical thinking ability increased.
[COG_06]: In this course, my ability to think creatively and take intellectual risks increased.
[COG_07]: The course materials were relevant and had a positive contribution to my learning.
[CLS_1]: Consider the departments responsible for the courses which you have taken this semester. Relative to other departments, my experience with the Department of Chemistry and Life Sciences was very satisfying.
[CLS_2]: The laboratory program was useful in supporting, reinforcing, or augmenting my understanding of course content.
[CLS_3]: The course text and references were very helpful in my gaining understanding of the course material.
[CLS_4]: The Course Goals and Lesson Objectives for this course gave me a clear understanding of what was expected of me in order to do well in this course.
[CLS_5]: The WPRs, block exams, and quizzes in this course were a reasonable evaluation of my understanding of the course material.



Watermark Code	CH350	Department - CBSE	Academy - USMA	ChemE Program
[ACAD_1]	4.40	4.68	4.66	4.71

[ACAD_2]	4.80	4.72	4.72	4.76
[ACAD_3]	4.40	4.66	4.63	4.58
[ACAD_4]	4.40	4.47	4.44	4.41
[ACAD_5]	4.20	4.25	4.25	4.45
[ACAD_6]	4.40	4.45	4.34	4.31
[ACAD_7]	4.40	4.56	4.36	4.48
[COG_1]	3.80	4.41	4.37	4.29
[COG_2]	4.00	4.47	4.46	4.45
[COG_3]	4.00	4.37	4.44	4.40
[COG_4]	4.00	4.24	4.15	4.22
[COG_5]	3.80	4.31	4.30	4.34
[COG_6]	3.80	4.17	4.2	4.28
[COG_7]	4.00	4.43	4.35	4.42
[CBSE_1]	3.60	4.29	N/A	4.42
[CBSE_2]	3.80	4.33	N/A	4.23
[CBSE_3]	3.60	4.11	N/A	4.02
[CBSE_4]	3.80	4.36	N/A	4.23
[CBSE_5]	3.80	4.29	N/A	4.20

SECTION II – ATTACHMENT 2

HANDWRITTEN SURVEY YEARLY COMPARISON

SECTION II – ATTACHMENT 3
STUDENT OUTCOMES WORKSHEETS

Student Outcomes (Outcomes are abbreviated)	← Categories Carried Forward from Grades Spreadsheet →															
On completion of the program, our students have demonstrated that they can:																
id, form & solve by appl mse	1	1	1	1	1	1	1.0	0	0	0.0	1	1.0	1	1.0	1	1.0
design to meet needs	0	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0	0	0.0
communicate	0	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0	0	0.0
professional ethical impact	0	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0	0	0.0
function on a team	0	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0	0	0.0
experiments	0	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0	0	0.0
new knowledge	0	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0	0	0.0
The program provides a thorough grounding and working knowledge of:																
chemistry	1	1	1	1	1	1	1.0	1	1	1.0	1	1.0	1	1.0	1	1.0
material & energy balances	0	1	0	1	0	0	0.3	0	0	0.0	1	1.0	1	1.0	1	1.0
safety & environmental	0	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0	0	0.0
phys. & chem. thermo	0	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0	0	0.0
heat, mass, & momentum transfer.	0	0	0	0	1	0	0.2	0	0	0.0	0	0.0	0	0.0	0	0.0
chemical reaction engineering	0	0	0	1	1	0	0.3	1	1	1.0	1	1.0	1	1.0	1	1.0
cont. & stage-wise separations	0	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0	0	0.0
process dynamics & control	0	0	0	0	1	0	0.2	0	0	0.0	0	0.0	0	0.0	0	0.0
modern exp. & comp. tech.	0	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0	0	0.0
process design.	0	0	0	0	0	1	0.2	0	0	0.0	0	0.0	0	0.0	1	1.0

Course Objectives																				
Solve problems by applying mse	0	0	0	1	1	0	0.3		1	1	1.0		1	1.0		1	1.0		1	1.0
Relate bioprocess eng. to other fields	0	0	0	0	1	0	0.2		1	1	1.0		0	0.0		0	0.0		0	0.0
Utilize bioengineerng principles	1	1	1	1	1	1	1.0		1	1	1.0		1	1.0		1	1.0		1	1.0
Communicate efficiently	0	0	0	0	0	0	0.0		1	1	1.0		0	0.0		0	0.0		0	0.0
Raw Grade Data	← Data Carried Forward from Grades Spreadsheet →																			
Average out of 100	98	##	##	92	92	##			92	95			92			91			90	
Do Not Change the Format of this Sheet!																				



TAB B AY2025 COURSE ASSESSMENT

Department of Chemistry and Life Science
United States Military Academy
West Point, New York 10996

MADN-CHM-LS

9 January 2025

MEMORANDUM FOR Chemical Engineering Program Director, Department of Chemistry and Life Science, United States Military Academy, West Point, New York 10996

SUBJECT: CH350 Chemical Reaction Engineering Course Assessment (AY25-1)

1. The course assessment meeting for CH350 was conducted 9 January 2025. Attendees: Dr. Simuck F. Yuk, Dr. Enoch Nagelli, Dr. Andrew Biaglow, LTC Corey James, and LTC Samuel V. Cowart. The topics covered included: 1) revision of major graded events; 2) reemphasis of general chemistry concepts related to bioprocess systems. Summary of recommended changes for AY26 are included in Section III.
2. The CH350 course assessment in accordance with the course director handbook from AY25 is attached.
3. POC is the undersigned.

CF: Dr. Biaglow, Dr. Nagelli, LTC James,
LTC Cowart

Dr. Simuck F. Yuk
Assistant Professor, Chemical Engineering
CH350 Course Director

SECTION I: COURSE DESCRIPTION

1. Current Red Book Description.

CH350 (Version: 2041 1) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	CREDIT HOURS
CH350	BIOPROCESS ENGINEERING	AY24	1	3.0 (BS=1.0, ET=2.0, MA=0.0)

SCOPE

This course is designed to develop an understanding of the field of biochemical engineering. The course covers application of engineering principles to the study of bioprocesses and biochemical reactor design. Topics include enzymes and enzyme kinetics, cell growth and cell growth kinetics, suspension and immobilization of cultures, bioreactor design, scale-up, and control, and recovery and purification technology. Mathematical modeling is an integral part of the course.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

None

CH350 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE-REQUISITE	CH102	2019	2	1	Y
	MA205	2023	2	1	Y
	PH202	2023	1	1	Y

2. Enrollment Numbers.

- AY 2022-1: 10
- AY 2023-1: 10
- AY 2024-1: 11
- AY 2025-1: 13

3. Current Course Objectives.

This course provides a basis for understanding the application of engineering principles to problems in bioprocessing and bioprocess design. Specifically, the objectives of the course are:

(1) to introduce the biochemical and bioprocess engineering and how it relates to other disciplines of engineering and science.

(2) the develop the ability to apply mathematics, science, and engineering to solve problems of biological and physiological significance.

(3) to develop an understanding of and ability to apply engineering principles to problems involving microbial and enzymatic processes.

(4) to understand current topics within the field of bioprocess engineering.

4. Survey Questions. Below are questions to the web-based survey of questions asked to CH350 cadets at the end of the AY23-1 semester:

Academy

Q1. My fellow students contributed to my learning in this course.

Q2. My motivation to learn and to continue learning has increased because of this course.

Q3. My personal schedule allows me enough time to reflect on the material I have learned in class.

Q4. My personal schedule allows me enough time to adequately prepare for my optimum academic performance.

Academy - I

Q5. This instructor encouraged students to be responsible for their own learning.

Q6. This instructor used effective techniques for learning, both in class and for out-of-class assignments.

Q7. My instructor cared about my learning in this course.

Q8. My instructor demonstrated respect for cadets as individuals.

5. Course GPA. The GPA for the course from the last 7 years follows below.

AY22-1 (Fall 2021)	3.60
AY23-1 (Fall 2022)	3.47
AY24-1 (Fall 2023)	3.58
AY25-1 (Fall 2024)	3.79

6. TEE Grade.

AY24-1 (Fall 2023)	3.57
AY25-1 (Fall 2024)	3.60

7.a. Current Textbook.

Shuler, Michael L., Kargi, Fikret, and DeLisa, Matthew P., *Bioprocess Engineering: Basic Concepts*, 3rd Edition, Prentice Hall, 2017, ISBN-13: 978-0137062706, ISBN-10: 0137062702.

K. D. Wittrup, B. Tidor, B. J. Hackel, and C. A. Sarkar., *Quantitative Fundamentals of Molecular and Cellular Bioengineering*, The MIT Press, 2019, ISBN: 978-0-262-04265-9.

7.b. Course Content.

Block I: Biomolecular Considerations
 Block II: Cellular Considerations
 Block III: Downstream Considerations

7.c. Lesson Sequence

LESSON	TOPICS
Block 1. Biomolecular Considerations	
1	Introduction to bioprocess engineering
2	Introduction to cells and biomolecules - Part 1
3	Introduction to cells and biomolecules - Part 2
4	Enzyme kinetics and immobilized enzyme systems - Part 1
5	Enzyme kinetics and immobilized enzyme systems - Part 2
6	Enzyme kinetics and immobilized enzyme systems - Part 3
7	Non-covalent binding interactions; Binding equilibria and kinetics
8	Central dogma of molecular biology
9	Genetic / Metabolic regulation - Part 1
10	Genetic / Metabolic regulation - Part 2
11	Recombinant DNA technology / Heterologous protein expression - Part 1
12	Recombinant DNA technology / Heterologous protein expression - Part 2
13	WPR I
Block 2. Cellular Considerations	
14	General concepts of cell growth - Part 1
15	General concepts of cell growth - Part 2
16	General concepts of cell growth - Part 3
17	Operational considerations for bioreactors: suspension and immobilized cultures
18	Bioreactor selection, scale-up and process control - Part 1
19	Bioreactor selection, scale-up and process control - Part 2
20	Block II review
21	WPR II
Block 3. Downstream Considerations	
22	Product recovery and separation - Part 1

23	Product recovery and separation - Part 2
24	Bioprocess considerations using animal cells
25	Bioprocess considerations using plant cells
26	Medical applications of bioprocess engineering
27	Traditional industrial bioprocesses
28	Guest lecture
29	Course review
30	Capstone presentation

7.d. Summary of Graded Requirements

Requirement	Number	Points	Total	%
Problem Sets	6	50	300	20%
Instructor Points	---	100	100	7%
Capstone Design Presentation	1	100	100	7%
Capstone Design Paper	1	200	200	12%
WPR	2	200	400	27%
TEE	1	400	400	27%
TOTAL			1500	100%

7.e. Areas of Special Emphasis.

Course Director's Area of Special Emphasis

My primary area of interest was Bioprocess Engineering. This course emphasized applying chemical engineering principles to biological systems. A significant portion of the material focused on designing various types of bioreactors. I aimed to help cadets establish a strong connection between chemical and bioprocess engineering by encouraging them to apply these principles when solving problems on PSs, WPRs, and Capstone design project.

The second area of focus was the design of reactors utilizing biological components. In-class problem-solving activities centered on designing reactor systems that incorporate material balances for cells, substrates, and product concentrations. The course introduced various bioreactor designs, highlighting the differences in reactor conditions for batch versus continuous cultivation.

My third area of emphasis was on effective communication skills. As part of the capstone project, cadets were required to individually submit a technical abstract and a final research paper. A portion of the grade for the written report was allocated to evaluating their communication skills. Cadets were assessed on their technical writing and oral presentation abilities, with detailed feedback provided on different aspects of their capstone projects.

The fourth area of emphasis involved the use of modern computational techniques. Some of the kinetic problems in the course required advanced calculus and the use of an ordinary differential equation (ODE) solver, such as Mathematica, to develop design equations and produce quantitative kinetic results. For example, solving Michaelis-Menten kinetics for multiple species in a chemostat was approached using computational codes, emphasizing the role of numerical solutions in problem-solving.

8. Contribution to Student Outcomes

The following list contains the Chemical Engineering Student Outcomes. The phrases in **bold-faced red font** are the topics that this course is specifically asked to accomplish by the program. These bullets are also indicated with **red dashed boxes** in Section II.8

The Chemical Engineering Program provides the graduate with a thorough grounding and working knowledge of ...

- General, organic, and physical chemistry.
- **Material and energy balances as well as safety and environmental factors.**
- Thermodynamics of physical and chemical equilibria.
- Heat, mass, and momentum transfer.
- **Chemical reaction engineering.**
- Continuous and staged separation operations.
- Process dynamics and control.
- Modern experimental and computing techniques.
- Process design.

9. Resources and Laboratories.

9a. Laboratories. There were no wet labs conducted in this course.

9b. Computer Labs. All class lectures met in BH630. No computer labs were executed in CH350.

9c. Physical Models & Demos. There are no physical models or demos used in the course.

9d. Technician Support. IT support from the CLS IT office for maintenance of computers in BHB42 was timely and addressed any issues that arose. In future considerations for distance learning, a scanner, Epson elmo, and a second monitor are important to have at designated location of teaching.

9e. Supplies. Mathematica 11 and Microsoft Office (Word, PowerPoint, Excel, and Explorer).

9f. Additional Facilities. None.

9g. Unfunded Requests. None.

10. Recommendations from AY24-1 and Implementation in AY5-1:

<i>Section I.</i>	<i>Recommendation (AY24-1)</i>	<i>Implementation (AY25-1)</i>
1. Redbook	No change.	Implemented as recommended (IAR)
2. Enrollment	No change.	IAR

3. Course Objectives	No change.	IAR
3.A. Coverage	To assess and reinforce the Cadets' understanding on kinetics, multiple in-class and after-class problems were incorporated to show how the ordinary differential equations (ODEs) can be coupled with the reaction mechanisms.	Additional in-class activities were developed to reinforce the Cadets' understanding on kinetics and other critical bioengineering knowledges. Deriving the critical design equations on boards led to multiple discussions between Cadets and faculty.
3.B. Performance	Additional focus was given on the organic chemistry concepts given in the Block I.	Additional in-class and after-class problems should be provided to further deepen cadets' understanding of how organic chemistry connects to bioengineering concepts.
4. Survey Questions	No change.	IAR
5. Course QPA	No change.	IAR
6. TEE Grades	No change.	IAR
7. Course Processes		
7.A. Textbook	No change.	IAR
7.B. Course Content	In AY24-1, the number of PSs were increased to 6 and TEE was introduced to better access cadets' understanding of bioengineering Los introduced in the textbook.	IAR
7.C. Lessons and Labs	No change.	IAR
7.D. Graded Events	A new type of problem was introduced in WPRs, PSs, and the TEE to assess cadets' understanding of the course content. Continuously incorporating these new problem types will be essential to discouraging cadets from relying solely on old class materials.	IAR
7.E. Areas of Special Emphasis	No change.	IAR
8. Contribution to Student Outcomes		
8.A. Coverage	In AY24-1, the additional emphasis was given on some of bioprocesses in the Block 3 by introducing the PS #6.	IAR
8.B. Performance	No change.	IAR

9. Resources and Laboratories 9.A. Laboratories	No change.	IAR
9.B. Computer Labs	No change.	IAR
9.C. Physical Models & Demos	No change.	IAR
9.D. Technician Support	No change.	IAR
9.E. Supplies	No change.	IAR
9.F. Additional Facilities	No change.	IAR
9.G. Unfunded Requests	No change.	IAR

SECTION II - COURSE ASSESSMENT

1. Red Book Description. The Red Book description was followed verbatim.

2. Enrollment. The enrollment in the course is 13 for AY25-1, with the projected enrollment in AY25-1 to be 18. During Term 25-1, no cadets failed the course, and none were at serious risk of failing the course going into the Term End.

3. Course Objectives. The course objectives were covered thoroughly and achieved with proficiency. The course objectives cover the body of knowledge appropriately. The course objectives lend themselves to assessment.

3.b. Coverage

Coverage of course objectives is presented in Figure 1. As in the other courses in this program, coverage is assessed by the course director, who assigns a 1 or a 0 to a graded event or portion of a graded event. A score of 1 on a graded event or a portion of a graded event designates significant coverage, while a 0 designates an insignificant amount or no coverage. Coverage scores are then averaged across all similar graded events in the course (problem sets, exams, etc.). A score close to 1 indicates a relatively large portion of that type of graded event covered that objective. For example, a score of 0.6 on design problems under Objective 1 (mass/energy balances) indicates that 60% or 3 out of 5 of the problem sets contained coverage of that objective. This does not say that the other problem set had zero coverage of that objective, only that it was not counted by the course director for assessment. An objective with multiple bars shows that multiple types of graded events covered that objective. Raw data are included in the spreadsheet attached to the end of this document in Attachment 3. Because of this analysis, we conclude that course objectives were covered.

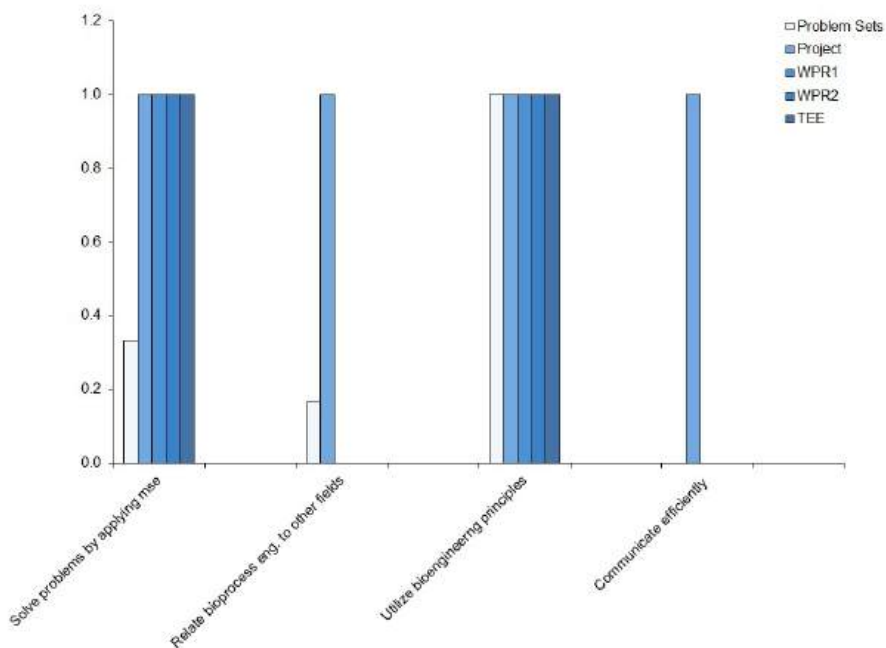


Figure 1. Course Objectives Coverage by Graded Event

3.b. Performance.

Performance on course objectives was assessed using grades taken from those graded events that included significant content in that objective area. The dotted red line shown below indicates the program standard for failure (67%). Some of the cadets still seemed to lack an understanding of how to set up and solve a series of differential or algebraic equations, which hampered their ability to get a numerical solution when analyzing the kinetic trends in biological environments, but overall, the class performed relatively well with all cadets passing the course.

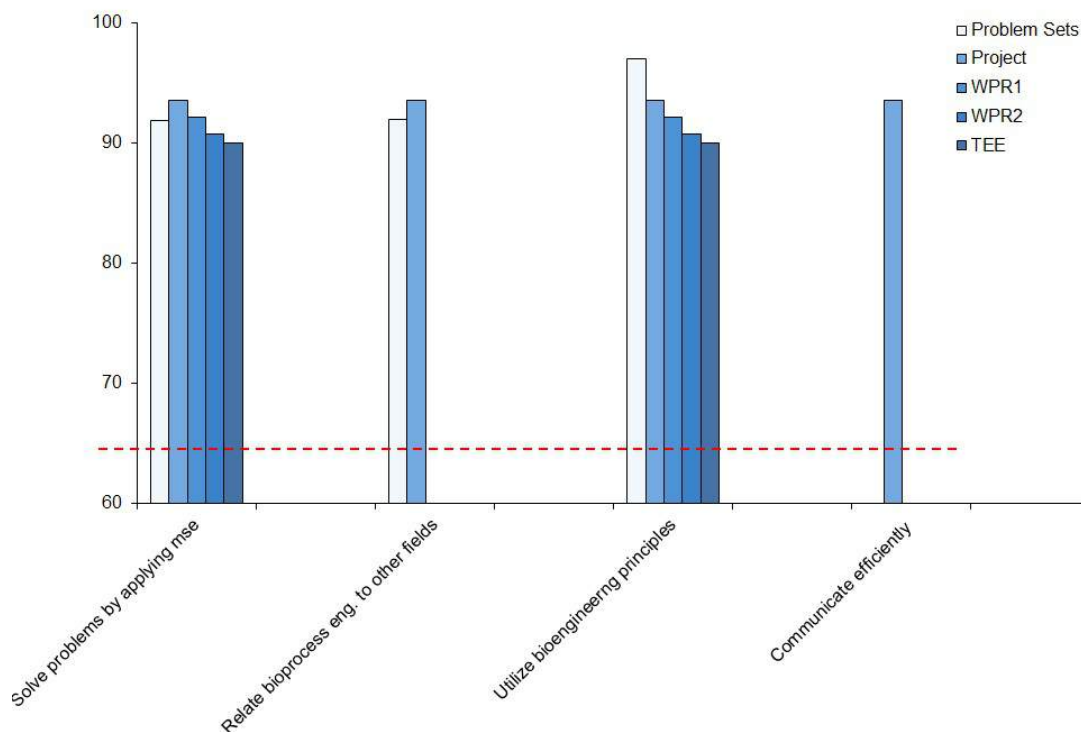


Figure 2. Course Objectives Performance by Graded Event.

4. Survey Questions

4a. Web-based:

The raw data from the web-based survey questions is captured in Section II – Attachment 1. The following graph is a summary of each of the web-based questions. For AY 25-1, the number of respondents was 6/13 cadets, or 46.15%.

4.b. Handwritten Surveys: The student outcomes survey is another method by which we assess student outcomes (SOs). No handwritten survey was conducted in AY25-1 for CH350.

5. Course QPA. The AY24-1 course QPA was 3.58. The AY25-1 course QPA was 3.79, which is higher than the AY24-1. One possible reason for the higher QPA is the additional inclusion of

qualitative content, particularly in Block 1, into PSs and WPR I. All cadets in AY25-1 CH350 are majoring in chemical engineering (CEN).

6. TEE Grade. The class average in TEE in AY25-1 was 90%.

7. Course Processes.

7.a. Textbook. The DeLisa Bioprocess Engineering textbook is relatively suitable for this course. DeLisa's book is up to date in terms of math and engineering contents. My assessment is that most of the material is presented in a method that is not too difficult for the cadets to follow or understand. One of the noticeable disadvantages of the text is that DeLisa introduces some of core concepts without giving enough background information for cadets to follow. Also, the unique way to expressing a variety of units throughout the textbook often confuses cadets.

Also, the textbook doesn't offer any example question requiring coding. Additional recourses may be needed if the instructors want to show how the core design equations learned in CH350 can be efficiently solved using the computer programs. Mathematica and ChemCad will be good problem candidates for demonstrating coding examples in the class.

7.b. Course Content. The course content and lesson blocks were appropriate for the course. Background reading and homework problems were assigned prior to each class. After each discussion, the instructor would plan to guide the students through an in-class problem or have the students work an in-class problem.

One major request received by cadets is to perform the in-class problem demonstration. Showing how to solve some of complex problems (from problem sets) step by step has helped cadets to understand how to efficiently tackle and produce the appropriate solutions. Sharing the partial solution further enhances cadets' problem-solving techniques. Most of in-class problems were mainly derived from the examples from the textbook.

All WPRs administered in CH350 were divided into two different sections. The fundamental questions were placed, so cadets can give in-depth conceptual perspectives on the biological systems. The quantitative questions were implemented, so cadets can have the opportunities of applying the appropriate types of design equations to the various types of biological systems.

Often, cadets completed their assigned readings before class, but lesson assignments and problem sets were designed to force them into the text. Cadets generally worked problem sets a day before they were due or, at worst, on the day they were due (by observing the submission site on Teams). Some of them often relied heavily on AI, their smarter colleagues, or any available external sources to solve the problem sets, which potentially degraded the learning of some cadets. This is the general trend that I observed not only from CH350, but for other STEM courses offered in the department.

More emphasize can be placed on the usage of computer program in the class. Since the textbook doesn't present any actual coding problems, additional sources might be required to design the coding demo problems. Again, Mathematica and ChemCad can be used to construct the coding problems as part of problem set or major graded events.

7.c. Lessons and Labs. The lessons are appropriate for this course. The sequence of lessons is arranged in a way so that cadets can see the course contents moving from molecular scale to reactor scale. No labs were conducted in AY25-1 for CH350.

7.d. Summary of Graded Requirements. The number and type of graded requirements was appropriate for the course. A large bank of problems should be developed so instructors can rotate problems from year to year. Also, I can see that the number of problem sets assigned could be increased to further reinforce the core concepts learned in the class.

7.e. Areas of Special Emphasis. The areas of special emphasis were appropriate for the course.

8. Contribution to Student Outcomes

8.a. Coverage. Contribution to student objectives was assessed using the same method as described in Section II.3. Figure 3 shows coverage of student curriculum outcomes. The spreadsheet containing the data for this analysis is attached to the end of this report, in the section labeled "Section II - Attachment 2 – Assessment Worksheet." The items marked with **red dashed boxes** in Figures 3 designate student directives from Section I.8.

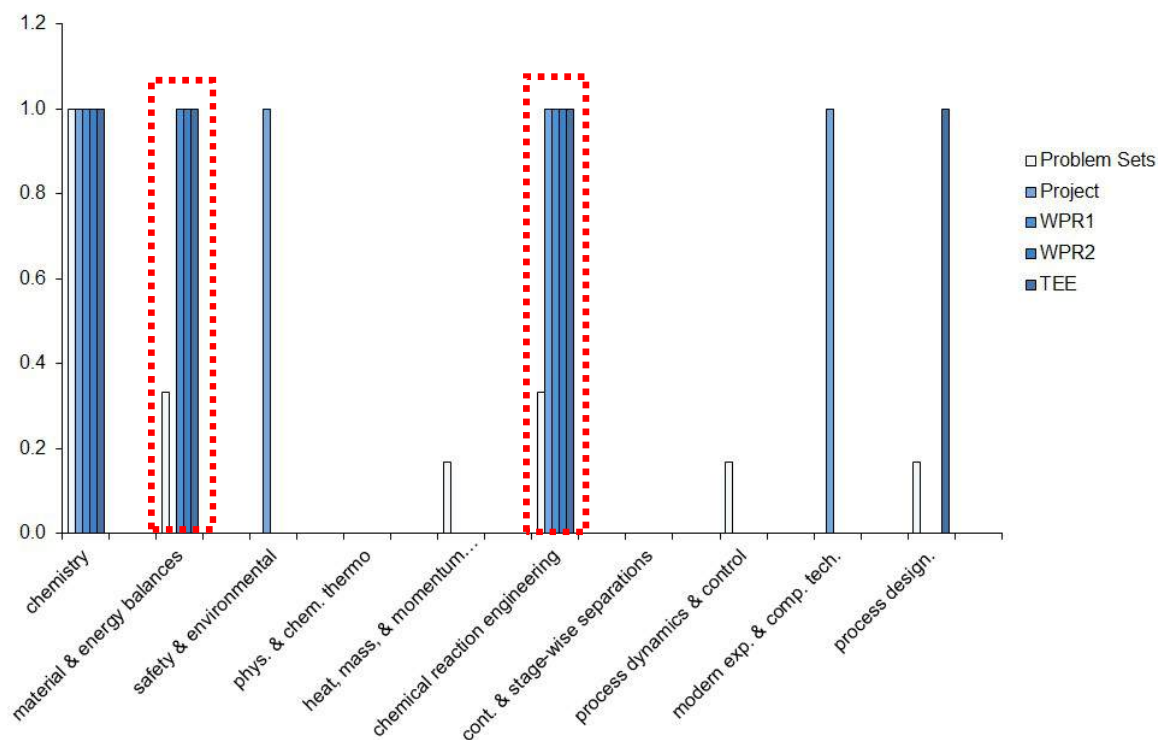


Figure 3. Student Curriculum Outcomes Coverage by Graded Event.

8.b. Performance

Performance on the student outcomes is assessed using the cadet scores from the indicators from the graded events that are outlined above. The results are shown in Figures 4 below. The average for all graded events met or exceeded the 67% standard set forth by USMA for a passing grade. Although none of cadets risked achieving less than 67% in any of area, some of them did struggle with the general chemistry-based problems (i.e., chemical equilibrium and/or kinetics).

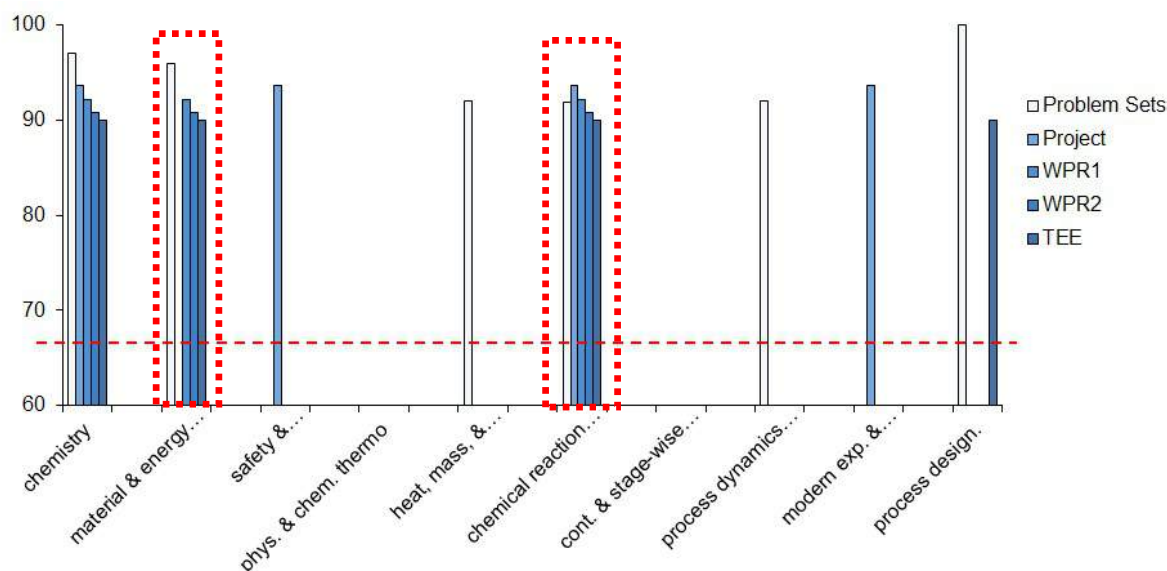


Figure 4. Student Curriculum Outcomes Performance by Graded Event.

9. Resources and Laboratories.

9a. Laboratories. There were no wet labs conducted in this course.

9b. Computer Labs. All class lectures met in BHB42. No computer labs were executed in CH350.

9c. Physical Models & Demos. There are no physical models or demos used in the course.

9d. Technician Support. IT support from CLS for maintenance of computers in BHB42. The IT support from CLS was adequate.

9e. Supplies. The supplies for the computer lab were adequate (no wet labs were conducted in CH350).

9f. Additional Facilities. None were used - adequate.

9g. Unfunded Requests. None - adequate.

SECTION III: RECOMMENDED CHANGES

1. Red Book Description. No change

2. Enrollment Numbers. Based on the projected enrollment of 18, the course should maintain 1 section and use both associated lab periods physically in the classroom.

3. Course Objectives. No change.

3.a. Coverage. To assess and reinforce the cadets' understanding of kinetics, additional in-class and after-class problems should be incorporated to demonstrate how ordinary differential equations (ODEs) can be coupled with reaction mechanisms. One lesson should be dedicated to revisiting the concepts of kinetics and chemical equilibria introduced in general chemistry, as many of the issues cadets struggled with during WPRs were closely related to these topics. Additional coverage should emphasize organic chemistry concepts, as Block 1 focuses on the qualitative aspects of various bioprocesses. Instructor points were placed to reflect cadets' contributions to in-class worksheets

3.b. Performance. No change.

4. Survey Questions. No change.

4.a. Survey Results. No change.

4.b. Survey Freeform Comments. Continue to include these as options.

5. Course QPA. No change.

6. TEE Grade. TEE was constructed based on the core problems from Block 1, 2, and 3, which are directly tied to the learning objectives of course. The problems from Block 1 were focused on calculating and quantitatively describing the enzyme kinetics (with and without) and metabolic pathways. Solving the material balance on different types of bioreactors was mainly associated with the contents from Block 2. The most of problems from Block 3 was designed to ask the cadets to consider various downstream conditions in bioprocess design. The average TEE score was 90%, showing that most of cadets have a good understanding of core concepts introduced in CH350.

7. Course Process

7.a. Current Textbook. Use of the 1) *Bioprocess Engineering: Basic Concepts*, by M. L. Shuler, F. Kargi, and M. DeLisa (published by Prentice-Hall in 2017) and 2) *Quantitative Fundamentals of Molecular and Cellular Bioengineering* by K. D. Wittrup, B. Tidor, B. J. Hackel, and C. A. Sarkar (published by the MIT Press in 2019) is a sustain. The proposed textbooks are comprehensive texts for cellular and bioprocess engineering and are widely used in the engineering community.

7.b. Course Content. To assess and reinforce the cadets' understanding on kinetics, multiple in-class and after-class problems were incorporated to show how the ordinary differential equations (ODEs) can be coupled with the reaction mechanisms. Also, in accordance with the developing the cadet's ability for technical writing, it will be beneficial to include the assignment for writing a technical abstract and developing figures/tables since these are the essential parts of constructing a scientific paper. Continued emphasis on IPRs to provide feedback for cadets on their assignments for development is always important for any type of project-based assignments. The capstone experiences still need to be fully converted into the project-based scenarios, so the cadets can spend additional time/efforts working on the engineering processes.

7.c. Lessons and Labs. Continue to have cadets work more in-class problems. Working a problem in every lesson is ideal and I recommend continuing to work through the full solution to the instructor problem to demonstrate to the cadets both setting the problem as well as the analytical solution with Mathematica. More time needs to be invested on fundamental reactor mass balances and data analysis. At the same time, cadets' heavy dependency on coding without attempting to access the problems contents need to be alleviated by introducing non-computer based graded events. As mentioned in Section 3.a., revisiting some of major contents from previous course (i.e., general chemistry) may be needed to reinforce the cadets' understanding of kinetic and chemical equilibrium in biological systems.

7.d. Summary of Graded Requirements. Problem sets need to be rewritten with new problems. Cadets may have access to previous years' solutions and online resources that may have direct answers to textbook questions (e.g., Chegg.com). Some of less-motivated cadets will do little more than copying the old solutions. As always, rewriting/adjusting the major portions of WPR/TEE is strongly recommended for each year.

7.e. Areas of Special Emphasis. No change.

8. Contribution to Student Outcomes

8.a. Coverage. No change.

8.b. Performance. No change.

9. Resources and Laboratories.

9a. Laboratories. No change.

9b. Computer Labs. No change.

9c. Physical Models & Demos. No change.

9d. Technician Support. No change.

9e. Supplies. No change

9f. Additional Facilities. No change.

9g. Unfunded Requests. No change.

SECTION IV: GRADED EVENTS AND SUPPLEMENTAL MATERIAL

- 1. Approved Solutions to all WPRS, Quizzes, and Labs.** Digital records are in the CH350 SharePoint graded events directory.
- 2. All supplemental material provided to cadets not included in their book issue purchase.** Digital records are in the CH350 directory.
- 3. A listing of any other resource that provided significant instructional material for cadet use. (Web sites, shared programs, etc.)**

None for CH350.

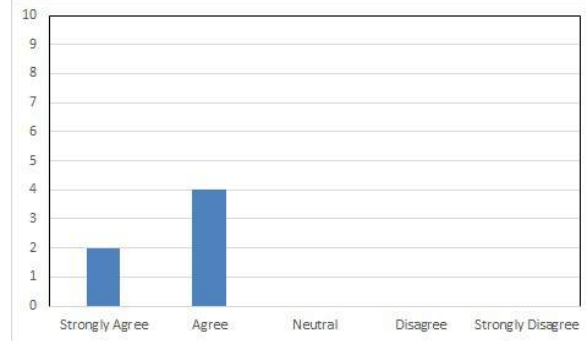
4. Other historical information

4.a. Course Director's Notes

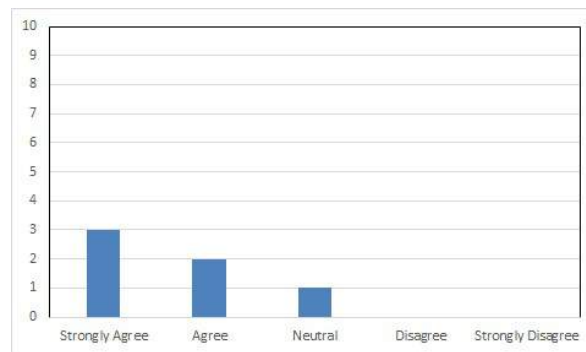
See course director's notebook.

SECTION II: ATTACHMENT 1 – WEB-BASED SURVEY RAW DATA

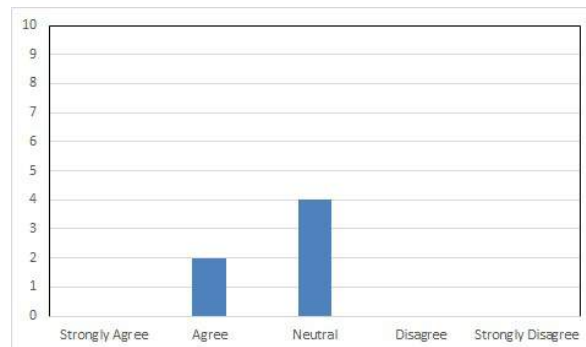
My fellow students contributed to my learning in this course:



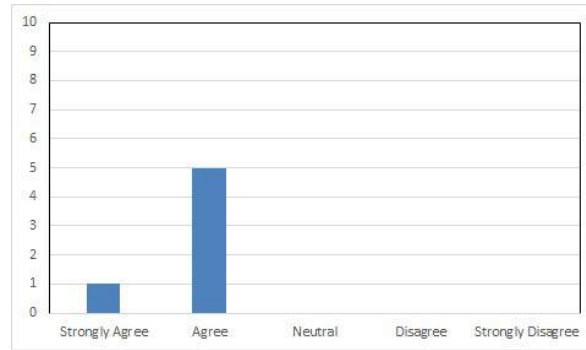
My motivation to learn and to continue learning has increased because of this course:



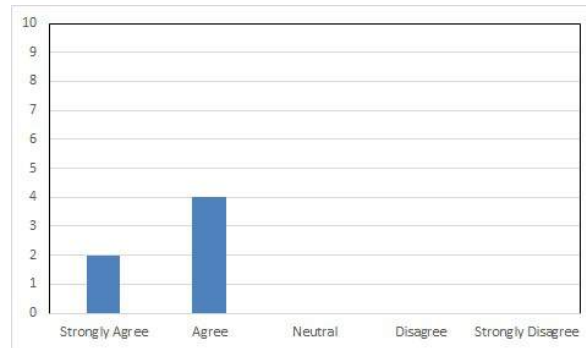
My personal schedule allows me enough time to reflect on the material I have learned in class:



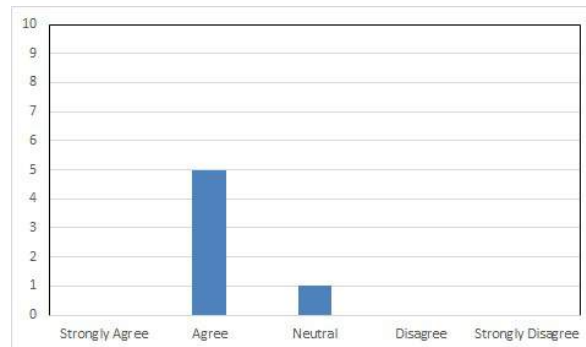
My personal schedule allows me enough time to adequately prepare for my optimum academic performance:



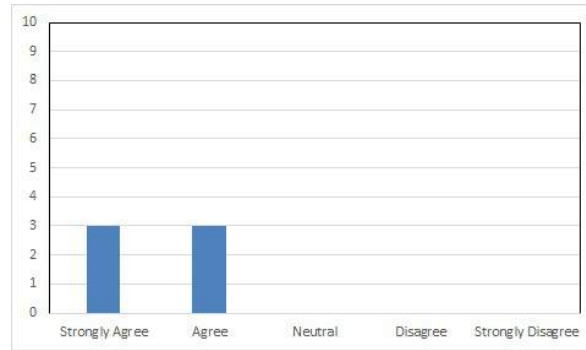
This instructor encouraged students to be responsible for their own learning:



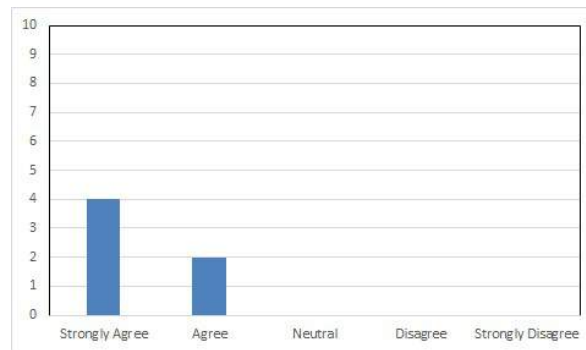
This instructor used effective techniques for learning, both in class and for out of class assignments:



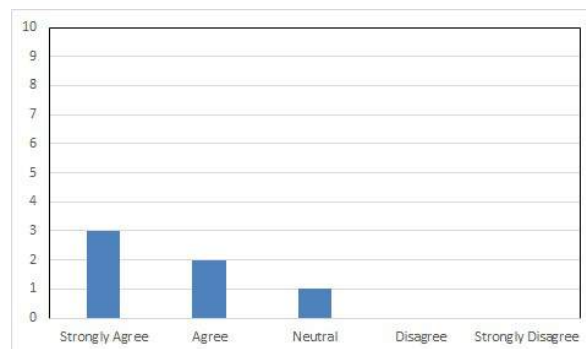
My instructor cared about my learning in this course:



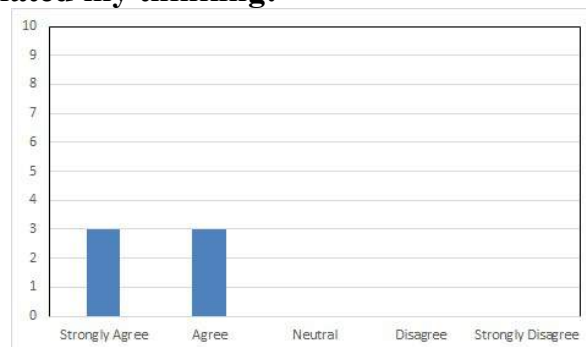
My instructor demonstrated respect for cadets as individuals:



In this course, my critical thinking ability increased:



This instructor stimulated my thinking:



SECTION II – ATTACHMENT 2

HANDWRITTEN SURVEY YEARLY COMPARISON

SECTION II – ATTACHMENT 3
STUDENT OUTCOMES WORKSHEETS

Student Outcomes		← Categories Carried Forward from Grades Spreadsheet →															
(Outcomes are abbreviated)																	
On completion of the program, our students have demonstrated that they can:																	
id, form & solve by appl mse		1	1	1	1	1	1	1.0	0	0	0.0	1	1.0	1	1.0	1	1.0
design to meet needs		0	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0	0	0.0
communicate		0	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0	0	0.0
professional ethical impact		0	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0	0	0.0
function on a team		0	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0	0	0.0
experiments		0	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0	0	0.0
new knowledge		0	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0	0	0.0
The program provides a thorough grounding and working knowledge of:																	
chemistry		1	1	1	1	1	1	1.0	1	1	1.0	1	1.0	1	1.0	1	1.0
material & energy balances		0	1	0	1	0	0	0.3	0	0	0.0	1	1.0	1	1.0	1	1.0
safety & environmental		0	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0	0	0.0
phys. & chem. thermo		0	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0	0	0.0
heat, mass, & momentum transfer.		0	0	0	0	1	0	0.2	0	0	0.0	0	0.0	0	0.0	0	0.0
chemical reaction engineering		0	0	0	1	1	0	0.3	1	1	1.0	1	1.0	1	1.0	1	1.0
cont. & stage-wise separations		0	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0	0	0.0
process dynamics & control		0	0	0	0	1	0	0.2	0	0	0.0	0	0.0	0	0.0	0	0.0
modern exp. & comp. tech.		0	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0	0	0.0
process design.		0	0	0	0	0	1	0.2	0	0	0.0	0	0.0	0	0.0	1	1.0

Course Objectives																				
Solve problems by applying mse	0	0	0	1	1	0	0.3		1	1	1.0		1	1.0		1	1.0		1	1.0
Relate bioprocess eng. to other fields	0	0	0	0	1	0	0.2		1	1	1.0		0	0.0		0	0.0		0	0.0
Utilize bioengineerng principles	1	1	1	1	1	1	1.0		1	1	1.0		1	1.0		1	1.0		1	1.0
Communicate efficiently	0	0	0	0	0	0	0.0		1	1	1.0		0	0.0		0	0.0		0	0.0
Raw Grade Data	← Data Carried Forward from Grades Spreadsheet →																			
Average out of 100	98	##	##	92	92	##			92	95			92			91			90	
Do Not Change the Format of this Sheet!																				



TAB C

AY2024 COURSE ASSESSMENT

Department of Chemistry and Life Science
United States Military Academy
West Point, New York 10996

MADN-CHM-LS

29 December 2023

MEMORANDUM FOR Chemical Engineering Program Director, Department of Chemistry and Life Science, United States Military Academy, West Point, New York 10996

SUBJECT: CH350 Chemical Reaction Engineering Course Assessment (AY24-1)

1. The course assessment meeting for CH350 was conducted 29 December 2023. Attendees: Dr. Simuck F. Yuk, Dr. Enoch Nagelli, Dr. Andrew Biaglow, LTC Corey James, and LTC Samuel V. Cowart. The topics covered included: 1) revision of major graded events; 2) reemphasis of general chemistry concepts related to bioprocess systems. Summary of recommended changes for AY25 are included in Section III.
2. The CH350 course assessment in accordance with the course director handbook from AY24 is attached.
3. POC is the undersigned.

CF: Dr. Biaglow, Dr. Nagelli, LTC James,
LTC Cowart

Dr. Simuck F. Yuk
Assistant Professor, Chemical Engineering
CH350 Course Director

SECTION I: COURSE DESCRIPTION

1. Current Red Book Description.

CH350 (Version: 2041 1) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	CREDIT HOURS
CH350	BIOPROCESS ENGINEERING	AY24	1	3.0 (BS=1.0, ET=2.0, MA=0.0)

SCOPE

This course is designed to develop an understanding of the field of biochemical engineering. The course covers application of engineering principles to the study of bioprocesses and biochemical reactor design. Topics include enzymes and enzyme kinetics, cell growth and cell growth kinetics, suspension and immobilization of cultures, bioreactor design, scale-up, and control, and recovery and purification technology. Mathematical modeling is an integral part of the course.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

None

CH350 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE-REQUISITE	CH102	2019	2	1	Y
	MA205	2023	2	1	Y
	PH202	2023	1	1	Y

2. Enrollment Numbers.

- AY 2022-1: 10
- AY 2023-1: 10
- AY 2024-1: 11

3. Current Course Objectives.

This course provides a basis for understanding the application of engineering principles to problems in bioprocessing and bioprocess design. Specifically, the objectives of the course are:

(1) to introduce the biochemical and bioprocess engineering and how it relates to other disciplines of engineering and science.

(2) the develop the ability to apply mathematics, science, and engineering to solve problems of biological and physiological significance.

(3) to develop an understanding of and ability to apply engineering principles to problems involving microbial and enzymatic processes.

(4) to understand current topics within the field of bioprocess engineering.

4. Survey Questions. Below are questions to the web-based survey of questions asked to CH350 cadets at the end of the AY23-1 semester:

Academy

Q1. My fellow students contributed to my learning in this course.

Q2. My motivation to learn and to continue learning has increased because of this course.

Q3. My personal schedule allows me enough time to reflect on the material I have learned in class.

Q4. My personal schedule allows me enough time to adequately prepare for my optimum academic performance.

Academy - I

Q5. This instructor encouraged students to be responsible for their own learning.

Q6. This instructor used effective techniques for learning, both in class and for out-of-class assignments.

Q7. My instructor cared about my learning in this course.

Q8. My instructor demonstrated respect for cadets as individuals.

5. Course GPA. The GPA for the course from the last 7 years follows below.

AY22-1 (Fall 2021)	3.60
AY23-1 (Fall 2022)	3.47
AY24-1 (Fall 2023)	3.58

6. TEE Grade.

AY24-1 (Fall 2023)	3.57
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7.a. Current Textbook.

Shuler, Michael L., Kargi, Fikret, and DeLisa, Matthew P., *Bioprocess Engineering: Basic Concepts*, 3rd Edition, Prentice Hall, 2017, ISBN-13: 978-0137062706, ISBN-10: 0137062702.

K. D. Wittrup, B. Tidor, B. J. Hackel, and C. A. Sarkar., *Quantitative Fundamentals of Molecular and Cellular Bioengineering*, The MIT Press, 2019, ISBN: 978-0-262-04265-9.

7.b. Course Content.

Block I: Biomolecular Considerations
 Block II: Cellular Considerations
 Block III: Downstream Considerations

7.c. Lesson Sequence

LESSON	TOPICS
Block 1. Biomolecular Considerations	
1	Introduction to bioprocess engineering
2	Introduction to cells and biomolecules - Part 1
3	Introduction to cells and biomolecules - Part 2
4	Enzyme kinetics and immobilized enzyme systems - Part 1
5	Enzyme kinetics and immobilized enzyme systems - Part 2
6	Enzyme kinetics and immobilized enzyme systems - Part 3
7	Non-covalent binding interactions; Binding equilibria and kinetics
8	Central dogma of molecular biology
9	Genetic / Metabolic regulation - Part 1
10	Genetic / Metabolic regulation - Part 2
11	Recombinant DNA technology / Heterologous protein expression - Part 1
12	Recombinant DNA technology / Heterologous protein expression - Part 2
13	WPR I
Block 2. Cellular Considerations	
14	General concepts of cell growth - Part 1
15	General concepts of cell growth - Part 2
16	General concepts of cell growth - Part 3
17	Operational considerations for bioreactors: suspension and immobilized cultures
18	Bioreactor selection, scale-up and process control - Part 1
19	Bioreactor selection, scale-up and process control - Part 2
20	Guest Lecture I
21	Guest Lecture II
22	WPR II
Block 3. Downstream Considerations	
23	Product recovery and separation - Part 1

24	Product recovery and separation - Part 2
25	Bioprocess considerations using animal cells
26	Bioprocess considerations using plant cells
27	Medical applications of bioprocess engineering
28	Traditional industrial bioprocesses
29	Capstone presentations I
30	Capstone presentations II

7.d. Summary of Graded Requirements

Requirement	Number	Points	Total	%
Problem Sets	6	50	300	20%
Instructor Points	---	100	100	7%
Capstone Design Presentation	1	100	100	7%
Capstone Design Paper	1	200	200	12%
WPR	2	200	400	27%
TEE	1	400	400	27%
TOTAL			1500	100%

7.e. Areas of Special Emphasis.

Course Director's Area of Special Emphasis

My primary area of interest was Bioprocess Engineering. This course emphasized the use of chemical engineering principles and how these concepts can be applied to the biological systems. Large portion of knowledges were geared toward building a different type of bioreactors. I wanted cadets to make a close connection between chemical and bioprocess engineering when developing the solution to the problems on PSs, WPRs, or the Capstone.

My second area of special emphasis was on the design of a reactor utilizing the biological components. The instructor problems solved during class were focused on designing a reactor system which incorporates the material balance on cell, substrate, and production concentration. Different designs of bioreactors were introduced through the course, emphasizing how the reactor conditions are different under batch and continuous cultivation.

My third area of special emphasis was on the ability to communicate effectively. As part of the capstone project, cadets are required to individually submit a technical abstract and subsequently produce a final research paper. Part of the grade on professionalism of the written report is devoted to their communication skills. Cadets are evaluated on their technical writing and oral presentation skills, along with the feedback on different aspects of capstone projects.

My fourth area of special emphasis was on the use of modern computing techniques. Some of the kinetic problems in this course involved higher level calculus that required the use of an ODE solver, i.e., Mathematica, to project the kinetic trends in efficient manners. Several problems, i.e., solving Michaelis Menten kinetics for different species presented in the chemostat, were introduced with the emphasis of using computing codes to approach the numerical solutions.

8. Contribution to Student Outcomes

The following list contains the Chemical Engineering Student Outcomes. The phrases in **bold-faced red font** are the topics that this course is specifically asked to accomplish by the program. These bullets are also indicated with **red dashed boxes** in Section II.8

The Chemical Engineering Program provides the graduate with a thorough grounding and working knowledge of ...

- General, organic, and physical chemistry.
- **Material and energy balances as well as safety and environmental factors.**
- Thermodynamics of physical and chemical equilibria.
- Heat, mass, and momentum transfer.
- **Chemical reaction engineering.**
- Continuous and staged separation operations.
- Process dynamics and control.
- Modern experimental and computing techniques.
- Process design.

9. Resources and Laboratories.

9a. Laboratories. There were no wet labs conducted in this course.

9b. Computer Labs. All class lectures met in BHB42. No computer labs were executed in CH350.

9c. Physical Models & Demos. There are no physical models or demos used in the course.

9d. Technician Support. IT support from the CLS IT office for maintenance of computers in BHB42 was timely and addressed any issues that arose. In future considerations for distance learning, a scanner, Epson elmo, and a second monitor are important to have at designated location of teaching.

9e. Supplies. Mathematica 11 and Microsoft Office (Word, PowerPoint, Excel, and Explorer).

9f. Additional Facilities. None.

9g. Unfunded Requests. None.

10. Recommendations from AY23-1 and Implementation in AY24-1:

<i>Section I.</i>	<i>Recommendation (AY22-1)</i>	<i>Implementation (AY23-1)</i>
1. Redbook	No change.	Implemented as recommended (IAR)
2. Enrollment	No change.	IAR

3. Course Objectives	No change.	IAR
3.A. Coverage	To assess and reinforce the cadets' understanding on kinetics, multiple in-class and after-class problems were incorporated to show how the ordinary differential equations (ODEs) can be coupled with the reaction mechanisms.	Additional in-class activities and PS #6 were developed to reinforce the cadets' understanding on kinetics and other critical bioengineering knowledges.
3.B. Performance	No change.	IAR
4. Survey Questions	The end-of-course survey administered through CoursEval had a response rate of 70%. This survey was not tied to course points, but cadets were reminded to complete the survey.	The end-of-course survey administered through CoursEval had a response rate of 100% since the extra instructor points were provided (AON) to the class. Thus, using this methodology will facilitate the cadets' response to CoursEval.
5. Course QPA	No change.	IAR
6. TEE Grades	No change.	IAR
7. Course Processes 7.A. Textbook	No change.	No change.
7.B. Course Content	In AY23-1, the number of PSs and WPRs was maintained at 5 and 2, respectively.	In AY24-1, the number of PSs were increased to 6 and TEE was introduced to better access cadets' understanding of bioengineering Los introduced in the textbook.
7.C. Lessons and Labs	No change.	IAR
7.D. Graded Events	Part of problems introduced in problem sets are revised to minimize the cadets' tendency to rely on the old solution. In addition, part of problems in WPRs were revised. The future goal will be the construction of large library with new problems, so it can be efficiently utilized by the future instructors.	A new type of problems was introduced in WPRs and PS, along with TEE to access the cadets' understanding of course contents. Again, continuously implementing the new types of problems will be key to combat cadets' tendency to rely on the old class materials.
7.E. Areas of Special Emphasis	No change.	IAR.
8. Contribution to Student Outcomes 8.A. Coverage	In AY23-1, the discussion on the enzyme kinetic was expanded, so the foundational understanding on the	In AY24-1, the additional emphasis was given on some of bioprocesses in the Block 3 by introducing the PS #6.

	material balance of bioreactor can be reinforced for the cadets.	
8.B. Performance	No change.	IAR
9. Resources and Laboratories 9.A. Laboratories	No change.	IAR
9.B. Computer Labs	No change.	IAR
9.C. Physical Models & Demos	No change.	IAR
9.D. Technician Support	No change.	IAR
9.E. Supplies	No change.	IAR
9.F. Additional Facilities	No change.	IAR
9.G. Unfunded Requests	No change.	IAR

SECTION II - COURSE ASSESSMENT

1. Red Book Description. The Red Book description was followed verbatim.

2. Enrollment. The enrollment in the course was 11 for AY24-1, with the projected enrollment in AY25-1 to be 18. During Term 24-1, no cadets failed the course, and none were at serious risk of failing the course going into the Term End.

3. Course Objectives. The course objectives were covered thoroughly and achieved with proficiency. The course objectives cover the body of knowledge appropriately. The course objectives lend themselves to assessment.

3.b. Coverage

Coverage of course objectives is presented in Figure 1. As in the other courses in this program, coverage is assessed by the course director, who assigns a 1 or a 0 to a graded event or portion of a graded event. A score of 1 on a graded event or a portion of a graded event designates significant coverage, while a 0 designates an insignificant amount or no coverage. Coverage scores are then averaged across all similar graded events in the course (problem sets, exams, etc.). A score close to 1 indicates a relatively large portion of that type of graded event covered that objective. For example, a score of 0.6 on design problems under Objective 1 (mass/energy balances) indicates that 60% or 3 out of 5 of the problem sets contained coverage of that objective. This does not say that the other problem set had zero coverage of that objective, only that it was not counted by the course director for assessment. An objective with multiple bars shows that multiple types of graded events covered that objective. Raw data are included in the spreadsheet attached to the end of this document in Attachment 3. Because of this analysis, we conclude that course objectives were covered.

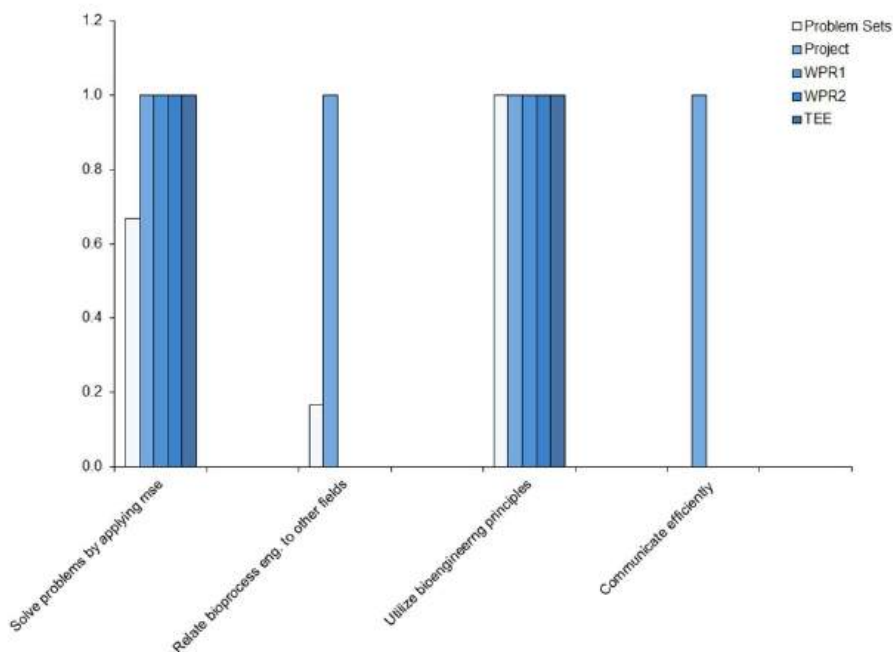


Figure 1. Course Objectives Coverage by Graded Event

3.b. Performance.

Performance on course objectives was assessed using grades taken from those graded events that included significant content in that objective area. The dotted red line shown below indicates the program standard for failure (67%). Some of the cadets still seemed to lack an understanding of how to set up and solve a series of differential or algebraic equations, which hampered their ability to get a numerical solution when analyzing the kinetic trends in biological environments, but overall, the class performed relatively well with all cadets passing the course.

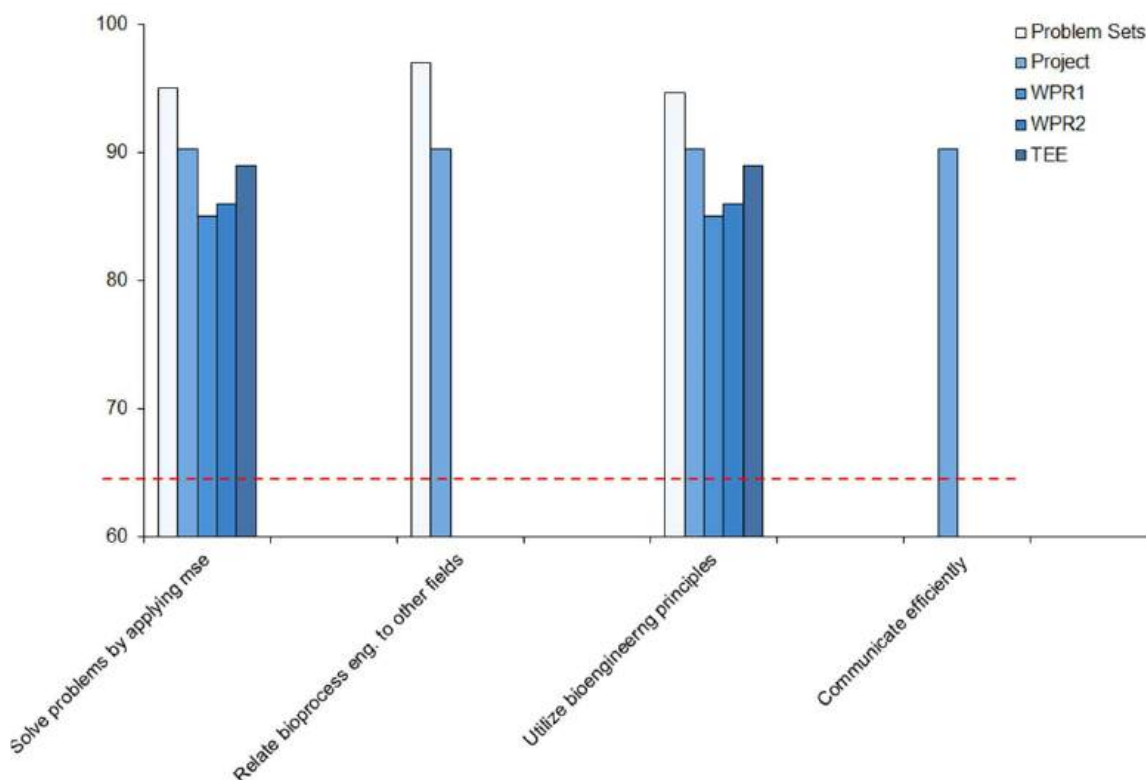


Figure 2. Course Objectives Performance by Graded Event.

4. Survey Questions

4a. Web-based:

The raw data from the web-based survey questions is captured in Section II – Attachment 1. For AY 24-1, the number of respondents was 11/11 cadets, or 100.00%.

4.b. Handwritten Surveys: The student outcomes survey is another method by which we assess student outcomes (SOs). No handwritten survey was conducted in AY24-1 for CH350.

5. Course QPA. The AY23-1 course QPA was 3.47. The AY24-1 course QPA was 3.58, which is slightly higher than the AY23-1. It is important to note that 1 of 11 cadets is majored in

chemistry, while the rest of cadet population is from chemical engineering program, like AY23-1. The chemistry cadet received the average percentage of 87%, showing that he/she possesses enough skills to navigate through the course materials without being majoring in engineering program. Additional data will be needed to make detailed assessment on the impact of non-engineering majors on the cadets' performance in CH350.

6. TEE Grade. The class average in TEE in AY24-1 was 89%.

7. Course Processes.

7.a. Textbook. The DeLisa Bioprocess Engineering textbook is relatively suitable for this course. DeLisa's book is up to date in terms of math and engineering contents. My assessment is that most of the material is presented in a method that is not too difficult for the cadets to follow or understand. One of the noticeable disadvantages of the text is that DeLisa introduces some of core concepts without giving enough background information for cadets to follow. Also, the unique way to expressing a variety of units throughout the textbook often confuses cadets.

Also, the textbook doesn't offer any example question requiring coding. Additional recourses may be needed if the instructors want to show how the core design equations learned in CH350 can be efficiently solved using the computer programs. Mathematica and ChemCad will be good problem candidates for demonstrating coding examples in the class.

7.b. Course Content. The course content and lesson blocks were appropriate for the course. Background reading and homework problems were assigned prior to each class. After each discussion, the instructor would plan to guide the students through an in-class problem or have the students work an in-class problem.

One major request received by cadets is to perform the in-class problem demonstration. Showing how to solve some of complex problems (from problem sets) step by step has helped cadets to understand how to efficiently tackle and produce the appropriate solutions. Sharing the partial solution further enhances cadets' problem-solving techniques. Most of in-class problems were mainly derived from the examples from the textbook.

All WPRs administered in CH350 were divided into two different sections. The fundamental questions were placed, so cadets can give in-depth conceptual perspectives on the biological systems. The quantitative questions were implemented, so cadets can have the opportunities of applying the appropriate types of design equations to the various types of biological systems.

Often, cadets completed their assigned readings before class, but lesson assignments and problem sets were designed to force them into the text. Cadets generally worked problem sets a day before they were due or, at worst, on the day they were due (by observing the submission site on Teams). Some of them often relied heavily on AI, their smarter colleagues, or any available external sources to solve the problem sets, which potentially degraded the learning of some cadets. This is the general trend that I observed not only from CH350, but for other STEM courses offered in the department.

More emphasize can be placed on the usage of computer program in the class. Since the textbook doesn't present any actual coding problems, additional sources might be required to design the coding demo problems. Again, Mathematica and ChemCad can be used to construct the coding problems as part of problem set or major graded events.

7.c. Lessons and Labs. The lessons are appropriate for this course. The sequence of lessons is arranged in a way so that cadets can see the course contents are moving from molecular scale to reactor scale. No labs were conducted in AY24-1 for CH350.

7.d. Summary of Graded Requirements. The number and type of graded requirements was appropriate for the course. A large bank of problems should be developed so instructors can rotate problems from year to year. Also, I can see that the number of problem sets assigned could be increased to further reinforce the core concepts learned in the class.

7.e. Areas of Special Emphasis. The areas of special emphasis were appropriate for the course.

8. Contribution to Student Outcomes

8.a. Coverage. Contribution to student objectives was assessed using the same method as described in Section II.3. Figure 3 shows coverage of student curriculum outcomes. The spreadsheet containing the data for this analysis is attached to the end of this report, in the section labeled "Section II - Attachment 2 – Assessment Worksheet." The items marked with **red dashed boxes** in Figures 3 designate student directives from Section I.8.

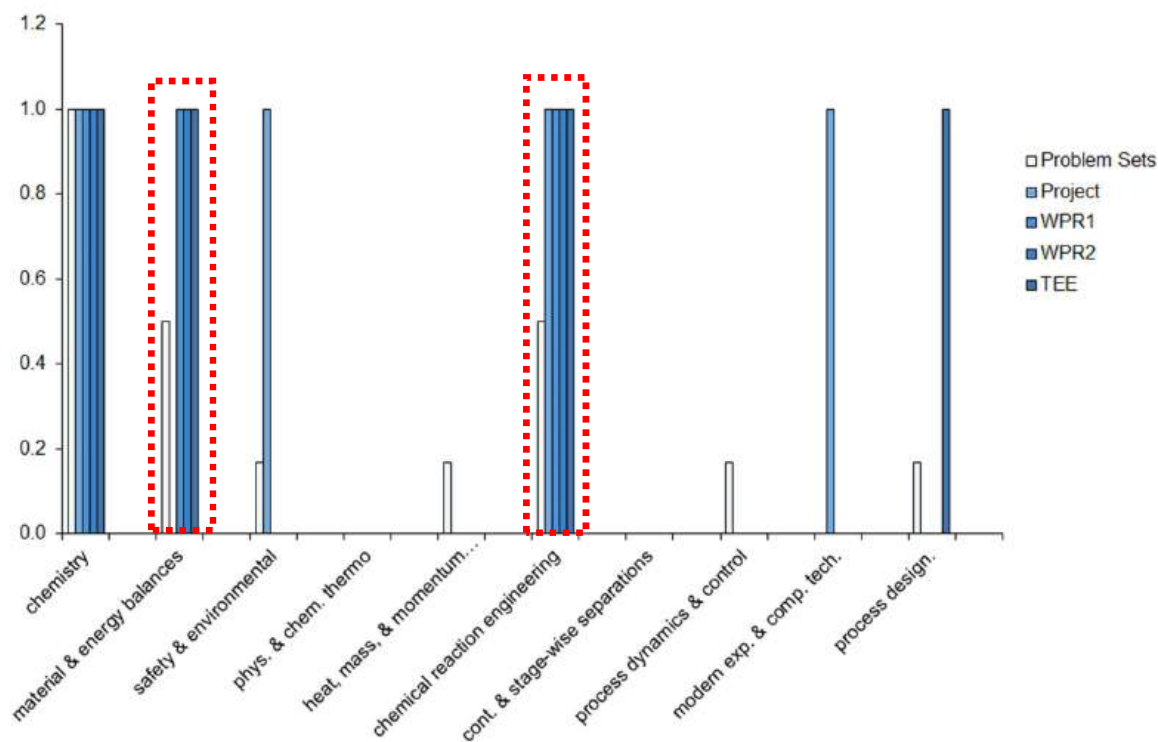


Figure 3. Student Curriculum Outcomes Coverage by Graded Event.

8.b. Performance

Performance on the student outcomes is assessed using the cadet scores from the indicators from the graded events that are outlined above. The results are shown in Figures 4 below. The average for all graded events met or exceeded the 67% standard set forth by USMA for a passing grade. Although none of cadets risked achieving less than 67% in any of area, some of them did struggle with the general chemistry-based problems (i.e., chemical equilibrium and/or kinetics).

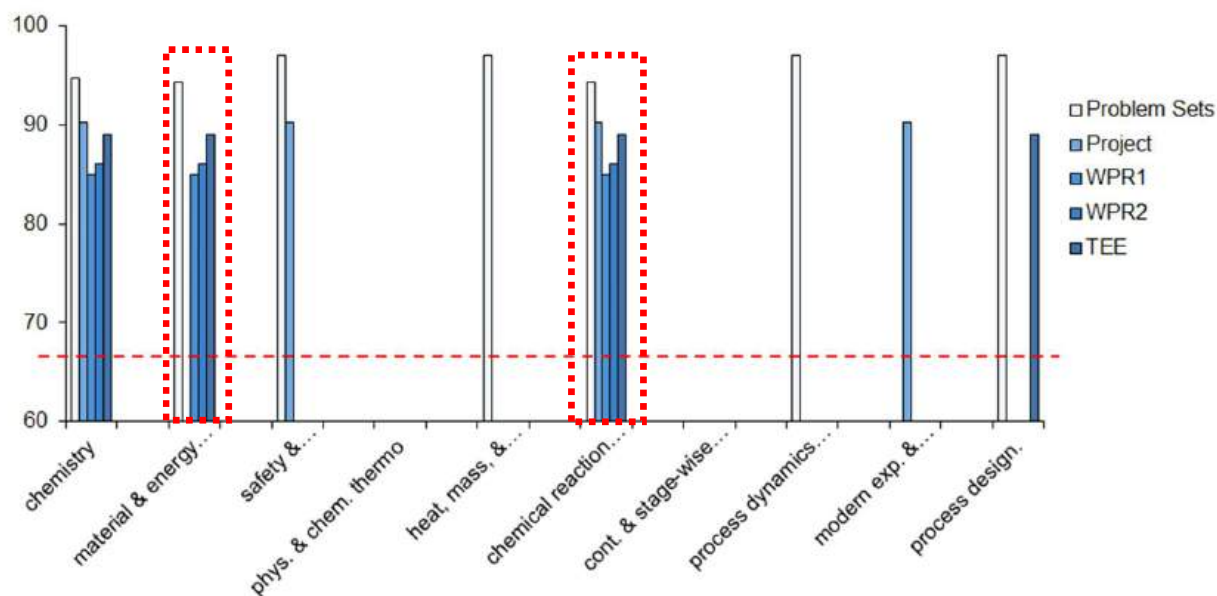


Figure 4. Student Curriculum Outcomes Performance by Graded Event.

9. Resources and Laboratories.

9a. Laboratories. There were no wet labs conducted in this course.

9b. Computer Labs. All class lectures met in BHB42. No computer labs were executed in CH350.

9c. Physical Models & Demos. There are no physical models or demos used in the course.

9d. Technician Support. IT support from CLS for maintenance of computers in BHB42. The IT support from CLS was adequate.

9e. Supplies. The supplies for the computer lab were adequate (no wet labs were conducted in CH350).

9f. Additional Facilities. None were used - adequate.

9g. Unfunded Requests. None - adequate.

SECTION III: RECOMMENDED CHANGES

1. Red Book Description. No change

2. Enrollment Numbers. Based on the projected enrollment of 18, the course should maintain 1 section and use both associated lab periods physically in the classroom.

3. Course Objectives. No change.

3.a. Coverage. To assess and reinforce the cadets' understanding on kinetics, multiple in-class and after-class problems needs to be further incorporated to show how the ordinary differential equations (ODEs) can be coupled with the reaction mechanisms. One lesson needs to be invested to reemphasize the knowledges associated with kinetics and chemical equilibria introduced in general chemistry since some of the problems that cadets struggled the most is closely related to chemical equilibrium and kinetics during WPRs. Instructor points were adapted to reflect the cadets' contribution towards in-class worksheets during the class.

3.b. Performance. No change.

4. Survey Questions. No change.

4.a. Survey Results. No change.

4.b. Survey Freeform Comments. Continue to include these as options.

5. Course QPA. No change.

6. TEE Grade. TEE was implemented to assess cadets' overall understandings of bioprocess engineering in AY24-1. TEE was constructed based on the core problems from Block 1, 2, and 3, which are directly tied to the learning objectives of course. The problems from Block 1 were focused on calculating and quantitatively describing the enzyme kinetics (with and without) and metabolic pathways. Solving the material balance on different types of bioreactors was mainly associated with the contents from Block 2. The most of problems from Block 3 was designed to ask the cadets to consider various downstream conditions in bioprocess design. The average TEE score was 89%, showing that most of cadets have a good understanding of core concepts introduced in CH350.

7. Course Process

7.a. Current Textbook. Use of the 1) *Bioprocess Engineering: Basic Concepts*, by M. L. Shuler, F. Kargi, and M. DeLisa (published by Prentice-Hall in 2017) and 2) *Quantitative Fundamentals of Molecular and Cellular Bioengineering* by K. D. Wittrup, B. Tidor, B. J. Hackel, and C. A. Sarkar (published by the MIT Press in 2019) is a sustain. The proposed textbooks are comprehensive texts for cellular and bioprocess engineering and are widely used in the engineering community.

7.b. Course Content. To assess and strengthen the cadets' understanding of kinetics, multiple in-class and after-class problems were incorporated, demonstrating how ordinary differential equations (ODEs) can be coupled with reaction mechanisms. In addition, to further develop the cadets' technical writing skills, it would be beneficial to include an assignment focused on writing a technical abstract and developing figures and tables, as these are essential components of a scientific paper. Continued emphasis on IPRs to provide feedback on cadet assignments is important for any project-based work. Finally, the capstone experiences still need to be fully converted into project-based scenarios to allow cadets to invest more time and effort in the engineering processes.

7.c. Lessons and Labs. Continue to have cadets work more in-class problems. Working a problem in every lesson is ideal and I recommend continuing to work through the full solution to the instructor problem to demonstrate to the cadets both setting the problem as well as the analytical solution with Mathematica. More time needs to be invested on fundamental reactor mass balances and data analysis. At the same time, cadets' heavy dependency on coding without attempting to access the problems contents need to be alleviated by introducing non-computer based graded events. As mentioned in Section 3.a., revisiting some of major contents from previous course (i.e., general chemistry) may be needed to reinforce the cadets' understanding of kinetic and chemical equilibrium in biological systems.

7.d. Summary of Graded Requirements. Problem sets need to be rewritten with new problems. Cadets may have access to previous years' solutions and online resources that may have direct answers to textbook questions (e.g., Chegg.com). Some of less-motivated cadets will do little more than copying the old solutions. As always, rewriting/adjusting the major portions of WPR/TEE is strongly recommended for each year.

7.e. Areas of Special Emphasis. No change.

8. Contribution to Student Outcomes

8.a. Coverage. No change.

8.b. Performance. No change.

9. Resources and Laboratories.

9a. Laboratories. No change.

9b. Computer Labs. No change.

9c. Physical Models & Demos. No change.

9d. Technician Support. No change.

9e. Supplies. No change

9f. Additional Facilities. No change.

9g. Unfunded Requests. No change.

SECTION IV: GRADED EVENTS AND SUPPLEMENTAL MATERIAL

- 1. Approved Solutions to all WPRS, Quizzes, and Labs.** Digital records are in the CH350 SharePoint graded events directory.
- 2. All supplemental material provided to cadets not included in their book issue purchase.** Digital records are in the CH350 directory.
- 3. A listing of any other resource that provided significant instructional material for cadet use. (Web sites, shared programs, etc.)**

None for CH350.

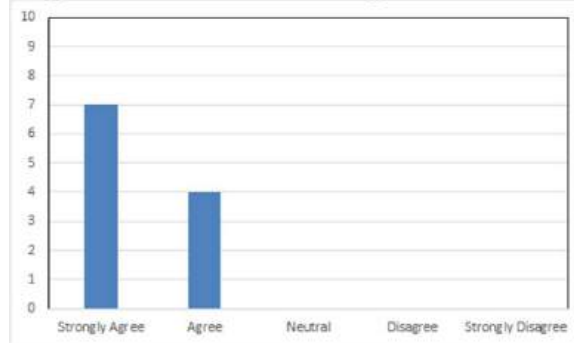
4. Other historical information

4.a. Course Director's Notes

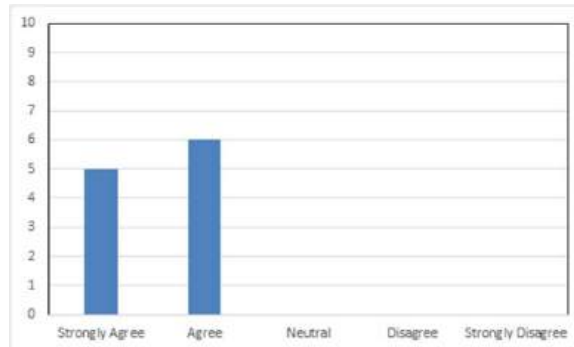
See course director's notebook.

SECTION II: ATTACHMENT 1 – WEB-BASED SURVEY RAW DATA

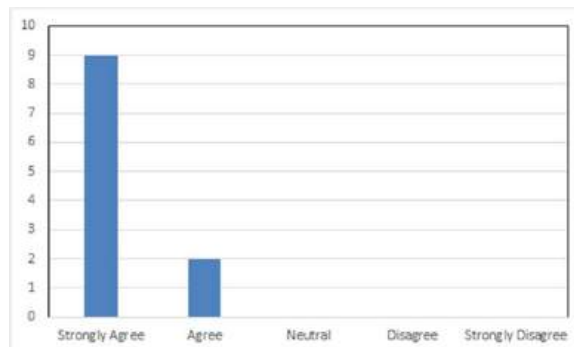
This instructor encouraged students to be responsible for their own learning:



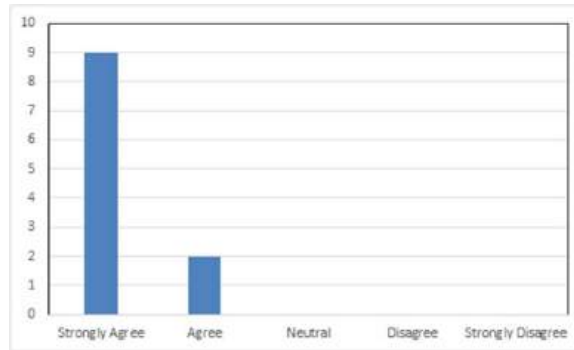
This instructor used effective techniques for learning, both in class and for out-of-class assignments:



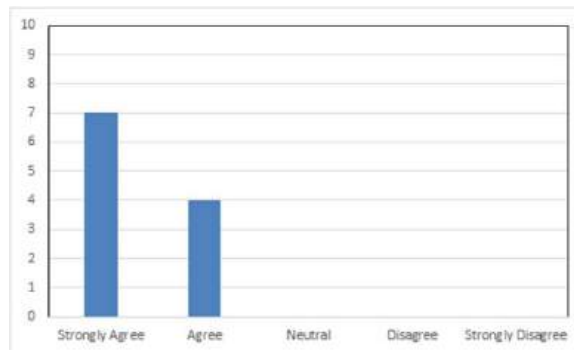
My instructor cared about my learning in this course:



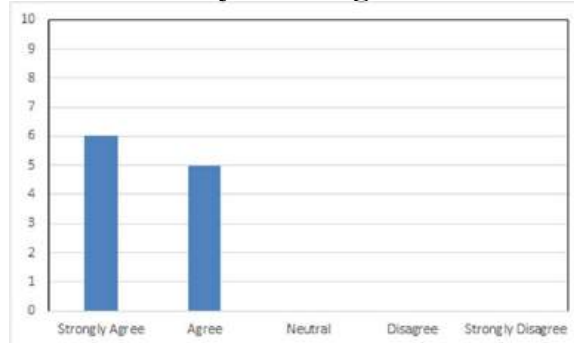
My instructor demonstrated respect for cadets as individuals:



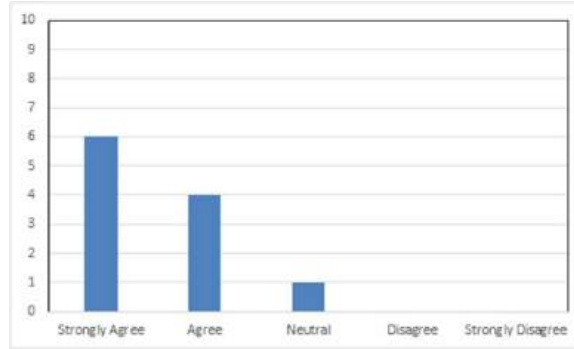
My instructor encouraged students to be responsible for their own learning:



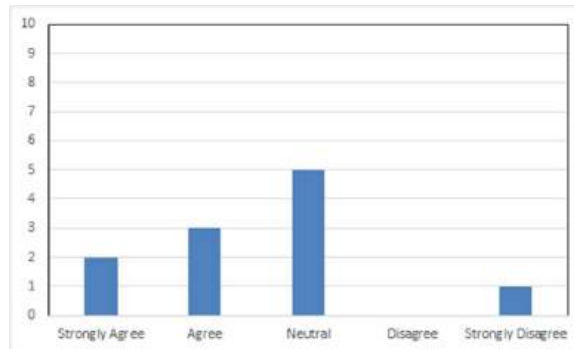
My fellow students contributed to my learning in this course:



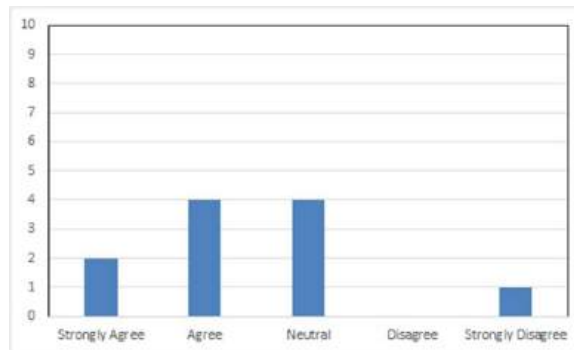
My motivation to learn and to continue learning has increased because of this course:



My personal schedule allows me enough time to reflect on the material I have learned in class:



My personal schedule allows me enough time to adequately prepare for my optimum academic performance:



SECTION II – ATTACHMENT 2

HANDWRITTEN SURVEY YEARLY COMPARISON

SECTION II – ATTACHMENT 3
STUDENT OUTCOMES WORKSHEETS

Coverage

Student Outcomes

(Outcomes are abbreviated)

On completion of the program, our students have demonstrated that they can:

id, form & solve by appl mse

PS1	PS2	PS3	PS4	PS5	PS6	PS	Capstone Present	Capstone Paper	Project	WPR1	WPR1	WPR2	WPR2	TEE	TEE
0	1	1	1	1	1	0.8	0	0	0.0	1	1.0	1	1.0	1	1.0
0	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0	0	0.0
0	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0	0	0.0
0	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0	0	0.0
0	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0	0	0.0
0	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0	0	0.0
1	0	0	0	0	0	0.2	1	1	1.0	0	0.0	0	0.0	0	0.0

design to meet needs

communicate

professional ethical impact

function on a team

experiments

new knowledge

The program provides a thorough grounding and working knowledge of:

chemistry

material & energy balances

safety & environmental

phys. & chem. thermo

1	1	1	1	1	1	1.0	1	1	1.0	1	1.0	1	1.0	1	1.0
0	1	0	1	0	1	0.5	0	0	0.0	1	1.0	1	1.0	1	1.0
1	0	0	0	0	0	0.2	1	1	1.0	0	0.0	0	0.0	0	0.0
0	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0	0	0.0

heat, mass, & momentum transfer.

0 0 0 0 1 0 0.2 0 0 0.0 0 0.0 0 0.0 0 0.0

chemical reaction engineering

0 1 0 1 1 0 0.5 1 1 1.0 1 1.0 1 1.0 1 1.0

cont. & stage-wise separations

0 0 0 0 0 0 0.0 0 0 0.0 0 0.0 0 0.0 0 0.0

process dynamics & control

0 0 0 0 1 0 0.2 0 0 0.0 0 0.0 0 0.0 0 0.0

modern exp. & comp. tech.

0 0 0 0 0 0 0.0 1 1 1.0 0 0.0 0 0.0 0 0.0

process design.

0 0 0 0 0 1 0.2 0 0 0.0 0 0.0 0 0.0 1 1.0

Course Objectives

Solve problems by applying mse

0 1 0 1 1 1 0.7 1 1 1.0 1 1.0 1 1.0 1 1.0

Relate bioprocess eng. to other fields

0 0 0 0 1 0 0.2 1 1 1.0 0 0.0 0 0.0 0 0.0

Utilize bioengineering principles

1 1 1 1 1 1 1.0 1 1 1.0 1 1.0 1 1.0 1 1.0

Communicate efficiently

0 0 0 0 0 0 0.0 1 1 1.0 0 0.0 0 0.0 0 0.0

Raw Grade Data

Average out of 100

← Data Carried Forward from Grades Spreadsheet →
97 95 91 91 97 97 91 90 85 86 89

Do Not Change the Format of this Sheet!



TAB D AY2023 COURSE ASSESSMENT

Department of Chemistry and Life Science
United States Military Academy
West Point, New York 10996

MADN-CHM-LS

24 January 2023

MEMORANDUM FOR Chemical Engineering Program Director, Department of Chemistry and Life Science, United States Military Academy, West Point, New York 10996

SUBJECT: CH350 Chemical Reaction Engineering Course Assessment (AY23-1)

1. The course assessment meeting for CH350 was conducted 24 January 2023. Attendees: Dr. Simuck F. Yuk, Dr. Enoch Nagelli, Dr. Andrew Biaglow, LTC Corey James, and LTC Samuel V. Cowart. The topics covered included: 1) revision of major graded events; 2) future adaptation of design-based scenarios into capstone experiences; 3) future inclusion of term end exam (TEE). Summary of recommended changes for AY24 are included in Section III.
2. The CH350 course assessment in accordance with the course director handbook from AY23 is attached.
3. POC is the undersigned.

CF: Dr. Biaglow, Dr. Nagelli, LTC James,
LTC Cowart

Dr. Simuck F. Yuk
Assistant Professor, Chemical Engineering
CH350 Course Director

SECTION I: COURSE DESCRIPTION

1. Current Red Book Description.

CH350 (Version: 2041 1) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	CREDIT HOURS
CH350	BIOPROCESS ENGINEERING	AY24	1	3.0 (BS=0.0, ET=Under Review, MA=0.0)

SCOPE

This course is designed to develop an understanding of the field of biochemical engineering. The course covers application of engineering principles to the study of bioprocesses and biochemical reactor design. Topics include enzymes and enzyme kinetics, cell growth and cell growth kinetics, suspension and immobilization of cultures, bioreactor design, scale-up, and control, and recovery and purification technology. Mathematical modeling is an integral part of the course (Note: Evaluation for Engineering Topics (ET) credit is currently under review by the USMA ABET Committee for 3.0 ET credit hours).

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

None

CH350 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE-REQUISITE	CH102	2019	2	1	Y
	MA205	2023	2	1	Y
	PH202	2023	1	1	Y

2. Enrollment Numbers.

- AY 2022-1: 10
- AY 2023-1: 10

3. Current Course Objectives.

This course provides a basis for understanding the application of engineering principles to problems in bioprocessing and bioprocess design. Specifically, the objectives of the course are:

(1) to introduce the biochemical and bioprocess engineering and how it relates to other disciplines of engineering and science.

(2) the develop the ability to apply mathematics, science, and engineering to solve problems of biological and physiological significance.

(3) to develop an understanding of and ability to apply engineering principles to problems involving microbial and enzymatic processes.

(4) to understand current topics within the field of bioprocess engineering.

4. Survey Questions. Below are questions to the web-based survey of questions asked to CH350 cadets at the end of the AY23-1 semester:

Academy

Q1. My fellow students contributed to my learning in this course.

Q2. My motivation to learn and to continue learning has increased because of this course.

Q3. My personal schedule allows me enough time to reflect on the material I have learned in class.

Q4. My personal schedule allows me enough time to adequately prepare for my optimum academic performance.

Academy - I

Q5. This instructor encouraged students to be responsible for their own learning.

Q6. This instructor used effective techniques for learning, both in class and for out-of-class assignments.

Q7. My instructor cared about my learning in this course.

Q8. My instructor demonstrated respect for cadets as individuals.

5. Course GPA. The GPA for the course from the last 7 years follows below.

AY22-1 (Fall 2021)	3.60
AY23-1 (Fall 2022)	3.47

6. TEE Grade.

No TEE implemented in CH350 for AY23-1.

7.a. Current Textbook.

Shuler, Michael L., Kargi, Fikret, and DeLisa, Matthew P., *Bioprocess Engineering: Basic Concepts*, 3rd Edition, Prentice Hall, 2017, ISBN-13: 978-0137062706, ISBN-10: 0137062702.

K. D. Wittrup, B. Tidor, B. J. Hackel, and C. A. Sarkar., *Quantitative Fundamentals of Molecular and Cellular Bioengineering*, The MIT Press, 2019, ISBN: 978-0-262-04265-9.

7.b. Course Content.

Block I: Biomolecular Considerations
 Block II: Cellular Considerations
 Block III: Downstream Considerations

7.c. Lesson Sequence

LESSON	TOPICS
Block 1. Biomolecular Considerations	
1	Introduction to bioprocess engineering
2	Introduction to cells and biomolecules - Part 1
3	Introduction to cells and biomolecules - Part 2
4	Enzyme kinetics and immobilized enzyme systems - Part 1
5	Enzyme kinetics and immobilized enzyme systems - Part 2
6	Enzyme kinetics and immobilized enzyme systems - Part 3
7	Non-covalent binding interactions; Binding equilibria and kinetics
8	Central dogma of molecular biology
9	Genetic / Metabolic regulation - Part 1
10	Genetic / Metabolic regulation - Part 2
11	Recombinant DNA technology / Heterologous protein expression - Part 1
12	Recombinant DNA technology / Heterologous protein expression - Part 2
13	WPR I
Block 2. Cellular Considerations	
14	General concepts of cell growth - Part 1
15	General concepts of cell growth - Part 2
16	General concepts of cell growth - Part 3
17	Operational considerations for bioreactors: suspension and immobilized cultures
18	Bioreactor selection, scale-up and process control - Part 1
19	Bioreactor selection, scale-up and process control - Part 2
20	Guest Lecture I
21	Guest Lecture II
22	WPR II
Block 3. Downstream Considerations	
23	Product recovery and separation - Part 1
24	Product recovery and separation - Part 2

25	Bioprocess considerations using animal cells
26	Bioprocess considerations using plant cells
27	Medical applications of bioprocess engineering
28	Traditional industrial bioprocesses
29	Capstone presentations I
30	Capstone presentations II

7.d. Summary of Graded Requirements

Requirement	Number	Points	Total	%
WPR	2	200	400	36%
Problem Sets (PSs)	5	50	250	23%
Capstone Paper	1	200	200	18%
Oral Presentation	1	100	100	9%
Discussion Grade	5	30	150	14%
TOTAL			1100	100%

7.e. Areas of Special Emphasis.

Course Director's Area of Special Emphasis

My primary area of interest was Bioprocess Engineering. This course emphasized the use of chemical engineering principles and how these concepts can be applied to the biological systems. Large portion of knowledges were geared toward building a different type of bioreactors. I wanted cadets to make a close connection between chemical and bioprocess engineering when developing the solution to the problems on PSs, WPRs, or the Capstone.

My second area of special emphasis was on the design of a reactor utilizing the biological components. The instructor problems solved during class were focused on designing a reactor system which incorporates the material balance on cell, substrate, and production concentration. Different designs of bioreactors were introduced through the course, emphasizing how the reactor conditions are different under batch and continuous cultivation.

My third area of special emphasis was on the ability to communicate effectively. As part of the capstone project, cadets are required to individually submit a technical abstract and subsequently produce a final research paper. Part of the grade on professionalism of the written report is devoted to their communication skills. Cadets are evaluated on their technical writing and oral presentation skills, along with the feedbacks on different aspects of capstone projects.

My fourth area of special emphasis was on the use of modern computing techniques. Some of the kinetic problems in this course involved higher level calculus that required the use of an ODE solver, i.e., Mathematica, to project the kinetic trends in efficient manners. Several problems, i.e., solving Michaelis Menten kinetics for different species presented in the chemostat, were introduced with the emphasis of using computing codes to approach the numerical solutions.

8. Contribution to Student Outcomes

The following list contains the Chemical Engineering Student Outcomes. The phrases in **bold-faced red font** are the topics that this course is specifically asked to accomplish by the program. These bullets are also indicated with **red dashed boxes** in Section II.8

The Chemical Engineering Program provides the graduate with a thorough grounding and working knowledge of ...

- General, organic, and physical chemistry.
- **Material and energy balances as well as safety and environmental factors.**
- Thermodynamics of physical and chemical equilibria.
- Heat, mass, and momentum transfer.
- **Chemical reaction engineering.**
- Continuous and staged separation operations.
- Process dynamics and control.
- Modern experimental and computing techniques.
- Process design.

9. Resources and Laboratories.

9a. Laboratories. There were no wet labs conducted in this course.

9b. Computer Labs. All class lectures met in BH340. No computer labs were executed in CH350.

9c. Physical Models & Demos. There are no physical models or demos used in the course.

9d. Technician Support. IT support from the CLS IT office for maintenance of computers in BH340 was timely and addressed any issues that arose. In future considerations for distance learning, a scanner, Epson elmo, and a second monitor are important to have at designated location of teaching.

9e. Supplies. Mathematica 11 and Microsoft Office (Word, PowerPoint, Excel, and Explorer).

9f. Additional Facilities. None.

9g. Unfunded Requests. None.

10. Recommendations from AY22-1 and Implementation in AY23-1:

<i>Section I.</i>	<i>Recommendation (AY22-1)</i>	<i>Implementation (AY23-1)</i>
1. Redbook	No change.	Implemented as recommended (IAR)
2. Enrollment	No change.	IAR
3. Course Objectives	No change.	IAR

3.A. Coverage	No change.	To assess and reinforce the cadets' understanding on kinetics, multiple in-class and after-class problems were incorporated to show how the ordinary differential equations (ODEs) can be coupled with the reaction mechanisms.
3.B. Performance	No change.	IAR
4. Survey Questions	No change.	The end-of-course survey administered through CoursEval had a response rate of 70%. This survey was not tied to course points, but cadets were reminded to complete the survey.
5. Course QPA	No change.	IAR
6. TEE Grades	No change.	IAR
7. Course Processes 7.A. Textbook	The course was taught mostly in-person, so the cadets had access to the hardcopy textbook this semester.	No change
7.B. Course Content	No change.	In AY23-1, the number of PSs and WPRs was maintained at 5 and 2, respectively.
7.C. Lessons and Labs	No change.	IAR
7.D. Graded Events	No change.	Part of problems introduced in problem sets are revised to minimize the cadets' tendency to rely on the old solution. In addition, part of problems in WPRs were revised. The future goal will be the construction of large library with new problems, so it can be efficiently utilized by the future instructors.
7.E. Areas of Special Emphasis	No change.	IAR.
8. Contribution to Student Outcomes 8.A. Coverage	No change.	In AY23-1, the discussion on the enzyme kinetic was expanded, so the foundational understanding on the material balance of bioreactor can be reinforced for the cadets.
8.B. Performance	No change.	IAR

9. Resources and Laboratories 9.A. Laboratories	No change.	IAR
9.B. Computer Labs	No change.	IAR
9.C. Physical Models & Demos	No change.	IAR
9.D. Technician Support	No change.	IAR
9.E. Supplies	No change.	IAR
9.F. Additional Facilities	No change.	IAR
9.G. Unfunded Requests	No change.	IAR

SECTION II - COURSE ASSESSMENT

1. Red Book Description. The Red Book description was followed verbatim.

2. Enrollment. The enrollment in the course was 10 for AY23-1, with the projected enrollment in AY24-1 to be 18. During Term 23-1, no cadets failed the course, and none were at serious risk of failing the course going into the Term End.

3. Course Objectives. The course objectives were covered thoroughly and achieved with proficiency. The course objectives cover the body of knowledge appropriately. The course objectives lend themselves to assessment.

3.b. Coverage

Coverage of course objectives is presented in Figure 1. As in the other courses in this program, coverage is assessed by the course director, who assigns a 1 or a 0 to a graded event or portion of a graded event. A score of 1 on a graded event or a portion of a graded event designates significant coverage, while a 0 designates an insignificant amount or no coverage. Coverage scores are then averaged across all similar graded events in the course (problem sets, exams, etc.). A score close to 1 indicates a relatively large portion of that type of graded event covered that objective. For example, a score of 0.6 on design problems under Objective 1 (mass/energy balances) indicates that 60% or 3 out of 5 of the problem sets contained coverage of that objective. This does not say that the other problem set had zero coverage of that objective, only that it was not counted by the course director for assessment. An objective with multiple bars shows that multiple types of graded events covered that objective. Raw data are included in the spreadsheet attached to the end of this document in Attachment 3. Because of this analysis, we conclude that course objectives were covered.

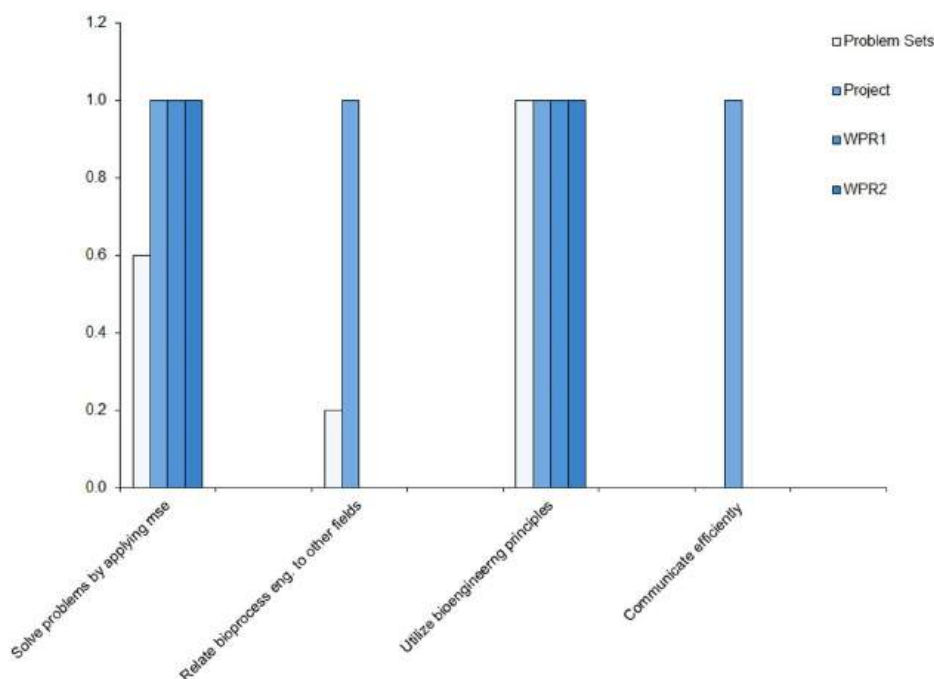


Figure 1. Course Objectives Coverage by Graded Event

3.b. Performance.

Performance on course objectives was assessed using grades taken from those graded events that included significant content in that objective area. The dotted red line shown below indicates the program standard for failure (67%). Some of the cadets still seemed to lack an understanding of how to set up and solve a series of differential or algebraic equations, which hampered their ability to get a numerical solution when analyzing the kinetic trends in biological environments, but overall, the class performed relatively well with all cadets passing the course.

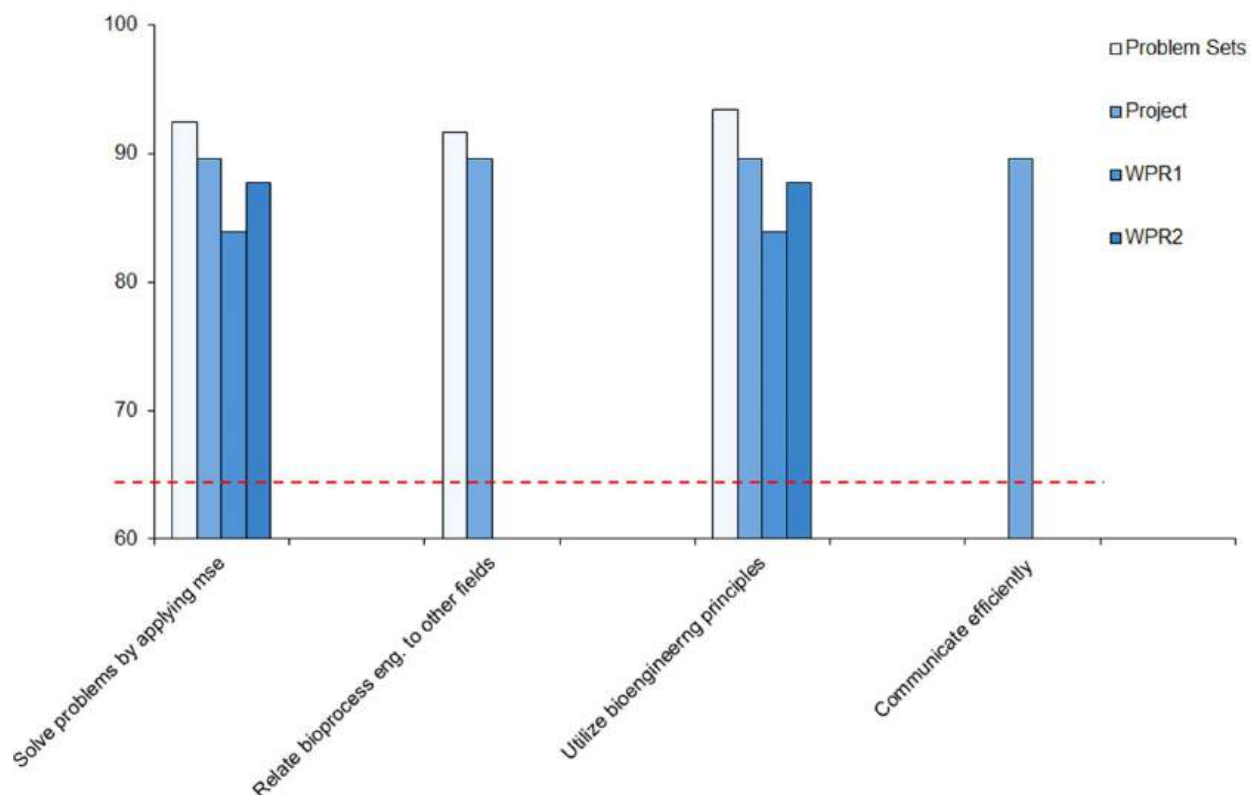
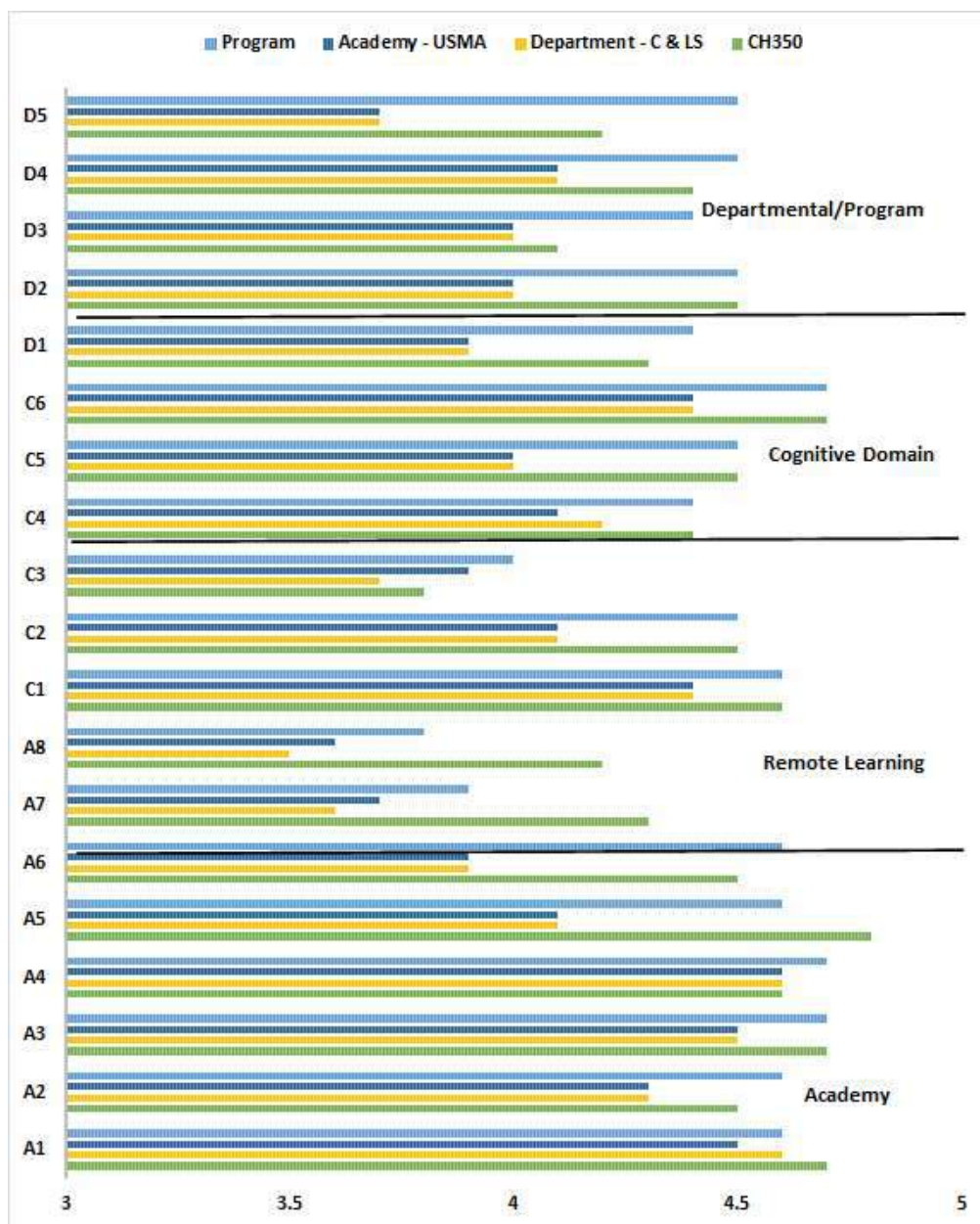


Figure 2. Course Objectives Performance by Graded Event.

4. Survey Questions

4a. Web-based:

The raw data from the web-based survey questions is captured in Section II – Attachment 1. The following graph is a summary of each of the web-based questions. For AY 23-1, the number of respondents was 7/10 cadets, or 70.00%. The error bars for the survey appear in the raw data in Attachment 1 and show that the responses for CH350 are within the standard deviations for all departmental, cognitive domain, and academy-wide groups.



4.b. Handwritten Surveys: The student outcomes survey is another method by which we assess student outcomes (SOs). No handwritten survey was conducted in AY23-1 for CH350.

5. Course QPA. The AY22-1 course QPA was 3.60. The AY23-1 course QPA was 3.47, which is slightly lower than the AY22-1. It is important to note that 1 of 10 cadets is majored in chemistry, while the rest of cadet population is from chemical engineering program. The chemistry cadet received the average percentage of 85%, showing that he/she possesses enough skills to navigate through the course materials without being majoring in engineering program. Additional data will be needed to make detailed assessment on the impact of non-engineering majors on the cadets' performance in CH350.

6. TEE Grade. N/A since TEE is not considered in AY23-1.

7. Course Processes.

7.a. Textbook. The DeLisa Bioprocess Engineering textbook is relatively suitable for this course. DeLisa's book is up to date in terms of math and engineering contents. My assessment is that most of the material is presented in a method that is not too difficult for the cadets to follow or understand. One of the noticeable disadvantages of the text is that DeLisa introduces some of core concepts without giving enough background information for cadets to follow. Also, the unique way to expressing a variety of units throughout the textbook often confuses cadets.

Also, the textbook doesn't offer any example question requiring coding. Additional recourses may be needed if the instructors want to show how the core design equations learned in CH350 can be efficiently solved using the computer programs. Mathematica and ChemCad will be good problem candidates for demonstrating coding demo in the class.

7.b. Course Content. The course content and lesson blocks were appropriate for the course. Background reading and homework problems were assigned prior to each class. After each discussion, the instructor would plan to guide the students through an in-class problem or have the students work an in-class problem.

One major request received by cadets is to perform the in-class problem demonstration. Showing how to solve some of complex problems (from problem sets) step by step has helped cadets to understand how to efficiently tackle and produce the appropriate solutions. Sharing the partial solution further enhances cadets' problem-solving techniques. Most of in-class problems were mainly derived from the examples from the textbook.

All WPRs administered in CH350 were divided into two different sections. The fundamental questions were placed, so cadets can give in-depth conceptual perspectives on the biological systems. The quantitative questions were implemented, so cadets can have the opportunities of applying the appropriate types of design equations to the various types of biological systems.

Often, cadets completed their assigned readings before class, but lesson assignments and problem sets were designed to force them into the text. Cadets generally worked problem sets a day before they were due or, at worst, on the day they were due (by observing the submission site on Teams). Some of them often relied heavily on AI, their smarter colleagues, or any available external sources to solve the problem sets, which potentially degraded the learning of some cadets. This is the general trend that I observed not only from CH350, but for other STEM courses offered in the department.

More emphasize can be placed on the usage of computer program in the class. Since the textbook doesn't present any actual coding problems, additional sources might be required to design the coding demo problems. Again, Mathematica and ChemCad can be used to construct the coding problems as part of problem set or major graded events.

7.c. Lessons and Labs. The lessons are appropriate for this course. The sequence of lessons is arranged in a way so that cadets can see the course contents are moving from molecular scale to reactor scale. No labs were conducted in AY23-1 for CH350.

7.d. Summary of Graded Requirements. The number and type of graded requirements was appropriate for the course. A large bank of problems should be developed so instructors can rotate

problems from year to year. Also, I can see that the number of problem sets assigned could be increased to further reinforce the core concepts learned in the class.

7.e. Areas of Special Emphasis. The areas of special emphasis were appropriate for the course.

8. Contribution to Student Outcomes

8.a. Coverage. Contribution to student objectives was assessed using the same method as described in Section II.3. Figure 3 shows coverage of student curriculum outcomes. The spreadsheet containing the data for this analysis is attached to the end of this report, in the section labeled "Section II - Attachment 2 – Assessment Worksheet." The items marked with **red dashed boxes** in Figures 3 designate student directives from Section I.8.

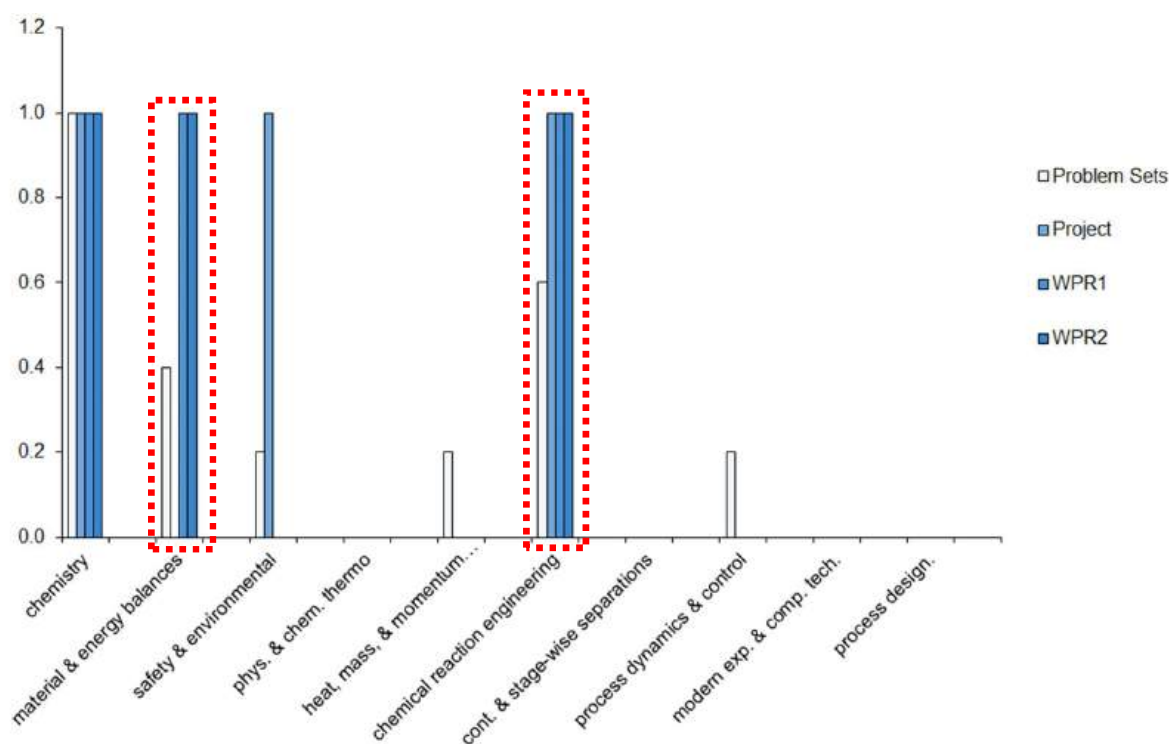


Figure 3. Student Curriculum Outcomes Coverage by Graded Event.

8.b. Performance

Performance on the student outcomes is assessed using the cadet scores from the indicators from the graded events that are outlined above. The results are shown in Figures 4 below. The average for all graded events met or exceeded the 67% standard set forth by USMA for a passing grade. However, some of the individual topics did not average to 67%. Cadets struggled with some of the Michaelis-Menten kinetic problems during major-graded events.

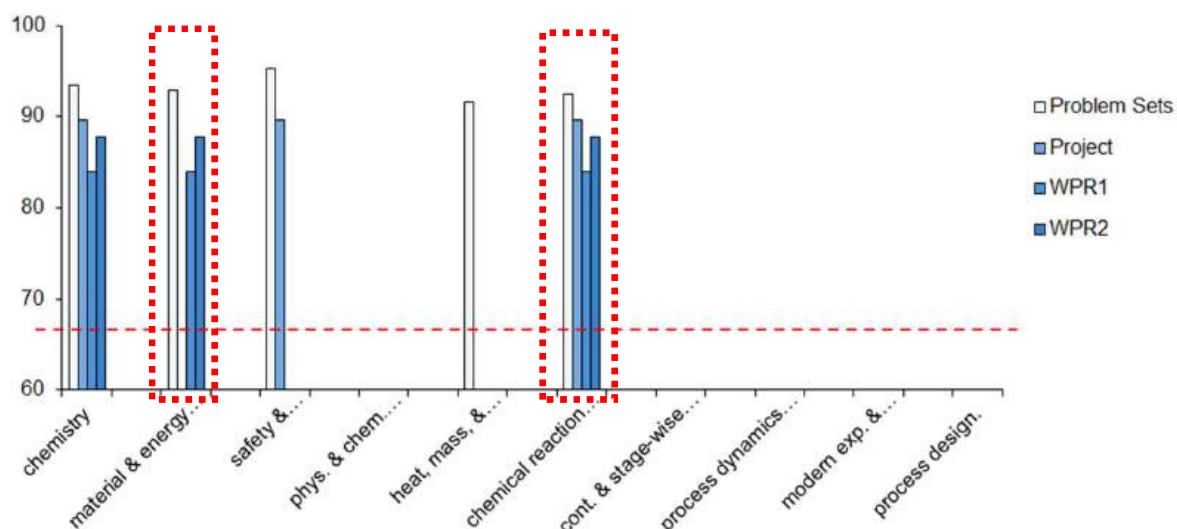


Figure 4. Student Curriculum Outcomes Performance by Graded Event.

9. Resources and Laboratories.

9a. Laboratories. There were no wet labs conducted in this course.

9b. Computer Labs. All class lectures met in BH340. No computer labs were executed in CH350.

9c. Physical Models & Demos. There are no physical models or demos used in the course.

9d. Technician Support. IT support from CLS for maintenance of computers in BH340. The IT support from CLS was adequate.

9e. Supplies. The supplies for the computer lab were adequate (no wet labs were conducted in CH350).

9f. Additional Facilities. None were used - adequate.

9g. Unfunded Requests. None - adequate.

SECTION III: RECOMMENDED CHANGES

1. Red Book Description. No change

2. Enrollment Numbers. Based on the projected enrollment of 27, the course should maintain two sections and use both associated lab periods physically in the classroom.

3. Course Objectives. No change.

3.a. Coverage. To assess and reinforce the cadets' understanding on kinetics, multiple in-class and after-class problems needs to be further incorporated to show how the ordinary differential equations (ODEs) can be coupled with the reaction mechanisms. One lesson can be invested to reemphasize the knowledges associated with kinetics and chemical equilibria introduced in general chemistry. Also, the points associated with discussion need to be redistributed to increase the individual efforts from the cadets for CH350.

3.b. Performance. No change.

4. Survey Questions. No change.

4.a. Survey Results. No change.

4.b. Survey Freeform Comments. Continue to include these as options.

5. Course QPA. No change.

6. TEE Grade. No TEE was implemented in AY23-1. However, the final assessment of cadets' understanding of CH350 course needs to be made and thus, new TEE will be constructed based on the core problems from Block 1, 2, and 3, which are directly tied to the learning objectives of course. The problems from Block 1 will be focused on calculating and quantitatively describing the enzyme kinetics (with and without) and metabolic pathways. Solving the material balance on different types of bioreactors will be mainly associated with the contents from Block 2. The most of problems from Block 3 will ask the cadets to consider various downstream conditions in bioprocess design.

7. Course Process

7.a. Current Textbook. Use of the 1) *Bioprocess Engineering: Basic Concepts*, by M. L. Shuler, F. Kargi, and M. DeLisa (published by Prentice-Hall in 2017) and 2) *Quantitative Fundamentals of Molecular and Cellular Bioengineering* by K. D. Wittrup, B. Tidor, B. J. Hackel, and C. A. Sarkar (published by the MIT Press in 2019) is a sustain. The proposed textbooks are comprehensive texts for cellular and bioprocess engineering and are widely used in the engineering community.

7.b. Course Content. To assess and reinforce the cadets' understanding on kinetics, multiple in-class and after-class problems were incorporated to show how the ordinary differential equations

(ODEs) can be coupled with the reaction mechanisms. Also, in accordance with the developing the cadet's ability for technical writing, it will be beneficial to include the assignment for writing a technical abstract and developing figures/tables since these are the essential parts of constructing a scientific paper. Continued emphasis on IPRs to provide feedback for cadets on their assignments for development is always important for any type of project-based assignments. The capstone experiences need to be fully converted into the project-based scenarios, so the cadets can spend additional time/efforts working on the engineering processes.

7.c. Lessons and Labs. Continue to have cadets work more in-class problems. Working a problem in every lesson is ideal and I recommend continuing to work through the full solution to the instructor problem to demonstrate to the cadets both setting the problem as well as the analytical solution with Mathematica. More time needs to be invested on fundamental reactor mass balances and data analysis. As mentioned in Section 3.a., revisiting some of major contents from previous course (i.e., general chemistry) may be needed to reinforce the cadets' understanding of kinetic and chemical equilibrium in biological systems.

7.d. Summary of Graded Requirements. Problem sets need to be rewritten with new problems. Cadets may have access to previous years' solutions and online resources that may have direct answers to textbook questions (e.g., Chegg.com). Some of less-motivated cadets will do little more than copying the old solutions. As always, rewriting/adjusting the major portions of WPR/TEE is strongly recommended for each year.

7.e. Areas of Special Emphasis. No change.

8. Contribution to Student Outcomes

8.a. Coverage. No change.

8.b. Performance. No change.

9. Resources and Laboratories.

9a. Laboratories. No change.

9b. Computer Labs. No change.

9c. Physical Models & Demos. No change.

9d. Technician Support. No change.

9e. Supplies. No change

9f. Additional Facilities. No change.

9g. Unfunded Requests. No change.

SECTION IV: GRADED EVENTS AND SUPPLEMENTAL MATERIAL

- 1. Approved Solutions to all WPRS, Quizzes, and Labs.** Digital records are in the CH350 SharePoint graded events directory.
- 2. All supplemental material provided to cadets not included in their book issue purchase.** Digital records are in the CH350 directory.
- 3. A listing of any other resource that provided significant instructional material for cadet use. (Web sites, shared programs, etc.)**

None for CH350.

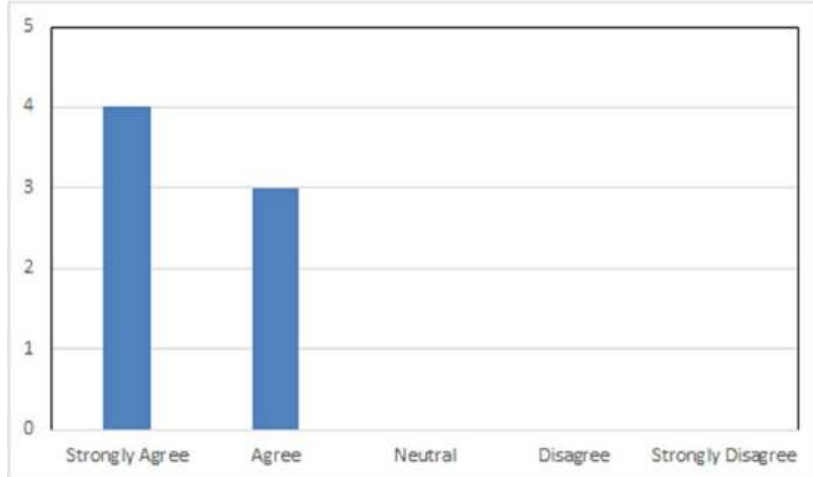
4. Other historical information

4.a. Course Director's Notes

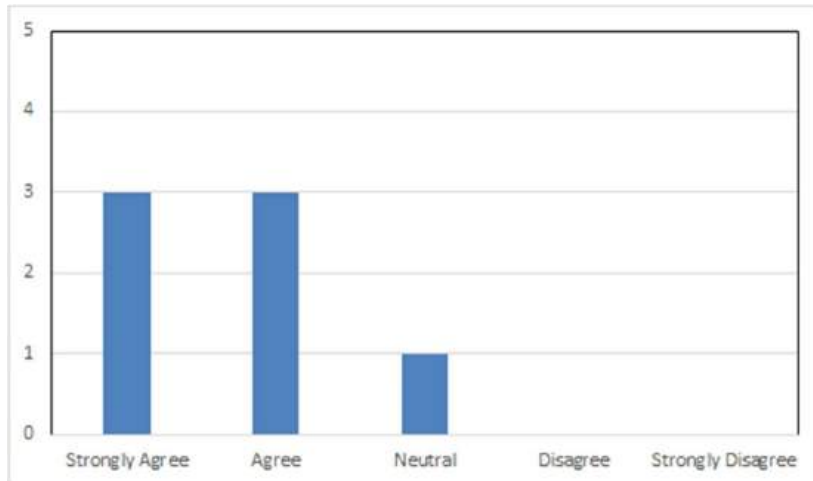
See course director's notebook.

SECTION II: ATTACHMENT 1 – WEB-BASED SURVEY RAW DATA

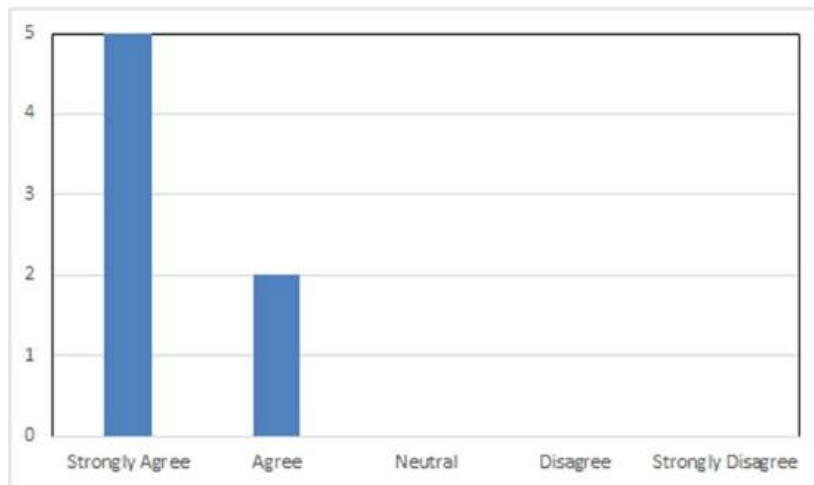
This instructor encouraged students to be responsible for their own learning:



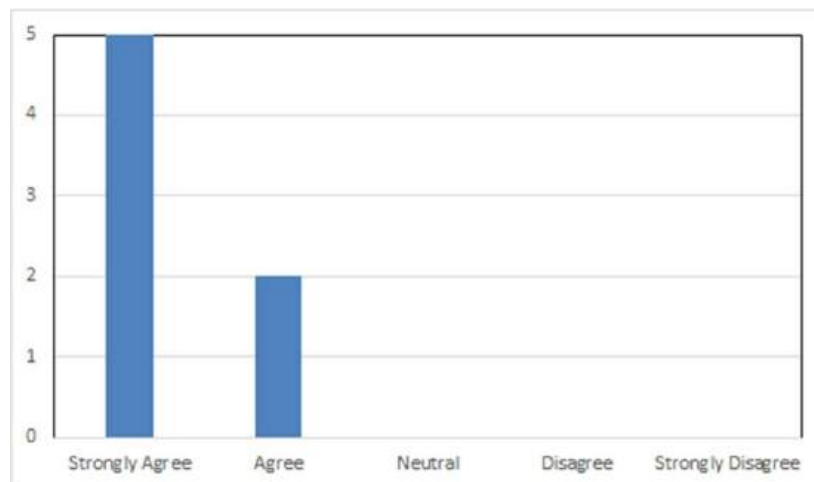
This instructor used effective techniques for learning, both in class and for out-of-class assignments:



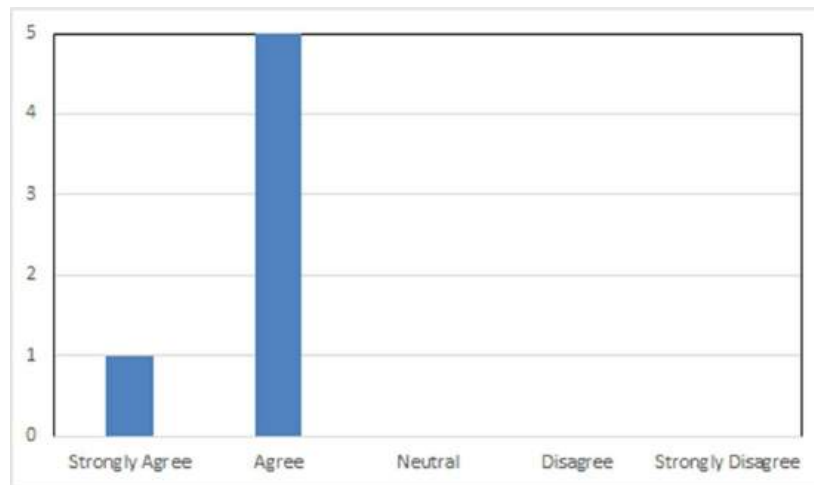
My instructor cared about my learning in this course:



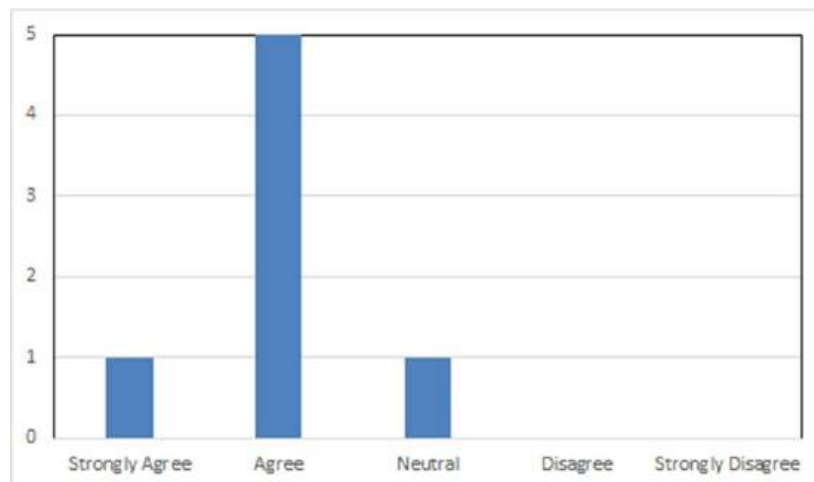
My instructor demonstrated respect for cadets as individuals



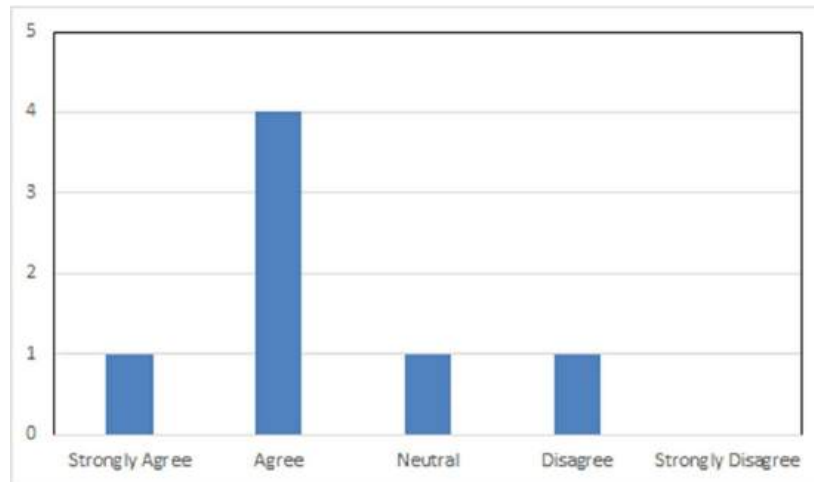
My fellow students contributed to my learning in this course



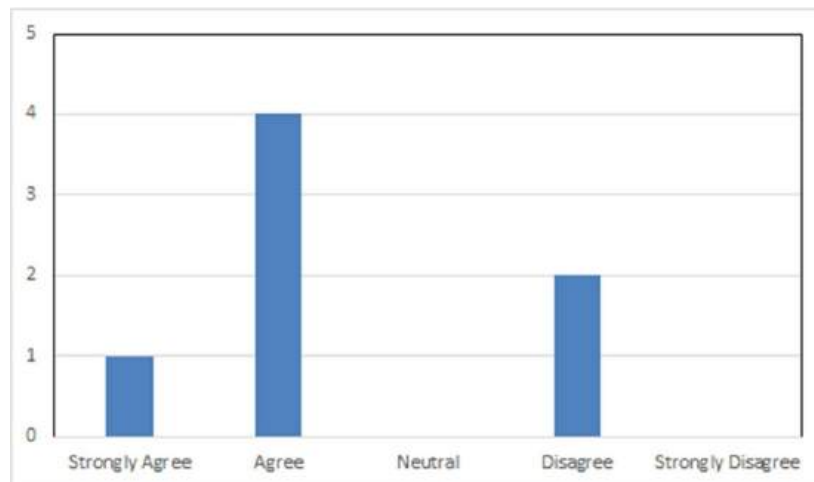
My motivation to learn and to continue learning has increased because of this course



My personal schedule allows me enough time to reflect on the material I have learned in class



My personal schedule allows me enough time to adequately prepare for my optimum academic performance



SECTION II – ATTACHMENT 2

HANDWRITTEN SURVEY YEARLY COMPARISON

SECTION II – ATTACHMENT 3
STUDENT OUTCOMES WORKSHEETS

Coverage	PS1	PS2	PS3	PS4	PS5	PS	Capstone Present	Capstone Paper	Project	WPQ1	WPQ1	WPQ2	WPQ2
Student Outcomes (Outcomes are abbreviated)	← Categories Carried Forward from Grades Spreadsheet →												
On completion of the program, our students have demonstrated that they can:													
Identify, form & solve by applying mse	0	1	1	1	1	0.8	0	0	0.0	1	1.0	1	1.0
design to meet needs	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0
communicate	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0
professional ethical impact	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0
function on a team	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0
experiments	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0
new knowledge	1	0	0	0	0	0.2	1	1	1.0	0	0.0	0	0.0
The program provides a thorough grounding and working knowledge of:													
chemistry	1	1	1	1	1	1.0	1	1	1.0	1	1.0	1	1.0
material & energy balances	0	1	0	1	0	0.4	0	0	0.0	1	1.0	1	1.0
safety & environmental	1	0	0	0	0	0.2	1	1	1.0	0	0.0	0	0.0
phys. & chem. thermo.	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0
heat, mass, & momentum transfer	0	0	0	0	1	0.2	0	0	0.0	0	0.0	0	0.0
chemical reaction engineering	0	1	0	1	1	0.6	1	1	1.0	1	1.0	1	1.0
cont. & stage-wise separations	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0
process dynamics & control	0	0	0	0	1	0.2	0	0	0.0	0	0.0	0	0.0
modern exp. & comp. tech.	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0
process design	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0
Course Objectives													
Solve problems by applying mse	0	1	0	1	1	0.6	1	1	1.0	1	1.0	1	1.0
Relate bioprocess eng. to other fields	0	0	0	0	1	0.2	1	1	1.0	0	0.0	0	0.0
Utilize bioengineering principles	1	1	1	1	1	1.0	1	1	1.0	1	1.0	1	1.0
Communicate efficiently	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0
Raw Grade Data	← Data Carried Forward from Grades Spreadsheet →												
Average out of 100	35	36	34	83	32		30	30		84		88	

	PS1	PS2	PS3	PS4	PS5	PS		Capstone Proposal	Capstone Paper	Project	WP01	WP01	WP02	WP02
Performance														
Student Outcomes														
(Outcomes are abbreviated)														
On completion of the program, our students have demonstrated that they can:														
identify, formulate & solve by applying mse	0	36	34	83	32	33		0	0	0	84	84	88	88
design to meet needs	0	0	0	0	0	0		0	0	0	0	0	0	0
communicate	0	0	0	0	0	0		30	30	30	0	0	0	0
professional ethical impact	0	0	0	0	0	0		30	30	30	0	0	0	0
function on a team	0	0	0	0	0	0		0	0	0	0	0	0	0
experiments	0	0	0	0	0	0		0	0	0	0	0	0	0
new knowledge	35	0	0	0	0	35		30	30	30	0	0	0	0
The program provides a thorough grounding and working knowledge of:														
chemistry	35	36	34	83	32	33		30	30	30	84	84	88	88
material & energy balances	0	36	0	83	0	33		0	0	0	84	84	88	88
safety & environmental	35	0	0	0	0	35		30	30	30	0	0	0	0
phys. & chem. thermo	0	0	0	0	0	0		0	0	0	0	0	0	0
heat, mass, & momentum transfer	0	0	0	0	32	32		0	0	0	0	0	0	0
chemical reaction engineering	0	36	0	83	32	32		30	30	30	84	84	88	88
cont. & stage-wise separations	0	0	0	0	0	0		0	0	0	0	0	0	0
process dynamics & control	0	0	0	0	32	8		0	0	0	0	0	0	0
modern exp. & comp. tech.	0	0	0	0	0	0		0	0	0	0	0	0	0
process design	0	0	0	0	0	0		0	0	0	0	0	0	0
Course Objectives														
Solve problems by applying mse	0	36	0	83	32	32		30	30	30	84	84	88	88
Relate bioprocess eng. to other fields	0	0	0	0	32	32		30	30	30	0	0	0	0
Utilize bioengineering principles	35	36	34	83	32	33		30	30	30	84	84	88	88
Communicate efficiently	0	0	0	0	0	0		30	30	30	0	0	0	0



TAB E AY2022 COURSE ASSESSMENT

Department of Chemistry and Life Science
United States Military Academy
West Point, New York 10996

MADN-CHM-LS

24 January 2022

MEMORANDUM FOR Chemical Engineering Program Director, Department of Chemistry and Life Science, United States Military Academy, West Point, New York 10996

SUBJECT: CH350 Chemical Reaction Engineering Course Assessment (AY22-1)

1. The course assessment meeting for CH350 was conducted 24 January 2022. Attendees: Dr. Simuck F. Yuk, Dr. Enoch Nagelli, Dr. Andrew Biaglow, LTC Matthew Armstrong, LTC Corey James, and LTC Samuel V. Cowart. The topics covered included: 1) revision of problem sets; 2) technical writing capstone experience; 3) future inclusion of Mathematica-based coding problems in graded events. Summary of recommended changes for AY23 are included in Section III.
2. The CH350 course assessment in accordance with the course director handbook from AY22 is attached.
3. POC is the undersigned.

CF: Dr. Biaglow, LTC Armstrong, Dr.
Nagelli, LTC James, LTC Cowart

Dr. Simuck F. Yuk
Assistant Professor, Chemical Engineering
CH350 Course Director

SECTION I: COURSE DESCRIPTION

1. Current Red Book Description.

CH350 (Version: 2022 1) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	CREDIT HOURS
CH350	BIOPROCESS ENGINEERING	2022	1	3.0 (BS=0.0, ET=Under Review, MA=0.0)

SCOPE

This course is designed to develop an understanding of the field of biochemical engineering. The course covers application of engineering principles to the study of bioprocesses and biochemical reactor design. Topics include enzymes and enzyme kinetics, cell growth and cell growth kinetics, suspension and immobilization of cultures, bioreactor design, scale-up, and control, and recovery and purification technology. Mathematical modeling is an integral part of the course (Note: Evaluation for Engineering Topics (ET) credit is currently under review by the USMA ABET Committee for 3.0 ET credit hours).

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

None

CH350 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE-REQUISITE	CH102	2019	2	1	Y
	MA205	2023	2	1	Y
	PH202	2023	1	1	Y

2. Enrollment Numbers.

- AY 2022-1: 10
- AY 2023-1: 9

3. Current Course Objectives.

This course provides a basis for understanding the application of engineering principles to problems in bioprocessing and bioprocess design. Specifically, the objectives of the course are:

- (1) to introduce the field and how it relates to other fields of engineering and science.
- (2) the develop the ability to apply mathematics, science, and engineering to solve problems.
- (3) to develop an understanding of and ability to apply engineering principles to problems involving microbial and enzymatic processes.

(4) to understand and communicate current topics within the field.

4. Survey Questions. Below are questions to the web-based survey of questions asked to CH350 cadets at the end of the AY22-1 semester:

Academy

Q1. My fellow students contributed to my learning in this course.

Q2. My motivation to learn and to continue learning has increased because of this course.

Q3. My personal schedule allows me enough time to reflect on the material I have learned in class.

Q4. My personal schedule allows me enough time to adequately prepare for my optimum academic performance.

Academy - I

Q5. This instructor encouraged students to be responsible for their own learning.

Q6. This instructor used effective techniques for learning, both in class and for out-of-class assignments.

Q7. My instructor cared about my learning in this course.

Q8. My instructor demonstrated respect for cadets as individuals.

5. Course GPA. The GPA for the course from the last 7 years follows below.

AY22-1 (Fall 2021) 3.60

6. TEE Grade.

No TEE implemented in CH350.

7.a. Current Textbook.

Shuler, Michael L., Kargi, Fikret, and DeLisa, Matthew P., *Bioprocess Engineering: Basic Concepts*, 3rd Edition, Prentice Hall, 2017, ISBN-13: 978-0137062706, ISBN-10: 0137062702.

7.b. Course Content.

Block I: Biomolecular Considerations
Block II: Cellular Considerations

Block III: Downstream Considerations

7.c. Lesson Sequence

LESSON	TOPICS
Block 1. Biomolecular Considerations	
1	Introduction to bioprocess engineering
2	Introduction to cells and biomolecules - Part 1
3	Introduction to cells and biomolecules - Part 2
4	Enzyme kinetics and immobilized enzyme systems - Part 1
5	Enzyme kinetics and immobilized enzyme systems - Part 2
6	Enzyme kinetics and immobilized enzyme systems - Part 3
7	Non-covalent binding interactions; Binding equilibria and kinetics
8	Central dogma of molecular biology
9	Genetic / Metabolic regulation - Part 1
10	Genetic / Metabolic regulation - Part 2
11	Recombinant DNA technology / Heterologous protein expression - Part 1
12	Recombinant DNA technology / Heterologous protein expression - Part 2
13	WPR I
Block 2. Cellular Considerations	
14	General concepts of cell growth - Part 1
15	General concepts of cell growth - Part 2
16	General concepts of cell growth - Part 3
17	Operational considerations for bioreactors: suspension and immobilized cultures
18	Bioreactor selection, scale-up and process control - Part 1
19	Bioreactor selection, scale-up and process control - Part 2
20	Guest Lecture I
21	Guest Lecture II
22	WPR II
Block 3. Downstream Considerations	
23	Product recovery and separation - Part 1
24	Product recovery and separation - Part 2
25	Bioprocess considerations using animal cells
26	Bioprocess considerations using plant cells
27	Medical applications of bioprocess engineering
28	Traditional industrial bioprocesses
29	Capstone presentations I

7.d. Summary of Graded Requirements

Requirement	Number	Points	Total	%
WPR	2	200	400	36%
Problem Sets (PSs)	5	50	250	23%
Capstone Paper	1	200	200	18%
Oral Presentation	1	100	100	9%
Discussion Grade	5	30	150	14%
TOTAL			1100	100%

7.e. Areas of Special Emphasis.

Course Director's Area of Special Emphasis

My primary area of interest was Bioprocess Engineering. This course emphasized the use of chemical engineering principles and how these concepts can be applied to the biological systems. Large portion of knowledges were geared toward building a different type of bioreactors. I wanted cadets to make a close connection between chemical and bioprocess engineering when developing the solution to the problems on PSs, WPRs, or the Capstone.

My second area of special emphasis was on the design of a reactor utilizing the biological components. The instructor problems solved during class were focused on designing a reactor system which incorporates the material balance on cell, substrate, and production concentration. Different designs of bioreactors were introduced through the course, emphasizing how the reactor conditions are different under batch and continuous cultivation.

My third area of special emphasis was on the ability to communicate effectively. As part of the capstone project, cadets are required to individually submit a technical abstract and subsequently produce a final research paper. Part of the grade on professionalism of the written report is devoted to their communication skills. Cadets are evaluated on their technical writing and oral presentation skills, along with the feedbacks on different aspects of capstone projects.

My fourth area of special emphasis was on the use of modern computing techniques. Some of the kinetic problems in this course involved higher level calculus that required the use of an ODE solver, i.e., Mathematica, to project the kinetic trends in efficient manners. Several problems, i.e., solving Michaelis Menten kinetics for different species presented in the chemostat, were introduced with the emphasis of using computing codes to approach the numerical solutions.

8. Contribution to Student Outcomes

The following list contains the Chemical Engineering Student Outcomes. The phrases in **bold-faced red font** are the topics that this course is specifically asked to accomplish by the program. These bullets are also indicated with **red dashed boxes** in Section II.8

The Chemical Engineering Program provides the graduate with a thorough grounding and working knowledge of ...

- General, organic, and physical chemistry.

- **Material and energy balances as well as safety and environmental factors.**
- Thermodynamics of physical and chemical equilibria.
- Heat, mass, and momentum transfer.
- **Chemical reaction engineering.**
- Continuous and staged separation operations.
- Process dynamics and control.
- Modern experimental and computing techniques.
- Process design

9. Resources and Laboratories.

9a. Laboratories. There were no wet labs conducted in this course.

9b. Computer Labs. All class lectures met in BH340. No computer labs were executed in CH350.

9c. Physical Models & Demos. There are no physical models or demos used in the course.

9d. Technician Support. IT support from the CLS IT office for maintenance of computers in BH340 was timely and addressed any issues that arose. In future considerations for distance learning, a scanner, Epson elmo, and a second monitor are important to have at designated location of teaching.

9e. Supplies. Mathematica 11 and Microsoft Office (Word, PowerPoint, Excel, and Explorer).

9f. Additional Facilities. None

9g. Unfunded Requests. None.

10. Recommendations from AY21-1 and Implementation in AY22-1:

N/A since a first pilot run of CH350 is executed in AY22-1.

SECTION II - COURSE ASSESSMENT

1. Red Book Description. The Red Book description was followed verbatim.

2. Enrollment. The enrollment in the course was 10 for AY22-1, with the projected enrollment in AY23-1 to be 9. During Term 22-1, no cadets failed the course, and none were at serious risk of failing the course going into the Term End.

3. Course Objectives. The course objectives were covered thoroughly and achieved with proficiency. The course objectives cover the body of knowledge appropriately. The course objectives lend themselves to assessment.

3.b. Coverage

Coverage of course objectives is presented in Figure 1. As in the other courses in this program, coverage is assessed by the course director, who assigns a 1 or a 0 to a graded event or portion of a graded event. A score of 1 on a graded event or a portion of a graded event designates significant coverage, while a 0 designates an insignificant amount or no coverage. Coverage scores are then averaged across all similar graded events in the course (problem sets, exams, etc.). A score close to 1 indicates a relatively large portion of that type of graded event covered that objective. For example, a score of 0.6 on design problems under Objective 1 (mass/energy balances) indicates that 60% or 3 out of 5 of the problem sets contained coverage of that objective. This does not say that the other problem set had zero coverage of that objective, only that it was not counted by the course director for assessment. An objective with multiple bars shows that multiple types of graded events covered that objective. Raw data are included in the spreadsheet attached to the end of this document in Attachment 3. Because of this analysis, we conclude that course objectives were covered.

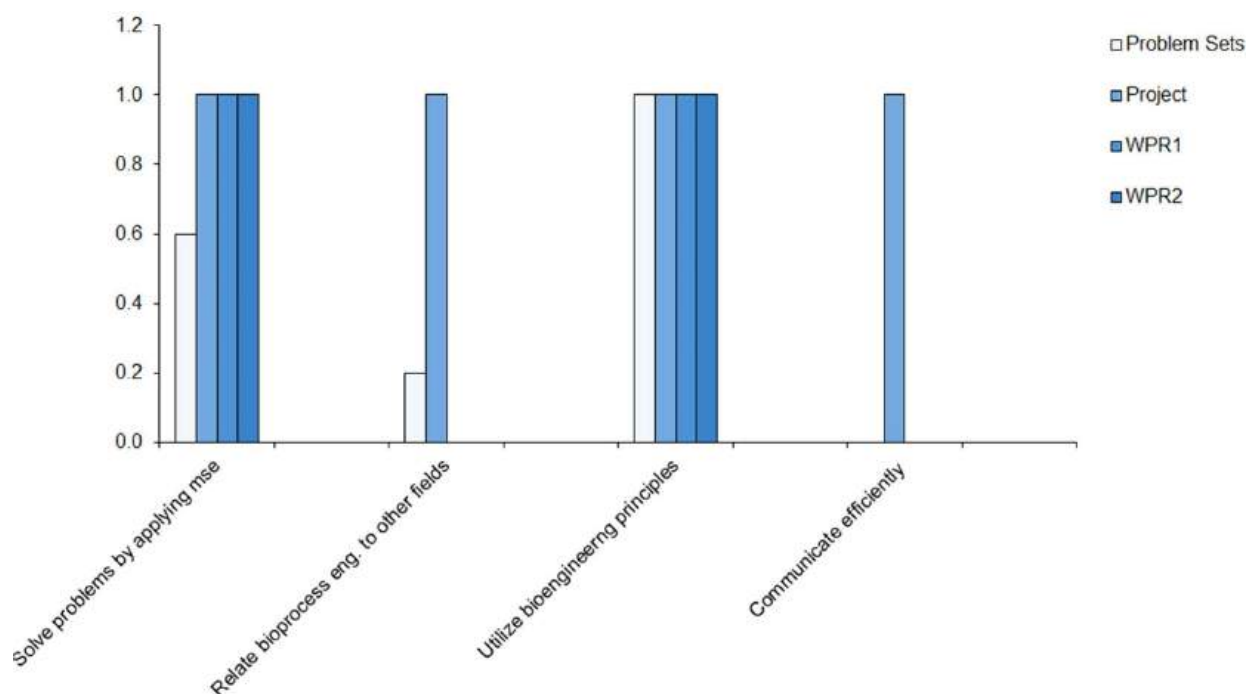


Figure 1. Course Objectives Coverage by Graded Event

3.b. Performance.

Performance on course objectives was assessed using grades taken from those graded events that included significant content in that objective area. The dotted red line shown below indicates the program standard for failure (67%). Some of the cadets still seemed to lack an understanding of how to set up and solve a series of differential or algebraic equations, which hampered their ability to get a numerical solution when analyzing the kinetic trends in biological environments, but overall, the class performed relatively well with all cadets passing the course.

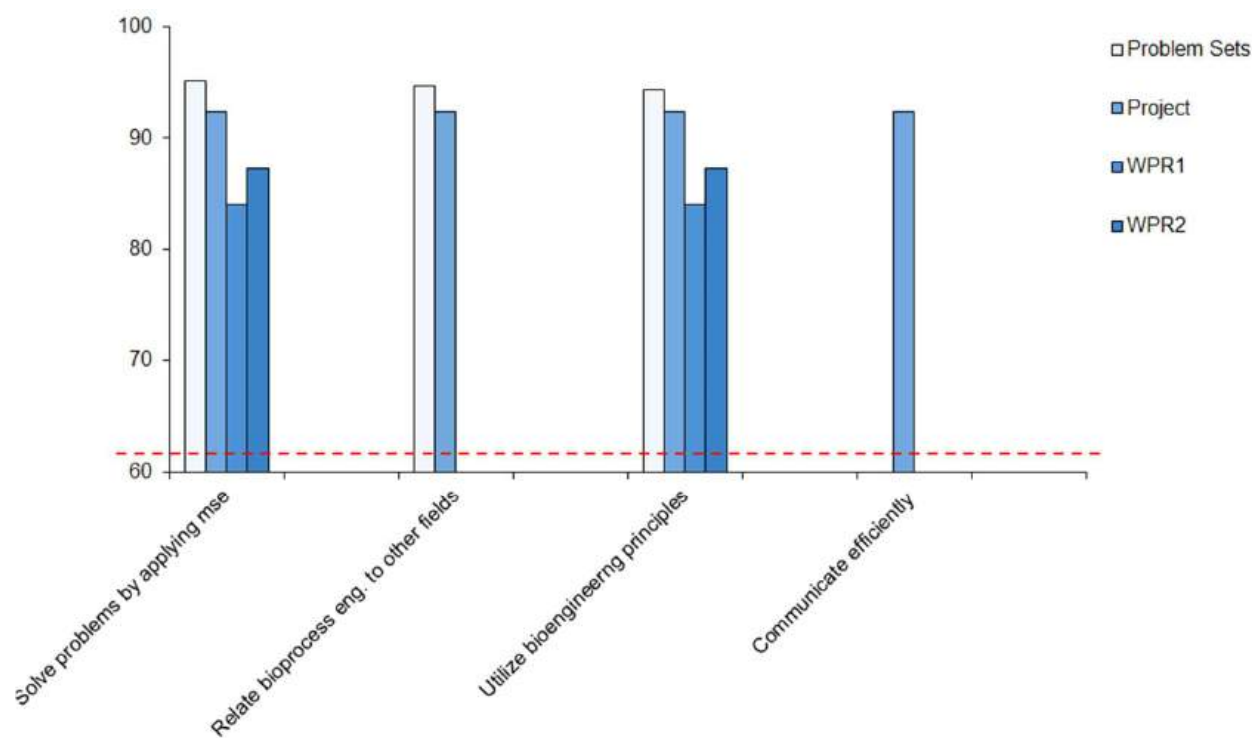
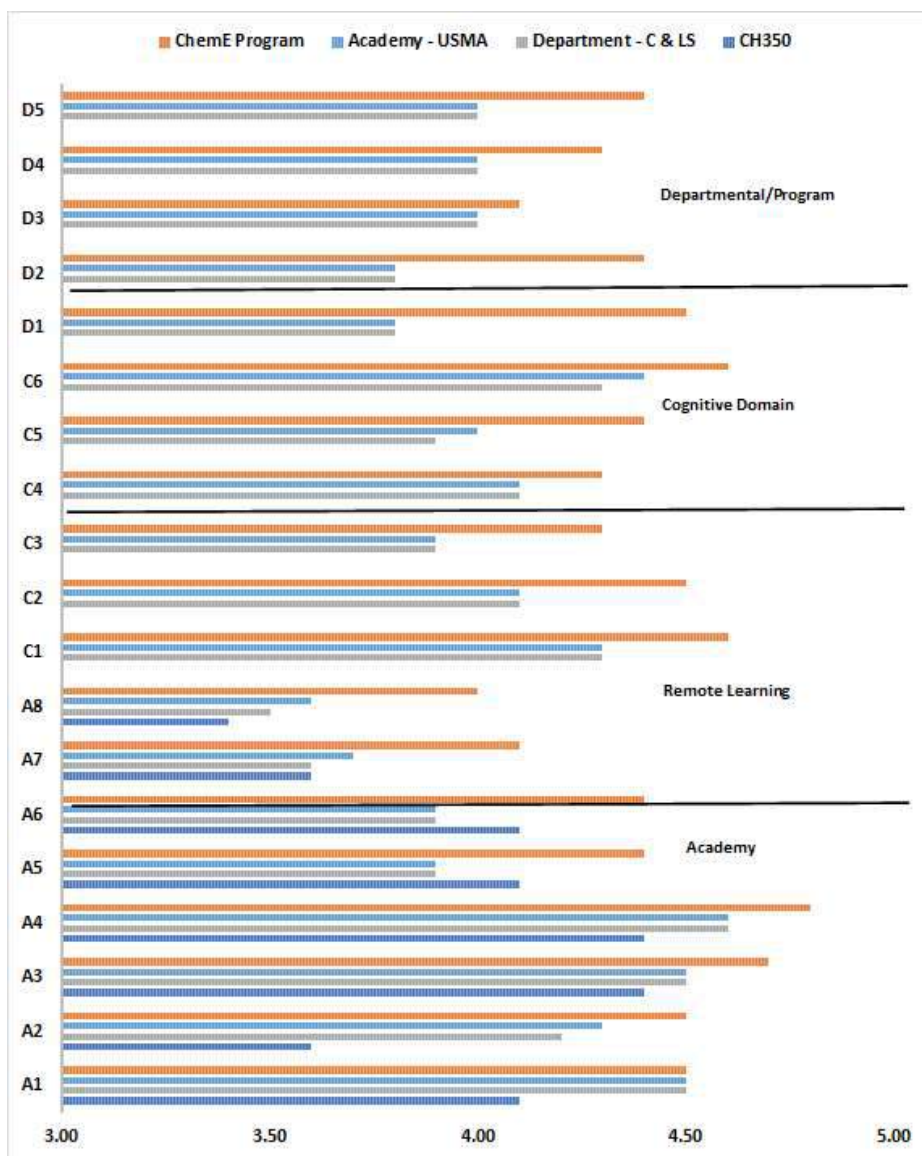


Figure 2. Course Objectives Performance by Graded Event.

4. Survey Questions

4a. Web-based:

The raw data from the web-based survey questions is captured in Section II – Attachment 1. The following graph is a summary of each of the web-based questions. For AY 22-1, the number of respondents was 7/10 cadets, or 70.00%. The error bars for the survey appear in the raw data in Attachment 1 and show that the responses for CH350 are within the standard deviations for all departmental, cognitive domain, and academy-wide groups.



4.b. Handwritten Surveys: The student outcomes survey is another method by which we assess student outcomes (SOs). No handwritten survey was conducted in AY22-1 for CH350.

5. Course QPA. The AY22-1 course QPA was 3.597. No comparison can be made against the previous years since a first pilot run of CH350 is executed in AY22-1.

6. TEE Grade. N/A since TEE is not considered in AY22-1.

7. Course Processes.

7.a. Textbook. The DeLisa Bioprocess Engineering textbook is relatively suitable for this course. DeLisa's book is up to date in terms of math and engineering contents. My assessment is that most of the material is presented in a method that is not too difficult for the cadets to follow or understand. One of the noticeable disadvantages of the text is that DeLisa introduces some of core

concepts without giving enough background information for cadets to follow. Also, the unique way to expressing a variety of units throughout the textbook often confuses cadets.

Also, the textbook doesn't offer any example question requiring coding. Additional recourses may be needed if the instructors want to show how the core design equations learned in CH350 can be efficiently solved using the computer programs. Mathematica and ChemCad will be good problem candidates for demonstrating coding demo in the class.

7.b. Course Content. The course content and lesson blocks were appropriate for the course. Background reading and homework problems were assigned prior to each class. After each discussion, the instructor would plan to guide the students through an in-class problem or have the students work an in-class problem.

One major request received by cadets is to perform the in-class problem demonstration. Showing how to solve some of complex problems (from problem sets) step by step has helped cadets to understand how to efficiently tackle and produce the appropriate solutions. Sharing the partial solution further enhances cadets' problem-solving techniques. Most of in-class problems were mainly derived from the examples from the textbook.

All the WPRs administered in CH350 were divided into two different sections. The fundamental questions were placed, so cadets can give in-depth conceptual perspectives on the biological systems. The quantitative questions were implemented, so cadets can have the opportunities of applying the appropriate types of design equations to the various types of bioprocesses.

Often, cadets completed their assigned readings before class, but lesson assignments and problem sets were designed to force them into the text. Cadets generally worked problem sets a day before they were due or, at worst, on the day they were due (by observing the submission site on Teams). Some of them often relied heavily on AI, their smarter colleagues, or any available external sources to solve the problem sets, which potentially degraded the learning of some cadets. This is the general trend that I observed not only from CH350, but for other STEM courses offered in the department.

More emphasize can be placed on the usage of computer program in the class. Since the textbook doesn't present any actual coding problems, additional sources might be required to design the coding demo problems. Again, Mathematica and ChemCad can be used to construct the coding problems as part of problem set or major graded events.

7.c. Lessons and Labs. The lessons are appropriate for this course. The sequence of lessons is arranged in a way so that cadets can see the course contents are moving from molecular scale to reactor scale. No labs were conducted in AY22-1 for CH350.

7.d. Summary of Graded Requirements. The number and type of graded requirements was appropriate for the course. A large bank of problems should be developed so instructors can rotate problems from year to year. Also, I can see that the number of problem sets assigned could be increased to further reinforce the core concepts learned in the class.

7.e. Areas of Special Emphasis. The areas of special emphasis were appropriate for the course.

8. Contribution to Student Outcomes

8.a. Coverage. Contribution to student objectives was assessed using the same method as described in Section II.3. Figure 3 shows coverage of student curriculum outcomes. The spreadsheet containing the data for this analysis is attached to the end of this report, in the section labeled "Section II - Attachment 2 – Assessment Worksheet." The items marked with **red dashed boxes** in Figures 3 designate student directives from Section I.8.

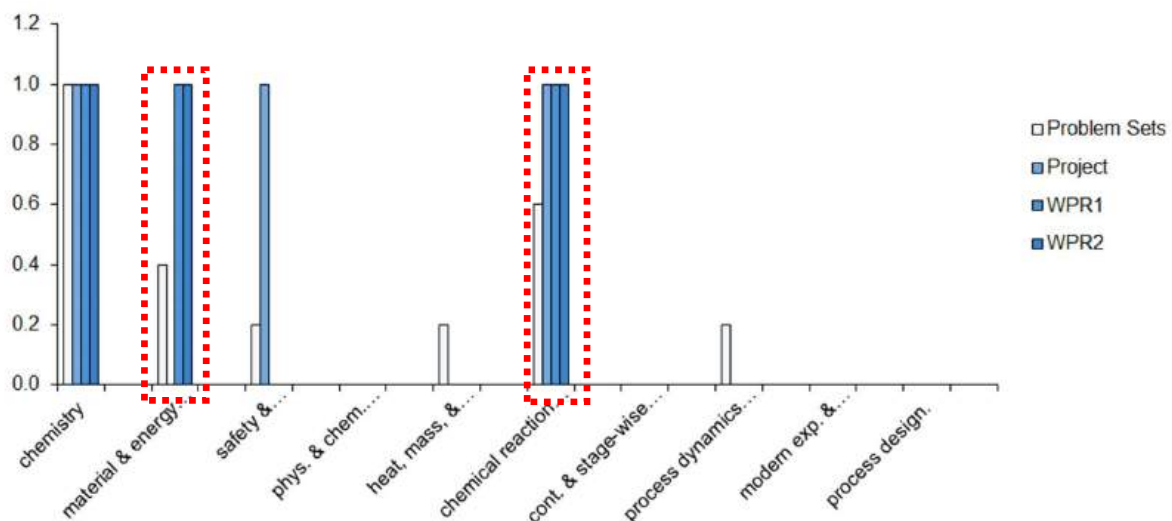


Figure 3. Student Curriculum Outcomes Coverage by Graded Event.

8.b. Performance

Performance on the student outcomes is assessed using the cadet scores from the indicators from the graded events that are outlined above. The results are shown in Figures 4 below. The average for all graded events met or exceeded the 67% standard set forth by USMA for a passing grade. However, some of the individual topics did not average to 67%. Cadets struggled with some of the Michaelis-Menten kinetic problems during major-graded events.

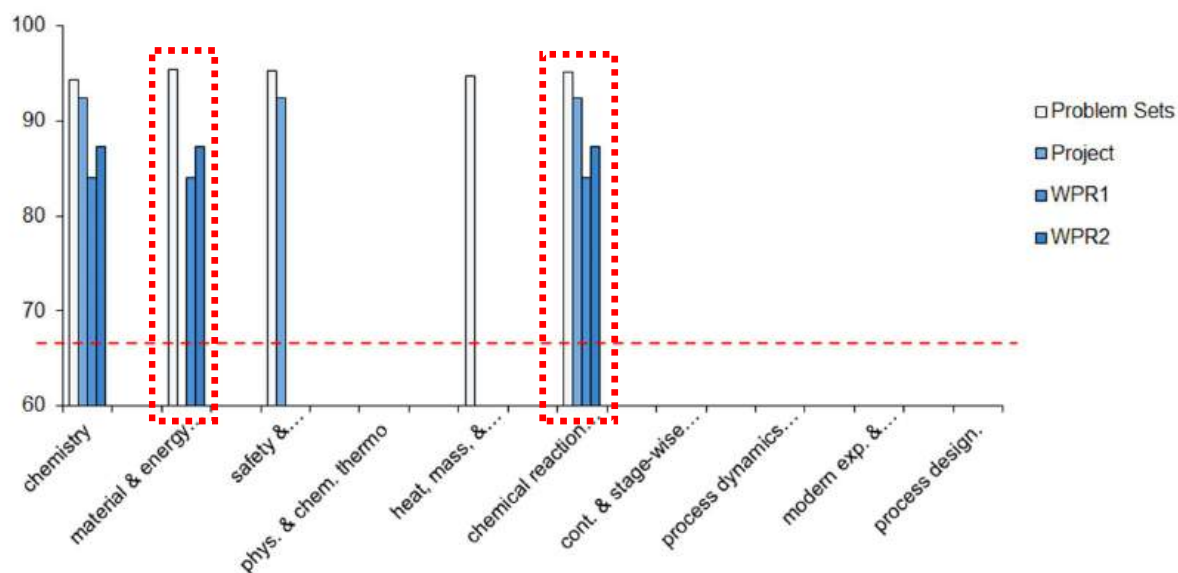


Figure 4. Student Curriculum Outcomes Performance by Graded Event.

9. Resources and Laboratories.

9a. Laboratories. There were no wet labs conducted in this course.

9b. Computer Labs. All class lectures met in BH340. No computer labs were executed in CH350.

9c. Physical Models & Demos. There are no physical models or demos used in the course.

9d. Technician Support. IT support from CLS for maintenance of computers in BH340. The IT support from CLS was adequate.

9e. Supplies. The supplies for the computer lab were adequate (no wet labs were conducted in that lab).

9f. Additional Facilities. None were used - adequate.

9g. Unfunded Requests. None - adequate.

SECTION III: RECOMMENDED CHANGES

1. Red Book Description. No change

2. Enrollment Numbers. Based on the projected enrollment of 27, the course should maintain two sections and use both associated lab periods both physically in the classroom and as distance learning.

3. Course Objectives. No change since a first pilot run of CH350 is executed in AY22-1.

3.a. Coverage. No change.

3.b. Performance. No change.

4. Survey Questions. No change.

4.a. Survey Results. No change.

4.b. Survey Freeform Comments. Continue to include these as options.

5. Course QPA. No change.

6. TEE Grade. No change since there is no TEE in CH350.

7. Course Process

7.a. Current Textbook. No change.

7.b. Course Content. No change.

7.c. Lessons and Labs. No change.

7.d. Summary of Graded Requirements. No change.

7.e. Areas of Special Emphasis. No change.

8. Contribution to Student Outcomes

8.a. Coverage. No change.

8.b. Performance. No change.

9. Resources and Laboratories.

9a. Laboratories. No change.

9b. Computer Labs. None since there is no computer lab for CH350.

9c. Physical Models & Demos. No change.

9d. Technician Support. No change.

9e. Supplies. No change

9f. Additional Facilities. No change.

9g. Unfunded Requests. No change.

SECTION IV: GRADED EVENTS AND SUPPLEMENTAL MATERIAL

- 1. Approved Solutions to all WPRS, Quizzes, and Labs.** Digital records are in the CH350 sharepoint graded events directory.
- 2. All supplemental material provided to cadets not included in their book issue purchase.** Digital records are in the CH350 directory.
- 3. A listing of any other resource that provided significant instructional material for cadet use. (Web sites, shared programs, etc.)**

None for CH350.

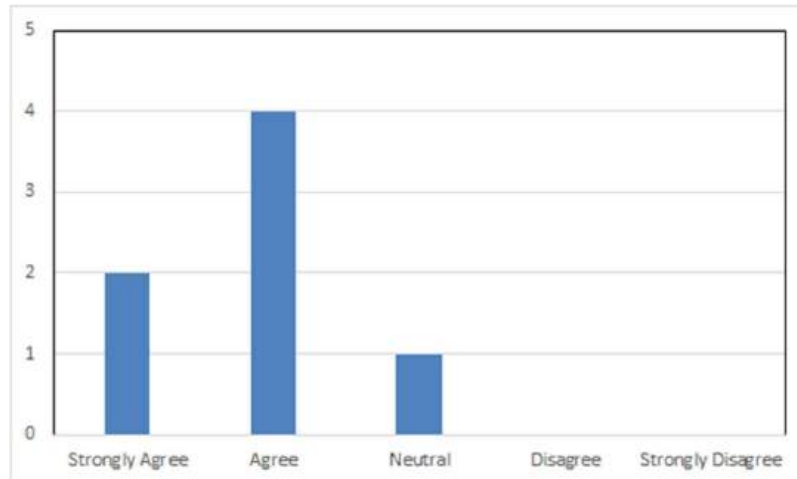
4. Other historical information

4.a. Course Director's Notes

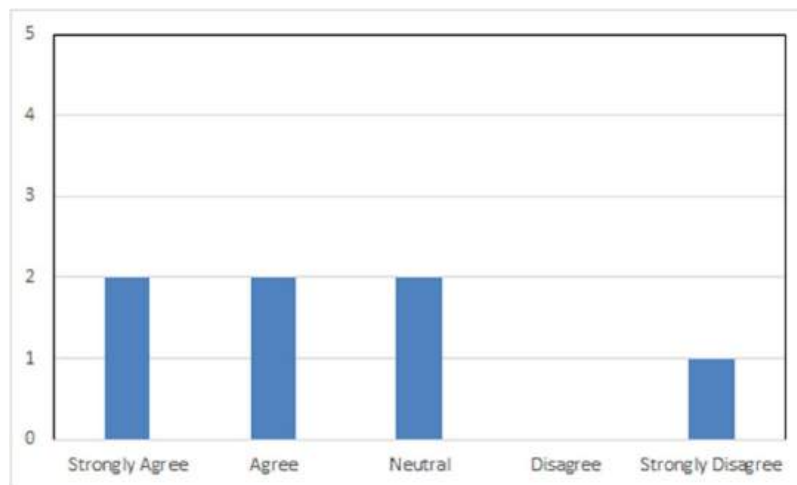
See course director's notebook.

SECTION II: ATTACHMENT 1 – WEB-BASED SURVEY RAW DATA

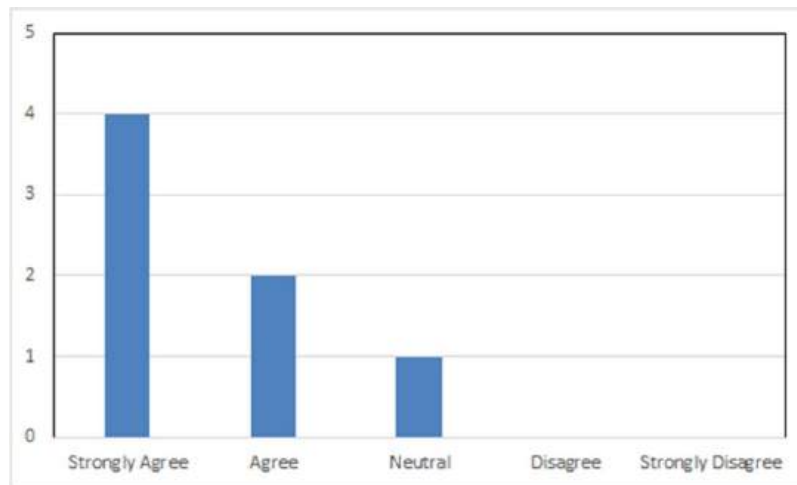
This instructor encouraged students to be responsible for their own learning:



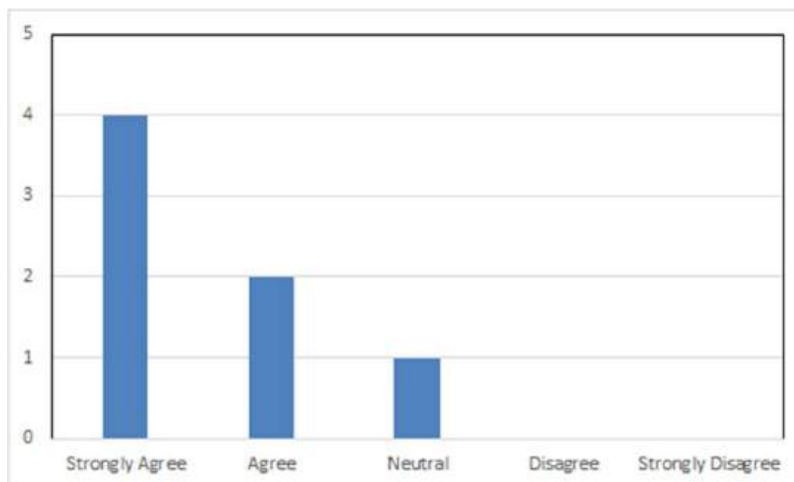
This instructor used effective techniques for learning, both in class and for out-of-class assignments:



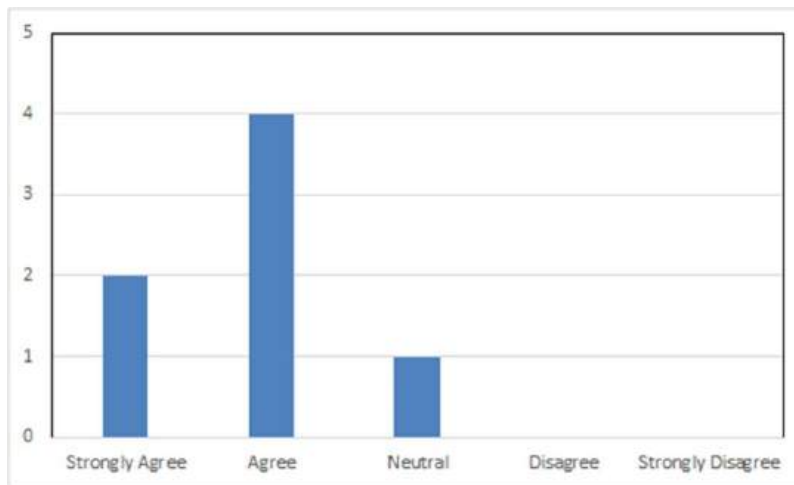
My instructor cared about my learning in this course:



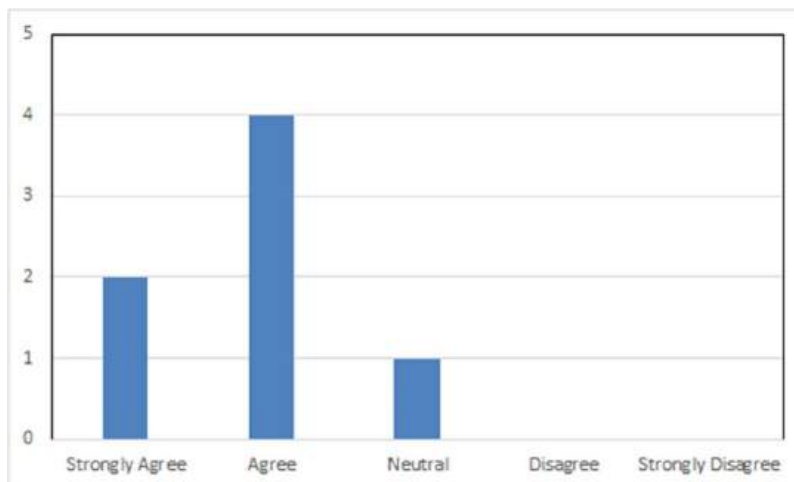
My instructor demonstrated respect for cadets as individuals



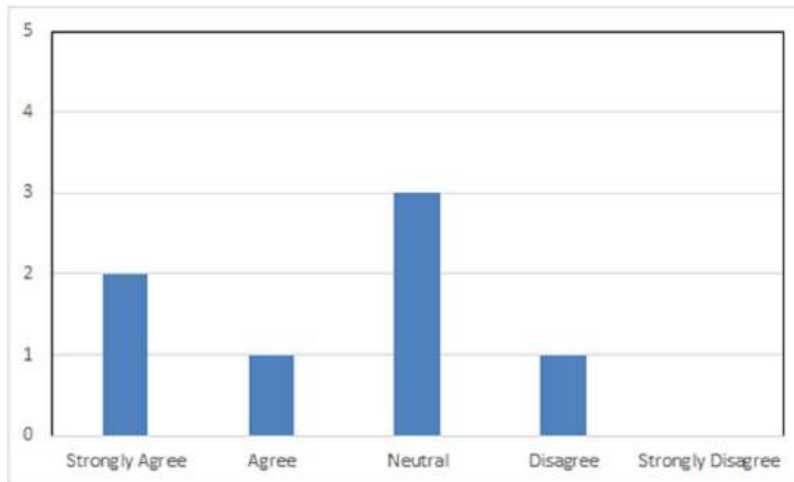
My fellow students contributed to my learning in this course



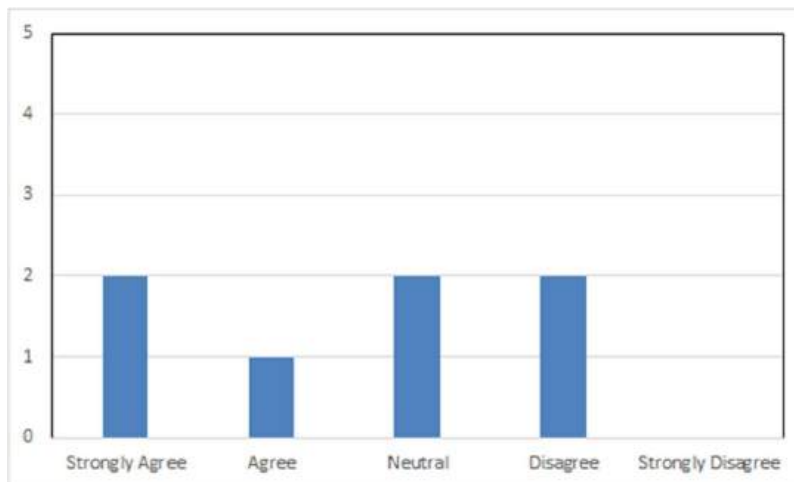
My motivation to learn and to continue learning has increased because of this course



My personal schedule allows me enough time to reflect on the material I have learned in class



My personal schedule allows me enough time to adequately prepare for my optimum academic performance



SECTION II – ATTACHMENT 2

HANDWRITTEN SURVEY YEARLY COMPARISON

SECTION II – ATTACHMENT 3
STUDENT OUTCOMES WORKSHEETS

Coverage	PS1	PS2	PS3	PS4	PS5	PS	Capstone Presentation	Capstone Paper	Project	WPR1	WPR1	WPR2	WPR2
Student Outcomes (Outcomes are abbreviated)	← Categories Carried Forward from Grades Spreadsheet →												
On completion of the program, our students have demonstrated that they can:													
identify, formulate & solve by applying mse	0	1	1	1	1	0.8	0	0	0.0	1	1.0	1	1.0
design to meet needs	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0
communicate	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0
professional ethical impact	1	0	0	0	0	0.2	1	1	1.0	0	0.0	0	0.0
function on a team	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0
experiments	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0
new knowledge	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0
The program provides a thorough grounding and working knowledge of:													
chemistry	1	1	1	1	1	1.0	1	1	1.0	1	1.0	1	1.0
material & energy balances	0	1	0	1	0	0.4	0	0	0.0	1	1.0	1	1.0
safety & environmental	1	0	0	0	0	0.2	1	1	1.0	0	0.0	0	0.0
phys. & chem. thermo	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0
heat, mass, & momentum transfer	0	0	0	0	1	0.2	0	0	0.0	0	0.0	0	0.0
chemical reaction engineering	0	1	0	1	1	0.6	1	1	1.0	1	1.0	1	1.0
cont. & stage-wise separations	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0
process dynamics & control	0	0	0	0	1	0.2	0	0	0.0	0	0.0	0	0.0
modern exp. & comp. tech.	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0
process design	0	0	0	0	0	0.0	0	0	0.0	0	0.0	0	0.0
Course Objectives													
Solve problems by applying mse	0	1	0	1	1	0.6	1	1	1.0	1	1.0	1	1.0
Relate bioprocess eng. to other fields	0	0	0	0	1	0.2	1	1	1.0	0	0.0	0	0.0
Utilize bioengineering principles	1	1	1	1	1	1.0	1	1	1.0	1	1.0	1	1.0
Communicate efficiently	0	0	0	0	0	0.0	1	1	1.0	0	0.0	0	0.0
Raw Grade Data Average out of 100	← Data Carried Forward from Grades Spreadsheet →												
	35	35	31	36	35		35	30		84		87	

	PS1	PS2	PS3	PS4	PS5	PS			Capstone Paper	Capstone Paper	Project		WPR1	WPR1		WPR2	WPR2
Performance																	
Student Outcomes																	
(Outcomes are abbreviated)																	
On completion of the program, our students have demonstrated that they can:																	
id, form & solve by appl mse	0	95	91	96	95	94			0	0	0		84	84		87	87
design to meet needs	0	0	0	0	0	0			0	0	0		0	0		0	0
communicate	0	0	0	0	0	0			95	90	92		0	0		0	0
professional ethical impact	95	0	0	0	0	95			95	90	92		0	0		0	0
function on a team	0	0	0	0	0	0			0	0	0		0	0		0	0
experiments	0	0	0	0	0	0			0	0	0		0	0		0	0
new knowledge	0	0	0	0	0	0			95	90	92		0	0		0	0
The program provides a thorough grounding and working knowledge of:																	
chemistry	95	95	91	96	95	94			95	90	92		84	84		87	87
material & energy balances	0	95	0	96	0	95			0	0	0		84	84		87	87
safety & environmental	95	0	0	0	0	95			95	90	92		0	0		0	0
phys. & chem. thermo	0	0	0	0	0	0			0	0	0		0	0		0	0
heat, mass, & momentum transfer	0	0	0	0	95	95			0	0	0		0	0		0	0
chemical reaction engineering	0	95	0	96	95	95			95	90	92		84	84		87	87
cont. & stage-wise separations	0	0	0	0	0	0			0	0	0		0	0		0	0
process dynamics & control	0	0	0	0	95	9			0	0	0		0	0		0	0
modern exp. & comp. tech.	0	0	0	0	0	0			0	0	0		0	0		0	0
process design	0	0	0	0	0	0			0	0	0		0	0		0	0
Course Objectives																	
Solve problems by applying mse	0	95	0	96	95	95			95	90	92		84	84		87	87
Relate bioprocess eng. to other fields	0	0	0	0	95	95			95	90	92		0	0		0	0
Utilize bioengineering principles	95	95	91	96	95	94			95	90	92		84	84		87	87
Communicate efficiently	0	0	0	0	0	0			95	90	92		0	0		0	0



UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION II

EXAMINATIONS



TAB F WRITTEN PARTIAL REVIEW #1

Handouts

MC: _____ / 120__
Short Answer: _____ / 80__
TOTAL: _____ / 200__

CH350 WPR I
AY26-1

Name: _____

Section: _____

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True _____

False _____

Multiple Choice (2.5 points each)

1. Which best explains why the FDA requires both product and process approval for a new biopharmaceutical?
 - A. The product defines safety, while the process has no regulatory role
 - B. Changes in process may alter safety or efficacy, even if the product looks the same
 - C. The process only impacts yield, not safety
 - D. Only the final product matters to patient outcomes

2. Which historical event most accelerated the development of large-scale bioprocess engineering?
 - A. The discovery of DNA double helix
 - B. WWII demand for penicillin
 - C. The Human Genome Project
 - D. Development of CRISPR-Cas9

3. Why is validation critical under Good Manufacturing Practice (GMP)?
 - A. It ensures documentation exists to prove consistent product quality
 - B. It guarantees higher product yield
 - C. It reduces the need for FDA approval
 - D. It makes laboratory testing unnecessary

4. A bioprocess engineer is designing a Taxol production system. Which host is most sustainable long-term?
 - A. Harvesting yew tree bark
 - B. Suspension plant cell cultures
 - C. Chemical synthesis from petroleum
 - D. Direct extraction from rainforest soils

5. Gram-negative bacteria are often harder to engineer for extracellular protein secretion because:
 - A. They lack ribosomes
 - B. Their thick peptidoglycan blocks protein diffusion
 - C. They possess an additional outer membrane with lipopolysaccharides
 - D. They lack cytoplasmic transport systems

6. Which observation best distinguishes archaea from bacteria?
 - A. Archaea cell walls lack peptidoglycan
 - B. Both reproduce by binary fission
 - C. Bacteria contain circular chromosomes
 - D. Archaea lack ribosomes

7. If your goal is rapid, low-cost protein production without post-translational modifications, the best host system is:
- A. Mammalian cells (CHO)
 - B. Yeast (*Saccharomyces*)
 - C. *E. coli*
 - D. Plant cells
8. Which structural feature explains why Gram-positive bacteria are often used for industrial secretion?
- A. They lack a thick cell wall
 - B. They lack an outer membrane, simplifying secretion
 - C. They contain more ribosomes
 - D. They divide more slowly than Gram-negatives
9. Which set of biomolecules is essential for a minimal synthetic cell to replicate and synthesize proteins?
- A. Proteins and lipids
 - B. Carbohydrates and lipids
 - C. Nucleic acids, proteins, lipids, and carbohydrates
 - D. Carbohydrates and nucleic acids
10. Which biopolymer is primarily responsible for catalysis in living systems?
- A. Lipids
 - B. Proteins
 - C. Carbohydrates
 - D. DNA
11. Which RNA molecule acts as the adapter translating mRNA codons into amino acids?
- A. mRNA
 - B. tRNA
 - C. rRNA
 - D. ncRNA
12. Which nutrient medium type provides precise reproducibility for FDA-regulated production?
- A. Complex medium
 - B. Enriched medium
 - C. Minimal medium
 - D. Defined medium

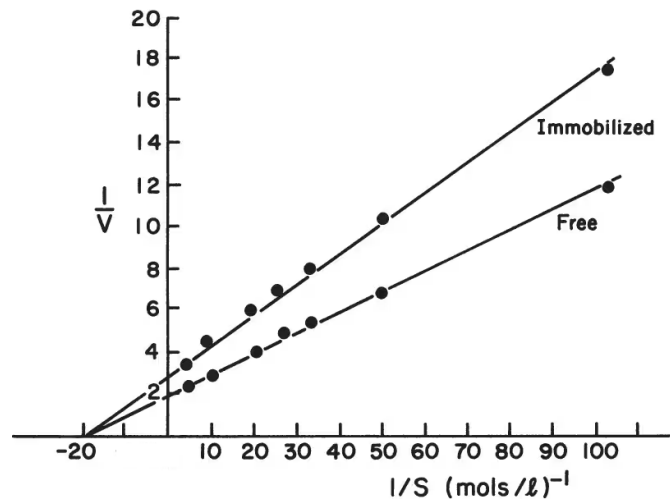
13. Which observation most strongly supports Michaelis–Menten kinetics rather than first-order kinetics?
- A. Doubling substrate always doubles rate
 - B. Rate approaches a maximum as substrate increases
 - C. Product forms linearly with time at all [S]
 - D. The enzyme concentration is irrelevant
14. Which model describes enzymes and substrate specificity based on complimentary three-dimensional structures?
- A. Induced-fit model
 - B. Lock-and-key model
 - C. Random collision model
 - D. Transition-state theory
15. Which type of plot best estimates $V_{\max}$ in enzyme kinetics?
- A. Lineweaver-Burk
 - B. Eadie-Hofstee
 - C. Hanes-Woolf
 - D. Scatchard
16. In enzyme catalysis, lowering activation energy is achieved by:
- A. Stabilizing transition states
 - B. Raising substrate energy
 - C. Producing more enzymes
 - D. Increasing product energy
17. If a Lineweaver–Burk plot shows inhibitor lines intersecting on the y-axis, this indicates:
- A. Substrate inhibition
 - B. Noncompetitive inhibition
 - C. Uncompetitive inhibition
 - D. Competitive inhibition
18. An enzyme shows identical $V_{\max}$ values with and without inhibitor, but K_m increases. What type of inhibition is this?
- A. Competitive
 - B. Noncompetitive
 - C. Uncompetitive
 - D. Irreversible
19. An allosteric enzyme shows a sigmoidal substrate-velocity curve because:
- A. Binding is cooperative between subunits
 - B. It has a single binding site
 - C. Its activity is unaffected by effectors
 - D. It follows zero-order kinetics

20. Which measure best reflects catalytic efficiency?
- A. K_m
 - B. k_{cat}
 - C. k_{cat}/K_m
 - D. V_{max}
21. Immobilized enzymes often show reduced observed activity compared to free enzymes primarily because:
- A. The active site is chemically destroyed
 - B. Mass transfer limitations restrict substrate diffusion
 - C. They cannot bind cofactors
 - D. Covalent bonds block activity
22. Which immobilization strategy traps enzymes inside a polymer matrix?
- A. Adsorption
 - B. Covalent bonding
 - C. Entrapment
 - D. Cross-linking
23. Which advantage is unique to immobilized enzyme systems?
- A. Higher intrinsic turnover number
 - B. Reuse of enzymes across multiple cycles
 - C. Ability to catalyze new reactions
 - D. Removal of inhibition
24. Which factor decreases the effectiveness factor (η) in porous immobilized systems?
- A. Decreasing particle size
 - B. Decreasing enzyme loading
 - C. Increasing particle diameter
 - D. Improving substrate diffusivity
25. Which binding change most likely increases affinity in a protein–ligand system?
- A. Higher ΔG
 - B. Lower ΔH and favorable ΔS
 - C. Positive ΔH and negative ΔS
 - D. Reduced hydrogen bonding
26. Which force primarily drives the hydrophobic effect?
- A. Hydrogen bonding
 - B. Exclusion of water molecules
 - C. Ionic interactions
 - D. Covalent bonds

27. Which experimental observation best indicates van der Waals interactions?
- A. Distance-dependent weak attractions between nonpolar atoms
 - B. Strong permanent ionic bonds
 - C. Release of water molecules
 - D. Formation of disulfide bridges
28. Which equilibrium constant defines ligand affinity?
- A. k_{cat}
 - B. K_m
 - C. V_{max}
 - D. K_d
29. A mutation that blocks RNA polymerase binding to DNA directly prevents:
- A. Replication
 - B. Transcription
 - C. Translation
 - D. Protein folding
30. Which type of RNA is the template for protein synthesis?
- A. mRNA
 - B. tRNA
 - C. rRNA
 - D. snRNA
31. Which posttranslational modification adds sugars to proteins?
- A. Phosphorylation
 - B. Glycosylation
 - C. Methylation
 - D. Ubiquitination
32. Which regulatory mechanism allows transcription to be switched on by disabling a repressor?
- A. Induction
 - B. Repression
 - C. Activation
 - D. Feedback inhibition
33. Which step in glycolysis consumes ATP rather than producing it?
- A. Phosphoenolpyruvate $\rightarrow$ Pyruvate
 - B. Glucose $\rightarrow$ Glucose-6-phosphate
 - C. 1,3-BPG $\rightarrow$ 3-PG
 - D. G3P $\rightarrow$ 1,3-BPG

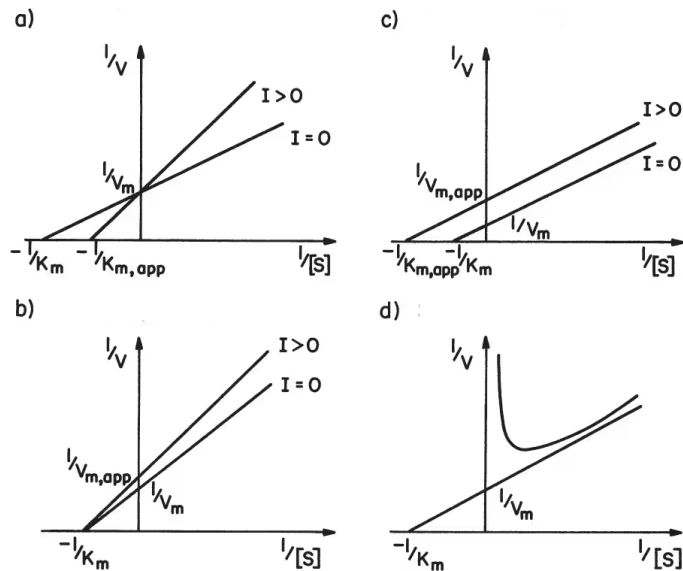
34. Which coenzyme carries reducing power during glycolysis and the TCA cycle?
- A. FAD
 - B. ATP
 - C. NADH
 - D. CoA
35. How many ATP molecules are produced (net) per glucose in glycolysis?
- A. 1
 - B. 2
 - C. 4
 - D. 36
36. Which cycle generates FADH₂, CO₂, and NADH?
- A. Calvin cycle
 - B. TCA cycle
 - C. Pentose phosphate pathway
 - D. Glyoxylate cycle
37. The Pasteur effect is observed when:
- A. Glycolysis accelerates in the presence of oxygen
 - B. Glycolysis slows because aerobic respiration provides more ATP per glucose
 - C. Fermentation rate increases under aerobic conditions
 - D. The TCA cycle halts in oxygen presence
38. Which pathway generates NADPH and ribose-5-phosphate?
- A. Glycolysis
 - B. TCA cycle
 - C. Pentose phosphate pathway
 - D. Electron transport chain
39. Which enzyme is the key regulatory site in glycolysis inhibited by high ATP?
- A. Hexokinase
 - B. Phosphofructokinase
 - C. Pyruvate kinase
 - D. Citrate synthase
40. Which phase of photosynthesis fixes CO₂ into carbohydrates?
- A. Light reactions
 - B. Photolysis
 - C. Electron transport
 - D. Dark reactions

41. D. Thornton and coworkers studied the hydrolysis of sucrose at pH = 4.5 and 25°C using crude invertase obtained from baker's yeast in free and immobilized form. The following initial velocity data were obtained with 408 units of crude enzyme (1 unit = quantity of enzyme hydrolyzing 1 μmol of sucrose/min when incubated with 0.29 M sucrose in a buffer at pH 4.5 and 25°C). Using the plot of their data below, what is the effect of immobilization on the K_m and V_m ?



- A. No change to either K_m and V_m
- B. K_m increases while V_m remains unchanged
- C. K_m remains unchanged while V_m decreases
- D. Both K_m and V_m increase

42. Which of the following Lineweaver-Burke plots indicate competitive inhibition?



43. What phase of the cell cycle does the genetic material of the cell double?
- A. Interphase
 - B. Anaphase
 - C. G2
 - D. Mitotic Phase
44. You want to produce a therapeutic protein requiring glycosylation. Which host system is most appropriate?
- A. *E. coli*
 - B. Yeast (*Saccharomyces*)
 - C. CHO mammalian cells
 - D. Gram-positive bacteria
45. A synthetic cell is constructed with DNA, ribosomes, and lipids, but no RNA polymerase. Which outcome is expected?
- A. DNA replication occurs normally
 - B. Transcription cannot occur, so no protein is produced
 - C. Translation proceeds without RNA
 - D. ATP cannot be generated
46. Two enzyme variants are compared: Enzyme A has $K_m = 1 \text{ mM}$, $V_{max} = 100 \text{ } \mu\text{mol/min}$; Enzyme B has $K_m = 10 \text{ mM}$, $V_{max} = 500 \text{ } \mu\text{mol/min}$. Which enzyme is more efficient at low substrate concentrations?
- A. Enzyme A
 - B. Enzyme B
 - C. Both equal
 - D. Cannot determine
47. Two ligands show the same ΔG for binding, but ligand A has a more favorable ΔS . Which is more likely to maintain binding at higher temperatures?
- A. Both are equal
 - B. Cannot determine
 - C. Ligand B
 - D. Ligand A
48. You need high-yield insulin production in *E. coli*. Which strategy best ensures consistent results?
- A. Use a plasmid with a strong inducible promoter
 - B. Grow cells in complex medium without monitoring
 - C. Rely on spontaneous expression
 - D. Replace *E. coli* with plant cells

Short Answer (10 points each)

1. Provide 3 differences between prokaryotic and eukaryotic cells in terms of structure and function.

2. Explain the light-dependent vs. light-independent reactions of photosynthesis (include reactants and products).

3. Describe the central dogma of molecular biology.

4. Explain the Michaelis-Menten model of enzyme kinetics and the significance of K_m and V_{max} . (Provide 3 key assumptions)

5. Name and describe the three major types of enzyme inhibition.

6. Explain the advantages and disadvantages of immobilized enzyme systems.

7. Name and describe the three main mechanisms of horizontal gene transfer in bacteria.

8. Describe the role of aerobic metabolism in energy production and its regulation in cells.

Answer Key

1. B
2. B
3. A
4. B
5. C
6. A
7. C
8. B
9. C
10. B
11. B
12. D
13. B
14. B
15. A
16. A
17. D
18. A
19. A
20. C
21. B
22. C
23. B
24. C
25. B
26. B
27. A
28. D
29. B
30. A
31. B
32. A
33. B
34. C
35. B
36. B
37. B
38. C
39. B
40. D
41. C
42. A
43. A
44. C
45. B
46. A
47. D
48. A

Instructor Solution

SOLUTION

MC: _____ / 120__
Short Answer: _____ / 80__
TOTAL: _____ / 200__

CH350 WPR I AY26-1

Name: _____

Section: _____

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True _____

False _____

Multiple Choice (2.5 points each)

1. Which best explains why the FDA requires both product and process approval for a new biopharmaceutical?
 - A. The product defines safety, while the process has no regulatory role
 - B. Changes in process may alter safety or efficacy, even if the product looks the same
 - C. The process only impacts yield, not safety
 - D. Only the final product matters to patient outcomes

2. Which historical event most accelerated the development of large-scale bioprocess engineering?
 - A. The discovery of DNA double helix
 - B. WWII demand for penicillin
 - C. The Human Genome Project
 - D. Development of CRISPR-Cas9

3. Why is validation critical under Good Manufacturing Practice (GMP)?
 - A. It ensures documentation exists to prove consistent product quality
 - B. It guarantees higher product yield
 - C. It reduces the need for FDA approval
 - D. It makes laboratory testing unnecessary

4. A bioprocess engineer is designing a Taxol production system. Which host is most sustainable long-term?
 - A. Harvesting yew tree bark
 - B. Suspension plant cell cultures
 - C. Chemical synthesis from petroleum
 - D. Direct extraction from rainforest soils

5. Gram-negative bacteria are often harder to engineer for extracellular protein secretion because:
 - A. They lack ribosomes
 - B. Their thick peptidoglycan blocks protein diffusion
 - C. They possess an additional outer membrane with lipopolysaccharides
 - D. They lack cytoplasmic transport systems

6. Which observation best distinguishes archaea from bacteria?
 - A. Archaea cell walls lack peptidoglycan
 - B. Both reproduce by binary fission
 - C. Bacteria contain circular chromosomes
 - D. Archaea lack ribosomes

7. If your goal is rapid, low-cost protein production without post-translational modifications, the best host system is:
- A. Mammalian cells (CHO)
 - B. Yeast (*Saccharomyces*)
 - C. *E. coli*
 - D. Plant cells
8. Which structural feature explains why Gram-positive bacteria are often used for industrial secretion?
- A. They lack a thick cell wall
 - B. They lack an outer membrane, simplifying secretion
 - C. They contain more ribosomes
 - D. They divide more slowly than Gram-negatives
9. Which set of biomolecules is essential for a minimal synthetic cell to replicate and synthesize proteins?
- A. Proteins and lipids
 - B. Carbohydrates and lipids
 - C. Nucleic acids, proteins, lipids, and carbohydrates
 - D. Carbohydrates and nucleic acids
10. Which biopolymer is primarily responsible for catalysis in living systems?
- A. Lipids
 - B. Proteins
 - C. Carbohydrates
 - D. DNA
11. Which RNA molecule acts as the adapter translating mRNA codons into amino acids?
- A. mRNA
 - B. tRNA
 - C. rRNA
 - D. ncRNA
12. Which nutrient medium type provides precise reproducibility for FDA-regulated production?
- A. Complex medium
 - B. Enriched medium
 - C. Minimal medium
 - D. Defined medium

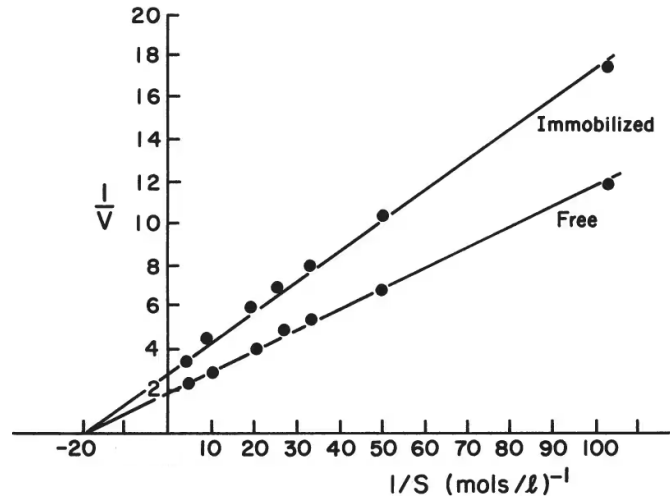
13. Which observation most strongly supports Michaelis–Menten kinetics rather than first-order kinetics?
- A. Doubling substrate always doubles rate
 - B. Rate approaches a maximum as substrate increases
 - C. Product forms linearly with time at all [S]
 - D. The enzyme concentration is irrelevant
14. Which model describes enzymes and substrate specificity based on complimentary three-dimensional structures?
- A. Induced-fit model
 - B. Lock-and-key model
 - C. Random collision model
 - D. Transition-state theory
15. Which type of plot best estimates $V_{\max}$ in enzyme kinetics?
- A. Lineweaver-Burk
 - B. Eadie-Hofstee
 - C. Hanes-Woolf
 - D. Scatchard
16. In enzyme catalysis, lowering activation energy is achieved by:
- A. Stabilizing transition states
 - B. Raising substrate energy
 - C. Producing more enzymes
 - D. Increasing product energy
17. If a Lineweaver–Burk plot shows inhibitor lines intersecting on the y-axis, this indicates:
- A. Substrate inhibition
 - B. Noncompetitive inhibition
 - C. Uncompetitive inhibition
 - D. Competitive inhibition
18. An enzyme shows identical $V_{\max}$ values with and without inhibitor, but K_m increases. What type of inhibition is this?
- A. Competitive
 - B. Noncompetitive
 - C. Uncompetitive
 - D. Irreversible
19. An allosteric enzyme shows a sigmoidal substrate-velocity curve because:
- A. Binding is cooperative between subunits
 - B. It has a single binding site
 - C. Its activity is unaffected by effectors
 - D. It follows zero-order kinetics

20. Which measure best reflects catalytic efficiency?
- A. K_m
 - B. k_{cat}
 - C. k_{cat}/K_m
 - D. V_{max}
21. Immobilized enzymes often show reduced observed activity compared to free enzymes primarily because:
- A. The active site is chemically destroyed
 - B. Mass transfer limitations restrict substrate diffusion
 - C. They cannot bind cofactors
 - D. Covalent bonds block activity
22. Which immobilization strategy traps enzymes inside a polymer matrix?
- A. Adsorption
 - B. Covalent bonding
 - C. Entrapment
 - D. Cross-linking
23. Which advantage is unique to immobilized enzyme systems?
- A. Higher intrinsic turnover number
 - B. Reuse of enzymes across multiple cycles
 - C. Ability to catalyze new reactions
 - D. Removal of inhibition
24. Which factor decreases the effectiveness factor (η) in porous immobilized systems?
- A. Decreasing particle size
 - B. Decreasing enzyme loading
 - C. Increasing particle diameter
 - D. Improving substrate diffusivity
25. Which binding change most likely increases affinity in a protein–ligand system?
- A. Higher ΔG
 - B. Lower ΔH and favorable ΔS
 - C. Positive ΔH and negative ΔS
 - D. Reduced hydrogen bonding
26. Which force primarily drives the hydrophobic effect?
- A. Hydrogen bonding
 - B. Exclusion of water molecules
 - C. Ionic interactions
 - D. Covalent bonds

27. Which experimental observation best indicates van der Waals interactions?
- A. Distance-dependent weak attractions between nonpolar atoms
 - B. Strong permanent ionic bonds
 - C. Release of water molecules
 - D. Formation of disulfide bridges
28. Which equilibrium constant defines ligand affinity?
- A. k_{cat}
 - B. K_m
 - C. V_{max}
 - D. K_d
29. A mutation that blocks RNA polymerase binding to DNA directly prevents:
- A. Replication
 - B. Transcription
 - C. Translation
 - D. Protein folding
30. Which type of RNA is the template for protein synthesis?
- A. mRNA
 - B. tRNA
 - C. rRNA
 - D. snRNA
31. Which posttranslational modification adds sugars to proteins?
- A. Phosphorylation
 - B. Glycosylation
 - C. Methylation
 - D. Ubiquitination
32. Which regulatory mechanism allows transcription to be switched on by disabling a repressor?
- A. Induction
 - B. Repression
 - C. Activation
 - D. Feedback inhibition
33. Which step in glycolysis consumes ATP rather than producing it?
- A. Phosphoenolpyruvate $\rightarrow$ Pyruvate
 - B. Glucose $\rightarrow$ Glucose-6-phosphate
 - C. 1,3-BPG $\rightarrow$ 3-PG
 - D. G3P $\rightarrow$ 1,3-BPG

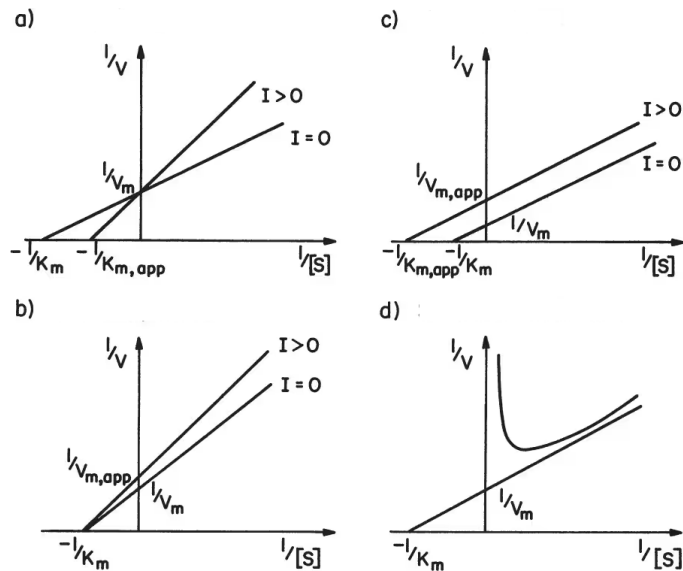
34. Which coenzyme carries reducing power during glycolysis and the TCA cycle?
- A. FAD
 - B. ATP
 - C. NADH
 - D. CoA
35. How many ATP molecules are produced (net) per glucose in glycolysis?
- A. 1
 - B. 2
 - C. 4
 - D. 36
36. Which cycle generates FADH₂, CO₂, and NADH?
- A. Calvin cycle
 - B. TCA cycle
 - C. Pentose phosphate pathway
 - D. Glyoxylate cycle
37. The Pasteur effect is observed when:
- A. Glycolysis accelerates in the presence of oxygen
 - B. Glycolysis slows because aerobic respiration provides more ATP per glucose
 - C. Fermentation rate increases under aerobic conditions
 - D. The TCA cycle halts in oxygen presence
38. Which pathway generates NADPH and ribose-5-phosphate?
- A. Glycolysis
 - B. TCA cycle
 - C. Pentose phosphate pathway
 - D. Electron transport chain
39. Which enzyme is the key regulatory site in glycolysis inhibited by high ATP?
- A. Hexokinase
 - B. Phosphofructokinase
 - C. Pyruvate kinase
 - D. Citrate synthase
40. Which phase of photosynthesis fixes CO₂ into carbohydrates?
- A. Light reactions
 - B. Photolysis
 - C. Electron transport
 - D. Dark reactions

41. D. Thornton and coworkers studied the hydrolysis of sucrose at pH = 4.5 and 25°C using crude invertase obtained from baker's yeast in free and immobilized form. The following initial velocity data were obtained with 408 units of crude enzyme (1 unit = quantity of enzyme hydrolyzing 1 μmol of sucrose/min when incubated with 0.29 M sucrose in a buffer at pH 4.5 and 25°C). Using the plot of their data below, what is the effect of immobilization on the K_m and V_m ?



- A. No change to either K_m and V_m
- B. K_m increases while V_m remains unchanged
- C. K_m remains unchanged while V_m decreases
- D. Both K_m and V_m increase

42. Which of the following Lineweaver-Burk plots indicate competitive inhibition?



43. What phase of the cell cycle does the genetic material of the cell double?
- A. Interphase
 - B. Anaphase
 - C. G2
 - D. Mitotic Phase
44. You want to produce a therapeutic protein requiring glycosylation. Which host system is most appropriate?
- A. *E. coli*
 - B. Yeast (*Saccharomyces*)
 - C. CHO mammalian cells
 - D. Gram-positive bacteria
45. A synthetic cell is constructed with DNA, ribosomes, and lipids, but no RNA polymerase. Which outcome is expected?
- A. DNA replication occurs normally
 - B. Transcription cannot occur, so no protein is produced
 - C. Translation proceeds without RNA
 - D. ATP cannot be generated
46. Two enzyme variants are compared: Enzyme A has $K_m = 1 \text{ mM}$, $V_{max} = 100 \text{ } \mu\text{mol/min}$; Enzyme B has $K_m = 10 \text{ mM}$, $V_{max} = 500 \text{ } \mu\text{mol/min}$. Which enzyme is more efficient at low substrate concentrations?
- A. Enzyme A
 - B. Enzyme B
 - C. Both equal
 - D. Cannot determine
47. Two ligands show the same ΔG for binding, but ligand A has a more favorable ΔS . Which is more likely to maintain binding at higher temperatures?
- A. Both are equal
 - B. Cannot determine
 - C. Ligand B
 - D. Ligand A
48. You need high-yield insulin production in *E. coli*. Which strategy best ensures consistent results?
- A. Use a plasmid with a strong inducible promoter
 - B. Grow cells in complex medium without monitoring
 - C. Rely on spontaneous expression
 - D. Replace *E. coli* with plant cells

Short Answer (10 points each)

1. Provide 3 differences between prokaryotic and eukaryotic cells in terms of structure and function.

Nucleus

- Prokaryotes: Lack a true nucleus; DNA is located in a nucleoid region.
- Eukaryotes: Have a membrane-bound nucleus containing linear chromosomes.

Organelles

- Prokaryotes: Do not contain membrane-bound organelles (e.g., mitochondria, ER).
- Eukaryotes: Contain membrane-bound organelles that compartmentalize functions.

Ribosomes and Complexity

- Prokaryotes: Contain 70S ribosomes; generally smaller and structurally simpler.
- Eukaryotes: Contain 80S ribosomes; cells are larger, more complex, and capable of multicellularity.

2. Explain the light-dependent vs. light-independent reactions of photosynthesis (include reactants and products).

Light-dependent reactions (occur in the thylakoid membranes):

- **Reactants:** Light energy, H_2O , $\text{ADP} + \text{P}_i$, NADP^+
- **Products:** ATP, NADPH, O_2
- Water is split (photolysis) to release oxygen, while the energy from sunlight drives the formation of ATP and NADPH, which carry chemical energy.

Light-independent reactions (Calvin cycle, occur in the stroma):

- **Reactants:** CO_2 , ATP, NADPH
- **Products:** Glucose (or other carbohydrates), $\text{ADP} + \text{P}_i$, NADP^+
- ATP and NADPH from the light reactions provide the energy and reducing power to fix carbon dioxide into sugars.

3. Describe the central dogma of molecular biology.

- DNA is replicated, then transcribed into RNA by RNA polymerase.
- mRNA carries information to ribosomes.
- Ribosomes translate codons into amino acids → polypeptide.
- Flow of information: $\text{DNA} \rightarrow \text{RNA} \rightarrow \text{Protein}$.

4. Explain the Michaelis-Menten model of enzyme kinetics and the significance of K_m and V_{max} . (Provide 3 key assumptions)

The Michaelis–Menten model describes how enzyme reaction velocity depends on substrate concentration. The curve shows an initial rapid increase in velocity with $[S]$, which then plateaus at V_{max} (maximum rate when all enzyme is saturated). The K_m is the substrate concentration at which the velocity is half of V_{max} , and it reflects the enzyme's apparent affinity for substrate (lower K_m = higher affinity).

Three key assumptions:

1. The reaction is studied under initial rate conditions, so product formation is negligible (no reverse reaction).
2. The enzyme–substrate complex reaches a steady state (its concentration remains constant over time).
3. Substrate concentration is much greater than enzyme concentration, so substrate is not significantly depleted during the reaction.

5. Name and describe the three major types of enzyme inhibition.

- **Competitive:** binds active site, $\uparrow K_m$, V_{max} unchanged.
- **Noncompetitive:** binds elsewhere, $\downarrow V_{max}$, K_m unchanged.
- **Uncompetitive:** binds ES complex, $\downarrow K_m$ and $\downarrow V_{max}$.

6. Explain the advantages and disadvantages of immobilized enzyme systems.

Immobilized enzyme systems offer several advantages:

- the enzymes can be reused across multiple cycles,
- they often show improved stability against temperature and pH changes,
- and product purification is easier since the enzyme remains fixed. They are also useful for continuous industrial processes.

However, disadvantages include:

- possible reduced catalytic activity due to conformational changes or steric hindrance,
- mass transfer limitations that slow substrate diffusion,
- and higher costs or complexity in preparing immobilized systems.

7. Name and describe the three main mechanisms of horizontal gene transfer in bacteria.

- **Transformation:** uptake of naked DNA from the environment; requires competence factors or artificial induction.
- **Transduction:** DNA transferred between bacteria by a bacteriophage (virus). Can be generalized or specialized.
- **Conjugation:** direct transfer of plasmid DNA from donor to recipient via pilus; often involves F plasmid.

8. Describe the role of aerobic metabolism in energy production and its regulation in cells.

Aerobic metabolism uses oxygen as the final electron acceptor in the electron transport chain, allowing cells to fully oxidize glucose through glycolysis, the TCA cycle, and oxidative phosphorylation. This yields far more ATP (~36–38 per glucose) than anaerobic pathways.

Key regulatory steps include phosphofructokinase (PFK) in glycolysis, which is activated by ADP/Pi and inhibited by high ATP or NADH/NAD⁺ ratios. This regulation ensures that ATP is produced efficiently when energy demand is high but slows when energy levels are already sufficient.

Answer Key

1. B
2. B
3. A
4. B
5. C
6. A
7. C
8. B
9. C
10. B
11. B
12. D
13. B
14. B
15. A
16. A
17. D
18. A
19. A
20. C
21. B
22. C
23. B
24. C
25. B
26. B
27. A
28. D
29. B
30. A
31. B
32. A
33. B
34. C
35. B
36. B
37. B
38. C
39. B
40. D
41. C
42. A
43. A
44. C
45. B
46. A
47. D
48. A



TAB G

WRITTEN PARTIAL REVIEW #2

Handouts

DEPARTMENT OF CHEMICAL AND BIOLOGICAL SCIENCES AND ENGINEERING

CH350, AY26-1
Written Partial Review II
04 November 2025

TEXT: Shuler *et al*, Bioprocess Engineering: Basic Concepts, 3rd Ed.
Wittrup *et al*, Quantitative Fundamentals of Molecular and Cellular Bioengineering.
SCOPE: Lessons 14-21, **75 minutes**

Authorized References

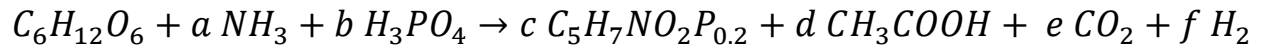
Textbooks (Shuler et al, 3rd Ed. and Wittrup et al), Calculator (TI-30X IIS, TI-36X Pro, or equivalent), 1-page handwritten notes (to be turned in), and Blank Mathematica Worksheet / Excel Sheet

- Do not mark on this exam or open it until "begin work" is given.
- Solve problems in the space provided. Be sure to do the following *to receive full credit*:
 - Print legibly.
 - Show all work (including any digital file, which is needed to be graded).**
 - Use complete sentences for all explanations.
 - Answer all questions and double underline answer for quantitative problems.
- There are **8 questions** in this WPR. Report any illegible or missing pages to the instructor. Write your name on every page in the spaces provided.

(TOTAL WEIGHT 200 POINTS)

THIS SPACE FOR INSTRUCTOR USE ONLY			
PAGE	MAX	CUT	POINTS
1	50		
2	30		
3	20		
4	20		
5	50		
6	10		
7	10		
8	10		
		TOTAL SCORE	/200

1. (50 pts) *Clostridium* sp. were used for anaerobic conversion of glucose ($C_6H_{12}O_6$) to acetic acid (CH_3COOH) and H_2 as described by the following reaction:



- a) Determine the stoichiometric coefficients by writing elemental balances if $Y_{x/s} = 0.1 C_5H_7NO_2P_{0.2} \text{ g}/C_6H_{12}O_6 \text{ g}$ (assuming the molar mass of C, O, H, N, and P as 12 g/mol, 16 g/mol, 1 g/mol, 14 g/mol, and 31 g/mol, respectively).

- b) Determine $\gamma_{C_6H_{12}O_6}$, $\gamma_{C_5H_7NO_2P_{0.2}}$, γ_{CH_3COOH} , and γ_{H_2} (γ stands for the degree of reduction).

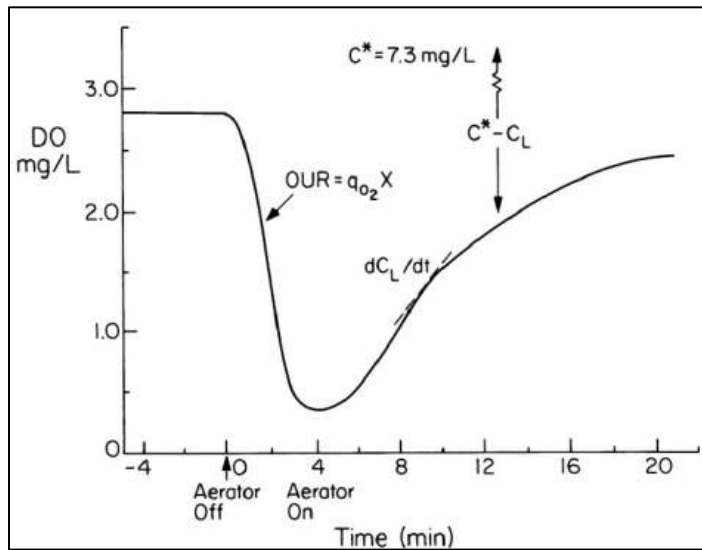
- c) Determine $Y_{p/s}$ for acetic acid (CH_3COOH) and hydrogen (H_2).

2. (30 pts) Two-stage chemostat system is used for fermentation of glucose ($C_6H_{12}O_6$) to ethanol (C_2H_5OH) using *Saccharomyces cerevisiae*. The feed flow rate and initial glucose concentrations are $F = 1$ L/hr and $S_0 = 40$ g/L, respectively. The following kinetic constants are known: $\mu_m = 0.139$ hr⁻¹, $K_s = 1$ g glucose/L, $Y_{x/s} = 0.1$ g cell/g glucose, $Y_{p/s} = 0.5$ g ethanol/g glucose. Note that the feed flow rate is same for the first and second stage of chemostat system.

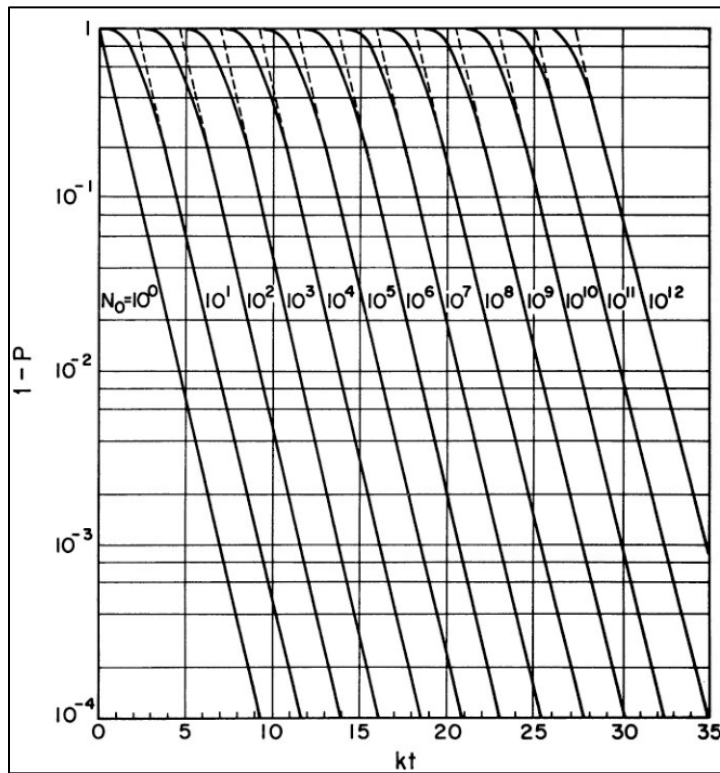
- a) Assuming 80% conversion in every reactor, determine the substrate and product concentrations in the effluent of reactors.
- b) Assuming 80% conversion in every reactor, determine the biomass concentration in both reactors.
- c) Assuming 80% conversion in every reactor, determine the volume of each reactor.

3. (20 pts) Discuss the effects of mixing on sterilization in a batch fermenter. Be sure to include information on thermal implications, common operating temperatures for thermal liquid sterilization processes and implications on sterilization times after scale-up.

4. (20 pts) Estimate $k_L a$ from Figure 10.5 of DeLisa, 2017 (shown below) if C^* is 35 mg/L due to the use of nearly pure oxygen rather than air.



5. (50 pts) An autoclave malfunctions, and the temperature reaches only 119.5 °C. The sterilization time at the maximum temperature was 20 min. The jar contains 10 L of complex medium that has 10^5 spores/L. At 121 °C, $k_d = 1.0 \text{ min}^{-1}$ and $E_{0d} = 90 \text{ kcal/mol}$. Using the Sterilization Chart from the text (shown below), what is the probability that the medium was sterile at 119.5 °C? Explain the broader significance of your answer. Be sure to use the gas constant of $1.99 \text{ cal}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$.



6. (10 pts) Draw a general growth curve for a batch bacterial culture. Be sure to list all important features.

7. (10 pts) Discuss three advantages of utilizing continuous cell cultivation.

8. (10 pts) List the three major categories of microbial products and provide an example of each category.

Instructor Solution

DEPARTMENT OF CHEMICAL AND BIOLOGICAL SCIENCES AND ENGINEERING

CH350, AY26-1
Written Partial Review II
04 November 2025

TEXT: Shuler *et al*, Bioprocess Engineering: Basic Concepts, 3rd Ed.
Wittrup *et al*, Quantitative Fundamentals of Molecular and Cellular Bioengineering.
SCOPE: Lessons 14-21, **75 minutes**

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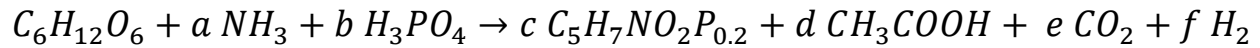
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1. (50 pts) *Clostridium* sp. were used for anaerobic conversion of glucose ($C_6H_{12}O_6$) to acetic acid (CH_3COOH) and H_2 as described by the following reaction:



- a) Determine the stoichiometric coefficients by writing elemental balances if $Y_{x/s} = 0.1 C_5H_7NO_2P_{0.2} \text{ g}/C_6H_{12}O_6 \text{ g}$ (assuming the molar mass of C, O, H, N, and P as 12 g/mol, 16 g/mol, 1 g/mol, 14 g/mol, and 31 g/mol, respectively).

Elemental (Stoichiometric) balances:

Carbon (C): $6 = 5c + 2d + e$

Hydrogen (H): $12 + 3a + 3b = 7c + 4d + 2f$

Oxygen (O): $6 + 4b = 2c + 2d + 2e$

Nitrogen (N): $a = c$

Phosphorous (P): $b = 0.2c$

+3 each for performing correct stoichiometric balance for each element (15 pts total).

+5 for correctly setting up $Y_{x/s}$ to get c .

+1 each for numerically correct stoichiometric coefficient (5 pts total).

Biomass: $c \times$

$C_5H_7NO_2P_{0.2}$

$\Rightarrow 119.2c \text{ g cell}$

Glucose:

$C_6H_{12}O_6$

$\Rightarrow 180 \text{ g substrate (since its stoichiometric coefficient is 1)}$

$Y_{x/s} = 119.2c / 180 = 0.1 \text{ g}$

$C_5H_7NO_2P_{0.2}$

/g

$C_6H_{12}O_6$

$a = 0.151$, $b = 0.0302$, $c = 0.151$, $d = 2.336$, $e = 0.574$, and $f = 1.072$

- b) Determine $\gamma_{C_6H_{12}O_6}$, $\gamma_{C_5H_7NO_2P_{0.2}}$, γ_{CH_3COOH} , and γ_{H_2} (γ stands for the degree of reduction).

$$\gamma_s = 6(4) + 12(1) + 6(-2) = 24,$$

$$\gamma_b = 5(4) + 7(1) + (-3) + 2(-2) + 5(0.2) = 21,$$

$$\gamma_{Ac} = 2(4) + 4(1) + 2(-2) = 8,$$

$$\gamma_{H_2} = 2$$

+4 each for correctly calculating for each degree of reduction (16 pts total).
+1 for numerical accuracy of answers (AON).

- c) Determine $Y_{p/s}$ for acetic acid (CH_3COOH) and hydrogen (H_2).

From part a: $C_6H_{12}O_6 + 0.151 NH_3 + 0.0302 H_3PO_4 \rightarrow 0.151 C_5H_7NO_2P_{0.2} + 2.336 CH_3COOH + 0.574 CO_2 + 1.072 H_2$

$$Y_{p/s(Acetic\ acid)} = 2.336(60)/180 = 0.778 \text{ g P/g S}$$

$$Y_{p/s(H_2)} = 1.072(2)/180 = 0.012 \text{ g P/g S}$$

+4 each for correctly calculating for $Y_{p/s}$ for acetic acid and hydrogen (8 pts total).

2. (30 pts) Two-stage chemostat system is used for fermentation of glucose ($C_6H_{12}O_6$) to ethanol (C_2H_5OH) using *Saccharomyces cerevisiae*. The feed flow rate and initial glucose concentrations are $F = 1$ L/hr and $S_0 = 40$ g/L, respectively. The following kinetic constants are known: $\mu_m = 0.139$ hr⁻¹, $K_s = 1$ g glucose/L, $Y_{x/s} = 0.1$ g cell/g glucose, $Y_{p/s} = 0.5$ g ethanol/g glucose. Note that the feed flow rate is same for the first and second stage of chemostat system.

- a) Assuming 80% conversion in every reactor, determine the substrate and product concentrations in the effluent of reactors.

1st reactor:

$$S_1 = 0.2 S_0 = 8 \text{ g/L}$$

$$P_1 = Y_{p/s} (S_0 - S_1) = 0.5(40 - 8) = 16 \text{ g/L}$$

+2 each for correctly calculating for substrate concentration in each reactor (4 pts total).

+3 each for correctly calculating for product concentration in each reactor (6 pts total).

2nd reactor:

$$S_2 = 0.2 S_1 = 1.6 \text{ g/L}$$

$$P_2 = P_1 + Y_{p/s} (S_1 - S_2) = 16 + 0.5(8 - 1.6) = 19.2 \text{ g/L}$$

- b) Assuming 80% conversion in every reactor, determine the biomass concentration in both reactors.

$$X_1 = Y_{x/s} (S_0 - S_1) = 0.1(40 - 8) = 3.2 \text{ g/L}$$

$$X_2 = X_1 + Y_{x/s} (S_1 - S_2) = 3.2 + 0.1(8 - 1.6) = 3.84 \text{ g/L}$$

+3 each for correctly calculating the biomass concentration in each reactor (6 pts total)

- c) Assuming 80% conversion in every reactor, determine the volume of each reactor.

$$\mu_m = 0.139 \text{ hr}^{-1}$$

$$F = 1 \text{ L/hr}$$

$$\mu_1 = \mu_m S_1 / (K_s + S_1) = 0.139(8) / (1 + 8) = 0.123 \text{ hr}^{-1} = F/V_1 = D_1$$

$$V_1 = 8.13 \text{ L}$$

+5 each for correctly setting up μ for each reactor (10 pts total).

+2 for correctly calculating the volume for each reactor (4 pts total).

$$\mu_2 = \mu_m S_2 / (K_s + S_2) = 0.139(1.6) / (1 + 1.6) = 0.0855 \text{ hr}^{-1}$$

$$\mu_2 = 0.0855 = D_2(1 - X_1/X_2) = D_2(1 - 3.2/3.84) = 0.167 D_2$$

$$D_2 = 0.51 \text{ hr}^{-1} = F/V_2$$

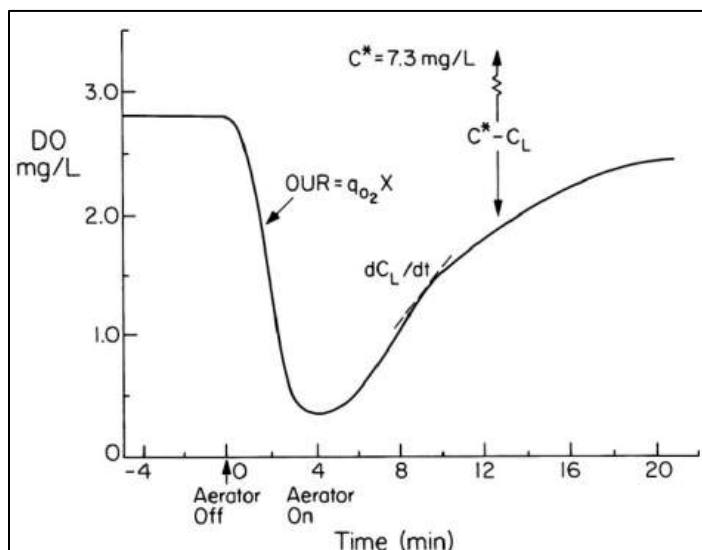
$$V_2 = 1.96 \text{ L}$$

3. (20 pts) Discuss the effects of mixing on sterilization in a batch fermenter. Be sure to include information on thermal implications, common operating temperatures for thermal liquid sterilization processes and implications on sterilization times after scale-up.

Achieving adequate mixing during batch sterilization is very important. Good mixing will allow the broth to both heat up and cool down quickly, thus minimizing the degradation of vitamins and proteins and the caramelization of sugars. Good mixing will ensure that all the broth reaches 121 °C and that the broth is maintained at 121 °C. If there is incomplete mixing, parts of the broth may not reach 121 °C and a longer sterilization time will be needed to ensure sterility. Scale-up alters the mixing, thus the heat-up, sterilization and cool-down times will change with scale-up. Some industrial air lift bioreactors are equipped with small turbine impellers to facilitate good mixing during sterilization.

+5 each for correctly explaining the effects in terms of thermal implication, operating temperatures, scale-up and time.

4. (20 pts) Estimate k_{LA} from Figure 10.5 of DeLisa, 2017 (shown below) if C^* is 35 mg/L due to the use of nearly pure oxygen rather than air.



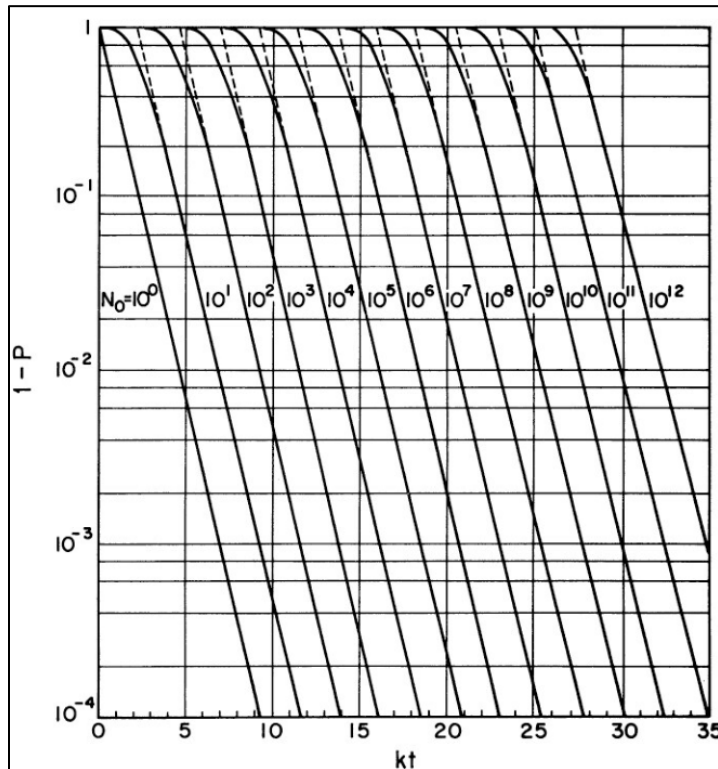
+4 for correctly using the k_{LA} expression.
 +4 for correctly choosing dC_L/dt .
 +4 for correctly choosing OUR.
 +4 for correctly calculating $C^* - C_L$.
 +4 for numerically correct k_{LA} .

$$k_{LA} = \frac{0.25 \frac{\text{mg}}{\text{L} \cdot \text{min}} + 0.8 \frac{\text{mg}}{\text{L} \cdot \text{min}}}{(35 - 1) \frac{\text{mg}}{\text{L}}} = 0.031 \text{ min}^{-1}$$

The approach is identical to that in example 10.1 (see pg. 336-337 of DeLisa, 2017) except $C^* = 7.3 \text{ mg/l}$ is replaced by $C^* = 35 \text{ mg/L}$. Thus, (at $C = 1 \text{ mg/L}$).

$C^* = 35 \text{ mg/L}$ would correspond to a case where the gas used to aerate the fermenter is highly enriched in O_2 .

5. (50 pts) An autoclave malfunctions, and the temperature reaches only 119.5 °C. The sterilization time at the maximum temperature was 20 min. The jar contains 10 L of complex medium that has 10^5 spores/L. At 121 °C, $k_d = 1.0 \text{ min}^{-1}$ and $E_{0d} = 90 \text{ kcal/mol}$. Using the Sterilization Chart from the text (shown below), what is the probability that the medium was sterile at 119.5 °C? Explain the broader significance of your answer. Be sure to use the gas constant of $1.99 \text{ cal}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$.



- +5 for correctly finding N_0 .
- +6 for setting up k_d .
- +5 for finding α .
- +8 for correctly calculating k_d at 119.5 °C.
- +8 for finding $k_d t$ at 20 min.
- +8 for setting up the probability expression.
- +5 for numerically correct P.
- +5 for providing a correct explanation based on the numerical answer of P.

$$N_0 = 10^5 \text{ spores/L} \cdot 10 \text{ L} = 10^6 \text{ spores}$$

$$k_d = \alpha e^{-E_{0d}/RT}$$

$$\text{At } 121^\circ\text{C}, k_d = 1 \text{ min}^{-1} = \alpha e^{-90\,000/(1.99 \times 394.16^\circ\text{K})}$$

$$\Rightarrow \alpha = 6.78 \times 10^{49} \text{ min}^{-1}$$

$$\text{At } 119.5^\circ\text{C}, k_d = 6.78 \times 10^{49} e^{-90\,000/(1.99 \times 392.65^\circ\text{K})} = 0.64 \text{ min}^{-1}$$

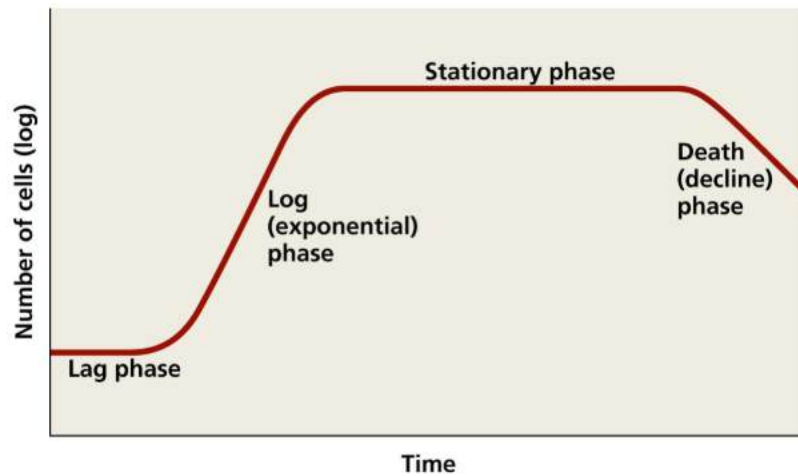
$$\text{If } t = 20 \text{ min}, k_d t = 0.64 \text{ min}^{-1} \cdot 20 \text{ min} = 13$$

The sterilization chart is hard to read in this region, but we know:

$$1 - P = 1 - [1 - e^{-k_d t}]^{N_0} = 1 - [1 - e^{-13}]^{10^6} = 0.896$$

Probability of success ≈ 0.10 . Overall, our answers demonstrate how sensitive system is towards small ΔT .

6. (10 pts) Draw a general growth curve for a batch bacterial culture. Be sure to list all important features.



7. (10 pts) Discuss three advantages of utilizing continuous cell cultivation.

1. Provides constant environmental conditions for growth and product formation.
2. Supplies uniform-quality product.
3. Determines the response of microorganisms to their environment and helps to create the desired products under optimal conditions.

+1 for providing the basic explanation.

+3 each for providing advantage of using continuous cell cultivation (9 pts total).

8. (10 pts) List the three major categories of microbial products and provide an example of each category.

1. Growth-associated products: the production of a constitutive enzyme
2. Non-growth associated products: secondary metabolites such as antibiotics
3. Mixed-growth-associated products: lactic acid fermentation, xanthan gum, and other secondary metabolites from cell cultures.

+1 for providing the basic explanation.

+1 each for stating the major category of microbial products (3 pts total).

+2 each for providing the examples for each category. (6 pts total).



TAB H TERM END EXAMINATION

Handouts

CADET _____ SECTION/HR _____

DEPARTMENT OF CHEMISTRY AND LIFE SCIENCE

CH350, AY26-1
Term End Exam
DEC 16 2025

TEXT: Shuler *et al*, Bioprocess Engineering: Basic Concepts, 3rd Ed.
Wittrup *et al*, Quantitative Fundamentals of Molecular and Cellular Bioengineering.
SCOPE: Lessons 1-30, **210 minutes**

Authorized References

Textbooks (Shuler et al, 3rd Ed. and Wittrup et al), Calculator (TI-30X IIS, TI-36X Pro, or equivalent), Single, 1-Sided Note Page, and Blank Mathematica Worksheet

1. Do not mark on this exam or open it until "begin work" is given.
2. Solve problems in the space provided. Be sure to do the following *to receive full credit*:
 - a. Print legibly.
 - b. **Show all work (including any digital file, which is needed to be graded).**
 - c. Use complete sentences for all explanations.
 - d. Answer all questions and double underline answer for quantitative problems.
3. There are 8 numbered pages in this TEE. Report any illegible or missing pages to the instructor. Write your name on every page in the spaces provided.

(TOTAL WEIGHT 400 POINTS)

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8	40		
9	50		
10	30		
11	25		
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13	35		
14	20		
15	40		
TOTAL SCORE			/400

1. (20 pts) During routine maintenance, you discovered that the performance of bioreactor was significantly influenced by the generation of foaming during ethanol fermentation. Provide your solution to reduce foaming and any potential tradeoff of your solution. (refer to pg. 324-329 of DeLisa, 2017)

2. (15 pts) What would be the crucial operational constraints if you want to design the bioreactor using animal cells? Be sure to discuss information on shear sensitivity and microenvironment presented in the reactor. (refer to pg. 443 of DeLisa, 2017)

3. (10 pts) Draw a general growth curve for a batch bacterial culture. Be sure to list all important features in each phase. (refer to pg. 175 of DeLisa, 2017)

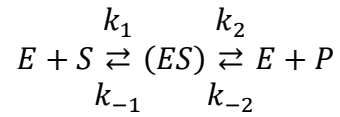
4. (15 pts) For a chemostat without recycling stream, write the mass balance on the limiting substrate. Be sure to define all the terms written in the equation. Assume no substrate is consumed due to maintenance metabolism. (refer to pg. 212 of DeLisa, 2017)

5. (20 pts) Explain why maintaining the height-to-diameter ratio would be problematic when scaling up a bioreactor. (refer to pg. 337 of DeLisa, 2017)

6. (15 pts) You wish to increase the oxygen uptake rate by the biomass in the stirred-tank bioreactor. Discuss at least three operating factors that could affect the volumetric oxygen transfer coefficient (k_La). (refer to pg. 324-336 of DeLisa, 2017)

7. (15 pts) Propose the type of bioreactors you would use if obtaining high-values products created from animal cell cultures is required. Provide your reasonings in terms of regulatory and economic perspectives. (refer to pg. 330-331 of DeLisa, 2017)

8. (40 pts) Consider the following reversible product-formation reaction in an enzyme-catalyzed bioreaction: (refer to pg. 66-67 of DeLisa, 2017)



Develop a rate expression for product-formation using the quasi-steady-state approximation and show that

$$v = \frac{d[P]}{dt} = \frac{\left(\frac{V_s}{K_m}\right)[S] - \left(\frac{V_p}{K_p}\right)[P]}{1 + \frac{[S]}{K_m} + \frac{[P]}{K_p}}$$

where $K_m = \frac{k_{-1}+k_2}{k_1}$ and $K_p = \frac{k_{-1}+k_2}{k_{-2}}$, along with $V_s = k_2[E_0]$ and $V_p = k_{-1}[E_0]$.

9. (50 pts) Two-stage chemostat system is used for fermentation of glucose to ethanol using *Saccharomyces cerevisiae*. The feed flow rate and glucose concentrations are set as 1 L/hr and $S_0 = 40$ g/L, respectively. The following kinetic parameters are for used for such chemostat system: $\mu_m = 0.139 \text{ hr}^{-1}$, $K_s = 1$ g/L, $Y_{X/S} = 0.1$ g X/g S, $Y_{P/S} = 0.5$ g EtOH/g glucose. After 80% of conversion is reached in each reactor, you wish to determine the productivity of biomass and product in each reactor. (refer to pg. 210-212 and pg.282-283 of DeLisa, 2017)

A) First, determine the substrate and product concentrations in the effluent from each reactor.

B) Also, find the biomass concentration in both reactors.

C) Next, calculate the volume of each reactor.

D) Finally, find the productivity of biomass (DX) and product (DP) in each reactor.

10. (30 pts) An industrial wastewater stream is fed to a stirred-tank reactor continuously, and the cells are recycled back to the reactor from the bottoms of the sedimentation tank placed after the reactor. The following are given for the reactor system: $F = 100 \text{ L/hr}$, $S_0 = 5000 \text{ mg/L}$, $\mu_m = 0.25 \text{ hr}^{-1}$, $K_s = 200 \text{ mg/L}$, α (recycle ratio) $= 0.6$, C (cell concentration factor) $= 2$, and $Y_{X/S}^M = 0.4$. You wish the effluent concentration to be 100 mg/L . (refer to pg. 278-279 of DeLisa, 2017)

A) Determine the required reactor volume.

B) Determine the cell concentration in the reactor and in the recycle stream.

11. (25 pts) An activated carbon adsorption column is used for removal of phenolic compounds from wastewater. The feed and desired effluent adsorbate concentrations are $C_0 = 100 \text{ mg/L}$ and $C_e = 1 \text{ mg/L}$, and the flow rate is $F = 10 \text{ m}^3/\text{hr}$. Adsorption column contains 3 m^3 activated carbon with 30% void fraction and the column diameter is $D_0 = 0.5 \text{ m}$. Adsorption capacity and rate for activated carbon are $N_0 = 60 \text{ kg adsorbate/m}^3 \text{ bed}$ and $K = 0.8 \text{ m}^3/\text{kg}\cdot\text{hr}$. Your task is to determine the adsorption column height and service time. (refer to pg. 407-408 of DeLisa, 2017)

12. (50 pts) A solute protein is to be separated from a liquid phase in a chromatographic column. The adsorption isotherm is given by

$$C_s = kC_L^2$$

where C_s is the solute concentration in solid phase (mg solute/mg adsorbent) and C_L is the liquid phase concentration of solute (mg solute/mL liquid). The following operation parameters are given: $k = 0.4$, $\varepsilon = 0.3$, $A = 25 \text{ cm}^2$, and $M = 10 \text{ g ads/100 mL column} = 100 \text{ mg/mL}$. (refer to pg. 412-413 of DeLisa, 2017)

A) For $V = 400 \text{ mL}$ and $X = 25 \text{ cm}$, determine the equilibrium solute concentrations in liquid and solid phase.

B) Determine the ratio of travel distances of solute to solvent, R_f .

13. (35 pts) A stirred-tank reactor is to be scaled down from 10 m^3 to 0.1 m^3 . The dimensions of the large tank are $D_t = 2 \text{ m}$; $D_i = 0.5 \text{ m}$; $N = 100 \text{ rpm}$. (refer to pg. 344-345 of DeLisa, 2017)

A) Determine the dimensions of the small tank (D_t , D_i , H) by using geometric similarity.

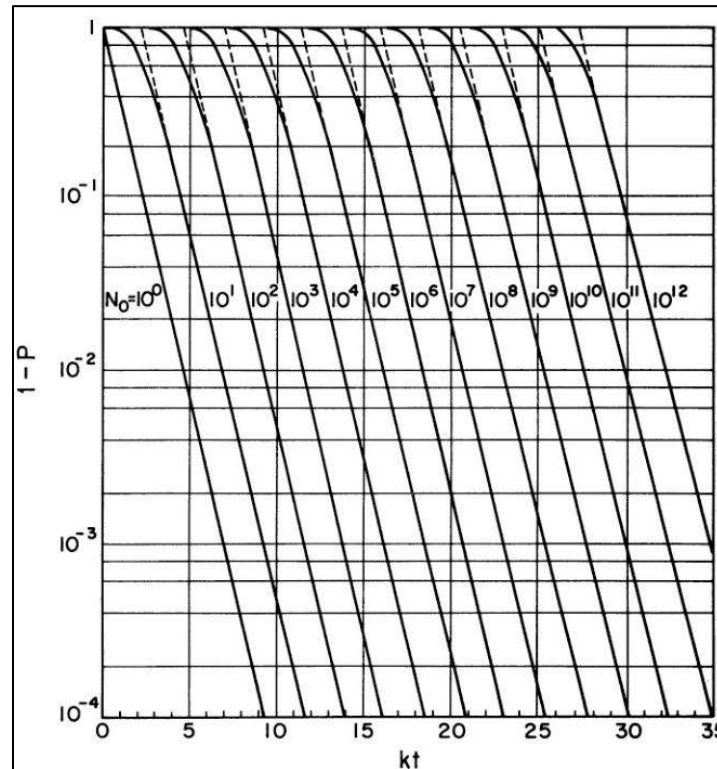
B) What would be the required rotational speed of the impeller in the small tank if constant impeller number (Re) is used?

14. (20 pts) *E. coli* has a maximum respiration rate, $q_{O_2, \max}$, of about $240 \text{ mg O}_2/\text{g dry wt}\cdot\text{hr}$. It is desired to achieve a cell mass of 20 g dry wt/L . The $k_L a$ is 120 hr^{-1} in a 1000 L reactor (800 L working volume). A gas stream enriched in oxygen is used, which gives a value of $C^* 28 \text{ mg/L}$. If oxygen becomes limiting,

$$q_{O_2} = \frac{q_{O_2, \max} C_L}{0.2 \frac{mg}{L} + C_L} X$$

Here C_L is the dissolved oxygen concentration in the fermenter. What is C_L when the cell mass (X) is at 20 g/L ? (refer to pg. 334-335 of DeLisa, 2017)

15. (40 pts) A medium containing a vitamin is to be sterilized. Assume that the number of spores initially present is $10^5/\text{L}$. The values of the pre-Arrhenius constant (α) and E_{0d} for the spores are 10^{36} min^{-1} and $65 \text{ kcal/g}\cdot\text{mol}$, respectively. For the inactivation of the vitamin, the pre-Arrhenius constant (α) and E_{0d} are 10^4 min^{-1} and $10 \text{ kcal/g}\cdot\text{mol}$, respectively. The initial concentration of the vitamin is 30 mg/L . Compare the amount of active vitamin in the sterilized medium for 10 L and $10,000 \text{ L}$ fermenters when both are sterilized at 121°C when the probability of an unsuccessful fermentation in both cases is required to be 0.001 . Ignore the effects of the heat-up and cool-down periods. Remember to use the Sterilization chart from the text (provided below). (refer to pg. 356-360 of DeLisa, 2017)



Instructor Solution

APPROVED SOLUTION

CADET **APPROVED SOLUTION** SECTION/HR _____

DEPARTMENT OF CHEMISTRY AND LIFE SCIENCE

CH350, AY26-1
Term End Exam
DEC 16 2025

TEXT: Shuler *et al*, Bioprocess Engineering: Basic Concepts, 3rd Ed.
Wittrup *et al*, Quantitative Fundamentals of Molecular and Cellular Bioengineering.
SCOPE: Lessons 1-30, **210 minutes**

Authorized References

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APPROVED SOLUTION

1. (20 pts) During routine maintenance, you discovered that the performance of bioreactor was significantly influenced by the generation of foaming during ethanol fermentation. Provide your solution to reduce foaming and any potential tradeoff of your solution. (refer to pg. 324-329 of DeLisa, 2017)

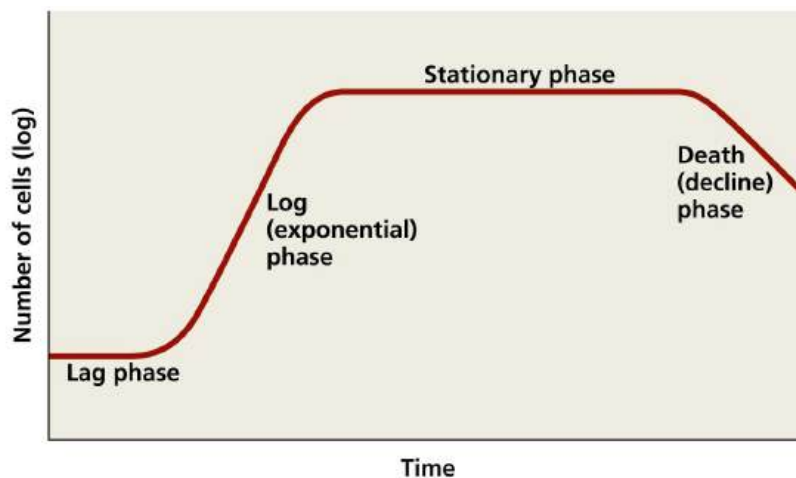
Answers can be varied. The generation of foaming can be reduced with the installation of mechanical foam breaker or the addition of surface-active chemical agents into the broth. The potential issue with the suggested solution would be the decrease in the k_{La} values, thereby hampering the reactor's capacity to supply oxygen or any other gaseous nutrients to the cells. The cell growth can even be inhibited depending on the conditions.

2. (15 pts) What would be the crucial operational constraints if you want to design the bioreactor using animal cells? Be sure to discuss information on shear sensitivity and microenvironment presented in the reactor. (refer to pg. 443 of DeLisa, 2017)

The following constraints need to be considered when designing animal cell bioreactors:

- 1) The reactor should be gently aerated and agitated due to shear sensitivity of animal cells.
- 2) Well-controlled homogeneous environment conditions are preferred (in terms of T, pH, DO, and redox potential).
- 3) A relatively large support material with surface to volume ratios are recommended for anchorage-dependent cells.
- 4) The removal of toxic products, such as lactic acids and ammonium, are necessary during cell cultivation.

3. (10 pts) Draw a general growth curve for a batch bacterial culture. Be sure to list all important features in each phase. (refer to pg. 175 of DeLisa, 2017)



APPROVED SOLUTION

4. (15 pts) For a chemostat without recycling stream, write the mass balance on the limiting substrate. Be sure to define all the terms written in the equation. Assume no substrate is consumed due to maintenance metabolism. (refer to pg. 212 of DeLisa, 2017)

$$FS_0 - FS - \frac{V_R u_g X}{Y_X^M \frac{S}{S}} - \frac{V_R q_p X}{Y_P^M \frac{S}{S}} = V_R \frac{dS}{dt}$$

where F is the flow rate of nutrient solution (L/hr), V_R is the culture volume (L), X is the cell concentration (g/L), u_g is growth rate constants (1/hr), S_0 is the feed substrate concentration (g/L), S is the effluent substrate concentration (g/L), and q_p is the specific rate of extracellular product formation (g P/g X·hr). Y_X^M (g X/g S) and Y_P^M (g P/g S) are the yield coefficient for cell and product formation, respectively.

5. (20 pts) Explain why maintaining the height-to-diameter ratio would be problematic when scaling up a bioreactor. (refer to pg. 337 of DeLisa, 2017)

Maintaining the height-to-diameter ratio decreases the surface area-to-volume ratio significantly during scale-up. This might be an issue in fermentation where wall growth of biomass may occur. In this scenario, the data obtained from small fermenters will be unreliable to be used for large fermenters.

6. (15 pts) You wish to increase the oxygen uptake rate by the biomass in the stirred-tank bioreactor. Discuss at least three operating factors that could affect the volumetric oxygen transfer coefficient (k_{La}). (refer to pg. 324-336 of DeLisa, 2017)

Answers can be varied:

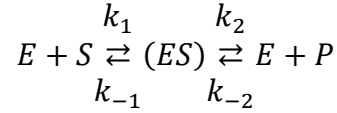
- 1) Bubble size
- 2) Bubble coalescence in the tank
- 3) Gas flow rates from sparger
- 4) Addition of surfactants/salts
- 5) Agitation/dispersion rate by mechanical agitator

7. (15 pts) Propose the type of bioreactors you would use if obtaining high-values products created from animal cell cultures is required. Provide your reasonings in terms of regulatory and economic perspectives. (refer to pg. 330-331 of DeLisa, 2017)

The single-use bioreactors would be desirable in this scenario because sterilization/cleaning procedures and their validation by FDA are not required. Such reactor design also leads to a great process flexibility and responsiveness to market changes, reduced costs, and improvements in biosafety.

APPROVED SOLUTION

8. (40 pts) Consider the following reversible product-formation reaction in an enzyme-catalyzed bioreaction: (refer to pg. 66-67 of DeLisa, 2017)



Develop a rate expression for product-formation using the quasi-steady-state approximation and show that

$$v = \frac{d[P]}{dt} = \frac{\left(\frac{V_s}{K_m}\right)[S] - \left(\frac{V_p}{K_p}\right)[P]}{1 + \frac{[S]}{K_m} + \frac{[P]}{K_p}}$$

where $K_m = \frac{k_{-1}+k_2}{k_1}$ and $K_p = \frac{k_{-1}+k_2}{k_{-2}}$, along with $V_s = k_2[E_0]$ and $V_p = k_{-1}[E_0]$.

$$\frac{d(ES)}{dt} = r_1(E)(S) + r_{-2}(E)(P) - r_{-1}(ES) - r_2(ES) \approx 0 \quad (1)$$

$$\text{rate} = \frac{d(P)}{dt} = r_2(ES) - r_{-2}(E)(P) \quad (2)$$

$$(E_0) = (E) + (ES), \therefore (E) = (E_0) - (ES) \quad (3)$$

$$\text{from (1): } (E) = \frac{r_{-1}+r_2}{r_1(S)+r_{-2}(P)}(ES) \quad (4)$$

$$\text{Combine (3) + (4), } (E_0) = \left(\frac{r_1(S)+r_{-2}(P)+r_{-1}+r_2}{r_1(S)+r_{-2}(P)}\right)(ES) \text{ and } (ES) = \frac{E_0(r_1(S)+r_{-2}(P))}{r_1(S)+r_{-2}(P)+r_{-1}+r_2} \quad (5)$$

$$\text{Substitute (3) and (5) into (2): rate} = \frac{r_1 r_2 E_0(S) + r_2 r_{-2} E_0(P)}{r_{-1} + r_2 + r_1(S) + r_{-2}(P)} - r_{-2}(E_0 - (ES))(P)$$

$$\text{Substituting (ES) in terms of } E_0: \text{rate} = \frac{r_1 r_2 E(S) - r_{-1} r_{-2} E_0(P)}{r_{-1} + r_2 + r_1(S) + r_{-2}(P)}$$

Let $V_s = r_2 E_0$ and $V_p = r_{-1} E_0$:

$$\text{rate} = \frac{r_1 V_s [S] - r_{-2} V_p [P]}{r_{-1} + r_2 + r_1 [S] + r_{-2} [P]}$$

Divide top and bottom by $(r_{-1} + r_2)$ and let $\frac{1}{K_m} = \frac{r_1}{r_{-1} + r_2}$ and $\frac{1}{K_p} = \frac{r_{-2}}{r_{-1} + r_2}$:

$$\text{rate} = \frac{\left(\frac{V_s}{K_m}\right)[S] - \left(\frac{V_p}{K_p}\right)[P]}{1 + \frac{[S]}{K_m} + \frac{[P]}{K_p}}$$

APPROVED SOLUTION

9. (50 pts) Two-stage chemostat system is used for fermentation of glucose to ethanol using *Saccharomyces cerevisiae*. The feed flow rate and glucose concentrations are set as 1 L/hr and $S_0 = 40$ g/L, respectively. The following kinetic parameters are for used for such chemostat system: $\mu_m = 0.139 \text{ hr}^{-1}$, $K_s = 1$ g/L, $Y_{X/S} = 0.1$ g X/g S, $Y_{P/S} = 0.5$ g EtOH/g glucose. After 80% of conversion is reached in each reactor, you wish to determine the productivity of biomass and product in each reactor. (refer to pg. 210-212 and pg.282-283 of DeLisa, 2017)

A) First, determine the substrate and product concentrations in the effluent from each reactor.

$$1^{\text{st}} \text{ reactor: } S_1 = 0.2S_0 = 8 \text{ g/L}$$

$$2^{\text{nd}} \text{ reactor: } S_2 = 0.2S_1 = 1.6 \text{ g/L}$$

$$P_1 = Y_{P/S} (S_0 - S_1) = 0.5 (40 - 8) = 16 \text{ g/L}$$

$$P_2 = P_1 + Y_{P/S} (S_1 - S_2) = 16 + 0.5(8 - 1.6) = 19.2 \text{ g/L}$$

B) Also, find the biomass concentration in both reactors.

$$X_1 = Y_{X/S} (S_0 - S_1) = 0.1 (40 - 8) = 3.2 \text{ g/L}$$

$$X_2 = X_1 + Y_{X/S} (S_1 - S_2) = 3.2 + 0.1 (8 - 1.6) = 3.84 \text{ g/L}$$

C) Next, calculate the volume of each reactor.

$$\mu_1 = \mu_m S_1 / (K_s + S_1) = 0.139(8) / (1+8) = 0.123 \text{ hr}^{-1} = F/V_1 = D_1$$

$$V_1 = 8.13 \text{ L}$$

$$\mu_2 = \mu_m S_2 / (K_s + S_2) = 0.139(1.6) / (1+1.6) = 0.0855 \text{ hr}^{-1}$$

$$\mu_2 = 0.0855 = D_2(1 - X_1/X_2) = D_2 (1 - 3.2/3.84) = 0.167 D_2$$

$$D_2 = 0.51 \text{ h}^{-1} = F/V_2$$

$$V_2 = 1.96 \text{ L}$$

D) Finally, find the productivity of biomass (DX) and product (DP) in each reactor.

$$(DX)_1 = D_1 X_1 = 0.123(3.2) = 0.394 \text{ gX/L} \cdot \text{hr}$$

$$(DX)_2 = D_2 X_2 = 0.51 (3.84) = 1.96 \text{ gX/L} \cdot \text{hr}$$

$$(DP)_1 = D_1 P_1 = 0.123 (16) = 1.968 \text{ gP/L} \cdot \text{hr}$$

$$(DP)_2 = D_2 P_2 = 0.51(19.2) = 9.79 \text{ gP/L} \cdot \text{hr}$$

APPROVED SOLUTION

10. (30 pts) An industrial wastewater stream is fed to a stirred-tank reactor continuously, and the cells are recycled back to the reactor from the bottoms of the sedimentation tank place after the reactor. The following are given for the reactor system: $F = 100$ L/hr, $S_0 = 5000$ mg/L, $\mu_m = 0.25$ hr⁻¹, $K_s = 200$ mg/L, α (recycle ratio) = 0.6, C (cell concentration factor) = 2, and $Y_{X/S}^M = 0.4$. You want the effluent substrate concentration to be 100 mg/L. (refer to pg. 278-279 of DeLisa, 2017)

A) Determine the required reactor volume.

$$S = \frac{K_s D (1 + \alpha - \alpha C)}{\mu_m - D (1 + \alpha - \alpha C)} ; 100 \text{ mg/L} = \frac{200 D (1.6 - 0.6(2))}{0.25 - D (1.6 - 0.6(2))}$$

$$0.5 = \frac{0.4 D}{0.25 - 0.4 D}$$

$$D = 0.208 \text{ hr}^{-1} = F/V$$

$$V = \frac{100 \text{ L/hr}}{0.208 \text{ hr}^{-1}} = 481 \text{ L}$$

B) Determine the cell concentration in the reactor and in the recycle stream.

$$X_1 = \frac{y(S_0 - S_1)}{1 + \alpha - \alpha C} = \frac{0.4 (5000 - 100)}{[1.6 - 0.6(2)]} = 4900 \text{ mg/L} = 4.9 \text{ g/L}$$

$$X_r = c X_1 = 2 (4900) = 9800 \text{ mg/L} = 9.8 \text{ g/L}$$

11. (25 pts) An activated carbon adsorption column is used for removal of phenolic compounds from wastewater. The feed and desired effluent adsorbate concentrations are $C_0 = 100$ mg/L and $C_e = 1$ mg/L, and the flow rate is $F = 10$ m³/hr. Adsorption column contains 3 m³ activated carbon with 30% void fraction and the column diameter is $D_0 = 0.5$ m. Adsorption capacity and rate for activated carbon are $N_0 = 60$ kg adsorbate/m³ bed and $K = 0.8$ m³/kg·hr. Your task is to determine the adsorption column height and service time. (refer to pg. 407-408 of DeLisa, 2017)

$$V_{AC} = V_b (1 - \varepsilon), 3 = V_b (1 - 0.3)$$

$$V_b = 4.3 \text{ m}^3$$

$$V_b = (3.14/4) D_0^2 H, 4.3 = (3.14/4) (0.5)^2 H$$

$$H = 21.8 \text{ m}$$

$$V_0 = F/A = 10 \text{ m}^3/\text{h} / ((3.14/4)(0.5)^2) \text{ m}^2 = 50.9 \text{ m/h}$$

$$a = N_0 / C_0 V_0 = 60 / (0.1 \times 50.9) = 11.8$$

$$b = -\ln(C_0/C_e - 1) / KC_0 = -\ln(100/1 - 1) / (0.8 \times 0.1) = -57.4$$

$$t = a H + b = 11.8 (21.8) - 57.4 = 200 \text{ h} = 8.3 \text{ days}$$

APPROVED SOLUTION

12. (50 pts) A solute protein is to be separated from a liquid phase in a chromatographic column. The adsorption isotherm is given by

$$C_s = kC_L^2$$

where C_s is the solute concentration in solid phase (mg solute/mg adsorbent) and C_L is the liquid phase concentration of solute (mg solute/mL liquid). The following operation parameters are given: $k = 0.4$, $\varepsilon = 0.3$, $A = 25 \text{ cm}^2$, and $M = 10 \text{ g ads/100 mL column} = 100 \text{ mg/mL}$. (refer to pg. 412-413 of DeLisa, 2017)

A) For $V = 400 \text{ mL}$ and $X = 25 \text{ cm}$, determine the equilibrium solute concentrations in liquid and solid phase.

$$\Delta X = \frac{\Delta V}{A(\varepsilon + Mf'(C_L))}$$
$$(C_L) = kC_L^2$$
$$f'(C_L) = 2kC_L$$

$$\therefore \Delta V = \Delta X A (\varepsilon + M2kC_L)$$

$$\text{or } C_L = \frac{\frac{\Delta V}{\Delta X A} - \varepsilon}{M2k} = \frac{\frac{400 \text{ cm}^3}{25 \text{ cm} \cdot 25 \text{ cm}^2} - 0.3}{2 \left(\frac{100 \text{ mg}}{\text{mL}} \right) 0.4 \frac{\text{mL}^2}{\text{mgs} \cdot \text{mg ads}}}$$

$$C_L = 0.00425 \frac{\text{mg}}{\text{mL}} \text{ solute}$$

$$C_s = kC_L^2 = 0.4(0.00425)^2 = 7.2 \times 10^{-6} \frac{\text{mg solute}}{\text{mg ads}}$$

B) Determine the ratio of travel distances of solute to solvent, R_f .

$$R_f = \frac{\varepsilon}{\varepsilon + MK} = \frac{0.3}{0.3 + 100 \frac{\text{mg}}{\text{ml}} K}$$

$$K = 2kC_L = 2 \cdot 0.4 \cdot (0.00425) = 0.0034$$

$$R_f = \frac{0.3}{0.64} = 0.47$$

APPROVED SOLUTION

13. (35 pts) A stirred-tank reactor is to be scaled down from 10 m³ to 0.1 m³. The dimensions of the large tank are $D_t = 2$ m; $D_i = 0.5$ m; $N = 100$ rpm. (refer to pg. 344-345 of DeLisa, 2017)

A) Determine the dimensions of the small tank (D_t , D_i , H) by using geometric similarity.

$$\begin{aligned}\frac{H_1}{H_2} &= \frac{D_{t1}}{D_{t2}} = \frac{D_{i1}}{D_{i2}} = 4.64 = \alpha \\ D_{t2} &= \frac{D_{t1}}{\alpha} = \frac{2}{4.64} = 0.43 \text{ m.} \\ D_{i2} &= \frac{D_{i1}}{\alpha} = \frac{0.5}{4.64} = 0.108 \text{ m} \\ H_1 &= \frac{4V_1}{\pi D_{t1}^2} = \frac{4(10)}{\pi (2)^2} = 3.185 \text{ m} \\ H_2 &= \frac{H_1}{\alpha} = \frac{3.185}{4.64} = 0.686 \text{ m.}\end{aligned}$$

B) What would be the required rotational speed of the impeller in the small tank if constant impeller number (Re) is used?

$$\begin{aligned}N_1 D_{i1}^2 &= N_2 D_{i2}^2 \\ N_2 &= N_1 \left(\frac{D_{i1}}{D_{i2}} \right)^2 = 100(\alpha)^2 = 100 (4.64)^2 \\ N_2 &= 2153 \text{ rpm.}\end{aligned}$$

14. (20 pts) *E. coli* has a maximum respiration rate, $q_{O_2, \max}$, of about 240 mg O₂/g·dry wt·hr. It is desired to achieve a cell mass of 20 g dry wt/L. The $k_L a$ is 120 hr⁻¹ in a 1000 L reactor (800 L working volume). A gas stream enriched in oxygen is used, which gives a value of C^* 28 mg/L. If oxygen becomes limiting,

$$q_{O_2} = \frac{q_{O_2, \max} C_L}{0.2 \frac{mg}{L} + C_L} X$$

Here C_L is the dissolved oxygen concentration in the fermenter. What is C_L when the cell mass (X) is at 20 g/L? (refer to pg. 334-335 of DeLisa, 2017)

If $dC_L/dt \approx 0$,

$$dC_L/dt \approx 0 = k_L a (C^* - C_L) - \frac{q_{O_2, \max} C_L}{K_{O_2} + C_L} X; k_L a (C^* - C_L) = \frac{q_{O_2, \max} C_L}{K_{O_2} + C_L} X$$

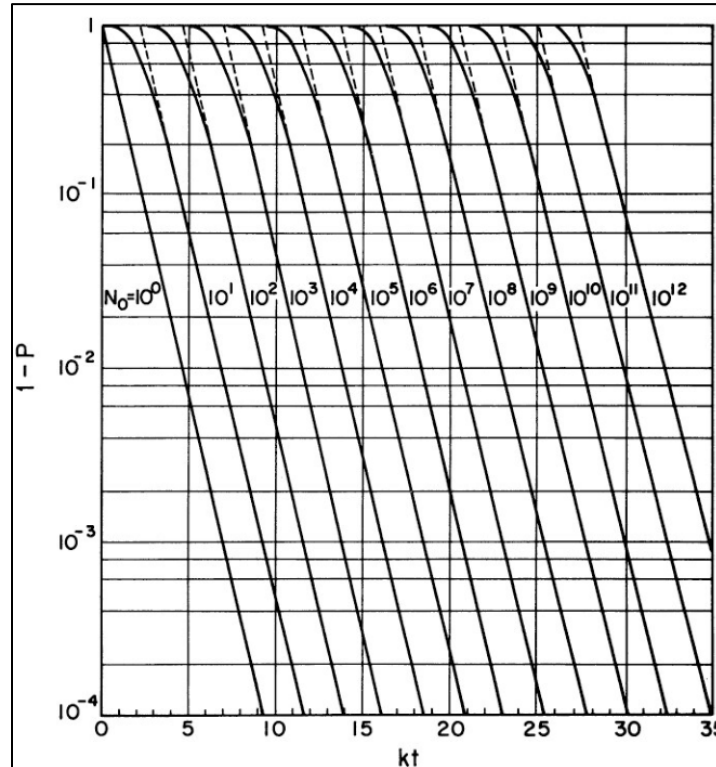
Then,

$$120 \text{ h}^{-1} (28 \text{ mg/L} - C_L) = \frac{240 \text{ mg O}_2/\text{g-cells-h} \cdot 20 \text{ g/l} \cdot C_L}{0.2 \text{ mg/l} + C_L}$$

Solve for C_L ; $C_L = 0.44$ mg/L

APPROVED SOLUTION

15. (40 pts) A medium containing a vitamin is to be sterilized. Assume that the number of spores initially present is $10^5/\text{L}$. The values of the pre-Arrhenius constant (α) and E_{0d} for the spores are 10^{36} min^{-1} and $65 \text{ kcal/g}\cdot\text{mol}$, respectively. For the inactivation of the vitamin, the pre-Arrhenius constant (α) and E_{0d} are 10^4 min^{-1} and $10 \text{ kcal/g}\cdot\text{mol}$, respectively. The initial concentration of the vitamin is 30 mg/L . Compare the amount of active vitamin in the sterilized medium for 10 L and $10,000 \text{ L}$ fermenters when both are sterilized at 121°C when the probability of an unsuccessful fermentation in both cases is required to be 0.001 . Ignore the effects of the heat-up and cool-down periods. Remember to use the Sterilization chart from the text (provided below). (refer to pg. 356-360 of DeLisa, 2017)



$$N_0/V = 10^5 \text{ L}^{-1} \quad C_{\text{vit},0} = 30 \text{ mg/L}$$

$$V = 10 \text{ L and } 10^4 \text{ L} \quad 1 - P_0(t) = 0.001$$

Calculating values of k_d at 121°C :

$$k_{d,S} = 1 \times 10^{36} \text{ min}^{-1} \exp \left[- \frac{6 \times 10^4 \text{ cal/mol}}{1.987 \frac{\text{cal}}{\text{mol K}} (394.15 \text{ K})} \right] = 0.9027 \text{ min}^{-1}$$

$$k_{d,V} = 1 \times 10^4 \text{ min}^{-1} \exp \left[- \frac{1 \times 10^4 \text{ cal/mol}}{1.987 \frac{\text{cal}}{\text{mol K}} (394.15 \text{ K})} \right] = 0.0285 \text{ min}^{-1}$$

Calculating sterilization time for 10 L fermenter:

$$N_0 = 10^5 \text{ L}^{-1} (10 \text{ L}) = 10^6 \text{ spores} \quad 1 - P_0(t) = 0.001$$

APPROVED SOLUTION

From the chart, $k_{d,st} = 20.78$

$$\therefore t = \frac{20.78}{k_{d,S}} = \frac{20.78}{0.9027 \text{ min}^{-1}} = 23.0 \text{ min}$$

Assuming first order degradation of vitamin:

$$C_{vit} = C_{vit,0} e^{-k_{d,v}t} = 30 \frac{\text{mg}}{\text{L}} e^{-0.0285 \text{ min}^{-1} (23 \text{ min})} = 15.6 \text{ mg/L}$$

$\therefore$ 52% of the vitamin is active in the 10 L fermenter after sterilization

For 10^4 L fermenter: $N_0 = 10^5 \text{ L}^{-1} (10^4 \text{ L}) = 109 \text{ spores}$

From the chart, $k_{d,st} = 27.8$

$$\therefore t = \frac{27.8}{k_{d,S}} = \frac{27.8}{0.9027 \text{ min}^{-1}} = 30.8 \text{ min}$$

$$C_{vit} = C_{vit,0} e^{-k_{d,v}t} = 30 \frac{\text{mg}}{\text{L}} e^{-0.0285 \text{ min}^{-1} (30.8 \text{ min})} = 12.5 \frac{\text{mg}}{\text{L}}$$

$\therefore$ 42% of the vitamin is active in the 10^4 L fermenter after sterilization (20% less than in the 10 L fermenter).



TAB I **FINAL COURSE GRADES**

Student	Cadet ID	Email	Section
Badger, James	C96705817	james.badger@westpoint.edu	CH350 G2 1 - HUMMEL, S
Baisted, Benjamin	C26124070	benjamin.baisted@westpoint.edu	CH350 G2 1 - HUMMEL, S
Ebiuwhe, Emmanuella	C26297842	emmanuella.ebiuwhe@westpoint.edu	CH350 G2 1 - HUMMEL, S
Janat, Tsegaye	C50740772	tsegaye.janat@westpoint.edu	CH350 G2 1 - HUMMEL, S
Leja, Gabriel	C35196169	gabriel.leja@westpoint.edu	CH350 G2 1 - HUMMEL, S
Littlehales, Andrew	C03910139	andrew.littlehales@westpoint.edu	CH350 G2 1 - HUMMEL, S
Mantooth, Jeanine	C19140069	jeanine.mantooth@westpoint.edu	CH350 G2 1 - HUMMEL, S
Muteteke, Colette	C73279534	colette.muteteke@westpoint.edu	CH350 G2 1 - HUMMEL, S
Rutherford, Trinity	C05439054	trinity.rutherford@westpoint.edu	CH350 G2 1 - HUMMEL, S

Major	Cadet	Section	Current Pts	Possible Pts	% Complete	Average
CEN	Mantooth, Jeanine	E2	1326.5	1500	100%	88.43
CEN	Littlehales, Andrew	E2	1323	1500	100%	88.20
CEN	Janat, Tsegaye	E2	1298	1500	100%	86.53
CEN	Baisted, Benjamin	E2	1279	1500	100%	85.27
CHM	Muteteke, Colette	E2	1261	1500	100%	84.07
BIO	Rutherford, Trinity	E2	1260.5	1500	100%	84.03
CEN	Badger, James	E2	1260	1500	100%	84.00
CEN	Leja, Gabriel	E2	1254	1500	100%	83.60
CEN	Ebiuwhe, Emmanuella	E2	1244	1500	100%	82.93



UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION III

ASSIGNMENTS



TAB J **PROBLEM SETS**



PROBLEM SET #1

Handouts

CH350 Problem Set 1

A Problem Set Submitted to:

Instructor/Section: LTC Hummel / Section _____

In Partial Fulfillment of

Requirements for CH350

By

Cadet: _____

West Point, New York

Date: _____ Time: _____

DO NOT WRITE IN THIS SPACE

PROBLEM	VALUE	CUT
1.1	5	
1.2	5	
1.3	5	
1.4	5	
1.5	5	
1.6	5	
1.7	5	
1.8	5	
1.9	5	
1.10	5	
TOTAL GRADE	50	

_____ MY DOCUMENTATION ATTRIBUTES ALL SOURCES USED AND ASSISTANCE
RECEIVED IN COMPLETING THIS ASSIGNMENT

_____ NO SOURCES WERE USED OR ASSISTANCE RECEIVED IN COMPLETING THIS
ASSIGNMENT

SIGNATURE: _____

- 1. Why is it important for regulatory bodies like the FDA to assess the manufacturing process in addition to the final biopharmaceutical product? (5 pts)**
- 2. Compare and contrast prokaryotic and eukaryotic cells in terms of organelles and compartmentalization. (5 pts)**
- 3. Describe the four levels of protein structure and explain the impact of replacing a reactive amino acid (e.g., cysteine) with a bulkier, non-reactive one (e.g., tyrosine) at the active site. (5 pts)**
- 4. Identify four key structural and functional differences between DNA and RNA. (5 pts)**
- 5. For industrial production of a therapeutic protein, would a defined or complex medium be more appropriate, and why? (5 pts)**
- 6. Evaluate the following biological statements. Identify if they are true or false and justify your answer: (5 pts)**
 - [a] Facultative organisms require oxygen to grow.**
 - [b] Yeasts are prokaryotic.**
 - [c] Bacteriophages are bacterial symbionts.**
 - [d] Growth media should be supplemented with D-amino acids.**
- 7. List and describe the four types of molecular interactions that influence protein folding in aqueous environments. (5 pts)**
- 8. Provide approximate dimensions for the following cell types and explain why size differences matter biologically: E. coli, yeast, liver cell, plant cell. (5 pts)**

9. What are the two main nutrient types required by cells, and what are their typical roles and examples? (5 pts)

10. Proteins are multifunctional molecules. List five diverse biological roles they play, with examples. (5 pts)

Instructor Solution

CH350 Problem Set 1

A Problem Set Submitted to:

Instructor/Section: LTC Hummel / Section _____

In Partial Fulfillment of

Requirements for CH350

By

Cadet: _____

West Point, New York

Date: _____ Time: _____

DO NOT WRITE IN THIS SPACE

PROBLEM	VALUE	CUT
1.1	5	
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1.4	5	
1.5	5	
1.6	5	
1.7	5	
1.8	5	
1.9	5	
1.10	5	
TOTAL GRADE	50	

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ASSIGNMENT

SIGNATURE: _____

1. Why is it important for regulatory bodies like the FDA to assess the manufacturing process in addition to the final biopharmaceutical product? (5 pts) (refer to pg. 9-10 of DeLisa, 2017)

Regulatory assessment of the manufacturing process is essential because even minor changes can introduce contaminants or alter product efficacy. Trace impurities might be undetectable unless specifically targeted, posing safety risks. By approving the process, regulators ensure consistency in raw materials and production methods, minimizing unforeseen side effects. However, it can hinder innovation, as any modifications require costly and lengthy re-approvals.

2. Compare and contrast prokaryotic and eukaryotic cells in terms of organelles and compartmentalization. (5 pts) (refer to pg. 18-23 of DeLisa, 2017)

Eukaryotic cells have membrane-bound organelles, such as a nucleus, mitochondria, and Golgi apparatus, enabling specialized internal environments. Prokaryotes lack these features but share some components, like ribosomes and plasma membranes. Both cell types perform protein synthesis and energy conversion, but their structural organization differs significantly.

3. Describe the four levels of protein structure and explain the impact of replacing a reactive amino acid (e.g., cysteine) with a bulkier, non-reactive one (e.g., tyrosine) at the active site. (5 pts) (refer to pg. 33-35 of DeLisa, 2017)

Protein structure includes:

- **Primary:** amino acid sequence.
- **Secondary:** hydrogen-bonded helices and sheets.
- **Tertiary:** 3D folding from distant R-group interactions.
- **Quaternary:** assembly of multiple polypeptides.

Substituting cysteine with tyrosine at the active site can abolish catalytic function by removing reactive thiol groups or altering geometry. If away from the active site, it may disrupt structural stability, especially if disulfide bonds are affected or if polarity and size differences impact folding.

4. Identify four key structural and functional differences between DNA and RNA. (5 pts) (refer to pg. 45-51 of DeLisa, 2017)

- **Strand Type:** DNA is double-stranded; RNA is single-stranded.
- **Sugar:** DNA uses deoxyribose; RNA uses ribose.
- **Bases:** DNA has thymine; RNA has uracil.
- **Functionality:** DNA stores genetic information; RNA performs various roles (mRNA, tRNA, rRNA) in gene expression.

5. For industrial production of a therapeutic protein, would a defined or complex medium be more appropriate, and why? (5 pts) (refer to pg. 55 of DeLisa, 2017)

A chemically defined medium is preferred because it allows precise control over composition, ensuring reproducibility and easier downstream purification. This reduces variability and enhances regulatory compliance, which is critical for high-value therapeutic proteins.

6. Evaluate the following biological statements. Identify if they are true or false and justify your answer: (5 pts) (refer to pg. 14-34 of DeLisa, 2017)

- [a] Facultative organisms require oxygen to grow.
- [b] Yeasts are prokaryotic.
- [c] Bacteriophages are bacterial symbionts.
- [d] Growth media should be supplemented with D-amino acids.

- [a] **False** – Facultative organisms can grow with or without oxygen.
- [b] **False** – Yeasts are eukaryotes.
- [c] **False** – Bacteriophages are viruses that infect bacteria, not symbionts.
- [d] **False** – L-amino acids should be used; D-forms are uncommon and biologically inactive in most cases.

7. List and describe the four types of molecular interactions that influence protein folding in aqueous environments. (5 pts) (refer to pg. 33-35 of DeLisa, 2017)

- Hydrophobic effect:** Drives nonpolar residues inward to avoid water, stabilizing the core.
- Disulfide bonds:** Covalent linkages between cysteine residues that stabilize structure.
- Hydrogen bonds:** Maintain secondary structures like α -helices and β -sheets.
- Van der Waals interactions:** Weak attractions contributing to overall shape.

8. Provide approximate dimensions for the following cell types and explain why size differences matter biologically: E. coli, yeast, liver cell, plant cell. (5 pts) (refer to pg. 18-27 of DeLisa, 2017)

- **E. coli:** $\sim 0.5\text{-}0.3\text{ }\mu\text{m}$
- **Yeast:** $\sim 5\text{ }\mu\text{m}$
- **Liver cell:** $\sim 10\text{ }\mu\text{m}$
- **Plant cell:** $\sim 20\text{ }\mu\text{m}$

Larger cells have more complex internal structures and functions. Smaller cells (like E. coli) rely on surface area for efficient nutrient uptake and replication.

9. What are the two main nutrient types required by cells, and what are their typical roles and examples? (5 pts) (refer to pg. 51 of DeLisa, 2017)

Macronutrients: Needed in high amounts; include C, H, O, N, S, P, K^+ , Mg^{2+} ; provide building blocks and energy.

Micronutrients: Needed in trace amounts; include Fe, Zn, Mo, vitamins; often act as cofactors or regulatory elements.

10. Proteins are multifunctional molecules. List five diverse biological roles they play, with examples. (5 pts) (refer to pg. 29-30 of DeLisa, 2017)

- 1) Structural (e.g., collagen, elastic, keratin)
- 2) Catalytic (e.g., amylase, pepsin, catalase)
- 3) Transport (e.g., hemoglobin, serum albumin, aquaporin)
- 4) Regulatory (e.g., transcription factor, insulin sigma factor)
- 5) Protective / Immune defense (e.g., antibodies, thrombin)



PROBLEM SET #2

Handouts

CH350 Problem Set 2

A Problem Set Submitted to:

Instructor/Section: LTC Stephen Hummel / Section _____

In Partial Fulfillment of

Requirements for CH350

By

Cadet: _____

West Point, New York

Date: _____ Time: _____

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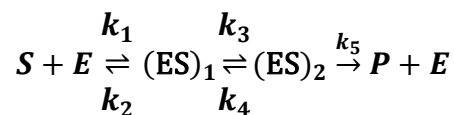
PROBLEM	VALUE	CUT
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2	10	
3 (3.6)	10	
4 (3.13)	10	
5 (3.18)	10	
TOTAL GRADE	50	

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RECEIVED IN COMPLETING THIS ASSIGNMENT

_____ NO SOURCES WERE USED OR ASSISTANCE RECEIVED IN COMPLETING THIS
ASSIGNMENT

SIGNATURE: _____

1. (10 pts) Consider the following reaction sequence:



Develop a suitable rate expression for production formation [$v = k_5(ES)_2$] by using (a) the equilibrium approach, and (b) the quasi-steady-state approach.

2. (10 pts) Data on enzymatic degradation of dichlorophenol at different concentrations are in the following table:

So (mg/l)	30	60	90	120	150	200	250	300
Rate (mg P/l-h)	2.1	3.8	5.7	7.4	8.5	7.1	4.5	2.5

Assuming the substrate-inhibited kinetics is given by the following equation, determine the constants V_m , K_m , and K_{si} .

$$V = \frac{V_m}{(1 + K_m / S)(1 + S / K_{si})}$$

3. (10 pts) Hydrogenase enzyme responsible for hydrogen gas synthesis in photoheterotrophic bacteria is activated by the addition of sodium molybdate into the medium. Hydrogen formation rate varies with the Mo concentration as follows:

Rate (ml H ₂ / h)	0.1	0.2	0.3	0.5	0.7	0.9	1.2
Mo (μg /l)	0	10	30	50	80	100	125

a. Assuming that the activator only affects the maximum reaction rate (V_m), determine the activation constant (K_A).

b. What would be the reaction rate at Mo = 60 μg /l?

4. (10 pts) The following data were obtained for an enzyme-catalyzed reaction. Determine V_{max} and K_m by inspection. Plot the data using the Eadie-Hofstee method and determine these constants graphically. Explain the discrepancy in your two determinations. The initial rate data for the enzyme-catalyzed reaction are as follows:

[S] mol / l	v μ mol / min
5.0×10^{-4}	125
2.0×10^{-4}	125
6.0×10^{-5}	121
4.0×10^{-5}	111
3.0×10^{-5}	96.5
2.0×10^{-5}	62.5
1.6×10^{-5}	42.7
1.0×10^{-5}	13.9
8.0×10^{-6}	7.50

5. (10 pts) Uric acid is degraded by uricase enzyme immobilized in porous Ca-alginate beads. Experiments conducted with different beads sizes result in the following rate data:

Bead diameter, Dp (cm)	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8
Rate, v (mg/l-h)	200	198	180	140	100	70	50	30

a. Determine the effectiveness factor for particle sizes $D_p = 0.5$ cm and $D_p = 0.7$ cm.

b. The following data were obtained for $D_p = 0.5$ cm at different bulk uric acid concentrations. Assuming negligible liquid film resistance, calculate V_m and K_s for the enzyme. Assume no substrate or product inhibition.

S_0 (mg UA / l)	10	25	50	100	200	250
v (mg UA /l-h)	10	20	30	40	45	46

Instructor Solution

CH350 Problem Set 2

A Problem Set Submitted to:

Instructor/Section: LTC Stephen Hummel / Section _____

In Partial Fulfillment of

Requirements for CH350

By

Cadet: APPROVED SOLUTION

West Point, New York

Date: _____ Time: _____

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PROBLEM	VALUE	CUT
1 (3.1)	10	
2 (3.4)	10	
3 (3.6)	10	
4 (3.13)	10	
5 (3.18)	10	
TOTAL GRADE	50	

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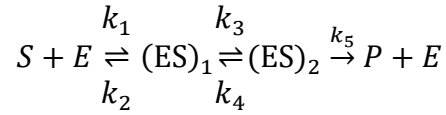
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ASSIGNMENT

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APPROVED SOLUTION

Problem 3.1. (10 pts) (also refer to p.65-67 of DeLisa, 2017)

Consider the following reaction sequence:



Develop a suitable rate expression for production formation [$v = k_5(ES)_2$] by using (a) the equilibrium approach, and (b) the quasi-steady-state approach.

a) Equilibrium approach

$$k_1(S)(E) = k_2(ES)_1 \quad (1)$$

$$k_3(ES)_1 = k_4(ES)_2 \quad (2)$$

$$(E_0) = (E) + (ES)_1 + (ES)_2 \quad (3)$$

$$\text{from (2): } (ES)_1 = \frac{k_4}{k_3}(ES)_2 \quad (4)$$

$$\text{from (1) + (4): } (E) = \frac{k_2 k_4}{k_1 k_3} \frac{(ES)_2}{(S)} \quad (5)$$

Substitute (4) and (5) into (3):

$$\begin{aligned} (E_0) &= \left(\frac{k_2 k_4 + k_1 k_4(S) + k_1 k_3(S)}{k_1 k_3(S)} \right) (ES)_2 \\ \therefore (ES)_2 &= \frac{k_1 k_3 (E_0)(S)}{k_2 k_4 + k_1 k_4(S) + k_1 k_3(S)} \\ \text{rate} &= \frac{k_5 E_0 S}{K_{m2}(K_{m1} + S) + S}, \text{ where } K_{m1} = \frac{k_2}{k_1}, K_{m2} = \frac{k_4}{k_3} \end{aligned}$$

b) Quasi-steady-state approach

$$\frac{d(ES)_1}{dt} = k_1(E)(S) - k_2(ES)_1 + k_4(ES)_2 - k_3(ES)_1 \approx 0 \quad (6)$$

APPROVED SOLUTION

$$\frac{d(ES)_2}{dt} = k_3(ES)_1 - k_4(ES)_2 - k_5(ES)_2 \approx 0 \quad (7)$$

$$\text{from (7):} \quad (ES)_1 = \frac{k_4 + k_5}{k_3} (ES)_2 \quad (8)$$

$$\text{from (6) + (8): } (E) = \left(\frac{k_2 k_4 + k_2 k_5 + k_3 k_5}{k_1 k_3(S)} \right) (ES)_2 \quad (9)$$

substitute (8) and (9) into (3):

$$(E_0) = \left(\frac{k_2 k_4 + k_2 k_5 + k_3 k_5 + k_1 k_4(S) + k_1 k_5(S) + k_1 k_3(S)}{k_1 k_3(S)} \right) (ES)_2$$

$$(ES)_2 = \left(\frac{k_1 k_3 (E_0)(S)}{k_2 k_4 + k_2 k_5 + k_3 k_5 + k_1 k_4(S) + k_1 k_5(S) + k_1 k_3(S)} \right)$$

$$\therefore \text{rate} = \frac{k_5 E_0 S}{\left(K_{m2} + \frac{k_5}{k_3} \right) (K_{m1} + S) + \frac{k_5}{k_1} + S}$$

$$\text{where } K_{m1} = \frac{k_2}{k_1} \text{ and } K_{m2} = \frac{k_4}{k_3}.$$

Problem 3.4. (5 pts) (also refer to p.76-77 of DeLisa, 2017)

At low substrate concentrations, $S_0 < 150$ mg/l. Thus, substrate inhibition is negligible:

$$V_0 = V_{m0} S_0 / (K_m + S_0) \text{ or } 1/V_0 = 1/V_m + K_m/V_m (1/S_0)$$

For $S_0 < 150$ mg/l, plot $1/V_0$ versus $1/S_0$.

$$\text{Slope} = K_m/V_{m0} = 13.8 \text{ and y-intercept} = 1/V_m = 0.023$$

$$\text{Then, } V_m = 43.5 \text{ mg/l-h and } K_m = 600 \text{ mg/l}$$

At high substrate concentrations above 150 mg/l. Thus, substrate inhibition is significant:

APPROVED SOLUTION

$$V_0 = V_{m0}/(1 + S_0/K_{S1}) \text{ or } 1/V_0 = 1/V_{m0} + S_0/(V_{m0}K_{S1})$$

For $S_0 > 150 \text{ mg/l}$, plot $1/V_0$ versus S_0 .

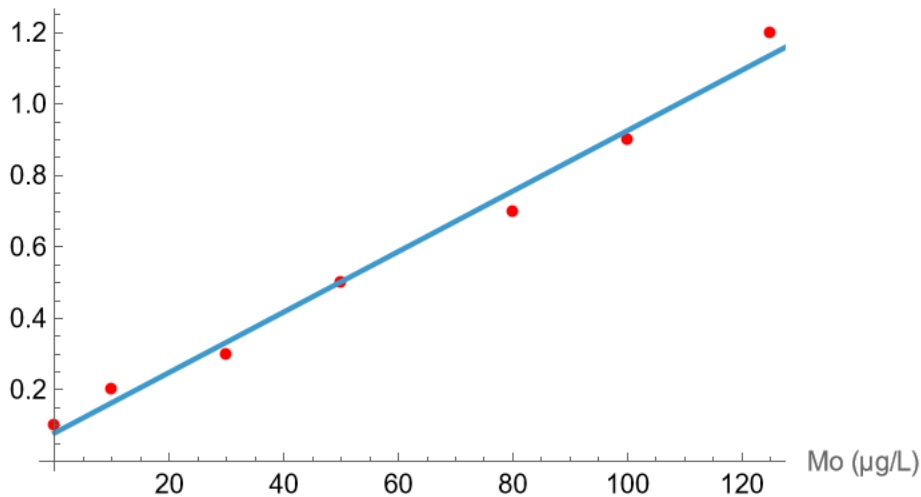
Slope = $1/(V_{m0}K_{S1}) = 2.59 \times 10^{-3}$. Then, $K_{S1} = 8.9 \text{ mg/l}$

Low K_{S1} indicates severe substrate inhibition.

Problem 3.6. (5 pts) (also refer to p.80 of DeLisa, 2017)

a) $V = V_m(1 + A/K_A)S/(K'_{m,app} + S)$

Rate (mL H₂/h)



Intercept (V_m) = 0.00799514

Slope = 0.00845656

$$K_A = (\text{intercept} / \text{slope}) = 9.45 \text{ ug /l}$$

~~Slope = $9.2 \times 10^{-3} = V_m/K_A$ and y intercept = $V_m = 0.04 \text{ ml/h}$~~

~~Then, $K_A = 4.35 \text{ ug/l}$~~

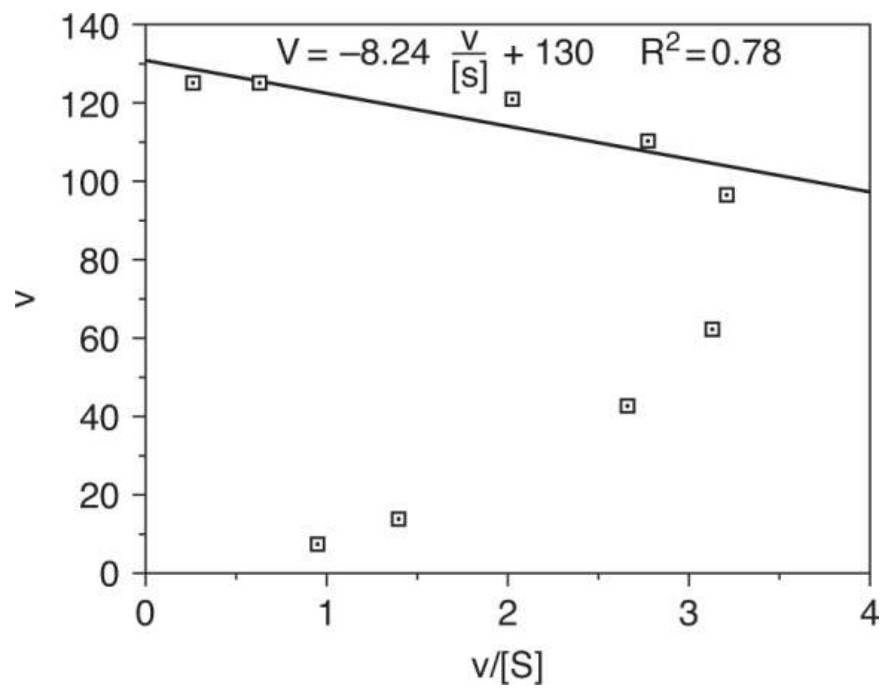
b) $V = V_m(1 + A/K_A) = 0.04(1 + 60/4.35) = 0.59 \text{ ml/h}$

APPROVED SOLUTION

Problem 3.13. (10 pts) (also refer to p.68-69 of DeLisa, 2017)

By inspection, $V_{\max} = 125 \mu\text{mol}/\text{min}$, $K_m = 20 \mu\text{mol}/\text{L}$

Plot V versus $V/[S]$ (Eadie-Hofstee), where y-int. = V_m , x-int. = V_m/K_m and slope = $-K_m$.



APPROVED SOLUTION

Plotting the data and using the data points for high substrate concentration,

$$V_m = 130 \mu\text{mol/min}, K_m = 8.24 \mu\text{mol/L}$$

From the plot, you can see that the data do not fit into Michaelis-Menten kinetics at low substrate concentrations.

Some of the error may be attributed to the method since both axes contain v . However, there are two possible explanations for the deviations from Michaelis-Menten kinetics:

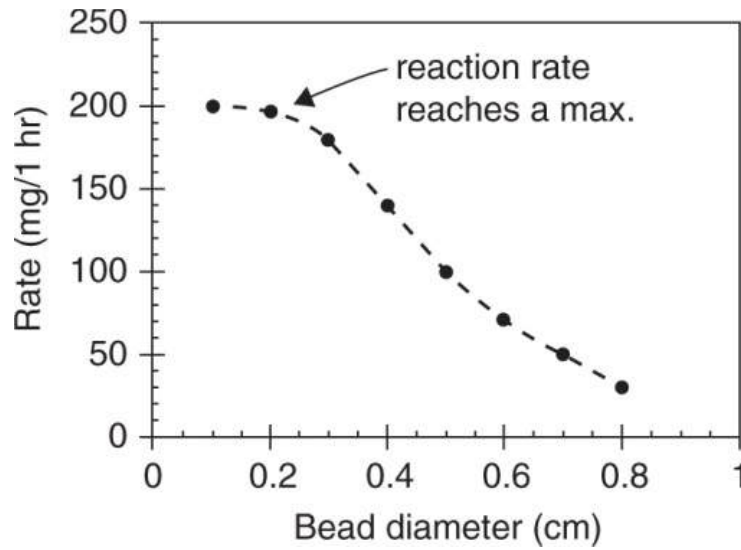
- (i) the enzyme is immobilized, thus diffusional effects become important.
- (ii) there might be unspecific binding of substrate to the enzyme thus requiring a critical substrate concentration for the reaction to follow Michaelis-Menten kinetics.

Problem 3.18. (10 pts) (also refer to p.93-96 of DeLisa, 2017)

$$\text{Effectiveness factor} = \frac{\text{reaction rate with diffusion limitation}}{\text{reaction rate with no diffusion limitation}}$$

We are given the reaction rate with diffusion limitation. To determine the reaction rate with no diffusion limitation, let's look at what happens to the rate as the particle gets smaller, and diffusional limitations are minimized. A plot of rate vs. particle size indicates that the rate reaches a maximum at small particle size (0.1 cm). At this size, there is no diffusion limitation and the rate is N 200 mg/l hr

APPROVED SOLUTION



a) Effectiveness factor at

$$D_p = 0.5 \text{ cm} \rightarrow \eta = \frac{100 \frac{\text{mg}}{\text{l} \cdot \text{hr}}}{\frac{200 \text{ mg}}{\text{l} \cdot \text{hr}}}$$

$$\underline{\eta = 0.50}$$

$$D_p = 0.7 \text{ cm} \rightarrow \eta = \frac{50 \frac{\text{mg}}{\text{l} \cdot \text{hr}}}{\frac{200 \text{ mg}}{\text{l} \cdot \text{hr}}}$$

$$\underline{\eta = 0.25}$$

b) Assume negligible film resistance, so $S_{\text{bulk}} = S_{\text{surface}}$ Determine rate u/o diffusional limitation from effectiveness factor:

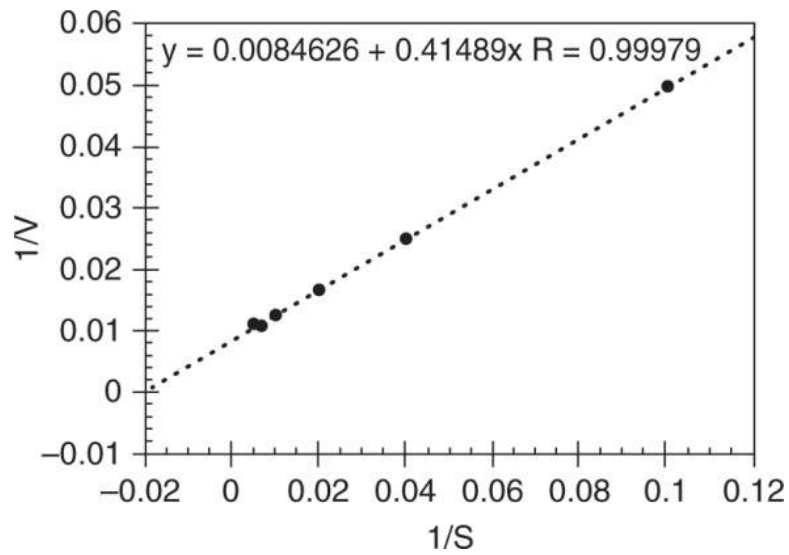
$$\text{rate} \left(\frac{w}{o} \text{ diff} \right) = \frac{\text{rate at } D_p = 0.5 \text{ cm}}{\eta \text{ at } D_p = 0.5 \text{ cm}}$$

Plot $1/V$ vs $1/S$ to obtain the following (Lineweaver-Burk plot):

S_0	10	25	50	100	200	250
V	10	20	30	40	45	46

APPROVED SOLUTION

$V_{\text{no diff}}$	20	40	60	80	90	92
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$1/V_m$ is at $x = 0$: $y = 0.0084626 \rightarrow \underline{V_m = 118.2 \text{ mg uA/l} \cdot \text{h}}$

$-1/K_m$ is at $y = 0$: $0 = 0.0084626 + 0.41489 (-1/K_m)$

$$\frac{-0.0084626}{0.41489} = \frac{-1}{K_m}$$

$$\underline{K_m = 49.0 \frac{\text{mg uA}}{\text{l}}}$$



PROBLEM SET #3

Handouts

CH350 Problem Set 3

A Problem Set Submitted to:

Instructor/Section: LTC Hummel / Section _____

In Partial Fulfillment of

Requirements for CH350

By

Cadet: _____

West Point, New York

Date: _____ Time: _____

DO NOT WRITE IN THIS SPACE

PROBLEM	VALUE	CUT
3.1	5	
3.2	5	
3.3	5	
3.4	5	
3.5	5	
3.6	5	
3.7	5	
3.8	5	
3.9	5	
3.10	5	
TOTAL GRADE	50	

_____ MY DOCUMENTATION ATTRIBUTES ALL SOURCES USED AND ASSISTANCE
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ASSIGNMENT

SIGNATURE: _____

1. Consider the aspartic acid pathway shown in Figure 4A.1 (in pg. 141 of *Shuler, Kargi, DeLisa, 2017*). Assume you have been asked to develop a high-lysine-producing mutant. What strategy would you pursue? (That is, which steps would you modify by removing feedback inhibitions, and what changes in medium composition would you make over a simple mineral salts-glucose base medium?). **(5 pts)**

2. Why is mRNA so unstable in most bacteria (half-life of about 1 min)? In many higher organisms, mRNA half-lives are much longer (> 1hr). Why? (5 pts)

3. Briefly explain the flow of genetic information in cells according to the central dogma of molecular biology. Include transcription and translation. **(5 pts)**

4. List the ATP-consuming and ATP-generating steps of glycolysis. What is the net ATP gain? **(5 pts)**

5. What are the two major functions of the TCA cycle in cellular metabolism? **(5 pts)**

6. Differentiate the Pasteur effect and the Crabtree effect in microbial metabolism. **(5 pts)**

7. What is the Gibbs free energy equation for protein-ligand binding, and how does it relate to enthalpic and entropic contributions? **(5 pts)**

8. Compare the light-dependent and light-independent (dark) reactions in photosynthesis. **(5 pts)**

9. Which is more likely to revert to wild-type function: a point mutation or a deletion? Explain. **(5 pts)**

10. Compare transformation and transduction as mechanisms for bacterial gene transfer. **(5 pts)**

Instructor Solution

APPROVED SOLUTION

CH350 Problem Set 3

A Problem Set Submitted to:

Instructor/Section: LTC Hummel / Section _____

In Partial Fulfillment of

Requirements for CH350

By

Cadet: APPROVED SOLUTION

West Point, New York

Date: _____ Time: _____

DO NOT WRITE IN THIS SPACE

PROBLEM	VALUE	CUT
3.1	5	
3.2	5	
3.3	5	
3.4	5	
3.5	5	
3.6	5	
3.7	5	
3.8	5	
3.9	5	
3.10	5	
TOTAL GRADE	50	

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ASSIGNMENT

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APPROVED SOLUTION

1. Consider the aspartic acid pathway shown in Figure 4A.1 (in pg. 141 of *Shuler, Kargi, DeLisa, 2017*). Assume you have been asked to develop a high-lysine-producing mutant. What strategy would you pursue? (That is, which steps would you modify by removing feedback inhibitions, and what changes in medium composition would you make over a simple mineral salts-glucose base medium?). **(5 pts)**

As shown in Figure 4A.1, the conversion of aspartic acid to aspartyl phosphate is mediated by three isofunctional enzymes. The aspartyl phosphate is further converted into lysine by five enzymes (m-q), which are subject to specific repression by the end product, L-lysine.

To develop a high-lysine-producing mutant, the first step of converting from aspartyl phosphate to L-lysine can be modified by removing feedback inhibitions on the enzyme through catalyzed process. Also, the changes in medium composition can be done to simplify the mineral salts-glucose base content, leading to the production of large amount of lysine.

2. Why is mRNA so unstable in most bacteria (half-life of about 1 min)? In many higher organisms, mRNA half-lives are much longer (> 1hr). Why? (5 pts)

The additional steps involved in eukaryotic mRNA maturation produce a molecule that is much more stable than a prokaryotic mRNA. The mRNA transcript is coated in RNA-stabilizing proteins to prevent it from degrading while it is processed and exported out of the nucleus. A special nucleotide “cap” is added to one end of the growing transcript, which also helps prevent degradation and helps the cell recognize this molecule as an mRNA that should be translated. Once transcription is complete, an enzyme adds a string of approximately 200 adenine residues to the end of the mRNA, called the poly-A tail. This modification further protects the pre-mRNA from degradation and signals to cellular factors that the transcript needs to be exported to the cytoplasm.

Such additional steps lead to the creation of eukaryotic mRNA metabolically stable compared to prokaryotic mRNA.

3. Briefly explain the flow of genetic information in cells according to the central dogma of molecular biology. Include transcription and translation. **(5 pts)**

The central dogma describes the flow:

DNA → RNA → Protein

- **Transcription:** DNA is used as a template to synthesize mRNA via RNA polymerase.
- **Translation:** Ribosomes decode mRNA to assemble amino acids into a polypeptide chain using the genetic code.
This process controls how genes are expressed and ultimately how cellular function is regulated.

APPROVED SOLUTION

4. List the ATP-consuming and ATP-generating steps of glycolysis. What is the net ATP gain? **(5 pts)**

ATP-consuming steps:

- *Glucose $\rightarrow$ Glucose-6-phosphate (hexokinase)*
- *Fructose-6-phosphate $\rightarrow$ Fructose-1,6-bisphosphate (PFK)*

ATP-generating steps:

- *1,3-bisphosphoglycerate $\rightarrow$ 3-phosphoglycerate*
- *Phosphoenolpyruvate $\rightarrow$ Pyruvate*

Net gain: 2 ATP per glucose.

5. What are the two major functions of the TCA cycle in cellular metabolism? **(5 pts)**

- ***Energy generation:** Produces NADH and FADH₂ for the electron transport chain*
- ***Biosynthetic intermediates:** Supplies carbon skeletons for amino acid and nucleotide synthesis*

6. Differentiate the Pasteur effect and the Crabtree effect in microbial metabolism. **(5 pts)**

- ***Pasteur Effect:** Under **aerobic conditions**, glycolysis slows due to efficient ATP generation via oxidative phosphorylation.*
- ***Crabtree Effect:** In **high-glucose aerobic conditions**, some cells (e.g., yeast) **prefer fermentation**, leading to ethanol production despite oxygen presence.*

7. What is the Gibbs free energy equation for protein-ligand binding, and how does it relate to enthalpic and entropic contributions? **(5 pts)**

The equation is:

$$\Delta G = \Delta H - T\Delta S$$

- *ΔG : Gibbs free energy change*
- *ΔH : Enthalpy change (bonding energy, hydrogen bonds, etc.)*
- *$T\Delta S$: Temperature $\times$ entropy change (disorder in the system)*

*A negative ΔG indicates favorable binding. Both **enthalpic gains** (favorable bonds) and **entropic effects** (e.g., release of water from hydrophobic pockets) contribute to this.*

8. Compare the light-dependent and light-independent (dark) reactions in photosynthesis. **(5 pts)**

APPROVED SOLUTION

- **Light-dependent:** Capture light to generate ATP and NADPH; split water to release O_2
- **Dark reactions (Calvin Cycle):** Use ATP/NADPH to fix CO_2 into carbohydrates (e.g., glucose)

9. Which is more likely to revert to wild-type function: a point mutation or a deletion? Explain. **(5 pts)**

- **Point mutation** is more likely to revert because only one nucleotide is affected. A **deletion** can cause frameshifts and remove essential residues, making reversal highly unlikely.

10. Compare transformation and transduction as mechanisms for bacterial gene transfer. **(5 pts)**

- **Transformation:** Uptake of free DNA from the environment
 - **Transduction:** Gene transfer via bacteriophages (viruses)
- Both are used in genetic engineering and natural bacterial evolution.



PROBLEM SET #4

Handouts

CH350 Problem Set 4

A Problem Set Submitted to:

Instructor/Section: LTC Hummel / Section _____

**In Partial Fulfillment of
Requirements for CH350**

By

Cadet: _____

West Point, New York

Date: _____ Time: _____

DO NOT WRITE IN THIS SPACE

PROBLEM	VALUE	CUT
1 (6.1)	4	
2 (6.3)	10	
3 (6.5)	6	
4 (6.9)	6	
5 (7.1)	4	
6 (7.4)	8	
7 (7.6)	6	
8 (7.8)	6	
TOTAL GRADE	50	

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THIS ASSIGNMENT

SIGNATURE: _____

1. By using the following information, determine the cell number concentration in a fermenter containing 1 g dry weight/l biomass:
 - Cell dimensions: $1 \times 2 \mu\text{m}$ cylindrical cells
 - Cell density: 1.05 g/cm^3
 - Water content of cells: 80%

2. A simple, batch fermentation of an aerobic bacterium growing on methanol gave the results shown in the following table. Calculate the following:
 - a. Maximum growth rate (μ_{max})
 - b. Yield on substrate ($Y_{X/S}$)
 - c. Mass doubling time (t_d)
 - d. Saturation constant (K_s)
 - e. Specific growth rate (μ_{net}) at $t = 10 \text{ h}$

Time (h)	X (g / l)	S (g / l)
0	0.2	9.23
2	0.211	9.21
4	0.305	9.07
8	0.98	8.03
10	1.77	6.8
12	3.2	4.6
14	5.6	0.92
16	6.15	0.077
18	6.2	0

3. The following data were obtained for the effect of temperature on the fermentative production of lactic acid by a strain of *Lactobacillus delbrueckii*. From these data, calculate the value of the activation energy for this process. Is the value of the activation energy typical of this sort of biological conversion?

Temperature (C)	Rate Constant (mol / l – h)
40.4	0.0140
36.8	0.0112
33.1	0.0074
30.0	0.0051
25.1	0.0036

4. *Pseudomonas* sp. has a mass doubling time of 2.4 h when grown on acetate. The saturation constant using this substrate is 1.3 g/l (which is unusually high), and cell yield on acetate is 0.46 g cell/g acetate. If we operate a chemostat on a feed stream containing 38 g/l acetate, find the following:
 - a. Cell concentration when the dilution rate is one-half of the maximum
 - b. Substrate concentration when the dilution rate is $0.8 D_{\max}$
 - c. Maximum dilution rate
 - d. Cell productivity at $0.8 D_{\max}$

5. Determine the amount of $(\text{NH}_4)_2\text{SO}_4$ to be supplied in a fermentation medium where the final cell concentration is 30 g/l in a 10^3 l culture volume. Assume that the cells are 12% nitrogen by weight and $(\text{NH}_4)_2\text{SO}_4$ is the only nitrogen source.

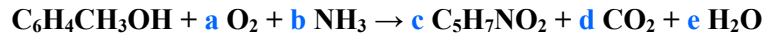
6. Aerobic growth of *S. cerevisiae* on ethanol is simply described by the following overall reaction:

$$\text{C}_2\text{H}_5\text{OH} + a \text{O}_2 + b \text{NH}_3 \rightarrow c \text{CH}_{1.704}\text{N}_{0.149}\text{O}_{0.408} + d \text{CO}_2 + e \text{H}_2\text{O}$$
 - a. Determine the coefficients a , b , c , and d , where $\text{RQ} = 0.66$.
 - b. Determine the biomass yield coefficient, $Y_{X/S}$, and oxygen yield coefficient, Y_{X/O_2} (gdw/g O_2).

7. Aerobic degradation of an organic compound by a mixed culture of organisms in waste water can be represented by the following reaction:

$$\text{C}_3\text{H}_6\text{O}_3 + a \text{O}_2 + b \text{NH}_3 \rightarrow c \text{C}_5\text{H}_7\text{NO}_2 + d \text{H}_2\text{O} + e \text{CO}_2$$
 - a. Determine a , b , c , d , and e if $Y_{X/S} = 0.4$ g X/g S.
 - b. Determine the yield coefficients Y_{X/O_2} and Y_{X/NH_3}
 - c. Determine the degree of reductions for the substrate, bacteria, and RQ for the organisms.

8. The following biological oxidation reaction takes place in biological treatment of an industrial wastewater:



- Determine the coefficients of the equation (a , b , c , d , and e) if $\text{RQ} = 0.5$.
- Determine $Y_{x/s}$, and Y_{x/O_2} .
- Determine Y_s and Y_b .

Instructor Solution

APPROVED SOLUTION

CH350 Problem Set 4

A Problem Set Submitted to:

Instructor/Section: LTC Hummel / Section _____

In Partial Fulfillment of

Requirements for CH350

By

Cadet: APPROVED SOLUTION

West Point, New York

Date: _____ Time: _____

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PROBLEM	VALUE	CUT
1 (6.1)	4	
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7 (7.6)	6	
8 (7.8)	6	
TOTAL GRADE	50	

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APPROVED SOLUTION

6.1. (4 pts) (refer to pg. 171 of DeLisa, 2017)

Volume of a single cell: $V_c = [(\pi) \times (1 \text{ um}/2)^2 \times (2 \text{ um})]/\text{cell} = 1.57 \text{ um}^3/\text{cell} = 1.57 \times 10^{-12} \text{ cm}^3/\text{cell}$ (using a simple cylinder volume equation of $\pi r^2 h$)

Mass of a single cell: $m_c = 1.05 \text{ g/cm}^3 \times 1.57 \times 10^{-12} \text{ cm}^3/\text{cell} = 1.65 \times 10^{-12} \text{ g/cell}$

Dry weight (dw) of a single cell: $m_{c,dw} = 0.20 \times 1.65 \times 10^{-12} \text{ g/cell} = 0.33 \times 10^{-12} \text{ g dw/cell}$

Number of cells in 1 g dry weight: $N = 1 \text{ g dw} / (0.33 \times 10^{-12} \text{ g dw/cell}) = 3 \times 10^{12} \text{ cell}$

Cell number density (per liter): $3 \times 10^{12} \text{ cells/liter}$

Cell dw density of 1 g/liter is approximately equal to $3 \times 10^{12} \text{ cells/liter}$ cell number density.

6.3. (10 pts) (refer to pg. 177-179 and 192-193 of DeLisa, 2017)

a. $\mu_{\max}$

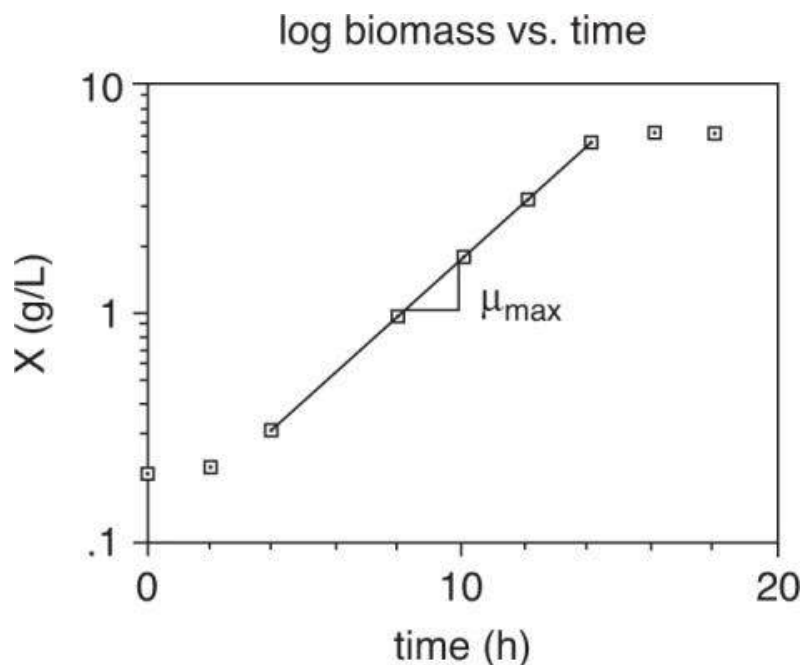
$$\mu_{\text{net}} = \frac{1}{X} \frac{dX}{dt}, \text{ when } \mu \text{ is constant} \rightarrow \ln X - \ln X_0 = \mu t$$

plot $\log X$ vs. t

The greatest slope occurs in linear region.

$$\mu_{\max} = \frac{\ln 3.2 - \ln 0.98}{(12 - 8) \text{ hr}}$$

$$\underline{\underline{\mu_{\max} = 0.296 \text{ hr}^{-1}}}$$



APPROVED SOLUTION

b. $Y_{X/S}^{App}$

$$Y_{X/S}^{App} = \frac{-\Delta X}{\Delta S} = \frac{-(6.2 - 0.2) \text{ g cells/L}}{(0 - 9.23) \text{ g} \frac{\text{substrate}}{\text{L}}} = 0.65 \frac{\text{g cells}}{\text{g substrate}}$$

c. t_d

$$\ln\left(\frac{X}{X_0}\right) = \mu t, t_d \text{ occurs when } X = 2X_0$$

$$\ln 2 = \mu t_d, t_d = \frac{\ln 2}{\mu}, t_d = \underline{\underline{2.34 \text{ hr}}}$$

d. K_s

$$\mu = \frac{\mu_m [S]}{K_s + [S]}$$

-up until $t = 4 \text{ hr}$ have lag phase, after that Monod kinetics can be applied

-between $t = 4 \text{ hr}$ & $t = 14 \text{ hr}$ $\mu = \mu_{\max}$

- at $[S] = 0$, $\mu = 0 \text{ hr}^{-1}$

- plot X vs. t , determine $\frac{dX}{dt}$, $\mu = \frac{1}{X} \frac{dX}{dt}$

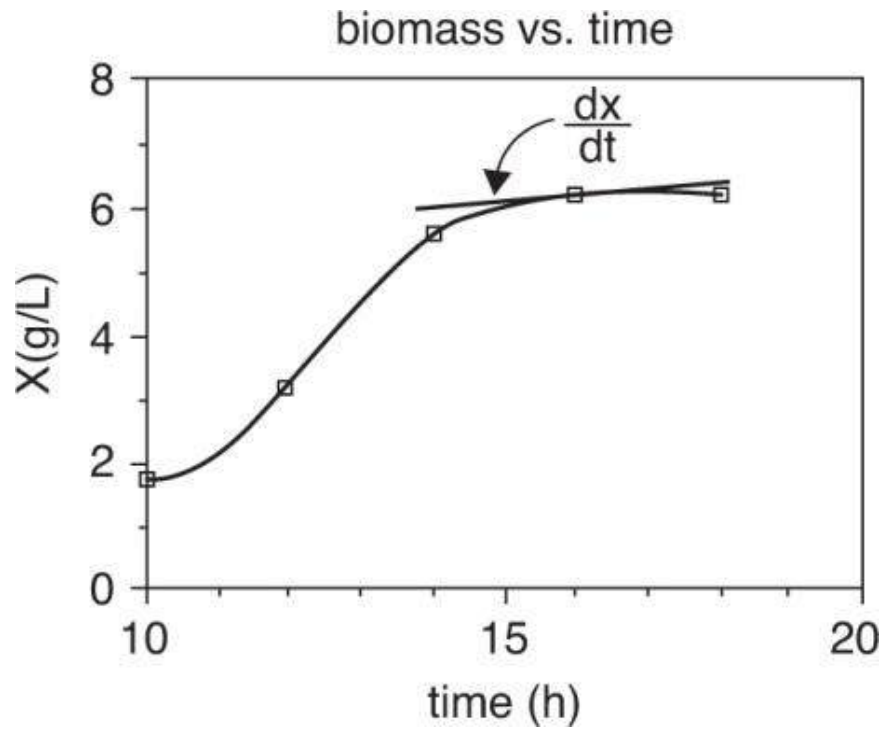
at $t = 16 \text{ hr}$

$$[S] = 0.077 \text{ g/L}$$

$$\frac{dX}{dt} = 0.123 \frac{\text{g}}{\text{L} \cdot \text{hr}}$$

$$\mu = 0.020 \text{ hr}^{-1}$$

APPROVED SOLUTION

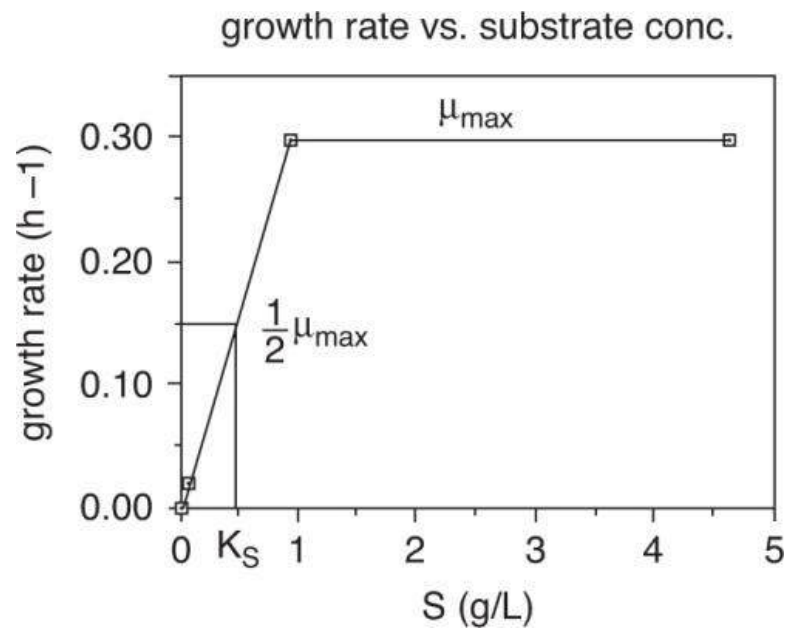


-plot μ vs. S

- $K_s = [S]$ at

$$\mu = \frac{1}{2} \mu_{\max}$$

$$\underline{\underline{K_s = 0.45 \text{ g/L}}}$$



e. at $t = 10$ hr, $\underline{\underline{\mu_{\text{net}} = \mu_{\max} = 0.296 \text{ hr}^{-1}}}$

APPROVED SOLUTION

6.5. (6 pts) (refer to pg. 185 of DeLisa, 2017)

$$k = Ae^{-E/RT}$$

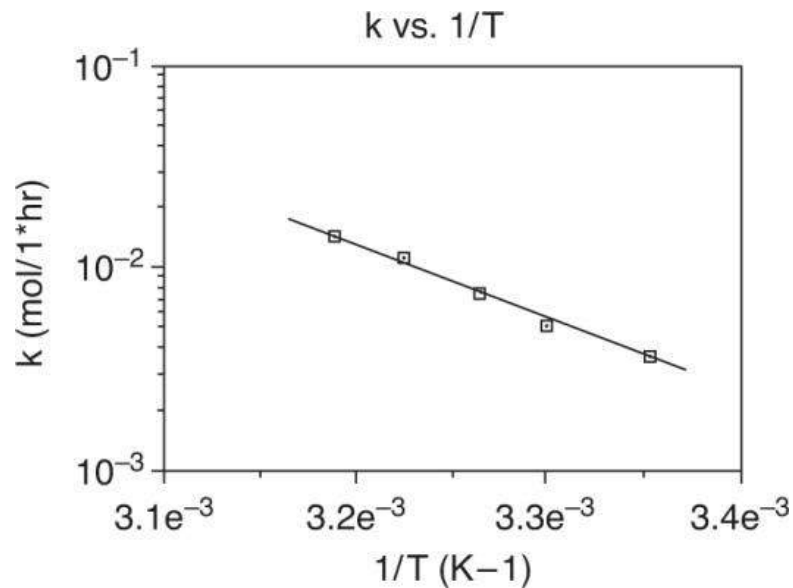
plot log k vs. 1/T

$$\text{slope} = \frac{-E}{R}$$

$$\frac{-E}{R} = \frac{\ln(.0036) - \ln(.0140)}{(3.353 - 3.189) \times 10^{-3} \text{ K}^{-1}}$$

$$\frac{-E}{R} = -8281 \text{ K}$$

$$\underline{\underline{E = 16.5 \text{ kcal/mol}}}$$



Biological conversions are on the order of 10 kcal/mol.

6.9. (6 pts) (refer to pg. 210 to 215 of DeLisa, 2017)

a. $\mu_{\max} = \frac{\ln 2}{t_d} = 0.289 \text{ hr}^{-1}$ where t_d is 2.4 hr.

$$D = \frac{1}{2} D_{\max} = \frac{1}{2} \mu_{\max}$$

$$X = Y_{X/S} \left[S_0 - \frac{K_s D}{\mu_m - D} \right] = Y_{X/S} (S_0 - K_s)$$

$$\underline{\underline{X = 16.9 \text{ g dry weight/L}}}$$

APPROVED SOLUTION

b. $D = 0.8D_{\max} = 0.8\mu_{\max}$

$$S = \frac{K_s D}{\mu_m - D} = 4K_s = \underline{\underline{5.2 \text{ g/L}}}$$

c. $\underline{\underline{D_{\max} = \mu_{\max} = 0.289 \text{ hr}^{-1}}}$

d. $DX = DY_{x/s} \left[S_0 - \frac{K_s D}{\mu_m - D} \right] \quad w/D = 0.8\mu_{\max}$

$$\underline{\underline{DX = 3.49 \frac{\text{g dry weight}}{\text{L} \cdot \text{hr}}}}$$

7.1. (4 pts)

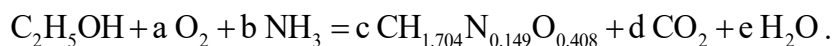
Total nitrogen in cells = (Nitrogen content of cells) $\times$ (Final cell concentration) $\times$ (Culture volume) = (0.12 g nitrogen/g cells) (30 g cell/liter) (1000 liters) = 3.6 kg nitrogen.

MW of $(\text{NH}_4)_2\text{SO}_4 = 132$; Nitrogen part is 28.

Amount of $(\text{NH}_4)_2\text{SO}_4$ needed = 3.6 kg nitrogen $\times$ (132 kg $(\text{NH}_4)_2\text{SO}_4$ /28 kg nitrogen) = 17.0 kg $(\text{NH}_4)_2\text{SO}_4$.

*Note: Remember that the stoichiometric conversion is based on the mole. You can use the mass to do conversion since $(\text{NH}_4)_2\text{SO}_4$ is the only nitrogen source for this fermentation process though.

7.4. (8 pts) (refer to pg. 230-235 of DeLisa, 2017)



Perform a material balance on each element based on the mole:

H-balance: $6 + 3b = 1.704c + 2e$

C-balance: $2 = c + d$

O-balance: $1 + 2a = 0.408c + 2d + e$

N-balance: $b = 0.149c$

RQ is given:

$$RQ = d/a = 0.66$$

Solve 5 equations for 5 unknowns: a, b, c, d, and e.

Result: $a=2.917$; $b=0.011$; $c=0.075$; $d=1.925$; $e=2.953$

Yield coefficients:

$Y_{x/s}$ = weight biomass / weight substrate

APPROVED SOLUTION

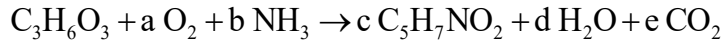
$$\begin{aligned}\text{Weight biomass} &= c \times (\text{MW } \text{CH}_{1.704}\text{N}_{0.149}\text{O}_{0.408}) \\ &= 0.075(12 + 1.704 + 14[0.149] + 16[0.408]) = 1.674 \\ \text{Weight substrate} &= 12(2) + 6 + 16 = 46\end{aligned}$$

$$Y_{x/s} = 1.674/46 = 0.364 \text{ gX / gS}$$

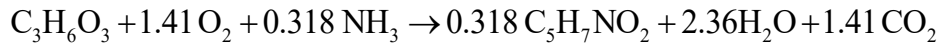
$$\begin{aligned}Y_{x/O_2} &= \text{Weight biomass} / \text{Weight } O_2 \text{ required} \\ \text{Weight } O_2 \text{ required} &= a \times \text{MW } O_2 = 2.917 (16) = 46.67\end{aligned}$$

$$Y_{x/O_2} = 1.674 / 46.67 = 0.359 \text{ gX / gO}_2$$

7.6. (6 pts) (refer to pg. 230-235 of DeLisa, 2017)



$$\begin{aligned}\text{a. C balance : } 3 &= 5c + e \\ \text{H balance : } 6 + 3b &= 7c + 2d \\ \text{O balance : } 3 + 2a &= 2c + d + 2e \\ \text{N balance : } b &= c \\ Y_{x/s} : Y_{x/s} &= \frac{C(60 + 7 + 14 + 32)}{(36 + 6 + 48)} = \frac{113 c}{90} = 0.4 \\ c &= 0.318 ; e = 1.41 \\ b &= 0.318 ; d = 2.364 \\ a &= 1.41\end{aligned}$$



$$\text{b. } Y_{x/O_2} = \frac{0.318 (60 + 7 + 14 + 32)}{1.41 (32)} = \frac{35.934}{45.12} \cong 0.8 \text{ gX / gO}_2$$

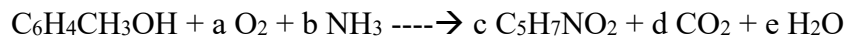
$$Y_{x/\text{NH}_3} = \frac{35.934}{0.318 (17)} = 6.65 \text{ gX / g NH}_3$$

$$\text{c. } \gamma'_b = 5(4) + 7(1) + 1(-3) + 2(-2) = 20 ; \gamma_b = \frac{20}{5} = 4$$

$$\gamma'_s = 3(4) + 6(1) + 3(-2) = 12 ; \gamma_s = \frac{12}{3} = 4$$

$$\text{RQ} = \frac{e}{a} = \frac{1.41}{1.41} = 1.$$

7.8. (6 pts) (refer to pg. 230-235 of DeLisa, 2017)



a. Elemental balances

APPROVED SOLUTION

Carbon: $7 = 5c + d$

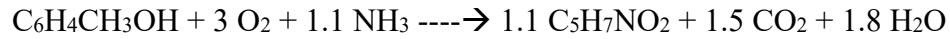
Hydrogen: $8 + 3b = 7c + 2e$

Oxygen: $1 + 2a = 2c + 2d + e$

Nitrogen: $b = c$

$RQ = d/a = 0.5$

Solution: $a = 3, b = 1.1, c = 1.1, d = 1.5, e = 1.8$



b. $Y_{x/s} = 1.1(113)/108 = 1.15 \text{ gX/gS}$

$$Y_{x/O_2} = 1.1(113)/96 = 1.295 \text{ gX/gO}_2$$

c. $\gamma, s = (7(4) + 8(1) + (-2)) / 7 = 34/7 = 4.86$

$$\gamma, b = (5(4) + 7(1) + (-3) + 2(-2)) / 5 = 20/5 = 4$$



PROBLEM SET #5

Handouts

CH350 Problem Set 5

A Problem Set Submitted to:

Instructor/Section: LTC Hummel / Section _____

In Partial Fulfillment of

Requirements for CH350

By

Cadet: _____

West Point, New York

Date: _____ Time: _____

DO NOT WRITE IN THIS SPACE

PROBLEM	VALUE	CUT
1	8	
2	8	
3	8	
4	8	
5	10	
6	8	
TOTAL GRADE	50	

_____ MY DOCUMENTATION ATTRIBUTES ALL SOURCES USED AND ASSISTANCE
RECEIVED IN COMPLETING THIS ASSIGNMENT

_____ NO SOURCES WERE USED OR ASSISTANCE RECEIVED IN COMPLETING THIS
ASSIGNMENT

SIGNATURE: _____

1. Consider a 1000 l CFSTR in which biomass is being produced with glucose as the substrate. The microbial system follows a Monod relationship with $\mu_m = 0.4 \text{ h}^{-1}$, $K_s = 1.5 \text{ g/l}$ (an unusually high value), and the yield factor $Y_{X/S} = 0.5 \text{ g/l biomass/g substrate consumed}$. Normal operation is with a sterile feed containing 10 g/l glucose at a rate of 100 l/h. **(8 pts)** (refer to pg. 278-281 of DeLisa, 2017)

a. What is the specific biomass production rate (g/l-h) at steady state?

b. If recycle is used with a recycle stream of 10 l/h and a recycle biomass concentration five times as large as that in the reactor exit, what would be the new specific biomass production rate?

c. Explain any difference between the values found in parts a and b.

2. In a two-stage chemostat system, the volumes of the first and second reactors are $V_1 = 500$ l and $V_2 = 300$ l, respectively. The first reactor is used for biomass production and the second is for a secondary metabolite formation. The feed flow rate to the first reactor is $F = 100$ l/h, and the glucose concentration in the feed is 5.0 g/l. Use the following constants for the cells: **(8 pts)** (refer to pg. 281-284 of DeLisa, 2017)

$$\mu_m = 0.3 \text{ h}^{-1}, K_s = 0.1 \text{ g/l}, Y_{X/S} = 0.4 \frac{\text{g dw cells}}{\text{g glucose}}$$

a. Determine cell and glucose concentration in the effluent of the first stage.

b) Assume that growth is negligible in the second stage and the specific rate of product formation is $q_p = 0.02$ g P/g cell h, and $Y_{P/S} = 0.6$ g P/g. Determine the product and substrate concentration in the effluent of the second reactor.

3. The bioconversion of glucose to ethanol is carried out in a packed-bed, immobilized cell bioreactor containing yeast cells entrapped in Ca-alginate beads. The rate-limiting substrate is glucose, and its concentration in the feed bulk liquid phase is $S_{0i} = 5$ g/l. The nutrient flow rate is $F = 2$ l/min. The particle size of Ca-alginate beads is $D_p = 0.5$ cm. The surface area of the alginate beads per unit volume of the reactor is $a = 25$ cm²/cm³, and the cross-sectional area of the bed is $A = 100$ cm². **(8 pts)** (refer to pg. 308-311 of DeLisa, 2017)

Following are the rate constants for this conversion:

$$r_m = 100 \text{ mg S/cm}^3 \cdot \text{h}$$

$$K_S = 10 \text{ mg S/cm}^3$$

for the following rate expression:

$$r_s = \frac{r_m S}{K_S + S}$$

Assuming a first-order reaction-kinetics (e.g., relatively low substrate concentrations), determine the required bed height for 80% conversion of glucose to ethanol at the exit stream. *Hint:* To calculate the effectiveness factor, we can use the following equations:

$$\eta = \frac{1}{\phi} \left[\frac{1}{\tanh 3\phi} - \frac{1}{3\phi} \right]$$

where

$$\phi = \frac{V_p}{A_p} \sqrt{\frac{k_1}{D_S}}$$

and

$$k_1 = \frac{r_m}{K_S}, \quad D_S = 10^{-6} \text{ cm}^2/\text{s}$$

4. A completely mixed bioreactor is used for ethanol fermentation of hydrolyzed starch solution containing 30 g/L total sugar. The following data are available: $\mu_m = 0.15 \text{ hr}^{-1}$, $K_s = 0.5 \text{ g/L}$, $F = 10 \text{ L/hr}$, $Y_{X/S} = 0.1 \text{ g X/g S}$, $Y_{P/S} = 0.5 \text{ g P/g S}$, $k_d = 0.1 \text{ day}^{-1}$. **(8 pts)** (refer to pg. 278-288 of DeLisa, 2017)

a. For 95% conversion of substrate to product, determine the required reactor volume and ethanol productivity when the system was operated without cell recycle.

b. For the same degree of conversion, determine the required reactor volume and ethanol productivity when the system was operated with cell recycle. $C = X_r/X = 3$ and $\alpha = 0.1$.

5. The air supply to a fermenter was turned off for a short period of time and then restarted. A value for C^* of 7.3 mg/l has been determined for the operating conditions. Use the tabulated measurements of dissolved oxygen (DO) values in the following table to estimate the oxygen uptake rate and $k_L a$ in this system. **(10 pts)** (refer to pg. 334-337 of DeLisa, 2017)

	Time (min)	DO (mg/l)
	-1	3.3
Air off	0	3.3
	1	2.4
	2	1.3
	3	0.3
	4	0.1
	5	0.0
Air on	6	0.0
	7	0.3
	8	1.0
	9	1.6
	10	2.0
	11	2.4
	12	2.7
	13	2.9
	14	3.0
	15	3.1
	16	3.2
	17	3.2

6. A lethal agent (KCN) was added to the fermentation medium in the presence of aeration using diffusors. Dissolved oxygen (DO) concentration increased with time as follows when $N = 100$ rev/min:

t (min)	0	1	2	3	5	7	8	10
DO (mg/L)	1	1.93	2.7	3.36	4.4	5.1	5.4	5.9

Saturation DO concentration in the fermentation medium is 7.0 mg/L. **(8 pts)** (refer to pg. 332-337 of DeLisa, 2017)

a. Determine oxygen transfer coefficient $k_L\alpha$.

b. Determine $k_L\alpha$ when N is increased to 200 rev/min.

Instructor Solution

APPROVED SOLUTION

CH350 Problem Set 5

A Problem Set Submitted to:

Instructor/Section: Dr. Yuk / Section _____

In Partial Fulfillment of

Requirements for CH350

By

Cadet: APPROVED SOLUTION

West Point, New York

Date: _____ Time: _____

DO NOT WRITE IN THIS SPACE

PROBLEM	VALUE	CUT
1 (9.1)	8	
2 (9.2)	8	
3 (9.7)	8	
4 (9.10)	8	
5 (10.1)	10	
6 (10.4)	8	
TOTAL GRADE	50	

_____ MY DOCUMENTATION ATTRIBUTES ALL SOURCES USED AND ASSISTANCE
RECEIVED IN COMPLETING THIS ASSIGNMENT

_____ NO SOURCES WERE USED OR ASSISTANCE RECEIVED IN COMPLETING THIS
ASSIGNMENT

SIGNATURE: _____

APPROVED SOLUTION

9.1. Consider a 1000 l CFSTR in which biomass is being produced with glucose as the substrate. The microbial system follows a Monod relationship with $\mu_m = 0.4 \text{ h}^{-1}$, $K_s = 1.5 \text{ g/l}$ (an unusually high value), and the yield factor $Y_{X/S} = 0.5 \text{ g/l biomass/g substrate consumed}$. Normal operation is with a sterile feed containing 10 g/l glucose at a rate of 100 l/h. **(8 pts)** (refer to pg. 278-281 of DeLisa, 2017)

a. What is the specific biomass production rate (g/l-h) at steady state?

Solution:

$$V = 1000 \text{ l} \quad K_s = 1.5 \text{ g/l} \quad S_0 = 10 \text{ g/l}$$

$$\mu_m = 0.4 \text{ h}^{-1} \quad Y_{X/S} = 0.5 \text{ g/g} \quad F = 100 \text{ l/h}$$

$$D = \frac{F}{V} = \frac{100 \text{ l/h}}{1000 \text{ l}} = 0.1 \text{ h}^{-1}$$

$$\text{At steady state: } \mu = D = 0.1 \text{ h}^{-1} = \mu_m \frac{S}{K_s + S}$$

$$\therefore 0.1 \text{ h}^{-1} = 0.4 \text{ h}^{-1} \left(\frac{S}{1.5 \text{ g/l} + S} \right)$$

Solving for S, $S = 0.5 \text{ g/l}$

$$X = Y_{X/S} (S_0 - S) = 0.5 \text{ g/g} (10 - 0.5 \text{ g/l}) = 4.75 \text{ g/l}$$

$$DX = 0.1 \text{ h}^{-1} (4.75 \text{ g/l}) = 0.475 \text{ g} / (\text{l} \cdot \text{h})$$

b. If recycle is used with a recycle stream of 10 l/h and a recycle biomass concentration five times as large as that in the reactor exit, what would be the new specific biomass production rate?

Solution:

$$X_{\text{recycle}} = 5 \cdot X_{\text{reactor}}$$

$$\alpha = \frac{\text{Recycle Flow Rate}}{\text{Feed Flow Rate}} = \frac{10 \text{ l/h}}{100 \text{ l/h}} = 0.1$$

$$C = \frac{X_{\text{recycle}}}{X_{\text{reactor}}} = 5$$

$$\begin{aligned} \mu &= [1 + \alpha(1 - C)]D \\ &= [1 + 0.1(1 - 5)](0.1 \text{ h}^{-1}) \quad (\text{equation 9.9}) \end{aligned}$$

$$\mu = 0.06 \text{ h}^{-1} \quad (\text{Note: } \mu < D)$$

APPROVED SOLUTION

As above: $0.06 \text{ h}^{-1} = 0.4 \text{ h}^{-1} \left(\frac{S}{1.5 \text{ g/l} + S} \right)$

Solving for S, $S = 0.265 \text{ g/l}$

$$X = Y_{X/S} (S_0 - S) \frac{D}{\mu} = 0.5 \text{ g/g} (10 - 0.265 \text{ g/l}) \left(\frac{0.1 \text{ h}^{-1}}{0.06 \text{ h}^{-1}} \right) = 8.11 \text{ g/l}$$

$$DX = 0.1 \text{ h}^{-1} (8.11 \text{ g/l}) = 0.811 \text{ g} / (\text{l} \cdot \text{h})$$

c. Explain any difference between the values found in parts a and b.

Solution:

Having a cell recycle increases the rate of substrate consumption (i.e. a lower S) resulting in a higher biomass concentration and in a higher biomass production rate.

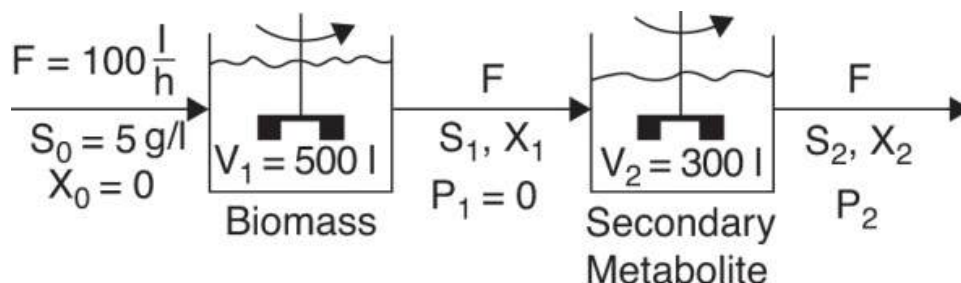
9.2. In a two-stage chemostat system, the volumes of the first and second reactors are $V_1 = 500 \text{ l}$ and $V_2 = 300 \text{ l}$, respectively. The first reactor is used for biomass production and the second is for a secondary metabolite formation. The feed flow rate to the first reactor is $F = 100 \text{ l/h}$, and the glucose concentration in the feed is 5.0 g/l . Use the following constants for the cells: **(8 pts)** (refer to pg. 281-284 of DeLisa, 2017)

$$\mu_m = 0.3 \text{ h}^{-1}, K_s = 0.1 \text{ g/l}, Y_{X/S} = 0.4 \frac{\text{g dw cells}}{\text{g glucose}}$$

a. Determine cell and glucose concentration in the effluent of the first stage.

Solution:

$$\mu_m = 0.3 \text{ h}^{-1} \quad K_s = 0.1 \text{ g/l} \quad Y_{X/S} = 0.4 \text{ g/g}$$



$$D_1 = \frac{F}{V_1} = \frac{100 \text{ l/h}}{500 \text{ l}} = 0.2 \text{ h}^{-1} = \mu, \text{ at steady state}$$

APPROVED SOLUTION

$$\therefore 0.2 \text{ h}^{-1} = \mu_m \frac{S_1}{K_s + S_1} = 0.3 \text{ h}^{-1} \left(\frac{S_1}{0.1 \text{ g/l} + S_1} \right)$$

Solving for S_1 : $S_1 = 0.2 \text{ g/l}$

$$X_1 = Y_{x/s} = (S_0 - S_1) = 0.4 \text{ g/g} (5 - 0.2 \text{ g/l}) = 1.92 \text{ g/l}$$

b) Assume that growth is negligible in the second stage and the specific rate of product formation is $q_p = 0.2 \text{ g P/g cell h}$, and $Y_{p/s} = 0.6 \text{ g P/g}$. Determine the product and substrate concentration in the effluent of the second reactor.

Solution:

$$\mu_2 = 0 \quad \therefore X_2 = X_1 \quad \begin{array}{l} q_p = 0.02 \text{ g / (g cell} \cdot \text{h)} \\ Y_{p/s} = 0.6 \text{ g/g} \end{array}$$

$$D_2 = \frac{F}{V_2} = \frac{100 \text{ l/h}}{300 \text{ l}} = 0.333 \text{ h}^{-1}$$

Substrate balance around second stage:

$$F = (S_1 - S_2) - \frac{q_p X_2 V_2}{Y_{p/s}} = V_2 \frac{dS_2}{dt} = 0 \text{ at steady state}$$

$$\text{Solving for } S_2: S_2 = S_1 - \frac{q_p X_2 V_2}{F Y_{p/s}} = S_1 - \frac{q_p X_2}{D_2 Y_{p/s}}$$

$$\therefore S_2 = 0.2 \text{ g/l} - \frac{0.02 \text{ g g}^{-1} \text{h}^{-1} (1.92 \text{ g/l})}{0.333 \text{ h}^{-1} (0.6 \text{ g/g})} = 0.008 \text{ g/l}$$

$$P_2 = Y_{p/s} (S_1 - S_2) = 0.6 \text{ g/g} (0.2 - 0.008 \text{ g/l}) = 0.115 \text{ g/l}$$

9.7. The bioconversion of glucose to ethanol is carried out in a packed-bed, immobilized cell bioreactor containing yeast cells entrapped in Ca-alginate beads. The rate-limiting substrate is glucose, and its concentration in the feed bulk liquid phase is $S_{0i} = 5 \text{ g/l}$. The nutrient flow rate is $F = 2 \text{ l/min}$. The particle size of Ca-alginate beads is $D_p = 0.5 \text{ cm}$. The surface area of the alginate beads per unit volume of the reactor is $a = 25 \text{ cm}^2/\text{cm}^3$, and the cross-sectional area of the bed is $A = 100 \text{ cm}^2$. **(8 pts)** (refer to pg. 308-311 of DeLisa, 2017)

Following are the rate constants for this conversion:

$$r_m = 100 \text{ mg S/cm}^3 \cdot \text{h}$$

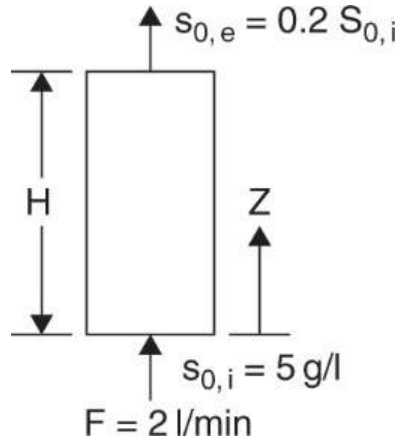
$$K_s = 10 \text{ mg S/cm}^3$$

for the following rate expression:

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$$r_s = \frac{r_m S}{K_s + S}$$

Solution:



$$D_p = 0.5 \text{ cm}$$

$$r_m = 100 \text{ mg} / (\text{cm}^3 \cdot \text{h})$$

$$K_s = 10 \text{ mg} / \text{cm}^3$$

$$r_s = \frac{r_m S}{K_s + S} \approx \frac{r_m}{K_s} S = k_L S \text{ (At low } S \text{)}$$

$$a = 25 \text{ cm}^2 / \text{cm}^3$$

$$A = 100 \text{ cm}^2$$

$$D_s = 10^{-6} \text{ cm}^2 / \text{s}$$

Calculating effectiveness factor, n :

$$\begin{aligned} \theta &= \frac{V_p}{A_p} \sqrt{\frac{r_m / K_s}{D_s}} = \frac{\frac{\pi}{6} D_p^3}{\pi D_p^2} \sqrt{\frac{r_m}{D_s K_s}} = \frac{D_p}{6} \sqrt{\frac{r_m}{D_s K_s}} \\ &= \frac{0.5 \text{ cm}}{6} \sqrt{\frac{100 \text{ mg cm}^{-3} \text{h}^{-1}}{10^{-6} \text{ cm}^2 \text{s}^{-1} (10 \text{ mg cm}^{-3}) (3600 \text{ S h}^{-1})}} \end{aligned}$$

$$\theta = 4.392$$

$$\eta = \frac{1}{\theta} \left(\frac{1}{\tanh 3\theta} - \frac{1}{3\theta} \right) = 0.210$$

Steady state substrate balance over element dz :

$$-F dS - \eta \frac{r_m}{K_s} S \frac{V_p}{A_p} a A dz = 0$$

APPROVED SOLUTION

Integrating from (0, $S_{0,i}$) to (H, $S_{0,e}$) :

$$-F \int_{S_{0,i}}^{S_{0,e}} \frac{dS}{S} = \frac{\eta r_m V_p a A}{K_s A_p} \int_0^H dz$$

$$-F \ln \frac{S_{0,e}}{S_{0,i}} = \frac{\eta r_m V_p a A}{K_s A_p} H$$

$$\begin{aligned} \therefore H &= \frac{K_s A_p F}{\eta r_m V_p a A} \ln \frac{S_{0,i}}{S_{0,e}} = \frac{6 K_s F}{\eta r_m D_p a A} \ln \frac{S_{0,i}}{S_{0,e}} \\ &= \frac{6 (10 \text{ mg cm}^{-3})(1.20 \times 10^5 \text{ cm}^3 \text{ h}^{-1})}{0.21 (100 \text{ mg cm}^{-3} \text{ h}^{-1})(0.5 \text{ cm})(25 \text{ cm}^2 \text{ cm}^{-3})(100 \text{ cm}^2)} \cdot \ln \left(\frac{5}{1} \right) \end{aligned}$$

$$H = 441 \text{ cm} = 4.4 \text{ m}$$

9.10. A completely mixed bioreactor is used for ethanol fermentation of hydrolyzed starch solution containing 30 g/L total sugar. The following data are available: $\mu_m = 0.15 \text{ hr}^{-1}$, $K_s = 0.5 \text{ g/L}$, $F = 10 \text{ L/hr}$, $Y_{X/S} = 0.1 \text{ g X/g S}$, $Y_{P/S} = 0.5 \text{ g P/g S}$, $k_d = 0.1 \text{ day}^{-1}$. **(8 pts)** (refer to pg. 278-288 of DeLisa, 2017)

a. For 95% conversion of substrate to product, determine the required reactor volume and ethanol productivity when the system was operated without cell recycle.

Solution:

$$E = 1 - S/S_0 = 0.95 \quad S = 1.5 \text{ g/L}$$

$$\mu_N = \mu_g - k_d = \mu_m S / (K_s + S) - k_d = D$$

$$D = F/V = 0.15(1.5)/(0.5 + 1.5) - 0.004 = 0.108 \text{ hr}^{-1}$$

$$V = F/D = 10/0.108 = 92.6 \text{ L}$$

$$P = Y_{p/s}(S_0 - S) = 0.5(30 - 1.5) = 14.25 \text{ g/L}$$

$$DP = 0.108(14.25) = 1.54 \text{ g/hr}$$

b. For the same degree of conversion, determine the required reactor volume and ethanol productivity when the system was operated with cell recycle. $C = X_r/X = 3$ and $\alpha = 0.1$.

Solution:

$$S = 1.5 \text{ g/L}, u_{\text{net}} = \mu_g - k_d = 0.108 \text{ hr}^{-1}$$

$$u_{\text{net}} = D(1 + \alpha(1 - C)) = D(1.0 + 0.1(1-3)) = 0.108 \text{ hr}^{-1}$$

$$D = 0.135 \text{ hr}^{-1}$$

APPROVED SOLUTION

$$V = F/D = 74.1 \text{ L}$$

$$P = Y_{p/s} \cdot (S_0 - S) = 14.25 \text{ g/L}$$

$$DP = 14.25 \cdot 0.135 = 1.92 \text{ g/hr}$$

Ethanol productivity increased by a factor of 1.25 and the required reactor volume decreased by a same factor of 1.25 using cell recycle.

10.1. The air supply to a fermenter was turned off for a short period of time and then restarted. A value for C^* of 7.3 mg/l has been determined for the operating conditions. Use the tabulated measurements of dissolved oxygen (DO) values in the following table to estimate the oxygen uptake rate and $k_L a$ in this system. **(10 pts)** (refer to pg. 334-337 of DeLisa, 2017)

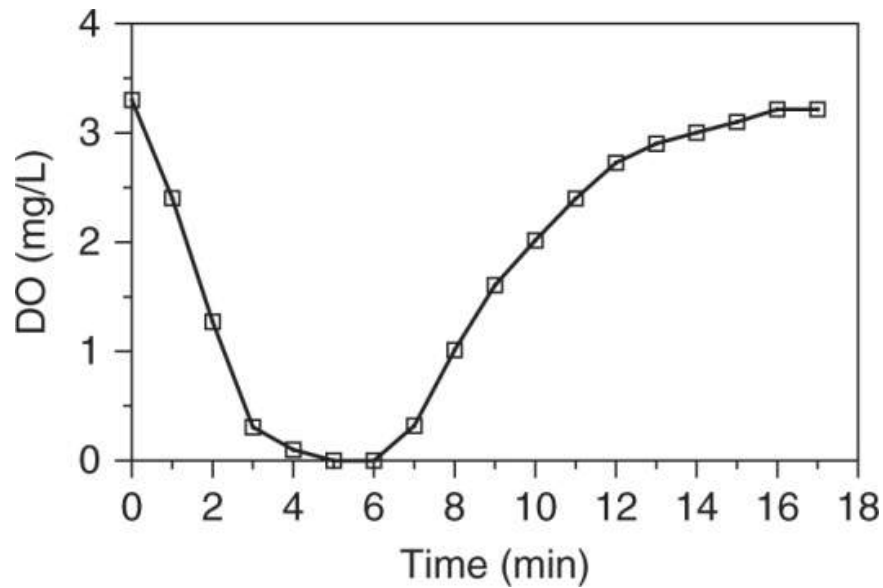
	Time (min)	DO (mg/l)
Air off	-1	3.3
	0	3.3
	1	2.4
	2	1.3
	3	0.3
Air on	4	0.1
	5	0.0
	6	0.0
	7	0.3
	8	1.0
	9	1.6
	10	2.0
	11	2.4
	12	2.7
	13	2.9
	14	3.0
	15	3.1
	16	3.2
	17	3.2

Solution:

$$C^* = 7.3 \text{ mg/l}$$

Plot DO versus t:

APPROVED SOLUTION



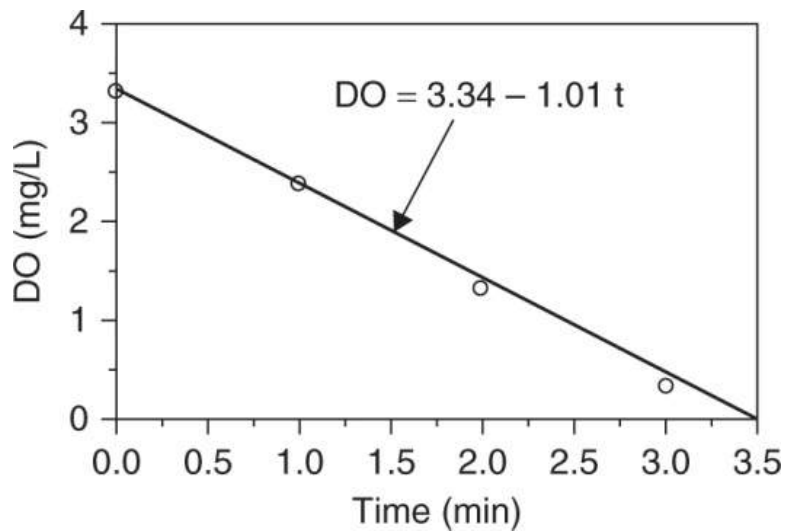
Determination of OUR:

$$\frac{dC_L}{dt} = k_L a (C^* - C_L) - q_{O_2} X$$

with air off: $k_L a (C^* - C_L) = 0 \quad \therefore \frac{dC_L}{dt} = -q_{O_2} X$

Integrating: $\int_{C_0}^C dC_L = -q_{O_2} X \int_0^t dt$
 $C - C_0 = -q_{O_2} X t$

$\therefore$ plotting C versus t should yield a straight line with slope = $-q_{O_2} X = \text{OUR}$



From least squares regression:

Slope = -1.01 mg/(L min)

APPROVED SOLUTION

$$\therefore \text{OUR} = 1.01 \text{ mg/(l min)}$$

Determination of $k_L a$: with air on

$$\begin{aligned}\frac{dC_L}{dt} &= k_L a (C^* - C_L) - q_{O_2} X \\ &= -k_L a C_L + \alpha\end{aligned}$$

$$\text{where } \alpha = k_L a C^* - q_{O_2} X$$

$\therefore$ plotting $\frac{dC_L}{dt}$ versus C_L should yield a straight line with slope $= -k_L a$

Least squares polynomial regression yields the following expression for C_L :

$$C_L = -8.0460 + 1.7279 t - 0.08463 t^2 + 0.001263 t^3$$

$$(\text{for } 8 < t < 15) \quad r^2 = 0.999$$

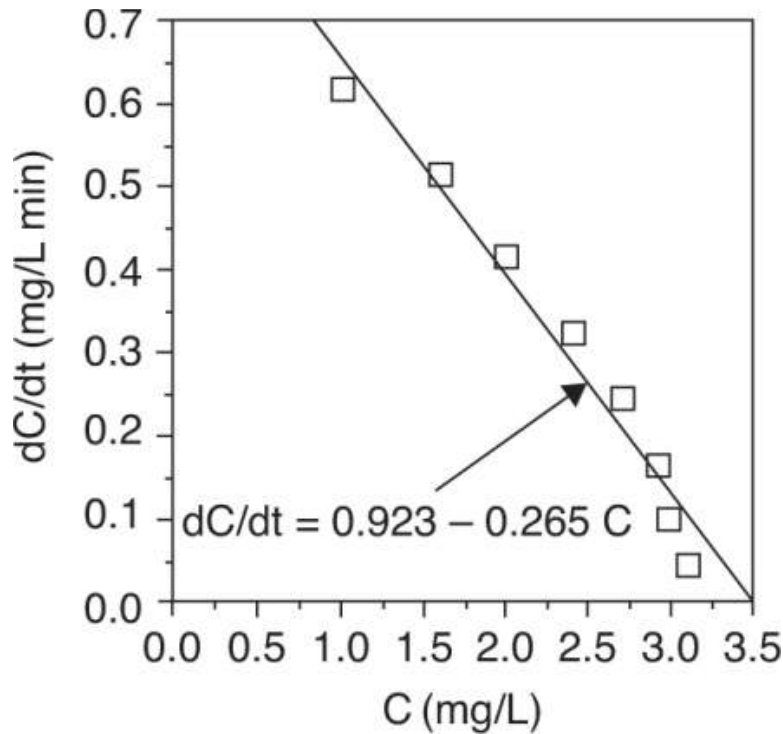
Taking the derivative:

$$\frac{dC_L}{dt} = 1.7279 - 0.16926t + 0.003789 t^2$$

Table of values:

t (h)	C_L (mg/l)	dC_L/dt (mg/l min)
8	1.0	0.616
9	1.6	0.512
10	2.0	0.414
11	2.4	0.324
12	2.7	0.242
13	2.9	0.168
14	3.0	0.101
15	3.1	0.042

APPROVED SOLUTION



From least squares regression: slope = -0.265 min^{-1}

$$\therefore k_L a = 0.265 \text{ min}^{-1}$$

Note: OUR can be calculated from α , i.e.

$$\text{OUR} = q_{O_2} \times$$

$$= k_L a C^* - \alpha$$

$$= 0.265 \text{ min}^{-1} (7.3 \text{ mg/l}) - 0.923 \text{ mg/l} \cdot \text{min}$$

OUR = $1.01 \text{ mg/l} \cdot \text{min}$ (This agrees with the value obtained above).

10.4. A lethal agent (KCN) was added to the fermentation medium in the presence of aeration using diffusors. Dissolved oxygen (DO) concentration increased with time as follows when $N = 100 \text{ rev/min}$:

t (min)	0	1	2	3	5	7	8	10
DO (mg/L)	1	1.93	2.7	3.36	4.4	5.1	5.4	5.9

Saturation DO concentration in the fermentation medium is 7.0 mg/L . **(8 pts)** (refer to pg. 332-337 of DeLisa, 2017)

a. Determine oxygen transfer coefficient $k_L a$.

Solution:

$$dC_L / dt = K_L a (C^* - C_L) - q_{O_2} X$$

APPROVED SOLUTION

Lethal agent addition, therefore, $q_{O_2} X = 0$

$$dC_L / (C^* - C_L) = K_L a \, dt$$

$$\text{Integration yields } \ln (C^* - C_L)/(C^* - C_{L0}) = - K_L a \, t$$

$$\ln (C^* - C_{L0}) = \ln 6 = 1.79$$

$$\ln(C^* - C_L) - 1.79 = - K_L a \, t$$

Plot $\ln(C^* - C_L)$ versus t .

$$\text{Slope} = K_L a = 10 \, \text{h}^{-1}$$

- b. Determine $k_L a$ when N_i is increased to 200 rev/min.

Solution:

$$K_L a = a (P/V)^{0.4} V_s^{0.5} N^{0.5} = a (N^3 D_i^2)^{0.4} V_s^{0.5} N^{0.5}$$

$$K_L a \text{ is proportional to } N^{1.7}$$

$$\text{Then, } K_{La2} = K_{La1} (N_2/N_1)^{1.7} = 10(200/100)^{1.7} = 32.5 \, \text{h}^{-1}$$



PROBLEM SET #6

Handouts

CH350 Problem Set 6

A Problem Set Submitted to:

Instructor/Section: LTC Hummel / Section _____

In Partial Fulfillment of

Requirements for CH350

By

Cadet: _____

West Point, New York

Date: _____ Time: _____

DO NOT WRITE IN THIS SPACE

PROBLEM	VALUE	CUT
6.1	10	
6.2	10	
6.3	10	
6.4	10	
6.5	10	
TOTAL GRADE	50	

_____ MY DOCUMENTATION ATTRIBUTES ALL SOURCES USED AND ASSISTANCE
RECEIVED IN COMPLETING THIS ASSIGNMENT

_____ NO SOURCES WERE USED OR ASSISTANCE RECEIVED IN COMPLETING THIS
ASSIGNMENT

SIGNATURE: _____

1. Yeast cells are recovered from a fermentation broth by using a tubular centrifuge 60% of the cells are recovered at a flow rate of 12 L/min with a rotational speed of 4000 rpm. Recovery % is inversely proportional to flow rate and directly proportional to the square of rotational speed. (10 pts) (refer to pg. 378-382 of DeLisa, 2017)

a. To increase the recovery of cells to 90 % at the same flow rate, what should be the rpm of the centrifuge?

B. Now, at a constant rpm of 4000 rpm, what would be the flow rate to result in 90 % cell recovery?

2. Consider the scale-up of a fermentation from a 50 L to 50,000 L vessel tank. The small fermenter has a height-to-diameter ratio of 5. The impeller diameter is 20 % of the tank diameter. Impeller speed is 300 rpm, and 3 Rushton impellers are used. Engineers decided that the scale-up factor to be the cube root of the ratio of vessel volumes. Assuming geometric similarity for scale-up process, determine the dimensions of the large fermenter (in meters) and impeller speed (in rpm) for the following conditions: **(10 pts)** (refer to pg. 344-345 of DeLisa, 2017)

*Hint: Refer to Table 10.2 of Shuler, Kargi, and DeLisa, 2017 for additional information.

A. Dimensions of the Large Fermenter (Including the diameter of the impeller)

B. Impeller Speed

1. At Constant P/V:

2. At Constant impeller tip speed:

3. At Constant Reynolds number:

3. Biomass present in a fermentation broth is to be separated by vacuum filtration. Following are the filter and broth characteristics: $A = 50 \text{ m}^2$, $\Delta P = 0.01 \text{ N/m}^2$, $C = 15 \text{ kg/m}^3$, $\mu = 2.5 \text{ kg/m} \cdot \text{sec}$, $\alpha = 2 \text{ m}^2/\text{kg}$. If rate of filtration has a constant value of $dV/dt = 50 \text{ L/min}$, determine the cake (r_c) and filter resistance (r_m) at $t = 30 \text{ min}$. **(10 pts)** (refer to pg. 375-376 of DeLisa, 2017)

4. Gentamycin crystals are filtered through a small test filter medium with a negligible filter medium resistance. The following data is available:

$t \text{ (sec)}$	10	20	30	40
$V \text{ (L)}$	0.6	0.78	0.95	1.1

The pressure drop in this test run was 1.8 times of that when water was used with a filter area of 100 cm^2 . The concentration of gentamycin in solution is 5 g/L . How long would it take (in hours) to filter 5000 L of gentamycin solution through a filter of 1.5 m^2 , assuming the pressure drop is constant? **(10 pts)** (refer to pg. 375-376 of DeLisa, 2017)

5. In a chromatographic separation column used for the adsorption of solute A onto an adsorbent solid B, the adsorption isotherm is given by

$$C_{ss} = U U_1 C_L^3 = f(C_L)$$

where C_{ss} is mg solute adsorbed/mg adsorbent, C_L is the solute concentration in liquid medium (mg solute/mL liquid), and $U U_1 = 0.2$ (mg solute adsorbed/mg adsorbent)/(mg solute/mL liquid)³. The porosity (void fraction) of the packed column is $\epsilon = 0.45$. The cross-sectional area of the column is 20 cm^2 and M is 5 g adsorbent per 100 mL column volume. The volume of the liquid added is $\Delta V = 250 \text{ mL}$. **(10 pts)** (refer to pg. 412-413 of DeLisa, 2017)

a. Determine the position (ΔX) of the solute band in the column when the solute concentration in the liquid phase at equilibrium is $C_L = 5 \times 10^{-2} \text{ mg/mL}$.

b. Now, find the ratio of the travel distance of solute A (L_A) to that of solvent B in the column (R_f) when $C_L = 5 \times 10^{-2} \text{ mg/mL}$.

Instructor Solution

APPROVED SOLUTION

CH350 Problem Set 6

A Problem Set Submitted to:

Instructor/Section: Dr. Yuk / Section _____

In Partial Fulfillment of

Requirements for CH350

By

Cadet: APPROVED SOLUTION

West Point, New York

Date: _____ Time: _____

DO NOT WRITE IN THIS SPACE

PROBLEM	VALUE	CUT
1	10	
2	10	
3	10	
4	10	
5	10	
TOTAL GRADE	50	

_____ MY DOCUMENTATION ATTRIBUTES ALL SOURCES USED AND ASSISTANCE
RECEIVED IN COMPLETING THIS ASSIGNMENT

_____ NO SOURCES WERE USED OR ASSISTANCE RECEIVED IN COMPLETING THIS
ASSIGNMENT

SIGNATURE: _____

APPROVED SOLUTION

1. Yeast cells are recovered from a fermentation broth by using a tubular centrifuge 60% of the cells are recovered at a flow rate of 12 L/min with a rotational speed of 4000 rpm. Recovery % is inversely proportional to flow rate and directly proportional to the square of rotational speed. **(10 pts)** (refer to pg. 378-382 of DeLisa, 2017)

a. To increase the recovery of cells to 90 % at the same flow rate, what should be the rpm of the centrifuge?

Solution:

Since $\frac{\text{Recovery } 1}{\text{Flow Rate } 1} \propto (\text{Rotational Speed } 1)^2$ and $\frac{\text{Recovery } 2}{\text{Flow Rate } 2} \propto (\text{Rotational Speed } 2)^2$, we need to use this

relationship to obtain the rotational speed at 90 % of recovery given that 60 % recovery is reached at 4000 rpm.

$$\frac{60\%}{90\%} = \frac{(4000 \text{ rpm})^2}{(\text{Rotational Speed } 2)^2}$$

Then, the rotational speed at 90 % recovery should be calculated as **4899 rpm**.

b. Now, at a constant rpm of 4000 rpm, what would be the flow rate to result in 90 % cell recovery?

Solution:

Since $\frac{\text{Recovery } 1}{\text{Flow Rate } 1} \propto (\text{Rotational Speed } 1)^2$ and $\frac{\text{Recovery } 2}{\text{Flow Rate } 2} \propto (\text{Rotational Speed } 2)^2$, this time, we need to use this inversely proportional relationship to obtain the flowrate at 90 % of recovery given that 60 % recovery is reached at the flowrate of 12 L/min.

$$\frac{60\%}{90\%} = \frac{12 \text{ L/min}}{\text{Flow Rate } 2}$$

Then, the flowrate at 90 % recovery should be calculated as **8 L/min**.

APPROVED SOLUTION

2. Consider the scale-up of a fermentation from a 50 L to 50,000 L vessel tank. The small fermenter has a height-to-diameter ratio of 5. The impeller diameter is 20 % of the tank diameter. Impeller speed is 300 rpm, and 3 Rushton impellers are used. Engineers decided that the scale-up factor to be the cube root of the ratio of vessel volumes. Assuming geometric similarity for scale-up process, determine the dimensions of the large fermenter (in meters) and impeller speed (in rpm) for the following conditions: **(10 pts)** (refer to pg. 344-345 of DeLisa, 2017)

*Hint: Refer to Table 10.2 of Shuler, Kargi, and DeLisa, 2017 for additional information.

Dimensions (including the diameter of impeller):

Solution:

Assume the vessel is in cylindrical shape,

$$V = \frac{\pi D^2 H}{4}$$

Since $H = 5 D_t$, then

$$V = \frac{\pi}{4} 5 D^3 = 50,000 \text{ L}$$

Solving for D_t , H , and $D_i = 0.2 D_t$ will give

$$\begin{aligned} D_{aa,ssrraaRRRR} &= 23.4 \text{ m} \\ H_{ssrraaRRRR} &= 116.8 \text{ m} \\ D_{CC,ssrraaRRRR} &= 4.7 \text{ m} \end{aligned}$$

Remember that the scale-up factor is the cube root of the ratio of tank volumes:

$$\sqrt[3]{\frac{50,000 \text{ L}}{50 \text{ L}}} = 10$$

To maintain geometric similarity, the larger vessel will have its dimensions simply increased by a factor of 10:

$$\begin{aligned} D_{aa,RRaaRRUURR} &= 2.34 \text{ m} \\ H_{RRaaRRUURR} &= 11.68 \text{ m} \\ D_{CC,RRaaRRUURR} &= 0.47 \text{ m} \end{aligned}$$

Impeller Speed:

a. At Constant P/V

Solution:

APPROVED SOLUTION

$$NN_{RRaaRRRIURR} = NN_{ssrraaRRRR} \frac{DD_{CC,ssrraaRRRR}^{2/3}}{DD_{CC,RRaaRRRIURR}^{2/3}} = 300(1/10)^{2/3} = \boxed{65 \text{ RRrrrr}}$$

b. At Constant impeller tip speed

Solution:

$$NN_{RRaaRRRIURR} = NN_{ssrraaRRRR} \frac{DD_{CC,ssrraaRRRR}}{DD_{CC,RRaaRRRIURR}} = 300(1/10) = \boxed{30 \text{ RRrrrr}}$$

c. At Constant Reynolds number

Solution:

$$NN_{RRaaRRRIURR} = NN_{ssrraaRRRR} \frac{DD_{CC,ssrraaRRRR}^2}{DD_{CC,RRaaRRRIURR}^2} = 300(1/10)^2 = \boxed{3 \text{ RRrrrr}}$$

3. Biomass present in a fermentation broth is to be separated by vacuum filtration. Following are the filter and broth characteristics: $A = 50 \text{ m}^2$, $\Delta P = 0.01 \text{ N/m}^2$, $C = 15 \text{ kg/m}^3$, $\mu = 2.5 \text{ kg/m} \cdot \text{sec}$, $\alpha = 2 \text{ m}^2/\text{kg}$. If rate of filtration has a constant value of $dV/dt = 50 \text{ L/min}$, determine the cake (r_c) and filter resistance (r_m) at $t = 30 \text{ min}$. **(10 pts)** (refer to pg. 375-376 of DeLisa, 2017)

Solution:

$$RR_{rr} = \alpha \frac{CCVV}{AA} = \frac{rr}{UUkk} \frac{15 \frac{UUkk}{rr^3} V}{50 rr^2} = 0.6 \frac{VV}{rr^4} \quad (ffRRRRrr \text{ RReeUU } 11.2)$$

Since $\frac{CCdd}{RR} = \frac{50 \text{ LL}}{rr}$,

$CCaa \quad rrCCCC$

$$VV = \frac{30 rrCCCC}{0 rrCCCC} \frac{50 \text{ LL}}{rrmmUU} ddaa = 1500 \text{ LL} = 1.5 rr^3$$

Then, $\boxed{RR_{rr} = 0.6 \frac{1.5 rr^3}{rr^4} = \frac{0.9}{rr}}$

$$\text{Now, } \frac{CCaa}{RR} = \frac{\Delta PPPP}{(RR_{mm} + RR_{cc})\mu} = \frac{(0.01 \frac{NN}{mm})(50 rr^2)}{(RR_{mm} + \frac{0.9}{rr})(\frac{2.5 kkkk}{mm})} = \frac{(0.01 \frac{kkk}{rr})(50 rr^2)}{(RR_{mm} + \frac{0.9}{rr})(\frac{2.5 kkkk}{mm})} = \frac{0.05 rr^3}{rrCCCC} \frac{1 rrCCCC}{60 ssRRRR} = \frac{0.000833}{ssRRRR} rr^3$$

$\frac{RR_{rr}}{rr} + \frac{0.9}{rr} = \frac{240}{rr}$ Then, $\boxed{RR_{rr} = \frac{239}{rr}}$

APPROVED SOLUTION

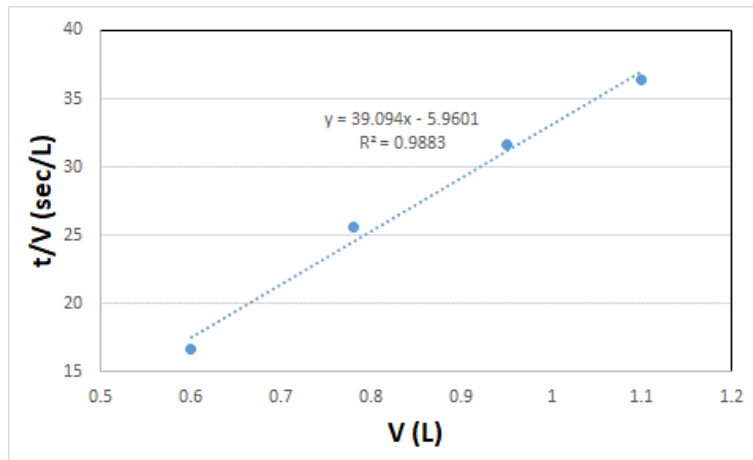
4. Gentamycin crystals are filtered through a small test filter medium with a negligible filter medium resistance. The following data is available:

t (sec)	10	20	30	40
V (L)	0.6	0.78	0.95	1.1

The pressure drop in this test run was 1.8 times of that when water was used with a filter area of 100 cm^2 . The concentration of gentamycin in solution is 5 g/L . How long would it take (in hours) to filter 5000 L of gentamycin solution through a filter of 1.5 m^2 , assuming the pressure drop is constant? **(10 pts)** (refer to pg. 375-376 of DeLisa, 2017)

Solution:

You will get a following plot of t/V vs. V based on the rearranged Ruth equation ($t/V = 1/K(V+2V_0)$, eqn. 11.6 of textbook), as shown below:



For the above plot, you will find a slope to be $\sim 39.1 \text{ sec/L}^2$. For constant pressure filtration, Ruth equation of $V^2 + 2VV_0 = Kt$ can be applied. If you recall from the textbook eqn. 11.5, $V_0 = (r_m/\alpha C)A$. However, a filter medium resistance is neglected in this problem. Thus, $V_0 = 0$ and $V^2 = Kt$.

Now using the obtained slope above, $K = 1/39.1 = 0.0256 \text{ L}^2/\text{sec}$ for a filter area (A) with 100 cm^2 .

Based on eqn. 11.5 of textbook, K is defined as $(2A^2/C\mu)\Delta p$. But Δp is also constant for this condition and C and μ should be constant as well for the same type of tank. Thus, the following relationship between K and A can be set up: $\frac{K_1 A_1^2}{K_2 A_2^2} = \frac{P_1}{P_2}$. Then,

$$\frac{K_1 A_1^2}{K_2 A_2^2} = \frac{P_1}{P_2} \Rightarrow \frac{0.0256 A_1^2}{K_2 A_2^2} = \frac{1.8}{1} \Rightarrow \frac{A_1^2}{A_2^2} = \frac{1.8}{0.0256} \Rightarrow \frac{A_1^2}{A_2^2} = 70.3125 \Rightarrow \frac{A_1}{A_2} = \sqrt{70.3125} \Rightarrow \frac{A_1}{A_2} = 8.385 \Rightarrow A_1 = 8.385 A_2$$

APPROVED SOLUTION

Since $V^2 = Kt$ for this condition, t can now be finally solved:

$$t = \frac{(5000 \text{ mL})^2}{575 \frac{\text{L}^2}{\text{s}}} \frac{1 \text{ hr}}{3600 \text{ s}} = 12 \text{ hr}$$

5. In a chromatographic separation column used for the adsorption of solute A onto an adsorbent solid B, the adsorption isotherm is given by

$$C_{ss} = U_1 C_L^3 = f(C_L)$$

where C_{ss} is mg solute adsorbed/mg adsorbent, C_L is the solute concentration in liquid medium (mg solute/mL liquid), and $U_1 = 0.2$ (mg solute adsorbed/mg adsorbent)/(mg solute/mL liquid)³. The porosity (void fraction) of the packed column is $\varepsilon = 0.45$. The cross-sectional area of the column is 20 cm² and M is 5 g adsorbent per 100 mL column volume. The volume of the liquid added is $\Delta V = 250$ mL. **(10 pts)** (refer to pg. 412-413 of DeLisa, 2017)

a. Determine the position (ΔX) of the solute band in the column when the solute concentration in the liquid phase at equilibrium is $C_L = 5 \times 10^{-2}$ mg/mL.

Solution:

$$C_L = 5 \times 10^{-2} \text{ mg/mL}$$

$$f(C_L) = U_1 C_L^3$$

$$\begin{aligned} f(C_L) &= 3 U_1 C_L^2 = 3 \times 0.2 \frac{\text{mg}}{\text{mg}} \frac{\text{L}^2}{\text{s}} \times 10^{-2} \frac{\text{mg}}{\text{mL}}^2 \\ &= 0.0015 \frac{\text{mg}}{\text{mg}} \frac{\text{L}^2}{\text{s}} \\ M &= \frac{5 \text{ g}}{100 \text{ mL}} = 50 \frac{\text{g}}{\text{mL}} \end{aligned}$$

$$\text{Then, } \Delta X = \frac{\Delta V}{PP(\varepsilon + M f(C_L))} = \frac{250 \text{ mL}}{(20 \text{ cm}^2)(0.45 + 50 \times 0.0015)} = 23.8 \text{ cm}$$

b. Now, find the ratio of the travel distance of solute A (L_A) to that of solvent B in the column (R_f) when $C_L = 5 \times 10^{-2}$ mg/mL.

Solution:

APPROVED SOLUTION

$$R_{ff} = \frac{\varepsilon}{\varepsilon + M f'(C)} = \frac{0.45}{0.45 + (0.0015 \frac{L}{L})} = 0.86$$



TAB K Capstone PROJECT

Handouts

CH350 Capstone Design Paper Guidance

This design paper is designed to give you an opportunity to explore and design a **bioreactor** of your choice and to understand how the fundamentals of bioengineering apply to this reaction system. Your study will lead you down a path to become a more proficient bioengineer to become more aware of the technical challenges existing in the multi-disciplinary growing field of engineering.

1. The basic format requires a title page with your name, class, paper topic and date of submission. Proper spelling, grammar, paragraph, and sentence construction is expected. Footnotes or endnotes, whether in MLA or APA format (**refer to reference style attachment on CH350 Canvas site**), are required, as is a complete bibliography. You will cite any source you use and will include in your bibliography all sources you looked at, even if you end up not using the source.
2. The paper shall have appropriate subsections (refer to the outline) to help you develop a clear and clearly seen division of ideas for the reader.
3. The paper should be clearly written, with an introduction that succinctly develops the paper's thesis, a descriptive and analytical progression that takes the reader through your argument and to your conclusions. Ideally, the paper should offer recommendations or design solutions when appropriate. It could be a separate subsection or part of your conclusion. Finally, the paper should offer a clear conclusion, which depending on your writing style, could incorporate a very concise summary of the paper's main points, combined with your recommendations and conclusion.

General Paper Guidance:

Your paper should be at **least 3-4 pages**, double spaced. Your paper should consist of an introduction of your topic that is broad enough to gain the reader's interest. The body of the paper should consist of why the bioreactor you chose is of relevance in study within in the field of bioengineering. For example, if you chose a specific fermenter model for describing an anaerobic fermentation process, you would discuss why this topic is an important, relevant issue for today's bioengineer. Where relevant, tie in any social, political, or economic intricacies which make this a more relevant issue. The body of your paper should analyze the model of interest from the perspective of bioengineering, using the concepts you've learned throughout the course. Finally, your paper should have a well thought out conclusion and summary.

1. Research a bioreactor model and explain how it connects to the concepts learned this semester. Use references such as key textbooks, peer reviewed journal articles, and other professional online references as sources in your paper. Alert readers in the first few paragraphs of your draft to the key points that you will be making and the value they will get from reading your article.
2. Provide all background details that are necessary to support your points, but avoid extraneous background details -- for example, on the prior state of the art, experimental methodology, or derivations of equations. Instead, as appropriate, give citations for such information.
3. **Use active voice and third person (refer to CH350 Canvas site).** Construct sentences that use action verbs — it's better to tell the reader what happens than to describe a state of being.

4. Define all variables, including units (AIChE prefers SI units), immediately after the first equation that uses the variables as well as in a listing of nomenclature.
5. Clearly explain any limitations and tradeoffs for the bioreactor model introduced in the paper.
7. Illustrations can help you make your points effectively, while enhancing the appearance of your article. So, keep good graphics, such as clear drawings and graphs, sharp photographs, tables, and charts, in mind right from the start. Include a clear, descriptive explanatory caption for each graphic. Captions should be complete sentences that tell readers what they are looking at or what is happening in the figure.
8. Final Paper is worth 200 points and due **NLT Lesson 30**.
9. Grades will be assigned as follows:

- 50 points – Introduction and Literature Review

Why your bioreactor model is relevant to **bioengineering**, and why it is important to current industry and research and development. If your topic has socio-political consequences, then you should discuss it.

- 50 points – Methodology

a) Choose the system of bioreactor model you wish to explore. You must provide 3-4 sentences at the beginning of Methodology section on why such model is relevant to bioengineering and/or bioprocess engineering application.

b) State your assumptions for your model. You will not be able to consider all the operational parameters, so reasonable assumptions (backed by the scientific literatures) need to be made and specified in the paper.

d) Use the Mathematica (MMA) or other ordinary differential equation (ODE) solvers to quantitatively describe your results. Make sure to process your models based on the kinetic data provided to you.

e) Insert the overall coding structure of your model as a figure. Failure to do so will result in **losing 30 points** for Methodology section.

- 50 points – Results and Discussion

Present your results based on the bioreactor models for your choice and discuss the overall trends observed with the reactants and products. You must include **at least one result figure** in the Results and Discussion section.

- 30 points – Conclusion

Well thought out conclusion and summary of your topic and the relevant findings from the model.

- 20 points – References

The paper provides an opportunity to learn about any research topic of interest and motivates you to investigate and learn as much as you can about the topic. Below is an outline for the research paper.

Paper Outline

I. Introduction

What is the problem or topic you wish to investigate? How will you approach this problem? How can your bioreactor model be applied to answer the quantitative questions? This is your connection with the reader/audience. Here you say what you will be doing in the remainder of the paper so define your interests and the parameters (scope) of your topic. Also remember to keep the topic focused on aspects you plan to discuss more in depth. As you collect material and structure an outline, keep referring to this section.

II. Literature Review

First, include a summary of the major studies related to bioreactor models that have been done and published in literature. If you use online databases for peer-reviewed journal articles, build a thorough database through a literature review which addresses or covers key aspects pertinent to your research topic. This may be in a press release or news article as well as bioengineering textbooks.

Second, you need to identify and cite other sources that you found particularly interesting and useful for this paper. (Make sure you try to verify the reliability and quality of your sources)

III. Methodology

In this section, you need to discuss how you develop the quantitative model based on the knowledge you have learned in CH350. Choose and explain the bioreactor model you wish to explore for this design paper. You must provide 3-4 sentences at the beginning of the Methodology section on why such a model is relevant to bioengineering and/or bioprocess engineering application. Make sure to state your assumptions for the model you developed (i.e., steady state assumption or Monod kinetics for substrate consumption). You will most likely use the MMA or other ordinary differential equation (ODE) solvers to quantitatively project your results based on the given kinetic data experimentally obtained from the fermenter.

IV. Discussion and Body of Research Topic

Discuss the most common type of research done for a topic of this scope which basically lays out a narrative of what has been written about this topic. This narrative should be comparative (you compare what different researchers and writers have to say) and analytical (you should make your own comments and assessments of what the existing material reveals and what is missing from the material you have identified in your view). Also, you will need to discuss your key findings from the model of your biological system.

V. Conclusions

Summarize your major findings and the major implications to the field of bioengineering as well as the results of your quantitative model. Make certain that you directly connect this section to the introduction you wrote and to what you said you were going to do in this paper. Look at the introduction and make sure that you have clearly stated in that section what you intend to do in this work. Furthermore, you need to state the significance of results you produced from the model and compare against the similar results found in the bioengineering and/or bioprocess engineering literatures.

VI. References

List references using the preferred citation method suggested by **AICHE CEP Reference Style (refer to reference style attachment on CH350 Canvas site)** for ALL sources you consulted for this topic. You can use in text citations by using a **superscript for the reference number** listed in the “References” section at the end of the paper. The “References section will have all sources used in the following format: the reference number (followed by a period) and the author’s name in bold; there is a tab between the period and the name. Elements (information) are separated by commas. Dates appear in parentheses. Each citation ends with a period. References use a hanging indent format. Do not use automatic numbering; number and tab manually.

Other Tips:

Number Pages: You must **number all pages** since that is how others may cite your work.

Font: use **12 point** and **Times New Roman** font. Use conventional **1-inch margins** left right and at top and bottom. Use one and a half spaces unless otherwise told. It makes it easier to read and write corrections/comments by the instructor.

Tables/Charts/Figures: It is always helpful to have appropriate graphics such as **plots, tables, charts, or other graphic representations in the paper**. Figures and tables to present complicated information in a way that is accessible and understandable to your reader. to use data in the form of numbers, words, or images that will help you construct and support your argument. If you can present your results clearly in a sentence or two, a table or figure is probably unnecessary. If your data are numerous or complicated to be described adequately in this amount of space, figures and tables can be effective ways of conveying lots of information without cluttering up your text. Additionally, they serve as quick references for your reader and can reveal trends, patterns, or relationships that might otherwise be difficult to grasp.

CH350 Capstone Presentation Guidance.

This presentation is designed for you to **apply** all the concepts that you have learned throughout the semester in designing the fermenter model based on the experimental kinetic data. Your choice of fermenter model is critical as it should be interesting and push you to learn and understand the process beyond simple recall of facts to a thorough understanding of the reactor process from an engineering point of view. Your briefing **should last no more than 5 minutes**. At the conclusion of your briefing, there will be a **2-minute question and answer period** where you will field questions from your classmates and instructor. You will receive a grade based upon 6 criteria:

1. *Topic Overview/Background*
2. *Technical Content and Knowledge of Fermenter and its Applications*
3. *Description of Fermenter Model Design*
4. *Application to a relevant social, political, environmental, economic, or safety issue*
5. *Quality of Presentation (slide appearance, clarity, transitions, etc.)*

A portion of the grade will be assessed on the quality of the oral presentation (10-point briefing grade). This grade will be based upon your specific enthusiasm, motivation, knowledge, and presentation of your portion of the material. Additionally, there will be a quality of PowerPoint slides associated with this project in terms of legibility and clarity (10-point PowerPoint slides grade).

Grade Breakdown:

- 0 points – (GO/NO GO) rough draft outline/plan (design paper draft and IPR)
- 20 points – overview/background (includes 10 points for topic focus)
- 25 points – technical knowledge
- 25 points –description of biological modeling and quantitative results generated from the fermenter design model (must include **at least one result figure**)
- 10 points – relevant social, political, environmental etc. issues
- 10 points – quality of oral presentation (briefing technique and competence)
- 10 points – quality of PowerPoint slides (clarity, transitions, documentation etc.)

Timeline:

Lsn 14 – Capstone Announcement

Lsn 30 – Capstone Briefings I

General Guidance:

Each presentation **must have** the following content:

- A question or statement of what you plan on proving and illustrating during your presentation.
- A thorough background of the fermenter that you are interested in.
- Correct engineering language and show direct linkage between content that you learned this year and your fermenter design model.
- A **quantitative analysis** of your fermenter models for the reaction systems based on the collective knowledge covered in class/textbook and the given data.
- Show a linkage between your topic and today's society in terms of social impact, economic impact, or socio-political impact.
- Discuss current technologies in your process.
- References should include any **bioengineering** peer review journal articles, websites, textbooks etc.

Each presentation **should** also address:

- Safety
- Environmental considerations
- Ethics
- Economics and/or operating costs of your process, or the economic issues associated with your process.

- Document and provide an extensive list of references that you used for your presentation.

This should be the last slide of your presentation. Your slides will be collected before your briefing. Documentation must include all sources (as in Capstone Design Paper), and any other sources used, but make sure to comment on the validity in the works cited portion of the presentation.

- Submit your entire presentation to your instructor via email **NLT 0700 on the day of your briefing**. No further change can be made to the slides after this time. All briefings will be loaded onto the classroom computer the day of class, so no thumb drives will be used.